

Notes and exercises from *Linear Algebra and Multilinear Algebra*

John Peloquin

Introduction

This document contains notes and exercises from [1] and [2].

Unless otherwise stated, Γ denotes a field of characteristic 0 over which all vector spaces are defined.

Linear Algebra

Chapter I

§ 1

Remark. The free vector space $C(X)$ is intuitively the space of all “formal linear combinations” of $x \in X$.

§ 2

Exercise (5 - Universal property of $C(X)$). Let X be a set and $C(X)$ the free vector space on X (subsection 1.7). Recall

$$C(X) = \{f : X \rightarrow \Gamma \mid f(x) = 0 \text{ for all but finitely many } x \in X\}$$

The inclusion map $i_X : X \rightarrow C(X)$ is defined by $a \mapsto f_a$ where f_a is the “characteristic function” of a : $f_a(a) = 1$ and $f_a(x) = 0$ for all $x \neq a$. For $f \in C(X)$, $f = \sum_{a \in X} f(a) f_a$.

- (i) If F is a vector space and $f : X \rightarrow F$, there is a unique *linear* $\varphi : C(X) \rightarrow F$ “extending f ” in the sense that $\varphi \circ i_X = f$:

$$\begin{array}{ccc} X & \xrightarrow{i_X} & C(X) \\ & \searrow f & \downarrow \varphi \\ & & F \end{array}$$

- (ii) If $\alpha : X \rightarrow Y$, there is a unique *linear* $\alpha_* : C(X) \rightarrow C(Y)$ which makes the following diagram commute:

$$\begin{array}{ccc} X & \xrightarrow{\alpha} & Y \\ i_X \downarrow & & \downarrow i_Y \\ C(X) & \xrightarrow{\alpha_*} & C(Y) \end{array}$$

If $\beta : Y \rightarrow Z$, then $(\beta \circ \alpha)_* = \beta_* \circ \alpha_*$.

- (iii) If E is a vector space, there is a unique linear map $\pi_E : C(E) \rightarrow E$ such that $\pi_E \circ i_E = \iota_E$ (where $\iota_E : E \rightarrow E$ is the identity map):

$$\begin{array}{ccc} E & \xrightarrow{i_E} & C(E) \\ & \searrow \iota_E & \vdots \pi_E \\ & & E \end{array}$$

- (iv) If E and F are vector spaces and $\varphi : E \rightarrow F$, then φ is linear if and only if $\pi_F \circ \varphi_* = \varphi \circ \pi_E$:

$$\begin{array}{ccc} E & \xrightarrow{\varphi} & F \\ \pi_E \uparrow & & \uparrow \pi_F \\ C(E) & \xrightarrow{\varphi_*} & C(F) \end{array}$$

- (v) Let E be a vector space and $N(E)$ the subspace of $C(E)$ generated by all elements of the form

$$f_{\lambda a + \mu b} - \lambda f_a - \mu f_b \quad (a, b \in E \text{ and } \lambda, \mu \in \Gamma)$$

Then $\ker \pi_E = N(E)$.

Proof.

- (i) By Proposition II, since $i_X(X)$ is a basis of $C(X)$.
(ii) By (i), applied to $i_Y \circ \alpha$. Note $\beta_* \circ \alpha_*$ is linear such that

$$(\beta_* \circ \alpha_*) \circ i_X = i_Z \circ (\beta \circ \alpha)$$

so $\beta_* \circ \alpha_* = (\beta \circ \alpha)_*$ by uniqueness:

$$\begin{array}{ccccc} X & \xrightarrow{\alpha} & Y & \xrightarrow{\beta} & Z \\ i_X \downarrow & & \downarrow i_Y & & \downarrow i_Z \\ C(X) & \xrightarrow{\alpha_*} & C(Y) & \xrightarrow{\beta_*} & C(Z) \end{array}$$

(iii) By (i), applied to ι_E .

(iv) If φ is linear, then $\varphi \circ \pi_E : C(E) \rightarrow F$ is linear and extends φ in the sense that $\varphi \circ \pi_E \circ i_E = \varphi \circ \iota_E = \varphi$. However, $\pi_F \circ \varphi_* : C(E) \rightarrow F$ is also linear and extends φ since

$$\pi_F \circ \varphi_* \circ i_E = \pi_F \circ i_F \circ \varphi = \iota_F \circ \varphi = \varphi$$

By uniqueness, these two maps must be equal. Conversely, if these two maps are equal, then φ is linear since $\pi_F \circ \varphi_*$ is linear and π_E is surjective.

(v) By (iii),

$$\begin{aligned} \pi_E(f_{\lambda a + \mu b} - \lambda f_a - \mu f_b) &= \pi_E(f_{\lambda a + \mu b}) - \lambda \pi_E(f_a) - \mu \pi_E(f_b) \\ &= \lambda a + \mu b - \lambda a - \mu b \\ &= 0 \end{aligned}$$

for all $a, b \in E$ and $\lambda, \mu \in \Gamma$. It follows that $N(E) \subseteq \ker \pi_E$ since $N(E)$ is the *smallest* subspace containing these elements and $\ker \pi_E$ is a subspace.

On the other hand, it follows from the fact that $N(E)$ is a subspace that

$$\sum \lambda_i f_{a_i} - f_{\sum \lambda_i a_i} \in N(E)$$

for all (finite) linear combinations. Now if $g = \sum_{a \in E} g(a) f_a \in \ker \pi_E$, then

$$0 = \pi_E(g) = \sum_{a \in E} g(a) \pi_E(f_a) = \sum_{a \in E} g(a) a$$

This implies $f_{\sum_{a \in E} g(a) a} = f_0 \in N(E)$. But by the above, $g - f_0 \in N(E)$, so $g \in N(E)$. Therefore also $\ker \pi_E \subseteq N(E)$. $\square$

Remark. Note (i) shows that $C(X)$ is a universal (initial) object in the category of “vector spaces with maps of X into them”. In this category, the objects are maps $X \rightarrow F$, for vector spaces F , and the arrows are *linear* (i.e. structure-preserving) maps $F \rightarrow G$ between the vector spaces which respect the mappings of X :

$$\begin{array}{ccc} X & \xrightarrow{\quad} & F \\ & \searrow & \downarrow \\ & & G \end{array}$$

By (i), every object $X \rightarrow F$ in this category can be obtained from the inclusion map $X \rightarrow C(X)$ in a unique way. This is why $C(X)$ is called “universal”. This is only possible because $C(X)$ is free from any nontrivial relations among the elements of X , so any relations among the images of those elements in F can be obtained starting from $C(X)$. This is why $C(X)$ is called “free”. It is immediate from the universal property that $C(X)$ is unique up to isomorphism: if $X \rightarrow U$ is also universal, then the composites $\psi \circ \varphi$ and $\varphi \circ \psi$ of the induced linear maps $\varphi : C(X) \rightarrow U$ and $\psi : U \rightarrow C(X)$ are linear and extend the inclusion maps, so must be the identity maps on $C(X)$ and U by uniqueness; that is, φ and ψ are mutually inverse and hence *isomorphisms*. In fact they are also unique by the universal property.

Now (ii) shows that we have a *functor* from the category of sets into the category of vector spaces, which sends sets X and Y to the vector spaces $C(X)$ and $C(Y)$, and which sends a set map $\alpha : X \rightarrow Y$ to the linear map $\alpha_* : C(X) \rightarrow C(Y)$. The functor preserves the category structure of composites of arrows.

In (iii), we are “forgetting” the linear structure of E when forming $C(E)$. For example, if $E = \mathbb{R}^2$, then $(1, 1) = (1, 0) + (0, 1)$ in E , but *not* in $C(E)$. The “formal” linear combination

$$(1, 1) - (1, 0) - (0, 1)$$

is not zero in $C(E)$ because the pairs are unrelated elements (symbols) which are *linearly independent*. Note π_E is surjective (since ι_E is), so E is a projection of $C(E)$. In (iv), we see that $\varphi : E \rightarrow F$ is linear if and only if it is a “projection” of $\varphi_* : C(E) \rightarrow C(F)$.

In (v), we see that π_E just recalls the linear structure of E that was forgotten in $C(E)$. In particular, $C(E)/N(E) \cong E$. In other words, if you start with E , then forget about its linear structure, then recall that linear structure, you just get E again.

§ 3

Remark. If

$$E_1 \oplus E_2 = E = F_1 \oplus F_2$$

are two direct sum decompositions of E with $E_i \subseteq F_i$ for $i = 1, 2$, then $E_i = F_i$ for $i = 1, 2$.

Proof. If $x \in F_i$, then $x = x_1 + x_2$ with $x_j \in E_j$. But then we must have $x = x_i$ since $x_j \in F_j$, so $x \in E_i$. □

Remark. If E_1 and E_2 are subspaces of E , then

$$\frac{E_1 + E_2}{E_1 \cap E_2} = \frac{E_1}{E_1 \cap E_2} \oplus \frac{E_2}{E_1 \cap E_2}$$

Proof. Let $E_{12} = E_1 \cap E_2$. Viewing both sides of the above equation as subspaces of E/E_{12} , it is clear that there is a sum since the canonical projection is linear. If $\overline{x_1} = \overline{x_2}$ with $x_1 \in E_1$ and $x_2 \in E_2$, then

$$\overline{x_1 - x_2} = \overline{x_1} - \overline{x_2} = 0$$

so $x_1 - x_2 \in E_{12}$, which implies that $x_1 = (x_1 - x_2) + x_2 \in E_{12}$, so $\overline{x_1} = 0$. It follows that $(E_1/E_{12}) \cap (E_2/E_{12}) = 0$, so the sum is direct. $\square$

§ 4

Exercise (9). Let E be a vector space and E_1, F_1 subspaces of finite codimension in decompositions

$$E_1 \oplus E_2 = E = F_1 \oplus F_2$$

There is a decomposition $E = H_1 \oplus H_2$ with $H_1 \subseteq E_1 \cap F_1$ of finite codimension and $H_2 \supseteq E_2 + F_2$. Moreover

$$(H_2 \cap E_1) \oplus E_2 = H_2 = (H_2 \cap F_1) \oplus F_2$$

Proof. Let $K = E_1 \cap F_1$ and $L = E_2 + F_2$. Then there is $H_1 \subseteq K$ with $K = (K \cap L) \oplus H_1$ and

$$\begin{aligned} \text{codim } H_1 &= \text{codim } K + \dim(K \cap L) \\ &\leq \text{codim } K + \dim L \\ &\leq 2(\text{codim } E_1 + \text{codim } F_1) < \infty \end{aligned}$$

Since $H_1 \cap L = 0$, there is $H_2 \supseteq L$ with $E = H_1 \oplus H_2$. The rest is immediate. $\square$

Exercise (11). Let E be a real vector space and E_1 a vector hyperplane in E (that is, a subspace of codimension 1). Define an equivalence relation on $E^1 = E - E_1$ as follows: for $x, y \in E^1$, $x \sim y$ if the segment

$$x(t) = (1 - t)x + ty \quad (0 \leq t \leq 1)$$

is disjoint from E_1 . Then there are precisely two equivalence classes.

Proof. Fix $e \in E^1$ with $E = E_1 \oplus \langle e \rangle$ and define $\alpha : E \rightarrow \mathbb{R}$ by $x - \alpha(x)e \in E_1$ for all $x \in E$. It is clear that α is linear, and $x \in E_1$ if and only if $\alpha(x) = 0$. For $x, y \in E^1$, it follows that $x \sim y$ if and only if

$$0 \neq \alpha(x(t)) = \alpha((1-t)x + ty) = (1-t)\alpha(x) + t\alpha(y)$$

for all $0 \leq t \leq 1$. But this is just equivalent to $\alpha(x)\alpha(y) > 0$.

Now if $x \in E^1$, then $\alpha(x) \neq 0$, so $\alpha(x)^2 > 0$ and $x \sim x$. If $x \sim y$, then $\alpha(y)\alpha(x) = \alpha(x)\alpha(y) > 0$, so $y \sim x$. If also $y \sim z$, then $\alpha(y)\alpha(z) > 0$, so $\alpha(x)\alpha(z) > 0$ and $x \sim z$. In other words, this is indeed an equivalence relation.

Note there are at least two equivalence classes since $\alpha(e) = 1$ and $\alpha(-e) = -1$, so $\alpha(e)\alpha(-e) = -1 < 0$ and $e \not\sim -e$. On the other hand, there are at most two classes since if $x \in E^1$, then either $\alpha(x) > 0$ and $x \sim e$ or $\alpha(x) < 0$ and $x \sim -e$. $\square$

Remark. This result shows that the hyperplane separates the vector space into two disjoint half-spaces.

Chapter II

§ 1

Remark. In subsection 2.4, intuitively the kernel of a linear function is “large” because it has codimension at most 1, so the intersection of the kernels of finitely many linear functions is also “large” in the sense that the codimension is at most the number of functions. However the dimension need not be high. For example in a finite-dimensional space the intersection of the kernels of the coordinate functions for a basis is zero.

§ 2

Remark. In subsection 2.11, in the second part of the proof of Proposition I, just let $\psi : E \leftarrow F$ be any linear mapping extending $\varphi_1^{-1} : E \leftarrow \text{Im } \varphi$.¹

§ 4

Remark. The direct sum $E \oplus F$ is a coproduct in the category of vector spaces in the following sense: if $\varphi : E \rightarrow G$ and $\psi : F \rightarrow G$ are linear maps, there is a unique linear map $\chi : E \oplus F \rightarrow G$ such that $\varphi = \chi \circ i_E$ and $\psi = \chi \circ i_F$, where i_E and i_F are the canonical injections:

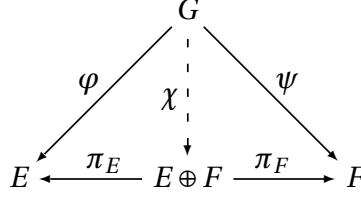
$$\begin{array}{ccccc}
 E & \xrightarrow{i_E} & E \oplus F & \xleftarrow{i_F} & F \\
 & \searrow \varphi & \downarrow \chi & \swarrow \psi & \\
 & & G & &
 \end{array}$$

Indeed, χ is given by $\chi(x + y) = \varphi(x) + \psi(y)$ for $x \in E$, $y \in F$. It is the unique linear map “extending” both φ and ψ . This property makes $E \oplus F$ unique up to a unique isomorphism.

Dually, $E \oplus F$ is a product in the following sense: if $\varphi : G \rightarrow E$ and $\psi : G \rightarrow F$ are linear maps, there is a unique linear map $\chi : G \rightarrow E \oplus F$ such that $\varphi = \pi_E \circ \chi$

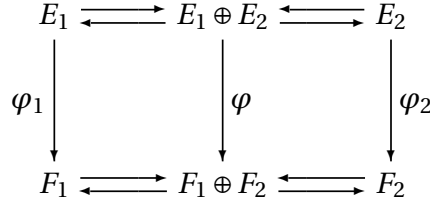
¹ See Corollary I to Proposition I in subsection 1.15.

and $\psi = \pi_F \circ \chi$:



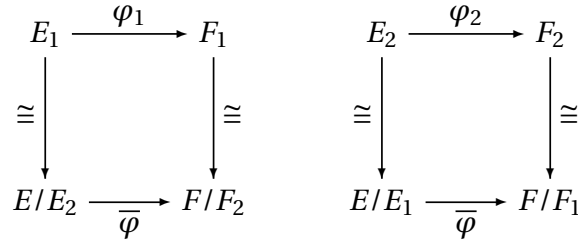
Indeed, χ is given by $\chi(x) = \varphi(x) + \psi(x)$, and “combines” φ and ψ . This property also makes $E \oplus F$ unique up to a unique isomorphism. An infinite direct sum is also a coproduct, but *not* a product, essentially because it has no infinite sums of elements.

In the proof of Proposition I, σ is the product map and τ is the coproduct map. If $\varphi_1 : E_1 \rightarrow F_1$ and $\varphi_2 : E_2 \rightarrow F_2$ are linear maps, then $\varphi = \varphi_1 \oplus \varphi_2$ is both a coproduct and product map:



The structure of φ is completely determined by the structures of φ_1 and φ_2 . In particular, φ is injective (surjective, bijective) if and only if φ_1 and φ_2 are.

Remark. If $E = E_1 \oplus E_2$ and $F = F_1 \oplus F_2$, and $\varphi : E \rightarrow F$ is defined by $\varphi = \varphi_1 \oplus \varphi_2$ where $\varphi_1 : E_1 \rightarrow F_1$ and $\varphi_2 : E_2 \rightarrow F_2$, then the following diagrams commute:



Here the isomorphisms are those induced by the canonical projections.

Remark. If E_1 and E_2 are subspaces of E , then there is a natural isomorphism²

$$\frac{E_1 + E_2}{E_1 \cap E_2} \cong \frac{E_1}{E_1 \cap E_2} \oplus \frac{E_2}{E_1 \cap E_2}$$

²Compare with the result for internal direct sums in chapter I, § 3 above.

Proof. We assume without loss of generality that $E = E_1 + E_2$, and we also write $E_{12} = E_1 \cap E_2$. Then we have the following commutative diagram:

$$\begin{array}{ccccccc}
 E & \xleftarrow{i_1} & E_1 & \xrightarrow{i_1} & E & \xleftarrow{i_2} & E_2 & \xrightarrow{i_2} & E \\
 \pi_2 \downarrow & & \rho_1 \downarrow & & \rho \downarrow & & \rho_2 \downarrow & & \pi_1 \downarrow \\
 E/E_2 & \xleftarrow{\cong} & E_1/E_{12} & \xrightarrow{\overline{i_1}} & E/E_{12} & \xleftarrow{\overline{i_2}} & E_2/E_{12} & \xrightarrow{\cong} & E/E_1
 \end{array}$$

The horizontal arrows on top are inclusions, the vertical arrows are canonical projections, and the horizontal arrows on bottom are induced. Note $\overline{i_1}$ and $\overline{i_2}$ are also just inclusions. Define

$$f = \overline{((\pi_2 \overline{i_1})^{-1} \pi_2, (\pi_1 \overline{i_2})^{-1} \pi_1)} : E/E_{12} \rightarrow (E_1/E_{12}) \oplus (E_2/E_{12})$$

where the bars denote induced maps and the parentheses denote pairing, and

$$g = [\overline{i_1}, \overline{i_2}] : (E_1/E_{12}) \oplus (E_2/E_{12}) \rightarrow E/E_{12}$$

where the brackets denote copairing. It is easy to verify that $f \circ g = \iota$ and $g \circ f = \iota$, so f and g are isomorphisms. $\square$

Remark. If $E = \bigoplus_{i=1}^n E_i$ and $F = \bigoplus_{j=1}^m F_j$, then there is a natural isomorphism

$$L(E; F) \cong \bigoplus_{i,j} L(E_i; F_j)$$

given by the mapping

$$\varphi \mapsto \sum_{i,j} \pi_j \circ \varphi \circ i_i$$

where $i_i : E_i \rightarrow E$ is the canonical injection and $\pi_j : F \rightarrow F_j$ is the canonical projection.

§ 5

Remark. The definition of dual space is fundamentally *symmetrical* between E and E^* , as is the definition of dual mapping between φ and φ^* . This symmetry often allows us to use bidirectional reasoning and derive two theorems from one proof. For example, (2.48) actually follows from (2.47) by symmetry of φ

and φ^* . The proof of Proposition I in subsection 2.23 exploits symmetry, as do other proofs in the book.

Many other books simply *define* the dual space of E to be $L(E)$ (no doubt in light of Proposition I of this subsection), at the expense of this symmetry. Such books make use of *reflexivity*.³

Remark. The definition of duality can be generalized to modules. Let R be a ring (with unit), M a *left* R -module, and N a *right* R -module. A binary mapping $\Phi : M \times N \rightarrow R$ is *bilinear* if

$$\Phi(\lambda x_1 + \mu x_2, y) = \lambda \Phi(x_1, y) + \mu \Phi(x_2, y)$$

and

$$\Phi(x, y_1 \lambda + y_2 \mu) = \Phi(x, y_1) \lambda + \Phi(x, y_2) \mu$$

for all $x, x_1, x_2 \in M$, $y, y_1, y_2 \in N$ and $\lambda, \mu \in R$. The map is *nondegenerate* if

$$\{x \in M \mid \Phi(x, y) = 0 \text{ for all } y \in N\} = 0$$

and

$$\{y \in N \mid \Phi(x, y) = 0 \text{ for all } x \in M\} = 0$$

Now M is *dual* to N if there is a nondegenerate bilinear map $\langle -, - \rangle : M \times N \rightarrow R$. Note this relation is not in general symmetric unless R is commutative.

Remark. The results in subsection 2.23 show that quotient spaces are dual to subspaces.

Remark. Let E, E^* be a pair of dual vector spaces. If $x_1, \dots, x_m \in E$ and $x_i \neq 0$ for all $1 \leq i \leq m$, then there is $x^* \in E^*$ with $\langle x^*, x_i \rangle \neq 0$ for all $1 \leq i \leq m$.

Proof. Since $x_i \neq 0$, $\langle x_i \rangle^\perp$ is a proper subspace of E^* , so $\bigcup_i \langle x_i \rangle^\perp \neq E^*$.⁴ Choose $x^* \in E^* - \bigcup_i \langle x_i \rangle^\perp$. $\square$

Remark. If E, E^* and F, F^* are pairs of dual spaces and $\varphi : E \rightarrow F$ is linear, then $\varphi^* : E^* \leftarrow F^*$ is dual to φ if and only if the following diagram commutes:

$$\begin{array}{ccc} F^* \times E & \xrightarrow{\varphi^* \times \iota_E} & E^* \times E \\ \downarrow \iota_{F^*} \times \varphi & & \downarrow \langle , \rangle \\ F^* \times F & \xrightarrow{\langle , \rangle} & \Gamma \end{array}$$

³See the remark in § 6 below.

⁴See problem 12 in chapter I, § 4.

Remark. Let E be a vector space and $(x_\alpha)_{\alpha \in A}$ be a basis of E . For each $x \in E$, write $x = \sum_{\alpha \in A} f_\alpha(x) x_\alpha$. Then $f_\alpha \in L(E)$ for each $\alpha \in A$. The function f_α is called the α -th coordinate function for the basis.

Coordinate functions can be used in an alternative proof of Proposition IV. If E_1 is a subspace of E , let B_1 be a basis of E_1 and extend it to a basis B of E . For each $x_\alpha \in B - B_1$, we have $f_\alpha \in E_1^\perp$. If $x \in E_1^{\perp\perp}$, then $f_\alpha(x) = \langle f_\alpha, x \rangle = 0$ for all such α , so $x \in E_1$. In other words, $E_1^{\perp\perp} \subseteq E_1$.

Remark. In the corollary to Proposition V, for $f \in L(E)$ let $f_k = f \circ i_k \circ \pi_k$ where $i_k : E_k \rightarrow E$ is the k -th canonical injection and $\pi_k : E \rightarrow E_k$ is the k -th canonical projection. Then $f = \sum_k f_k$ and $f_k \in F_k^\perp$ for all k , so $L(E) = \sum_k F_k^\perp$. The sum is direct since if $f \in F_k^\perp \cap \sum_{j \neq k} F_j^\perp$, then f kills $\sum_{j \neq k} E_j$ and E_k , so $f = 0$. A scalar product is induced between $E_k, F_k^\perp$ since $E_k \cap F_k = 0$ and $F_k^\perp \cap E_k^\perp = 0$.⁵ The induced injection $F_k^\perp \rightarrow L(E_k)$ is surjective since every linear function on E_k can be extended to a linear function on E which kills F_k .

Remark. Let E, E^* be a pair of dual finite-dimensional vector spaces. If F is a subspace of E , then Propositions V and VI together show that a subspace H of E is supplementary⁶ to F in E if and only if H is dual to $F^\perp$ in E^* . In other words, *the supplements of a subspace are the duals of its complement*, or equivalently, *the duals of a subspace are the supplements of its complement*.

Remark. For $\varphi : E \rightarrow F$ a linear map, let $L(\varphi) : L(E) \leftarrow L(F)$ be the dual map given by $L(\varphi)(f) = f \circ \varphi$ (2.50). Then L linearly embeds $L(E; F)$ in $L(L(F); L(E))$, by (2.43) and (2.44). Also, $L(\psi \circ \varphi) = L(\varphi) \circ L(\psi)$ and $L(\iota_E) = \iota_{L(E)}$. This shows that L is a contravariant functor in the category of vector spaces. This functor preserves exactness of sequences (see 2.29), and finite direct sums, which are just (co)products in the category (see 2.30), among other things.

Exercise (10). If $\varphi : E \rightarrow F$ is a linear map with restriction $\varphi_1 : E_1 \rightarrow F_1$ and dual map $\varphi^* : E^* \leftarrow F^*$, then φ^* can be restricted to $F_1^\perp, E_1^\perp$ and the induced map $\overline{\varphi^*} : E^*/E_1^\perp \leftarrow F^*/F_1^\perp$ is dual to φ_1 .

Proof. If $y^* \in F_1^\perp$ and $x \in E_1$, then

$$\langle \varphi^* y^*, x \rangle = \langle y^*, \varphi x \rangle = 0$$

⁵See subsection 2.23.

⁶Greub says *complementary*, but this risks confusion with the orthogonal complement.

so φ^* maps $F_1^\perp$ into $E_1^\perp$. We know that the pairs $E_1, E^*/E_1^\perp$ and $F_1, F^*/F_1^\perp$ are dual under the induced scalar products. For $\overline{y^*} \in F^*/F_1^\perp$ and $x \in E_1$,

$$\begin{aligned}\langle \overline{\varphi^* y^*}, x \rangle &= \langle \overline{\varphi^* y^*}, x \rangle \\ &= \langle \varphi^* y^*, x \rangle \\ &= \langle y^*, \varphi_1 x \rangle \\ &= \langle \overline{y^*}, \varphi_1 x \rangle\end{aligned}\quad \square$$

Remark. This result shows that quotient maps are dual to restriction maps. The examples in subsections 2.24 and 2.27 are special cases.

§ 6

Remark. If E is finite-dimensional, then since E is dual to $L(E)$ and $L(E)$ is dual to $L(L(E))$, it follows from Corollary II that there is a natural isomorphism $E \rightarrow L(L(E))$ given by

$$x \mapsto (f \mapsto f(x))$$

This is called *reflexivity*, and means that the vector x may be naturally *identified* with the linear function “evaluation at x ”.

Remark. If E is finite-dimensional, then every basis of a dual space E^* is a dual basis. Indeed, if $f_1, \dots, f_n$ is a basis of E^* , let $e_1, \dots, e_n$ be its dual basis in E . Then $\langle f_i, e_j \rangle = \delta_{ij}$ by (2.62), so $f_1, \dots, f_n$ is the dual basis of $e_1, \dots, e_n$, again by (2.62).

Alternatively, with $E^* = L(E)$, let $f_1^*, \dots, f_n^*$ be the dual basis of $f_1, \dots, f_n$ in $E^{**} = L(L(E))$, so $\langle f_j^*, f_i \rangle = \delta_{ij}$ by (2.62). Let $e_1, \dots, e_n \in E$ be defined by $\langle f_j^*, f \rangle = \langle f, e_j \rangle$ for all $f \in E^*$.⁷ Then $\langle f_i, e_j \rangle = \langle f_j^*, f_i \rangle = \delta_{ij}$, so $f_1, \dots, f_n$ is the dual basis of $e_1, \dots, e_n$.

The first proof here uses the symmetry between E and E^* , while the second uses reflexivity. In either case, we see that the relationship between dual bases is fundamentally symmetrical.

Remark. In Proposition III,

$$\Phi(x^*, x) = \langle x^*, \varphi x \rangle = \langle \varphi^* x^*, x \rangle$$

so there is also a natural isomorphism $B(E^*, E) \cong L(E^*; E^*)$. We see that φ^* is “left-dual” to Φ while φ is “right-dual” to Φ . It might be said that Φ, φ, φ^* enter into a holy trinity.⁸

⁷See problem 3 in § 5 or the prior remark.

⁸No one says this.

Exercise (9). If E and F are finite-dimensional, then the mapping

$$\Phi : L(E; F) \rightarrow L(F^*; E^*)$$

defined by $\varphi \mapsto \varphi^*$ is a linear isomorphism.

Proof. By the remark in § 5 above, and the fact that $\varphi^{**} = \varphi$. □

Chapter III

Warning. Greub's notational choices in this chapter are insane. In particular, although he uses left-hand mapping notation (writing φx instead of $x\varphi$, and $\varphi\psi$ to mean φ after ψ), and follows the usual “row-by-column” convention for matrix multiplication, his convention for the matrix of a linear mapping is the transpose of that normally used in these circumstances. This has the following unfortunate consequences:

- The matrix of the linear mapping naturally associated with a system of linear equations has the coefficients from each equation appear *vertically in columns*.
- If $M(x)$ is the *column vector* representing x , then $M(\varphi x) = M(\varphi)^* M(x)$, and if $M(x)$ is the *row vector* representing x , then $M(\varphi x) = M(x)M(\varphi)$.
- $M(\varphi\psi) = M(\psi)M(\varphi)$

Compounding the insanity, Greub (inspired by tensor notation) indexes over columns instead of rows when working in dual spaces. This further increases the risk of confusion and error, as we see below. Greub says that “it would be very undesirable...to agree once and for all to always let the subscript count the rows”, but we couldn't disagree more in this context.

§ 3

Remark. In subsection 3.13, although $\beta_v^\varrho = \check{\alpha}_v^\varrho$, we must remember that v indexes the *columns* of the matrix of the dual basis transformation $x^{*v} \mapsto \bar{x}^{*v}$ by (3.24), whereas v indexes the *rows* of the matrix of the basis transformation $x_v \mapsto \bar{x}_v$ by (3.22). In other words, the matrix of the dual basis transformation $x^{*v} \mapsto \bar{x}^{*v}$ is the *transpose* of the inverse of the matrix of the basis transformation $x_v \mapsto \bar{x}_v$, contrary to what the book says.⁹ It's easier to remember that the matrix of $x^{*v} \leftarrow \bar{x}^{*v}$ (arrow reversed!) is the transpose of the matrix of $x_v \mapsto \bar{x}_v$.

Remark. In subsection 3.13, we see that if a basis transformation is effected by τ , then the corresponding *coordinate* transformation is effected by τ^{-1} . The coordinates of a vector are transformed “exactly in the same way” as the vectors of the dual basis, despite the previous remark, because of Greub's notational choices.

⁹Compare this to the proof of equation (3.4) in subsection 3.3.

Chapter IV

Remark. In this chapter, it is implicitly assumed that all vector spaces have dimension $n \geq 1$, except in the definition of intersection number (subsection 4.31) where $n = 0$. Here we summarize results for the case $n = 0$:

- For a set X , $X^0 = \{\emptyset\}$. Therefore maps $\Phi : X^0 \rightarrow Y$ can be identified with elements of Y .
- For vector spaces E and F , a map $\Phi : E^0 \rightarrow F$ is vacuously 0-linear. Since the only permutation in S_0 is the identity $\iota = \emptyset$, Φ is also trivially skew symmetric.

In particular if $E = 0$, the following results hold:

- Determinant functions in E are just scalars in Γ , and dual determinant functions are just reciprocal scalars.
- The only transformation of E is the zero transformation, which is also the identity transformation. It has determinant 1, trace 0, and constant characteristic polynomial 1. It has no eigenvalues or eigenvectors. Its adjoint is also the zero transformation. Its matrix on the empty basis is empty.
- If E is real ($\Gamma = \mathbb{R}$), the orientations in E are represented by the scalars ± 1 , and determine whether the empty basis is positive or negative. The zero transformation is orientation preserving. The empty basis is deformable into itself.

§ 1

Remark. To see why (4.1) holds, observe by definition of $\tau(\sigma\Phi)$ that

$$(\tau(\sigma\Phi))(x_1, \dots, x_p) = (\sigma\Phi)(x_{\tau(1)}, \dots, x_{\tau(p)})$$

Let $y_i = x_{\tau(i)}$. Then by definition of $\sigma\Phi$ and $(\tau\sigma)\Phi$,

$$\begin{aligned} (\sigma\Phi)(x_{\tau(1)}, \dots, x_{\tau(p)}) &= (\sigma\Phi)(y_1, \dots, y_p) \\ &= \Phi(y_{\sigma(1)}, \dots, y_{\sigma(p)}) \\ &= \Phi(x_{\tau(\sigma(1))}, \dots, x_{\tau(\sigma(p))}) \\ &= \Phi(x_{(\tau\sigma)(1)}, \dots, x_{(\tau\sigma)(p)}) \\ &= ((\tau\sigma)\Phi)(x_1, \dots, x_p) \end{aligned}$$

Therefore $\tau(\sigma\Phi) = (\tau\sigma)\Phi$.

Remark. By Proposition I(iii) and Proposition II, a determinant function $\Delta \neq 0$ “determines” linear independence in the sense that $\Delta(x_1, \dots, x_n) \neq 0$ if and only if $x_1, \dots, x_n$ are linearly independent. By (4.8), it follows that $\det \varphi$ “determines” whether a linear transformation φ preserves linear independence, i.e. whether or not φ is invertible.

Geometrically, $\Delta(x_1, \dots, x_n)$ measures the oriented (signed) volume of the n -dimensional parallelepiped determined by the vectors $x_1, \dots, x_n$. Therefore $\det \varphi$ is the factor by which φ changes oriented volume. Since a small change in the vectors $x_1, \dots, x_n$ results in a small change in the oriented volume, Δ is continuous.

Remark. We provide an alternative proof of Proposition IV. First note

$$(-1)^{j-1} \Delta(x, x_1, \dots, \widehat{x_j}, \dots, x_n) = \Delta(x_1, \dots, x, \dots, x_n)$$

where x is in the j -th position on the right.¹⁰ Therefore

$$\sum_{j=1}^n (-1)^{j-1} \Delta(x, x_1, \dots, \widehat{x_j}, \dots, x_n) x_j = \Delta(x, x_2, \dots, x_n) x_1 + \dots + \Delta(x_1, \dots, x_{n-1}, x) x_n$$

Viewing this as a function of $x_1, \dots, x_n$ (that is, a set map from $E^n \rightarrow L(E; E)$), it is obviously multilinear and skew symmetric (by Proposition I(ii)). Therefore if $x_1, \dots, x_n$ are linearly dependent, it is zero and hence equal to $\Delta(x_1, \dots, x_n) x$ (by Proposition I(iii)). On the other hand if $x_1, \dots, x_n$ are linearly independent (and hence a basis), then viewing it as a function of x , its value at x_i is just $\Delta(x_1, \dots, x_n) x_i$ (by Proposition I(ii)), so it agrees on a basis with $\Delta(x_1, \dots, x_n) x$ and hence is equal to it. Note this result shows that $\text{ad } \iota = \iota$.

Remark. In relation to the 2×2 determinant $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$, consider the “bordered” determinant

$$\begin{vmatrix} x & a & b \\ x & a & b \\ y & c & d \end{vmatrix}$$

It must be zero since the first two rows are equal. Expanding along the first row, we obtain

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} x = \begin{vmatrix} x & b \\ y & d \end{vmatrix} a + \begin{vmatrix} a & x \\ c & y \end{vmatrix} b$$

¹⁰ $\widehat{x_j}$ denotes deletion of x_j from the sequence on the left.

A similar argument shows

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} y = \begin{vmatrix} x & b \\ y & d \end{vmatrix} c + \begin{vmatrix} a & x \\ c & y \end{vmatrix} d$$

In vector form, we have

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{vmatrix} x & b \\ y & d \end{vmatrix} \begin{bmatrix} a \\ c \end{bmatrix} + \begin{vmatrix} a & x \\ c & y \end{vmatrix} \begin{bmatrix} b \\ d \end{bmatrix}$$

This is a more classical 2×2 version of Proposition IV.¹¹

Remark. Let E be a vector space with $\dim E = n > 1$ and E_1 a subspace with $\dim E_1 = 1$. Let Δ be a determinant function in E with $\Delta(e_1, \dots, e_n) = 1$ where $e_1 \in E_1$. Then Δ induces a determinant function Δ_1 in E/E_1 by

$$\Delta_1(\overline{x_2}, \dots, \overline{x_n}) = \Delta(e_1, x_2, \dots, x_n)$$

with $\Delta_1(\overline{e_2}, \dots, \overline{e_n}) = 1$. Define $D : E^n \rightarrow \Gamma$ by

$$D(x_1, \dots, x_n) = \sum_{j=1}^n (-1)^{j-1} \Delta_1(\overline{x_1}, \dots, \widehat{\overline{x_j}}, \dots, \overline{x_n}) \cdot \pi_1(x_j)$$

where $\pi_1 : E \rightarrow \Gamma$ is the coordinate function for e_1 . Then D is skew symmetric and n -linear with $D(e_1, \dots, e_n) = 1$, so $D = \Delta$ by uniqueness (Proposition III). Therefore

$$\Delta(x_1, \dots, x_n) = \sum_{j=1}^n (-1)^{j-1} \Delta_1(\overline{x_1}, \dots, \widehat{\overline{x_j}}, \dots, \overline{x_n}) \cdot \pi_1(x_j)$$

This result, which can also be obtained by applying π_1 to both sides of (4.6) with $x = e_1$, expresses a fundamental relationship between an n -dimensional determinant function and an $(n-1)$ -dimensional one. The cofactor expansion formulas for the determinant (subsection 4.15) follow immediately.

Note that this relationship can also be exploited in reverse to *define* an n -dimensional determinant function in terms of an $(n-1)$ -dimensional one. In fact, if $\Phi : E^{n-1} \rightarrow \Gamma$ is a skew symmetric $(n-1)$ -linear function in E , then there is a vector $a \in E$ such that

$$\Phi(x_1, \dots, x_{n-1}) = \Delta(x_1, \dots, x_{n-1}, a) \quad (1)$$

¹¹See also the remark in chapter 5, § 7 of [2] below.

To see this, define $\varphi : E^n \rightarrow E$ by

$$\varphi(x_1, \dots, x_n) = \sum_{j=1}^n (-1)^{n-j} \Phi(x_1, \dots, \widehat{x_j}, \dots, x_n) \cdot x_j$$

Then φ is skew symmetric and n -linear, so $\varphi = \Delta a$ where $a = \varphi(e_1, \dots, e_n)$. This means

$$\Delta(x_1, \dots, x_n) a = \sum_{j=1}^n (-1)^{n-j} \Phi(x_1, \dots, \widehat{x_j}, \dots, x_n) \cdot x_j \quad (2)$$

If $x_1, \dots, x_{n-1}$ are linearly dependent, then (1) holds trivially, so we may assume that they are linearly independent. If a is not in their span, then (1) follows from (2) with $x_n = a$; if a is in the span, then (1) follows from (2) with x_n chosen so that $\Delta(x_1, \dots, x_n) = 1$.

Geometrically, this result expresses a relationship between n -dimensional oriented volume and $(n-1)$ -dimensional oriented volume. We also see from (1) that the latter is just the former measured *relative to a fixed vector*. The fact that the volume of an n -dimensional parallelepiped is equal to the product of the volume of any $(n-1)$ -dimensional “base” and the corresponding “height” (subsection 7.15) is a special case.

§ 2

Remark. In subsection 4.6, we want a transformation ψ with $\psi\varphi = (\det \varphi)\iota$. We can choose a basis $x_1, \dots, x_n$ in E with $\Delta(x_1, \dots, x_n) = 1$, for which we want

$$\begin{aligned} \psi(\varphi x_i) &= (\psi\varphi)x_i = (\det \varphi)x_i \\ &= (\det \varphi)\Delta(x_1, \dots, x_n)x_i \\ &= \Delta(\varphi x_1, \dots, \varphi x_n)x_i \end{aligned}$$

To obtain this, we can define

$$\psi(x) = \sum_{j=1}^n \Delta(\varphi x_1, \dots, x, \dots, \varphi x_n) x_j$$

where x is in the j -th position on the right.¹² Then ψ obviously satisfies the above properties, by multilinearity and skew symmetry of Δ .

¹²See the remark on Proposition IV above.

To obtain ψ in a “coordinate-free” manner (without choosing a basis), we observe that the construction on the right is multilinear and skew symmetric in $x_1, \dots, x_n$ when viewed as a mapping $\Phi : E^n \rightarrow L(E; E)$. By the universal property of Δ (Proposition III), there is a unique $\psi \in L(E; E)$ satisfying the above; this ψ is also seen to be independent of the choice of Δ .

Remark. Alternatively, fix a vector x and define $\Phi : E^{n-1} \rightarrow \Gamma$ by

$$\Phi(x_1, \dots, x_{n-1}) = \Delta(\varphi x_1, \dots, \varphi x_{n-1}, x)$$

By a remark above, there is a unique vector x' with

$$\Phi(x_1, \dots, x_{n-1}) = \Delta(x_1, \dots, x_{n-1}, x')$$

for all vectors $x_1, \dots, x_{n-1}$. Writing $x' = \psi x$, we obtain

$$\Delta(\varphi x_1, \dots, \varphi x_{n-1}, x_n) = \Delta(x_1, \dots, x_{n-1}, \psi x_n) \quad (1)$$

for all vectors $x_1, \dots, x_n$. It follows from (1) that $\psi : E \rightarrow E$ is linear and uniquely determined by φ . It is easy to verify that $\psi = \text{ad } \varphi$.¹³

Remark. In subsection 4.7, observe that

$$\Delta(x_1, \dots, x_p, y_1, \dots, y_q)$$

induces a determinant function in E_2 when $x_1, \dots, x_p \in E$ are fixed, and induces a determinant function in E_1 when $y_1, \dots, y_q \in E$ are fixed. Now let $a_1, \dots, a_p$ be a basis of E_1 , so $a_1, \dots, a_p, b_1, \dots, b_q$ is a basis of E . Then by (4.8),

$$\begin{aligned} \det \varphi \cdot \Delta(a_1, \dots, a_p, b_1, \dots, b_q) &= \Delta(\varphi_1 a_1, \dots, \varphi_1 a_p, \varphi_2 b_1, \dots, \varphi_2 b_q) \\ &= \det \varphi_1 \cdot \Delta(a_1, \dots, a_p, \varphi_2 b_1, \dots, \varphi_2 b_q) \\ &= \det \varphi_1 \cdot \det \varphi_2 \cdot \Delta(a_1, \dots, a_p, b_1, \dots, b_q) \end{aligned}$$

Since $\Delta(a_1, \dots, a_p, b_1, \dots, b_q) \neq 0$, it follows that $\det \varphi = \det \varphi_1 \cdot \det \varphi_2$. Note this result shows that

$$\det(\varphi_1 \oplus \varphi_2) = \det \varphi_1 \cdot \det \varphi_2$$

Exercise (2). Let $\varphi : E \rightarrow E$ be linear with E_1 a stable subspace. If $\varphi_1 : E_1 \rightarrow E_1$ and $\bar{\varphi} : E/E_1 \rightarrow E/E_1$ are the induced maps, then

$$\det \varphi = \det \varphi_1 \cdot \det \bar{\varphi}$$

¹³See also the remark in chapter 7, § 4 of [2] below.

Proof. Let $e_1, \dots, e_n$ be a basis of E where $e_1, \dots, e_p$ is a basis of E_1 . Let $\Delta \neq 0$ be a determinant function in E . First observe that

$$\Delta_1(x_1, \dots, x_p) = \Delta(x_1, \dots, x_p, \varphi e_{p+1}, \dots, \varphi e_n) \quad (1)$$

is a determinant function in E_1 and

$$\Delta_2(\overline{x_{p+1}}, \dots, \overline{x_n}) = \Delta(e_1, \dots, e_p, x_{p+1}, \dots, x_n) \quad (2)$$

is a well-defined determinant function in E/E_1 . Now

$$\det \overline{\varphi} \cdot \Delta_2(\overline{x_{p+1}}, \dots, \overline{x_n}) = \Delta_2(\overline{\varphi x_{p+1}}, \dots, \overline{\varphi x_n}) = \Delta_2(\overline{\varphi x_{p+1}}, \dots, \overline{\varphi x_n}) \quad (3)$$

It follows from (2) and (3) that

$$\det \overline{\varphi} \cdot \Delta(e_1, \dots, e_p, x_{p+1}, \dots, x_n) = \Delta(e_1, \dots, e_p, \varphi x_{p+1}, \dots, \varphi x_n) \quad (4)$$

Now

$$\begin{aligned} \det \varphi \cdot \Delta(e_1, \dots, e_n) &= \Delta(\varphi e_1, \dots, \varphi e_n) \\ &= \Delta_1(\varphi_1 e_1, \dots, \varphi_1 e_p) && \text{by (1)} \\ &= \det \varphi_1 \cdot \Delta_1(e_1, \dots, e_p) \\ &= \det \varphi_1 \cdot \det \overline{\varphi} \cdot \Delta(e_1, \dots, e_n) && \text{by (1), (4)} \end{aligned}$$

Since $\Delta(e_1, \dots, e_n) \neq 0$, the result follows. $\square$

Remark. It follows that if A is an $n \times n$ matrix of the form

$$A = \begin{pmatrix} A_1 & \\ * & A_2 \end{pmatrix}$$

where A_1 is $p \times p$ and A_2 is $(n-p) \times (n-p)$, then

$$\det A = \det A_1 \cdot \det A_2$$

Indeed, let E be an n -dimensional vector space and $\varphi : E \rightarrow E$ be defined by $M(\varphi; e_1, \dots, e_n) = A$, so $\det A = \det \varphi$. Let $E_1 = \langle e_1, \dots, e_p \rangle$ and $E_2 = \langle e_{p+1}, \dots, e_n \rangle$. Then $E = E_1 \oplus E_2$ and E_1 is stable under φ . If $\varphi_1 : E_1 \rightarrow E_1$ and $\overline{\varphi} : E/E_1 \rightarrow E/E_1$ are the induced maps, then $A_1 = M(\varphi_1)$ and $A_2 = M(\overline{\varphi})$, so $\det A_1 = \det \varphi_1$ and $\det A_2 = \det \overline{\varphi}$. The result now follows from the problem.

Exercise (7).

$$(i) \quad \text{ad}(\psi\varphi) = \text{ad}(\varphi)\text{ad}(\psi)$$

$$(ii) \quad \det \text{ad} \varphi = (\det \varphi)^{n-1}$$

Proof. For (i), by a remark above we have for all vectors $x_1, \dots, x_n$

$$\begin{aligned} \Delta(x_1, \dots, x_{n-1}, \text{ad}(\psi\varphi)x_n) &= \Delta(\psi\varphi x_1, \dots, \psi\varphi x_{n-1}, x_n) \\ &= \Delta(\varphi x_1, \dots, \varphi x_{n-1}, \text{ad}(\psi)x_n) \\ &= \Delta(x_1, \dots, x_{n-1}, \text{ad}(\varphi)\text{ad}(\psi)x_n) \end{aligned}$$

For (ii), since $\text{ad}(\varphi)\varphi = (\det \varphi)\iota$, we have

$$\det \text{ad} \varphi \cdot \det \varphi = \det(\text{ad}(\varphi)\varphi) = \det((\det \varphi)\iota) = (\det \varphi)^n$$

If $\det \varphi \neq 0$, then the result follows, so suppose $\det \varphi = 0$. If $\varphi = 0$, then $\text{ad} \varphi = 0$ and the result holds. If $\varphi \neq 0$, then since $\text{ad}(\varphi)\varphi = 0$, we must have $\det \text{ad} \varphi = 0$ and the result holds. $\square$

§ 5

Remark. Recall that the system (4.39) is equivalent to $\varphi x = y$ where $\varphi : \Gamma^n \rightarrow \Gamma^n$ is defined by $M(\varphi) = (\alpha_k^j) = A$, $x = (\xi^i)$, and $y = (\eta^j)$. If $\det A \neq 0$, then φ is invertible and

$$x = \varphi^{-1}y = \frac{1}{\det A} \text{ad}(\varphi)(y)$$

It follows from the analysis of the adjoint matrix in subsection 4.13 that

$$\xi^i = \frac{1}{\det A} \sum_j \text{cof}(\alpha_i^j) \eta^j$$

Moreover, it follows from (4.38) that $\sum_j \text{cof}(\alpha_i^j) \eta^j = \det A_i$ where A_i is the matrix obtained from A by replacing the i -th row with y .¹⁴ Therefore

$$\xi^i = \frac{\det A_i}{\det A}$$

Remark. In subsection 4.14, $\det B_i^j = \det S_i^j$ does *not* follow from (4.38), which only tells us that $\det B_i^j = \det B_i^j$. However, it follows from (4.16), or from our remarks in § 4 above.

¹⁴The cofactors of A_i and A along the i -th row agree since A_i and A agree on the other rows.

§ 6

Exercise (5). If $\varphi_1 : E_1 \rightarrow E_1$ and $\varphi_2 : E_2 \rightarrow E_2$ are linear, then

$$\chi_{\varphi_1 \oplus \varphi_2} = \chi_{\varphi_1} \chi_{\varphi_2}$$

where χ_φ denotes the characteristic polynomial of φ .

Proof. This follows from the result in subsection 4.7 and the fact that

$$\varphi_1 \oplus \varphi_2 - \lambda \iota = (\varphi_1 - \lambda \iota_{E_1}) \oplus (\varphi_2 - \lambda \iota_{E_2}) \quad \square$$

Exercise (6). Let $\varphi : E \rightarrow E$ be linear with E_1 a stable subspace. If $\varphi_1 : E_1 \rightarrow E_1$ and $\bar{\varphi} : E/E_1 \rightarrow E/E_1$ are the induced maps, then

$$\chi_\varphi = \chi_{\varphi_1} \chi_{\bar{\varphi}}$$

Proof. This follows from problem 2 in § 2, the fact that $\varphi - \lambda \iota_E$ restricted to E_1 is just $\varphi_1 - \lambda \iota_{E_1}$, and $\overline{\varphi - \lambda \iota_E} = \bar{\varphi} - \lambda \iota_{E/E_1}$. $\square$

Remark. Taking $E_1 = \ker \varphi$, we have $\chi_{\varphi_1} = \chi_{0_{E_1}} = (-\lambda)^p$ where $p = \dim E_1$, so $\chi_\varphi = (-\lambda)^p \chi_{\bar{\varphi}}$.

Exercise (7). A linear map $\varphi : E \rightarrow E$ is nilpotent if and only if $\chi_\varphi = (-\lambda)^n$.

Proof. If φ is nilpotent, we proceed by induction on k least such that $\varphi^k = 0$. If $k = 1$, the result is trivial. If $k > 1$, let $E_1 = \ker \varphi$ and $\bar{\varphi} : E/E_1 \rightarrow E/E_1$ the induced map. Then $\bar{\varphi}^{k-1} = 0$ since

$$\bar{\varphi}^{k-1}(\bar{x}) = \overline{\varphi^{k-1}(x)} = \overline{\varphi^{k-1}(x)} = 0$$

as $\varphi^{k-1}(x) \in E_1$. By the induction hypothesis, $\chi_{\bar{\varphi}} = (-\lambda)^{n-p}$ where $p = \dim E_1$, so by the previous problem,

$$\chi_\varphi = (-\lambda)^p (-\lambda)^{n-p} = (-\lambda)^n$$

Conversely, if $\varphi \neq 0$ and $\chi_\varphi = (-\lambda)^n$, then the constant term $\det \varphi = 0$, so $p > 0$ and by the previous problem $(-\lambda)^n = (-\lambda)^p \chi_{\bar{\varphi}}$, which implies $\chi_{\bar{\varphi}} = (-\lambda)^{n-p}$. By induction, $\bar{\varphi}$ is nilpotent. If $\bar{\varphi}^k = 0$, then $\varphi^{k+1} = 0$, so φ is nilpotent. $\square$

§ 7

Remark. It follows by symmetry from (4.61) that

$$\operatorname{tr} \varphi^* = \sum_i \langle \varphi^* e^{*i}, e_i \rangle = \sum_i \langle e^{*i}, \varphi e_i \rangle = \operatorname{tr} \varphi$$

where φ^* is dual to φ .

Exercise (5). If $f : L(E; E) \rightarrow \Gamma$ is linear, there is $\alpha \in L(E; E)$ unique with

$$f(\varphi) = \operatorname{tr}(\varphi \circ \alpha)$$

Proof. Since $L(E; E)$ is finite-dimensional and dual to itself under the scalar product

$$\langle \varphi, \psi \rangle = \operatorname{tr}(\psi \circ \varphi)$$

it follows from Proposition I in chapter II, § 6 that there is α unique with

$$f(\varphi) = \langle \alpha, \varphi \rangle = \operatorname{tr}(\varphi \circ \alpha) \quad \square$$

Exercise (6). If $f : L(E; E) \rightarrow \Gamma$ is linear with

$$f(\psi \circ \varphi) = f(\varphi \circ \psi)$$

there is $\lambda \in \Gamma$ with

$$f(\varphi) = \lambda \cdot \operatorname{tr} \varphi$$

Proof. By problem 5, there is α with

$$f(\varphi) = \operatorname{tr}(\varphi \circ \alpha) = \operatorname{tr}(\alpha \circ \varphi)$$

We claim $\alpha = \lambda \iota$ for some $\lambda \in \Gamma$, from which the result follows. By assumption,

$$\operatorname{tr}(\alpha \circ \varphi \circ \psi) = \operatorname{tr}(\alpha \circ \psi \circ \varphi) = \operatorname{tr}(\varphi \circ \alpha \circ \psi) \quad (1)$$

for all φ and ψ . Fix a basis of E . Let $A = M(\alpha)$, let R_l be the matrix with 1's in the l -th row and 0's elsewhere, and let C_k be the matrix with 1's in the k -th column and 0's elsewhere. By direct computation,

$$(C_k R_l A)_{ij} = 0 \quad (k \neq l)$$

while

$$(C_k A R_l)_{ij} = A_{kl}$$

It follows from (1) with $R_l = M(\varphi)$ and $C_k = M(\psi)$ that A is diagonal. Now let T_{kl} be the matrix with 1's in entries (k, l) and (l, k) and 0's elsewhere. It follows from (1) with $T_{kl} = M(\varphi)$ and $C_k = M(\psi)$ that

$$A_{ll} = \text{tr}(C_l A) = \text{tr}(C_k T_{kl} A) = \text{tr}(C_k A T_{kl}) = A_{kk}$$

Therefore the claim holds with $\lambda = A_{11}$. □

Exercise (10). If $A : L(E; E) \rightarrow L(E; E)$ is linear and “functorial” in that

$$A(\varphi \circ \psi) = A(\varphi) \circ A(\psi) \quad \text{and} \quad A(\iota) = \iota$$

then $\text{tr } A(\varphi) = \text{tr } \varphi$.

Proof. The function $\text{tr} \circ A : L(E; E) \rightarrow \Gamma$ is linear and

$$(\text{tr} \circ A)(\varphi \circ \psi) = \text{tr}(A(\varphi) \circ A(\psi)) = \text{tr}(A(\psi) \circ A(\varphi)) = (\text{tr} \circ A)(\psi \circ \varphi)$$

so $\text{tr} \circ A = \lambda \cdot \text{tr}$ for some $\lambda \in \Gamma$ by problem 6. But this implies

$$\lambda \cdot \text{tr } \iota = (\text{tr} \circ A)(\iota) = \text{tr } \iota$$

so $\lambda = 1$ and $\text{tr} \circ A = \text{tr}$. □

Exercise (12). If $\varphi_1 : E_1 \rightarrow E_1$ and $\varphi_2 : E_2 \rightarrow E_2$ are linear, then

$$\text{tr}(\varphi_1 \oplus \varphi_2) = \text{tr } \varphi_1 + \text{tr } \varphi_2$$

Proof. Immediate since

$$M(\varphi_1 \oplus \varphi_2) = \begin{pmatrix} M(\varphi_1) & \\ & M(\varphi_2) \end{pmatrix} \quad \square$$

§ 8

Remark. In (4.68), if instead we define

$$\Delta_1(x_1, \dots, x_p) = \Delta(x_1, \dots, x_p, e_{p+1}, \dots, e_n)$$

then Δ_1 represents the original orientation in E_1 . Indeed, in this case

$$\Delta_1(e_1, \dots, e_p) = \Delta(e_1, \dots, e_p, e_{p+1}, \dots, e_n) = \Delta_2(e_{p+1}, \dots, e_n) > 0$$

Chapter V

§ 1

Remark. An algebra A is a *zero algebra* if $xy = 0$ for all $x, y \in A$; this is equivalent to $A^2 = 0$. As an example, *the zero algebra* is the algebra $A = 0$. A zero algebra is unital if and only if it is the zero algebra.

Remark. Let $\varphi : A \rightarrow B$ be a homomorphism of algebras. If A_1 is a subalgebra of A and B_1 is a subalgebra of B and $\varphi(A_1) \subseteq B_1$, then the restriction $\varphi_1 : A_1 \rightarrow B_1$ of φ to A_1, B_1 is a homomorphism.

If A_1 and B_1 are *ideals*, then the induced linear map $\bar{\varphi} : A/A_1 \rightarrow B/B_1$ is also a homomorphism since

$$\bar{\varphi}(\bar{x}\bar{y}) = \bar{\varphi}(\overline{xy}) = \overline{\varphi(xy)} = \overline{\varphi(x)\varphi(y)} = \overline{\varphi(x)}\overline{\varphi(y)} = \bar{\varphi}(\bar{x})\bar{\varphi}(\bar{y})$$

In the problems below, E is a finite-dimensional vector space.

Exercise (12). The mapping

$$\Phi : A(E; E) \rightarrow A(E^*; E^*)^{\text{opp}}$$

defined by $\varphi \mapsto \varphi^*$ is an algebra isomorphism.

Proof. Φ is a linear isomorphism by problem 9 of chapter II, § 6, and preserves products since $(\varphi \circ \psi)^* = \psi^* \circ \varphi^*$. $\square$

Exercise (16). Every algebra automorphism $\Phi : A(E; E) \rightarrow A(E; E)$ is an *inner* automorphism; that is, there exists $\alpha \in GL(E)$ such that $\Phi(\varphi) = \alpha\varphi\alpha^{-1}$ for all $\varphi \in A(E; E)$.

Proof. First, observe that every basis (e_i) of E induces a basis (φ_{ij}) of $A(E; E)$ defined by $\varphi_{ij}(e_k) = \delta_{jk}e_i$. This basis satisfies

$$\varphi_{ij}\varphi_{lk} = \delta_{jl}\varphi_{ik} \quad \text{and} \quad \sum_i \varphi_{ii} = \iota \tag{1}$$

Conversely, every basis satisfying these properties is induced by a basis of E in this manner (see problem 14). Moreover, any two of these bases are conjugate to each other via the change of basis transformation between their inducing bases of E (see problem 15).

Now fix (e_i) and (φ_{ij}) as above. Since Φ is an automorphism, $(\Phi(\varphi_{ij}))$ is also a basis of $A(E; E)$ which satisfies (1), so there is $\alpha \in GL(E)$ with $\Phi(\varphi_{ij}) = \alpha\varphi_{ij}\alpha^{-1}$ for all i, j . It follows that $\Phi(\varphi) = \alpha\varphi\alpha^{-1}$ for all $\varphi \in A(E; E)$. $\square$

Remark. The result is true for any nonzero endomorphism Φ , since $A(E; E)$ is simple (see subsection 5.12).

§ 3

Remark. We see the following examples of change of coefficient field of a vector space:

- Taking the underlying real space of a complex space (5.16): for example changing from $\mathbb{C}$ over $\mathbb{C}$ to $\mathbb{C}$ over $\mathbb{R}$. In this case the dimension is doubled. Moreover, the underlying real space can be decomposed into “real” and “imaginary” parts of equal dimension (11.7).
- Complexifying a real space (2.16): for example changing from $\mathbb{R}$ over $\mathbb{R}$ to $\mathbb{R}^2$ over $\mathbb{C}$. In this case the dimension is preserved.
- Inducing complex structure on a real space (8.21): for example changing from $\mathbb{R}^2$ over $\mathbb{R}$ to $\mathbb{R}^2$ over $\mathbb{C}$. In this case the dimension is halved.

Remark. Let $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be counterclockwise rotation about the origin by $\pi/2$ radians. With respect to the standard basis over $\mathbb{R}$, the matrix of φ is

$$M_{\mathbb{R}}(\varphi) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

The characteristic polynomial of φ and $M_{\mathbb{R}}(\varphi)$ over $\mathbb{R}$ is $\chi_{\varphi}^{\mathbb{R}} = \lambda^2 + 1$.

If we induce the canonical complex structure on $\mathbb{R}^2$, then the matrix of φ with respect to the standard basis over $\mathbb{C}$ is

$$M_{\mathbb{C}}(\varphi) = (i)$$

The characteristic polynomial of φ and $M_{\mathbb{C}}(\varphi)$ over $\mathbb{C}$ is $\chi_{\varphi}^{\mathbb{C}} = -\lambda + i$. By contrast, the characteristic polynomial of $M_{\mathbb{R}}(\varphi)$ viewed as a matrix over $\mathbb{C}$ (representing a transformation $\mathbb{C}^2 \rightarrow \mathbb{C}^2$) is again $\lambda^2 + 1$.

This example shows that the characteristic polynomial of a *transformation* is not in general preserved under field extension, although the characteristic polynomial of a *matrix* is. The same is true of the minimum polynomial.¹⁵

Remark. Let $\Delta \subseteq \Gamma$ be a subfield and E be a Γ -vector space. A Δ -subspace F of E is a Γ -subspace if and only if F is stable under the Δ -linear transformations

$$\varepsilon_{\alpha} : x \mapsto \alpha x \quad \alpha \in \Gamma$$

This stability condition for subspaces is analogous to the commutativity condition for linear transformations in the proposition of subsection 5.19.

¹⁵See chapter XIII. In this example, $\mu_{\varphi}^{\mathbb{R}} = t^2 + 1$ while $\mu_{\varphi}^{\mathbb{C}} = t - i$.

Chapter VI

§ 1

Remark. The space of polynomials in one variable is positively graded by the degrees of monomials. More generally, the space of polynomials in p variables is p -graded by the multidegrees of monomials.

Remark. Let $E = \sum_{\alpha \in I} E_{\alpha}$ be a G -graded space with degree mapping $k : I \rightarrow G$. If $j : G \rightarrow H$ is an injective mapping into an abelian group H , then by (6.1) E is H -graded with the elements of E_{α} having degree $(j \circ k)(\alpha)$.

In particular if p is a fixed integer such that the mapping of G defined by $k \mapsto pk$ is injective, then we may consider the G -gradation of E under which the elements of E_{α} have degree $pk(\alpha)$. This is done for example in the construction of the exterior algebra over a graded space.¹⁶

Remark. If $E = \sum_{k \in G} E_k$ is a G -graded space and $F = \sum_{k \in G} F \cap E_k$ is a G -graded subspace, then $E/F = \sum_{k \in G} E_k / (F \cap E_k)$ is the G -graded factor space.¹⁷

Remark. The zero map between two G -graded vector spaces is homogeneous of every degree. A nonzero homogeneous map has a unique degree.

Remark. Let E and F be G -graded vector spaces. If $\varphi : E \rightarrow F$ is linear and homogeneous of degree k and φx is homogeneous of degree l , then we may assume without loss of generality that x is homogeneous of degree $l - k$. Indeed, writing $x = \sum_m x_m$ with $\deg x_m = m$, we have $\varphi x = \sum_m \varphi x_m$ with $\deg(\varphi x_m) = m + k$. Since $\deg(\varphi x) = l$, we must have $\varphi x_m = 0$ for $m \neq l - k$, so $\varphi x = \varphi x_{l-k}$.

Remark. If E is a finite-dimensional G -graded vector space and $\varphi : E \rightarrow E$ is linear and homogeneous with $\deg \varphi \neq 0$, then $\text{tr } \varphi = 0$.

Proof. Write $E = E_{k_1} \oplus \cdots \oplus E_{k_n}$ with $k_i \in G$ and $d_i = \dim E_{k_i} < \infty$. Let (e_{ij}) be a basis of E such that for each $1 \leq i \leq n$, (e_{ij}) is a basis of E_{k_i} for $1 \leq j \leq d_i$. Let Δ be a determinant function in E with $\Delta(e_{ij}) = 1$. Then

$$\text{tr } \varphi = \sum_{i,j} \Delta(e_{11}, \dots, e_{1d_1}, \dots, \varphi(e_{ij}), \dots, e_{n1}, \dots, e_{nd_n})$$

By assumption, $\varphi(e_{ij}) \in E_{k_l}$ for some $l \neq i$, so each term in this sum is zero, and hence $\text{tr } \varphi = 0$. □

¹⁶See subsection 5.21 of [2].

¹⁷See problem 2 in chapter II, § 4.

As an example, formal differentiation in the space of polynomials of degree at most n (graded by the degrees of monomials) is homogeneous of degree -1 , so has zero trace. This is also obvious from its matrix representation with respect to the standard basis.

Exercise (6). Let E, E^* and F, F^* be pairs of dual G -graded vector spaces and let $\varphi : E \rightarrow F$ and $\varphi^* : E^* \leftarrow F^*$ be dual linear maps. If φ is homogeneous of degree k , then φ^* is homogeneous of degree $-k$.

Proof. We have direct sum decompositions

$$E = \sum_{m \in G} E_m \quad E^* = \sum_{m \in G} E^{*m}$$

and

$$F = \sum_{n \in G} F_n \quad F^* = \sum_{n \in G} F^{*n}$$

where the pairs E_m, E^{*m} and F_n, F^{*n} are dual for all m, n under the restrictions of the scalar products between E, E^* and F, F^* , respectively (see subsection 6.5).

We also have $\varphi E_m \subseteq F_{m+k}$ for all m . We must prove $\varphi^* F^{*n} \subseteq E^{*n-k}$ for all n .

Let $y^* \in F^{*n}$ and $x \in E$. Write $x = \sum_m x_m$ where $x_m \in E_m$. Then

$$\langle \varphi^* y^*, x \rangle = \langle y^*, \varphi x \rangle = \sum_m \langle y^*, \varphi x_m \rangle = \langle y^*, \varphi x_{n-k} \rangle = \langle \varphi^* y^*, x_{n-k} \rangle$$

which implies

$$\langle \varphi^* y^*, x - \pi_{n-k} x \rangle = 0 \tag{1}$$

where $\pi_{n-k} : E \rightarrow E_{n-k}$ is the canonical projection. Now write $\varphi^* y^* = \sum_m x^{*m}$ where $x^{*m} \in E^{*m}$. We claim $x^{*m} = 0$ for all $m \neq n-k$. Indeed, for $m \neq n-k$ and $x \in E_m$ we have $\pi_{n-k} x = 0$, so by (1)

$$\langle x^{*m}, x \rangle = \sum_p \langle x^{*p}, x \rangle = \langle \varphi^* y^*, x \rangle = 0$$

Therefore $x^{*m} = 0$. It follows that $\varphi^* y^* = x^{*n-k} \in E^{*n-k}$, as desired. $\square$

Remark. It follows that the restrictions $\varphi_m : E_m \rightarrow F_{m+k}$ and $\varphi_{m+k}^* : E_m^* \leftarrow F_{m+k}^*$ are dual.

Exercise (8). Let E, E^* be a pair of almost finite dual G -graded vector spaces. If F is a G -graded subspace of E , then $F^\perp$ is a G -graded subspace of E^* and $F^{\perp\perp} = F$.

Proof. We have direct sums $E = \sum_{m \in G} E_m$ and $E^* = \sum_{m \in G} E^{*m}$ where the pairs E_m, E^{*m} are dual under the restrictions of the scalar product between E, E^* and $\dim E_m = \dim E^{*m} < \infty$ for all m . By assumption, $F = \sum_{m \in G} F \cap E_m$. We must prove

$$F^\perp = \sum_{m \in G} F^\perp \cap E^{*m} \quad (1)$$

Let $x^* \in F^\perp$ and write $x^* = \sum_m x^{*m}$ where $x^{*m} \in E^{*m}$. We claim $x^{*n} \in F^\perp$ for all n . Indeed, if $x \in F$, write $x = \sum_m x_m$ where $x_m \in F \cap E_m$. Then

$$\langle x^{*n}, x \rangle = \sum_m \langle x^{*n}, x_m \rangle = \langle x^{*n}, x_n \rangle = \sum_m \langle x^{*m}, x_n \rangle = \langle x^*, x_n \rangle = 0$$

This establishes (1). By symmetry, we have

$$F^{\perp\perp} = \sum_{m \in G} F^{\perp\perp} \cap E_m \quad (2)$$

We claim $F^{\perp\perp} \cap E_n \subseteq F \cap E_n$ for all n . To prove this, we first show

$$F^{\perp\perp} \cap E_n \subseteq (F \cap E_n)^{\perp_n \perp_n} \quad (3)$$

where $\perp_n$ is taken relative to the scalar product between E_n, E^{*n} . Indeed, let $x \in F^{\perp\perp} \cap E_n$ and $x^* \in (F \cap E_n)^{\perp_n \perp_n} \subseteq E^{*n}$. If $y \in F$, write $y = \sum_m y_m$ where $y_m \in F \cap E_m$. Then

$$\langle x^*, y \rangle = \sum_m \langle x^*, y_m \rangle = \langle x^*, y_n \rangle = 0$$

This implies $x^* \in F^\perp$, which implies $\langle x^*, x \rangle = 0$, which in turn implies (3). Now $(F \cap E_n)^{\perp_n \perp_n} = F \cap E_n$ since $\dim E_n < \infty$, which establishes the claim. Finally, it follows from (2) that $F^{\perp\perp} = F$. $\square$

§ 2

Remark. Let $A = \sum_k A_k$ be a positively graded algebra. If $b \in A_q$ with $b \neq 0$ and

$$b = \sum_i u_i a_i \quad (u_i \in A, a_i \in A_p)$$

then $q \geq p$ and we may assume that $u_i \in A_{q-p}$ for all i .

Proof. By bilinearity of the algebra multiplication, we may assume that each u_i is homogeneous, say $u_i \in A_{k_i}$. Since $\deg b = q$ and $\deg(u_i a_i) = k_i + p$, it follows that $\sum_{k_i + p \neq q} u_i a_i = 0$. On the other hand since $b \neq 0$, there is at least one i with $k_i + p = q$, so $q \geq p$. $\square$

Exercise (1). Let A be a G -graded algebra. If $x \in A$ is an invertible element homogeneous of degree k , then x^{-1} is homogeneous of degree $-k$. If A is nonzero and positively graded, then $k = 0$.

Proof. Write $A = \sum_{m \in G} A_m$ and $x^{-1} = \sum_m x_m$ with $x_m \in A_m$. Then

$$e = xx^{-1} = \sum_m xx_m$$

Since $\deg e = 0$ and $\deg(xx_m) = m + k$, it follows that $xx_m = 0$ for all $m \neq -k$. Therefore $e = xx_{-k}$ and $x^{-1} = x_{-k}$, so x^{-1} is homogeneous of degree $-k$.

If $A \neq 0$, then $x \neq 0$ and $x^{-1} \neq 0$, so $A_k \neq 0$ and $A_{-k} \neq 0$. If A is positively graded, this forces $k = 0$. $\square$

Exercise (4). Let E be a G -graded vector space. Then the subspace $A_G(E; E)$ of $A(E; E)$ generated by homogeneous linear transformations of E forms a G -graded subalgebra of $A(E; E)$.

Proof. First observe that $A_G(E; E)$ is naturally graded as a vector space by the degrees of homogeneous transformations (see problem 3). If $\varphi, \psi \in A_G(E; E)$ are homogeneous with $\deg \varphi = m$ and $\deg \psi = n$, then it is obvious that $\varphi\psi$ is homogeneous with $\deg(\varphi\psi) = m + n$. It follows from this that $A_G(E; E)$ is a G -graded subalgebra. $\square$

Exercise (7). Let E, E^* be a pair of almost finite dual G -graded vector spaces. Then the mapping

$$\Phi : A_G(E; E) \rightarrow A_G(E^*; E^*)^{\text{opp}}$$

defined by $\varphi \mapsto \varphi^*$ is an algebra isomorphism.

Proof. Φ is well defined by problems 6 and 10 of § 1, and is an isomorphism by problem 12 of chapter V, § 1. $\square$

Chapter VII

In this chapter, all vector spaces are real.

§ 1

Remark. The Riesz representation theorem shows that for a finite-dimensional inner product space E , there is a *natural* isomorphism between E and its dual space $L(E)$ given by $x \mapsto (x, -)$. This is unlike for a finite-dimensional vector space, where the isomorphism is in general non-natural, and means we may naturally *identify* a vector x with its *dual vector* or *covector* $(x, -)$.

Remark. If E is a finite-dimensional inner product space and E_1 is a subspace of E , then E induces an inner product in E/E_1 by

$$(\bar{x}, \bar{y}) = (x, y)$$

where x, y are the unique representatives of $\bar{x}, \bar{y}$ in $E_1^\perp$ (see problem 5).

Proof. The only thing to check is bilinearity, which follows from the fact that $E_1^\perp$ is a subspace of E . $\square$

Exercise (5). If E is a finite-dimensional inner product space and E_1 is a subspace of E , then every element of E/E_1 has exactly one representative in $E_1^\perp$.

Proof. By duality, $\dim E = \dim E_1 + \dim E_1^\perp$, and by definiteness of the inner product, $E_1 \cap E_1^\perp = 0$, so $E = E_1 \oplus E_1^\perp$. For $x + E_1 \in E/E_1$, let $y = \pi(x) - x$ where π is the canonical projection onto $E_1^\perp$. Then $y \in E_1$ so $x + y \in x + E_1$ and $x + y \in E_1^\perp$, as desired. If $z \in E_1$ and $x + z \in E_1^\perp$, then

$$y - z = (y + x) - (x + z) \in E_1 \cap E_1^\perp = 0$$

so $y = z$, establishing uniqueness. $\square$

§ 2

Remark. Let E be an inner product space of finite dimension n . We provide an inductive proof of the existence of an orthonormal basis in E .

For $n = 0, 1$, the result is trivial. For $n > 1$, fix a unit vector $e_1 \in E$ and let $E_1 = \langle e_1 \rangle$. By induction, there is an orthonormal basis $\bar{e}_2, \dots, \bar{e}_n$ in the induced inner product space E/E_1 (see above). Letting $e_2, \dots, e_n$ be the representatives in $E_1^\perp$, it follows that $e_1, \dots, e_n$ is an orthonormal basis in E .

Remark. In the Gram-Schmidt process, we can just let $e_1 = a_1/|a_1|$ and

$$e_k = \frac{a_k - (a_k, e_1)e_1 - \cdots - (a_k, e_{k-1})e_{k-1}}{|a_k - (a_k, e_1)e_1 - \cdots - (a_k, e_{k-1})e_{k-1}|} \quad (k = 2, \dots, n)$$

At each step, we compute the difference between the current vector a_k and its orthogonal projection onto the subspace generated by the previous vectors, then normalize (compare (7.19)).

Remark. If E is a finite-dimensional inner product space and $\varphi : E \rightarrow E$ is linear, then the dual transformation $\varphi^* : E \rightarrow E$ satisfies

$$(\varphi^* x, y) = (x, \varphi y)$$

If φ preserves inner products, then it also preserves orthonormal bases and is invertible. In this case, φ^{-1} also preserves inner products, so

$$(\varphi^{-1} x, y) = (\varphi^{-1} x, \varphi^{-1} \varphi y) = (x, \varphi y)$$

and it follows that $\varphi^{-1} = \varphi^*$. Conversely if $\varphi^{-1} = \varphi^*$, then

$$(\varphi x, \varphi y) = (\varphi^* \varphi x, y) = (x, y)$$

so φ preserves inner products.

Such a φ is called an *orthogonal* transformation. Note φ is orthogonal if and only if $M(\varphi)$ is an orthogonal matrix relative to an orthonormal basis.

§ 3

Remark. A determinant function Δ in an inner product space E is normed if and only if $|\Delta(e_1, \dots, e_n)| = 1$ for any orthonormal basis $e_1, \dots, e_n$. If E is oriented and Δ is the normed determinant function representing the orientation, then $\Delta(e_1, \dots, e_n) = 1$ if $e_1, \dots, e_n$ is positive. Geometrically, this is just *the* oriented volume function in E . By comparing (4.26) and (7.23), we see that *the normed determinant functions are precisely the self-dual determinant functions*.

If $\Delta_0 \neq 0$ is a determinant function in E and Δ_0^* is its dual, then $\Delta_0^* = \alpha \Delta_0$ for some real number α by uniqueness of Δ_0 , so

$$\alpha \Delta_0(x_1, \dots, x_n) \Delta_0(y_1, \dots, y_n) = \det(x_i, y_j) \quad (x_i, y_i \in E)$$

Substituting $x_i = y_i = e_i$ shows that $\alpha > 0$. It follows that $\Delta_1 = \pm \sqrt{\alpha} \cdot \Delta_0$ is a normed determinant function.

Remark. It follows from (7.37) that if x and y are linearly independent, then

$$|x \times y| = \Delta(x, y, z)$$

where $z = (x \times y)/|x \times y|$. Since z is a unit vector orthogonal to x and y , it follows (subsection 7.15) that $\Delta(x, y, z)$ is just the area of the parallelogram determined by x and y . In other words,

$$|x \times y| = |x||y|\sin\theta$$

where $0 \leq \theta \leq \pi$ is the angle between x and y . This last equation obviously still holds if x and y are linearly dependent but nonzero.

Exercise (12). For vectors $x_1, \dots, x_p$,

$$G(x_1, \dots, x_p) \leq |x_1|^2 \cdots |x_p|^2 \quad (1)$$

Additionally

$$\begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix}^2 \leq \sum_{k=1}^n |a_{1k}|^2 \cdots \sum_{k=1}^n |a_{nk}|^2 \quad (2)$$

Proof. By induction with the “base times height” rule for volume (subsection 7.15), $V(u_1, \dots, u_p) \leq 1$ for linearly independent *unit* vectors $u_1, \dots, u_p$.

Now if $x_1, \dots, x_p$ are linearly dependent, then $G(x_1, \dots, x_p) = 0$ and (1) holds trivially. Otherwise, setting $u_i = x_i/|x_i|$ we have

$$\sqrt{G(x_1, \dots, x_p)} = V(x_1, \dots, x_p) = |x_1| \cdots |x_p| \cdot V(u_1, \dots, u_p) \leq |x_1| \cdots |x_p|$$

Squaring both sides yields (1). Since the determinant of a matrix is a (normed) determinant function of the rows, (2) follows. $\square$

Remark. These results both simply express an upper bound for the volume of a parallelepiped in terms of the lengths of the edges.

§ 4

Remark. The covariant components of a vector are just its coordinates in the dual space, which are also just the entries of its matrix (as a linear function). They are called “covariant” because they vary in the same way as basis vectors

in the original space under a change of basis (see subsections 3.13–14). This is unlike the regular components of the vector, which vary inversely to the basis vectors and may be called the *contravariant components*.

To see why the covariant components of a vector are its inner products with the basis vectors, observe from (7.42) and (7.49) that

$$(x, e_\nu) = \langle \tau x, e_\nu \rangle = \sum_\lambda \xi_\lambda \langle e^{*\lambda}, e_\nu \rangle = \xi_\nu$$

Dually, the contravariant components of a vector are its inner products with the dual basis vectors:

$$(\tau x, e^{*\nu}) = \xi^\nu$$

Remark. If e_ν is an orthonormal basis of E , then the dual basis $e^{*\nu} = \tau e_\nu$ is an orthonormal basis of E^* .

§ 5

Remark. In subsection 7.22, observe that Q is just the set of unit vectors under the norm defined in example 3 in subsection 7.20. Since norm functions are continuous under the natural topology, Q is closed under this topology, and Q is also clearly bounded under this topology. It follows that Q is compact under this topology since E is finite-dimensional.

§ 6

Remark. If $x = \lambda e + x_1$ and $y = \mu e + y_1$ are quaternions ($\lambda, \mu \in \mathbb{R}$ and $x_1, y_1 \in E_1$), then by the definition of quaternion multiplication

$$xy = (\lambda\mu - (x_1, y_1))e + \lambda y_1 + \mu x_1 + x_1 \times y_1$$

Remark. In subsection 7.24, in the proof of Lemma I, observe that the result holds trivially if $y = \pm x$ by taking $\lambda = \mp 1$. If $y \neq \pm x$, then $e \neq 0$. Suppose

$$\alpha x + \beta y + \gamma e = 0 \quad (\alpha, \beta, \gamma \in \mathbb{R})$$

If $\alpha \neq 0$, then $x = \beta_1 y + \gamma_1 e$ for $\beta_1, \gamma_1 \in \mathbb{R}$, so

$$-e = x^2 = (\beta_1 y + \gamma_1 e)^2 = 2\beta_1 \gamma_1 y + (\gamma_1^2 - \beta_1^2)e$$

which implies

$$2\beta_1 \gamma_1 y = (\beta_1^2 - \gamma_1^2 - 1)e$$

If $\beta_1 = 0$, it follows that $\gamma_1^2 = -1$, which is impossible. If $\gamma_1 = 0$, it follows that $\beta_1 = \pm 1$, so $x = \pm y$ contrary to assumption. Therefore $\beta_1 \gamma_1 \neq 0$, so $y = \delta e$ for $\delta \in \mathbb{R}$. But then $-e = y^2 = \delta^2 e$, so $\delta^2 = -1$, which is impossible. It follows that $\alpha = 0$. Similarly $\beta = 0$. Finally, $\gamma = 0$. This result shows that the vectors x, y, e are linearly independent.

Now $x + y$ and $x - y$ are roots of polynomials of degree 1 or 2, but the linear independence of x, y, e implies that these polynomials must have degree 2, so (7.60) and (7.61) follow.

Chapter VIII

In this chapter, all vector spaces are real and finite-dimensional.

Remark. In this chapter, it is useful to think intuitively of transformations like complex numbers, which induce transformations of the complex plane through multiplication. Under this analogy, adjoints correspond to complex conjugates; self-adjoint transformations, to real numbers; positive transformations, to non-negative real numbers; skew transformations, to purely imaginary numbers; and isometries, to complex numbers on the unit circle.

§ 1

Remark. In subsection 8.2, if the bases (x_ν) and (y_μ) are orthonormal, then they are self-dual,¹⁸ so $M(\tilde{\varphi}, y_\mu, x_\nu) = M(\varphi, x_\nu, y_\mu)^*$ by (3.4). It follows that

$$\tilde{\alpha}_\mu^\rho = M(\tilde{\varphi})_\mu^\rho = M(\varphi)_\rho^\mu = \alpha_\rho^\mu$$

Remark. In subsection 8.4, the natural isomorphism $B(E, E) \cong L(E; E)$ follows from Proposition III in chapter II, § 6. There is also a natural isomorphism

$$B(E, E) \cong L(E; L(E))$$

given by

$$\Phi \mapsto (x \mapsto (y \mapsto \Phi(x, y))) \quad (1)$$

Recall from subsection 7.18 the natural isomorphism $\tau : E \rightarrow L(E)$ given by $x \mapsto (x, -)$, which maps each vector to its dual vector. For a linear transformation $\varphi : E \rightarrow E$ and its corresponding bilinear function Φ defined by $\Phi(x, y) = (\varphi x, y)$, the linear map Φ' corresponding to Φ under the isomorphism (1) is $\tau\varphi$:

$$\begin{array}{ccc} x & \xrightarrow{\varphi} & \varphi x \\ & \searrow \Phi' & \downarrow \tau \\ & & (\varphi x, -) \end{array}$$

¹⁸See problem 2 in chapter VII, § 2.

In these ways, Φ is naturally dual to φ . Properties of Φ naturally correspond to those of φ ; for example, Φ is symmetric if and only if φ is symmetric (self-adjoint), and Φ is skew symmetric if and only if φ is skew symmetric. Since

$$\Phi(x, y) = (\varphi x, y) = (x, \tilde{\varphi} y)$$

we see that φ is “left-dual” to Φ , while $\tilde{\varphi}$ is “right-dual” to Φ .

Remark. In subsection 8.5, observe for a direct sum $E = \sum E_i$ and $\varphi = \sum \varphi_i$ with $\varphi_i : E_i \rightarrow E_i$, if E_i is stable under $\tilde{\varphi}$ then the restriction of $\tilde{\varphi}$ to E_i is the adjoint of φ_i . In other words, if $\tilde{\varphi}_i$ denotes the restriction, then

$$\tilde{\varphi}_i = \tilde{\varphi}_i$$

Indeed, for $x, y \in E_i$,

$$(\tilde{\varphi}_i x, y) = (\tilde{\varphi} x, y) = (x, \varphi y) = (x, \varphi_i y)$$

It follows that $\tilde{\varphi} = \sum \tilde{\varphi}_i$. If additionally the E_i are orthogonal, we see that φ is normal if and only if each φ_i is normal.

§ 2

Remark. Geometrically, a self-adjoint transformation just independently scales the axes in some system of orthogonal axes for the space.

Remark. For any transformation φ , the transformations

$$\varphi + \tilde{\varphi} \quad \varphi \tilde{\varphi} \quad \tilde{\varphi} \varphi$$

are self-adjoint.

Remark. If $E = E_1 \oplus E_2$ with $E_1 \perp E_2$, and $\varphi_1 : E_1 \rightarrow E_1$ and $\varphi_2 : E_2 \rightarrow E_2$ are self-adjoint, then $\varphi = \varphi_1 \oplus \varphi_2$ is self-adjoint.

Proof. For $x = x_1 + x_2$ and $y = y_1 + y_2$ with $x_i, y_i \in E_i$,

$$\begin{aligned} (\varphi x, y) &= (\varphi_1 x_1 + \varphi_2 x_2, y_1 + y_2) \\ &= (\varphi_1 x_1, y_1) + (\varphi_2 x_2, y_2) \\ &= (x_1, \varphi_1 y_1) + (x_2, \varphi_2 y_2) \\ &= (x_1 + x_2, \varphi_1 y_1 + \varphi_2 y_2) \\ &= (x, \varphi y) \end{aligned}$$

□

Remark. We provide an alternative proof of the existence of eigenvectors for self-adjoint transformations (subsection 8.6).

Let $\varphi: E \rightarrow E$ be a transformation. First observe that for $u \neq 0$, $\varphi u = \lambda u$ for some λ if and only if $\varphi u \in u^{\perp\perp}$, which is true if and only if $(\varphi u, v) = 0$ for all $v \in u^\perp$. This motivates us to study the bilinear function

$$\Phi(x, y) = (\varphi x, y)$$

We may assume without loss of generality that $|u| = |v| = 1$. Consider a unit vector w coincident with u and moving in the direction of v :

$$w(t) = (\cos t)u + (\sin t)v \quad 0 \leq t \leq \pi/2$$

Note $|w(t)| = 1$ and $w(0) = u$. Also

$$w'(t) = (-\sin t)u + (\cos t)v$$

so $w'(0) = v$. This means $\Phi(u, v) = \Phi(w(0), w'(0))$. If φ is self-adjoint, then Φ is symmetric and

$$2 \cdot \Phi(u, v) = \Phi(w'(0), w(0)) + \Phi(w(0), w'(0)) = (\Psi \circ w)'(0)$$

where $\Psi(x) = \Phi(x, x)$ is the quadratic function associated with Φ . Now we have $(\Psi \circ w)'(0) = 0$ if $t = 0$ minimizes (or maximizes) $\Psi \circ w$, which is true if $w(0) = u$ minimizes (or maximizes) Ψ . But we know such a u exists since the unit sphere is compact in E , and the argument above then shows that u is an eigenvector of φ . Note that

$$\Psi(u) = \Phi(u, u) = (\varphi u, u) = (\lambda u, u) = \lambda(u, u) = \lambda$$

is the eigenvalue, which is the minimum (resp. maximum) eigenvalue of φ . In summary: the quadratic function associated with a self-adjoint transformation attains its extreme values on the unit sphere at eigenvectors, where its values are the extreme eigenvalues.

Exercise (5). Every positive transformation φ has a unique positive square root (that is, a positive transformation ψ such that $\psi^2 = \varphi$).

Proof. There is an orthonormal basis $e_1, \dots, e_n$ of eigenvectors of φ , so that $\varphi e_i = \lambda_i e_i$ for $\lambda_i \in \mathbb{R}$. Now $\lambda_i = (e_i, \varphi e_i) \geq 0$, so setting $\psi e_i = \sqrt{\lambda_i} e_i$ it follows that ψ is positive with $\psi^2 = \varphi$.

If ψ_1 is positive with $\psi_1^2 = \varphi$, then the eigenvalues of ψ_1 must be $\sqrt{\lambda_i}$. Also, if $\psi_1 x = \sqrt{\lambda_i} x$, then $\varphi x = \lambda_i x$, so $E_{\psi_1}(\sqrt{\lambda_i}) \subseteq E_\varphi(\lambda_i)$. By (8.21), it follows that $E_{\psi_1}(\sqrt{\lambda_i}) = E_\varphi(\lambda_i)$, so $\psi_1 e_i = \sqrt{\lambda_i} e_i$ and $\psi_1 = \psi$. $\square$

§ 3

Remark. In subsection 2.19, we see that a linear transformation π is a projection operator (that is, $\pi^2 = \pi$) if and only if $\pi = 0_{\ker \pi} \oplus \iota_{\operatorname{Im} \pi}$. In subsection 8.11, we see that π is additionally an *orthogonal* projection if and only if $\ker \pi \perp \operatorname{Im} \pi$. In summary, for a projection operator π , the following are equivalent:

- $\ker \pi \perp \operatorname{Im} \pi$.
- π is self-adjoint.
- π is normal.

§ 4

Remark. Geometrically, a skew transformation, apart from possibly killing off part of the space, induces scaled 90-degree rotations¹⁹ on orthogonal stable planes in the space.

Remark. For any transformation φ , the transformation $\varphi - \tilde{\varphi}$ is skew. Also

$$\varphi = \frac{1}{2}(\varphi + \tilde{\varphi}) + \frac{1}{2}(\varphi - \tilde{\varphi})$$

uniquely represents φ as a sum of self-adjoint and skew transformations.

Remark. If φ is a skew transformation of E and F is a stable subspace of E , then $F^\perp$ is also stable.

Proof. If $x \in F$ and $y \in F^\perp$, then $(x, \varphi y) = -(\varphi x, y) = 0$. □

Remark. In subsection 8.16, the proof of the normal form (8.35) is incorrect because it is not true in general that the a_n defined form an orthonormal basis of the space. For example in $\mathbb{R}^4$, if we define the transformation ψ by

$$e_1 \mapsto e_2 \quad e_2 \mapsto -e_1 \quad e_3 \mapsto e_4 \quad e_4 \mapsto -e_3$$

where e_i is the i -th standard basis vector, then ψ is skew and $\varphi = \psi^2 = -\iota$ is diagonalized by the standard basis. If we follow the proof for this example, we get $a_1 = e_1$, $a_2 = \psi e_1 = e_2$, $a_3 = e_2$, and $a_4 = \psi e_2 = -e_1$, so the a_n do not form a basis of $\mathbb{R}^4$.

¹⁹No one calls these “scrotations”.

To prove the result, let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of $\varphi = \psi^2$ and write the orthogonal decomposition (8.21)

$$E = E_1 \oplus \dots \oplus E_r$$

where E_i is the eigenspace of φ corresponding to λ_i . Observe that E_i is stable under ψ , for if $x \in E_i$ then

$$\varphi(\psi x) = \psi(\varphi x) = \psi(\lambda_i x) = \lambda_i(\psi x)$$

so $\psi x \in E_i$. As in the book, the λ_i are negative or zero. If $\lambda_i < 0$, construct a basis for $F = E_i$ in the following way: let a_1 be an arbitrary unit vector in F and $a_2 = \kappa_i^{-1} \psi a_1$ where $\kappa_i = \sqrt{-\lambda_i}$. It is immediate that a_2 is a unit vector in F orthogonal to a_1 and $H = \langle a_1, a_2 \rangle$ is stable under ψ . Let $H^\perp$ be the orthogonal complement of H in F . If $H^\perp = 0$, take the basis a_1, a_2 . Otherwise, adjoin to a_1, a_2 the result of applying this procedure recursively to $F = H^\perp$ (recall that $H^\perp$ is stable under ψ by the remark above). If $\lambda_i = 0$, assume that $i = r$ and choose any orthonormal basis of E_r , noting that ψ is zero on E_r since

$$|\psi x|^2 = (\psi x, \psi x) = -(x, \varphi x) = 0$$

for $x \in E_r$. Combine the resulting bases of the E_i to obtain an orthonormal basis of E with respect to which the matrix of ψ has the form (8.35).²⁰

Remark. If E is an oriented Euclidean 3-space and $\varphi : E \rightarrow E$ is linear, then

$$\varphi x \times \varphi y = \widetilde{(\text{ad } \varphi)}(x \times y)$$

Moreover if φ is invertible, then

$$\varphi(x \times y) = \det \varphi \cdot \tilde{\varphi}^{-1} x \times \tilde{\varphi}^{-1} y$$

Proof. For all vectors z ,

$$\begin{aligned} (\varphi x \times \varphi y, z) &= \Delta(\varphi x, \varphi y, z) \\ &= \Delta(x, y, (\text{ad } \varphi)z) \\ &= (x \times y, (\text{ad } \varphi)z) \\ &= (\widetilde{(\text{ad } \varphi)}(x \times y), z) \end{aligned}$$

so the first equation follows by definiteness of the inner product. The second equation is proved in a similar way. $\square$

²⁰See <https://math.stackexchange.com/q/3402347>.

Exercise (1). If φ is a skew transformation of a plane, then

$$(\varphi x, \varphi y) = \det \varphi \cdot (x, y)$$

Proof. If $\varphi = 0$, then the result is trivial; if $\varphi \neq 0$, then φ is an automorphism since it has even rank. Now $\Delta(x, y) = (x, y)$ is a determinant function, so

$$(\varphi^2 x, \varphi y) = \Delta(\varphi x, \varphi y) = \det \varphi \cdot \Delta(x, y) = \det \varphi \cdot (\varphi x, y)$$

Substituting $\varphi^{-1}x$ for x yields the result. $\square$

Exercise (2 - Skew transformations of 3-space). Let E be an oriented Euclidean 3-space.

(i) For $a \in E$, $\varphi_a(x) = a \times x$ is a skew transformation of E .

(ii) For $a, b \in E$, $\varphi_{a \times b} = \varphi_a \varphi_b - \varphi_b \varphi_a$.

(iii) If φ is a skew transformation of E , then $\varphi = \varphi_a$ for a unique $a \in E$.

(iv) The vector a in (iii) is given by

$$a = \alpha_{23}e_1 + \alpha_{31}e_2 + \alpha_{12}e_3$$

where e_1, e_2, e_3 is a positive orthonormal basis of E and $(\alpha_{ij}) = M(\varphi; e_i)$.

(v) If $a \neq 0$, then $\ker \varphi_a = \langle a \rangle$ and $a^\perp$ is stable under φ_a .

(vi) If e_1, e_2 are normal such that e_1, e_2, a is a positive orthogonal basis of E , then

$$M(\varphi_a; e_1, e_2, a) = \begin{pmatrix} 0 & |a| & 0 \\ -|a| & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Proof.

(i) It is a transformation since the cross product is bilinear, and it is skew since $(x, a \times x) = 0$ for all $x \in E$.

(ii) By the vector triple product formula (7.41),

$$(a \times b) \times x = a \times (b \times x) - b \times (a \times x)$$

- (iii) For uniqueness, note that if $\varphi_a = \varphi_b$, then $a \times x = b \times x$ for all $x \in E$, so $(a - b) \times x = 0$ for all $x \in E$, so $a - b = 0$ and $a = b$.

For existence, note that the bilinear function $(\varphi x, y)$ is skew symmetric, and hence a determinant function in any plane in E . Define

$$\Phi(x, y, z) = (\varphi y, z)x + (\varphi z, x)y + (\varphi x, y)z$$

Then Φ is clearly trilinear and skew symmetric. By the universal property of determinants if Δ is the normed determinant function representing the orientation in E and e_1, e_2, e_3 is an orthonormal basis with $\Delta(e_1, e_2, e_3) = 1$, then $\Phi = \Delta a$ where $a = \Phi(e_1, e_2, e_3)$.²¹ By direct computation it is easily verified that $a \times e_i = \varphi e_i$, so $\varphi = \varphi_a$.

- (iv) By the proof of (iii), noting that $a = \Phi(e_1, e_2, e_3)$ and $\alpha_{ij} = (\varphi e_i, e_j)$.
(v) By the fact that $a \times x = 0$ if and only if a and x are linearly dependent.
(vi) By direct computation. □

Remark. The definition of Φ in (iii) is motivated by the fundamental relationship between n -dimensional and $(n - 1)$ -dimensional determinant functions discussed in § 1 of chapter IV above. If φ is skew, then its dual bilinear function $\Delta_1(x, y) = (\varphi x, y)$ is a determinant function in any plane in E . If $\varphi = \varphi_a$, then

$$\Delta_1(x, y) = (a \times x, y) = \Delta(a, x, y)$$

where Δ is the normed determinant function representing the orientation in E . In other words, Δ_1 is just the 2-dimensional oriented area function induced by the 3-dimensional oriented volume function Δ and the vector a . Therefore if we *define* a 3-dimensional oriented volume map Φ in terms of Δ_1 , we should expect that $\Phi = \Delta a$.

This result, which shows that any skew linear *transformation* of the space can be represented by a unique vector under the *cross* product, is analogous to the Riesz representation theorem, which shows that any linear *function* of the space can be represented by a unique vector under the *inner* product.

Exercise (3). If $\varphi \neq 0$ and ψ are skew transformations of an oriented Euclidean 3-space with $\ker \varphi \subseteq \ker \psi$, then $\psi = \lambda \varphi$ for some $\lambda \in \mathbb{R}$.

²¹See Proposition III in subsection 4.3.

Proof. By the previous problem, we can write $\varphi(x) = a \times x$ and $\psi(x) = b \times x$ with $a \neq 0$. By assumption, a and b are orthogonal to the same plane, so $b = \lambda a$ for some $\lambda \in \mathbb{R}$ and hence $\psi = \lambda\varphi$. $\square$

Exercise (4).

$$(a_1 \times a_2) \times a_3 = a_2(a_1, a_3) - a_1(a_2, a_3)$$

Proof. Without loss of generality, we may assume that a_1, a_2 are orthonormal. In particular, $a_1 \times a_2 \neq 0$. Define

$$\varphi(x) = (a_1 \times a_2) \times x \quad \text{and} \quad \psi(x) = a_2(a_1, x) - a_1(a_2, x)$$

Then $\varphi \neq 0$ is skew, and ψ is skew since $(x, \psi x) = 0$ for all x . The kernel of φ is the line determined by $a_1 \times a_2$, which is killed by ψ . By the previous problem, $\psi = \lambda\varphi$ for some $\lambda \in \mathbb{R}$. Substituting $x = a_1 + a_2$ into this equation, it follows that $a_2 - a_1 = \lambda(a_2 - a_1)$, so $\lambda = 1$ and $\psi = \varphi$. $\square$

Exercise (5). A linear transformation φ satisfies $\tilde{\varphi} = \lambda\varphi$ for $\lambda \in \mathbb{R}$ if and only if φ is self-adjoint or skew.

Proof. If $\varphi \neq 0$ and $\tilde{\varphi} = \lambda\varphi$ for some $\lambda \in \mathbb{R}$, then there exist vectors x, y such that $(\varphi x, y) \neq 0$ and

$$(\varphi x, y) = \lambda(x, \varphi y) = \lambda^2(\varphi x, y)$$

so $\lambda = \pm 1$ and φ is self-adjoint or skew, respectively. The rest is obvious. $\square$

Exercise (6). If Φ is a skew symmetric bilinear function in an oriented Euclidean 3-space, then there is a unique vector a such that

$$\Phi(x, y) = (a, x \times y)$$

Proof. By problem 2, the skew transformation φ dual to Φ can be written in the form $\varphi(x) = a \times x$ for some vector a . Then

$$\Phi(x, y) = (\varphi x, y) = (a \times x, y) = (a, x \times y)$$

Uniqueness is obvious. $\square$

Remark. This result shows that any function measuring 2-dimensional oriented area in planes in 3-space actually measures 3-dimensional oriented volume relative to some fixed vector in the space.

§ 5

Remark. Geometrically, a rotation (and more generally an isometry) preserves length and angle. A proper rotation additionally preserves orientation, whereas an improper rotation reverses it.

Exercise (2). A linear transformation φ is regular and preserves orthogonality (that is, $(\varphi x, \varphi y) = 0$ whenever $(x, y) = 0$) if and only if $\varphi = \lambda \tau$ where $\lambda \neq 0$ and τ is a rotation.

Proof. For the forward direction, let $e_1, \dots, e_n$ be an orthonormal basis. Then $\varphi e_1, \dots, \varphi e_n$ is an orthogonal basis. Also

$$|\varphi e_i|^2 - |\varphi e_j|^2 = (\varphi e_i - \varphi e_j, \varphi e_i + \varphi e_j) = (\varphi(e_i - e_j), \varphi(e_i + e_j)) = 0$$

since

$$(e_i - e_j, e_i + e_j) = 1 - 1 = 0$$

for all i, j . Let $\lambda = |\varphi e_i| > 0$. Then clearly $\tau = \lambda^{-1} \varphi$ is a rotation.

The reverse direction is trivial. □

Remark. This proof is motivated by the geometrical fact that a rectangle is a square if and only if its diagonals are orthogonal.

Exercise (5). If $\varphi : E \rightarrow E$ is a mapping such that $\varphi(0) = 0$ and

$$|\varphi x - \varphi y| = |x - y|$$

for all $x, y \in E$, then φ is linear.

Proof. Substituting $y = 0$, we have $|\varphi x| = |x|$ for all x , so

$$(\varphi x - \varphi y, \varphi x - \varphi y) = |x|^2 - 2(\varphi x, \varphi y) + |y|^2$$

On the other hand,

$$(\varphi x - \varphi y, \varphi x - \varphi y) = |\varphi x - \varphi y|^2 = |x - y|^2 = |x|^2 - 2(x, y) + |y|^2$$

Therefore

$$(\varphi x, \varphi y) = (x, y)$$

It now follows that

$$|\varphi(x + y) - \varphi x - \varphi y|^2 = (\varphi(x + y) - \varphi x - \varphi y, \varphi(x + y) - \varphi x - \varphi y) = 0$$

so $\varphi(x + y) = \varphi x + \varphi y$. Similarly $\varphi(\lambda x) = \lambda \varphi x$. □

§ 6

Remark. In subsection 8.21, j is called the canonical “complex structure” on E because it induces a complex vector space structure on the underlying set of E in which scalar multiplication is defined by

$$(\alpha + \beta i) \cdot x = \alpha x + \beta jx \quad \alpha, \beta \in \mathbb{R}$$

Remark. In subsection 8.21, to derive (8.40) from (8.39) and (8.41), first observe that $j\varphi = \varphi j$. Indeed, for $z \neq 0$, $(jz, z) = 0$, so $(\varphi jz, \varphi z) = 0$. On the other hand, $(j\varphi z, \varphi z) = 0$. Since E is a plane, it follows that $j\varphi z = \lambda \varphi jz$ for some $\lambda \in \mathbb{R}$. But

$$\begin{aligned} \lambda |z|^2 &= \lambda |\varphi jz|^2 \\ &= \lambda (\varphi jz, \varphi jz) \\ &= (j\varphi z, \varphi jz) \\ &= \Delta(\varphi z, \varphi jz) \\ &= \Delta(z, jz) \\ &= (jz, jz) \\ &= (z, z) = |z|^2 \end{aligned}$$

so $\lambda = 1$ and $j\varphi z = \varphi jz$.

From this and (8.39), it follows that

$$\varphi^{-1}x = x \cdot \cos(-\Theta) + jx \cdot \sin(-\Theta) = x \cdot \cos \Theta - jx \cdot \sin \Theta$$

where x and Θ are as in (8.39). In (8.41),

$$\Delta(x, \varphi y) = (jx, \varphi y) = (\varphi^{-1}jx, y) = (j\varphi^{-1}x, y) = \Delta(\varphi^{-1}x, y)$$

so

$$\Delta(\varphi x, y) + \Delta(x, \varphi y) = \Delta(\varphi x + \varphi^{-1}x, y) = 2 \cos \Theta \cdot \Delta(x, y)$$

and it follows that $\cos \Theta = \frac{1}{2} \operatorname{tr} \varphi$. Similar reasoning shows that $\sin \Theta = -\frac{1}{2} \operatorname{tr}(j\varphi)$, contrary to what the book says.

Remark. In subsection 8.21, to make sense of the “definition” in (8.43), fix $x \neq 0$ and let $0 \leq \Theta \leq \pi$ be the angle between x and φx . Fix an orientation in E so that Θ is the *oriented* angle between x and φx . Then (8.40) holds, so $\cos \Theta = \frac{1}{2} \operatorname{tr} \varphi$. Clearly Θ is independent of x and depends only on φ , so can be written as $\Theta(\varphi)$ and satisfies (8.43).

Remark. In subsection 8.22, why would we expect the rotation vector u in (8.45) to lie on the rotation axis? Since ψ is skew, $\Psi(x, y) = (\psi x, y)$ measures oriented area in planes in E , and since ψ kills off E_1 , $\Psi(x, y) = 0$ if $x \in E_1$ or $y \in E_1$. But we know from previous results that Ψ actually measures oriented *volume* relative to u , so it follows that u must lie in E_1 .

Note that ψ stabilizes $F = E_1^\perp$, and if ψ_1 denotes the restriction, then

$$\psi_1 = \frac{1}{2}(\varphi_1 - \widetilde{\varphi_1}) = \frac{1}{2}(\varphi_1 - \varphi_1^{-1})$$

where φ_1 denotes the restriction of φ . If F is oriented by E and by $u \neq 0$, and j is the induced canonical complex structure on F , then

$$\psi_1 = j \cdot \sin \Theta$$

where $0 < \Theta < \pi$ is the oriented angle of rotation. If $x \in F$ is any unit vector, then

$$|u| = |u \times x| = |\psi_1 x| = |jx| \sin \Theta = \sin \Theta$$

as expected.

Remark. In subsection 8.24, for the proof of the first part of Proposition I, let $q_i = p_i - \lambda_i e$ where $\lambda_i = (p_i, e)$ for $i = 1, 2$. Since q_1 and q_2 each generate the same axis of rotation, either $q_1 = 0 = q_2$ in which case $p_1 = \pm e$ and $p_2 = \pm e$ and the result holds, or else $q_1 \neq 0$ and $q_2 = \alpha q_1$ for some $\alpha \neq 0$. If $\alpha > 0$, then q_1 and q_2 induce the same orientation in their orthogonal plane (in E_1), so the oriented angle Θ of rotation in that plane is the same. It follows that

$$\lambda_1 = \cos \frac{\Theta}{2} = \lambda_2 \quad \text{and} \quad |q_1| = \sin \frac{\Theta}{2} = |q_2|$$

The second equation implies that $\alpha = 1$, so $q_1 = q_2$ and $p_1 = p_2$. On the other hand if $\alpha < 0$, then $-p_2$ is a unit quaternion inducing the same rotation as p_1 and its pure part $-q_2$ satisfies $-q_2 = (-\alpha)q_1$ with $-\alpha > 0$, so by the previous case $p_1 = -p_2$.

Exercise (16). If $p \neq \pm e$ is a unit quaternion, then the rotation vector induced by p is

$$u = 2\lambda(p - \lambda e) \quad \lambda = (p, e)$$

Proof. Let $q = p - \lambda e$. By assumption, $q \neq 0$ and $u = \alpha q$ since q and u both lie on the axis of rotation. If $\alpha > 0$, then q and u both induce the same orientation on the orthogonal plane and

$$\alpha|q| = |u| = \sin \Theta = 2\lambda|q|$$

where Θ is the oriented angle of rotation. Therefore $\alpha = 2\lambda$ and the result holds. If $\alpha < 0$, then $-p$ induces the same rotation vector and satisfies the hypotheses of the previous case since $u = (-\alpha)(-q)$ with $-\alpha > 0$, so $-\alpha = 2(-\lambda)$ and $\alpha = 2\lambda$ and the result holds. If $\alpha = 0$, then $u = 0$ and $\Theta = \pi$, so $\lambda = \cos(\pi/2) = 0$ and the result holds. $\square$

Exercise (17). Let $p \neq \pm e$ be a unit quaternion. If F denotes the plane generated by e and p , then the rotations $\varphi x = px$ and $\psi x = xp$ agree on F and stabilize F and $F^\perp$.

Proof. The rotations agree on e and p , hence on F . By definition of quaternion multiplication, $p^2 \in F$, so they also stabilize F and $F^\perp$. $\square$

Chapter XI

Remark. In parts of this chapter, it is implicitly assumed that unitary spaces have dimension $n \geq 1$.

Remark. A map $\varphi : E \rightarrow F$ of complex vector spaces E, F is *conjugate-linear* if

$$\varphi(\lambda x + \mu y) = \bar{\lambda}\varphi x + \bar{\mu}\varphi y$$

for all $x, y \in E$ and $\lambda, \mu \in \mathbb{C}$, where $\bar{\lambda}, \bar{\mu}$ denote the complex conjugates of λ, μ respectively.

§ 1

Remark. A sesquilinear function $\Phi(x, y)$ is linear in x and conjugate-linear in y .

§ 2

Remark. The identity map $\kappa : F \rightarrow \bar{F}$ is conjugate-linear. A map $\varphi : E \rightarrow F$ is conjugate-linear if and only if $\kappa\varphi : E \rightarrow \bar{F}$ is linear. This yields a natural bijective correspondence between conjugate-linear maps $E \rightarrow F$ and linear maps $E \rightarrow \bar{F}$.

Remark. If $z \mapsto \bar{z}$ is a conjugation in E , then for $\lambda = \alpha + i\beta$ with $\alpha, \beta \in \mathbb{R}$,

$$\overline{\lambda z} = \overline{\alpha z + i\beta z} = \alpha\bar{z} - i\beta\bar{z} = \bar{\lambda}\bar{z}$$

Therefore a conjugation is just a conjugate-linear involution. By the previous remark, a conjugation can also be viewed as a linear map $E \rightarrow \bar{E}$.

§ 3

Remark. If E is a unitary space, then there exists a conjugation in E . In fact, if $z_1, \dots, z_n$ is a basis of E and F is the real span of $z_1, \dots, z_n$, then F is a real form of E and the map $z \mapsto \bar{z}$ defined by $x + iy \mapsto x - iy$ for $x, y \in F$ is a conjugation in the *vector space* E . It is also a conjugation in the *unitary space* E since for $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ with $x_j, y_j \in F$,

$$(z_1, z_2) = (x_1, x_2) + (y_1, y_2) + i[(y_1, x_2) - (x_1, y_2)]$$

and

$$(\bar{z}_1, \bar{z}_2) = (x_1, x_2) + (y_1, y_2) - i[(y_1, x_2) - (x_1, y_2)]$$

so $(\overline{z_1}, \overline{z_2}) = \overline{(z_1, z_2)}$ as required.²²

Remark. If E, F are unitary spaces and $\overline{E}, \overline{F}$ the corresponding conjugate spaces, then E is dual to $\overline{E}$ under the scalar product

$$\langle x, x^* \rangle = (x, \kappa_E^{-1} x^*)$$

where $\kappa_E : E \rightarrow \overline{E}$ is the identity map; similarly F is dual to $\overline{F}$ under the scalar product

$$\langle y, y^* \rangle = (y, \kappa_F^{-1} y^*)$$

where $\kappa_F : F \rightarrow \overline{F}$ is the identity map.

If $\varphi : E \rightarrow F$ is a linear map, then the dual map $\varphi^* : \overline{E} \leftarrow \overline{F}$ satisfies

$$(\varphi x, \kappa_F^{-1} y^*) = \langle \varphi x, y^* \rangle = \langle x, \varphi^* y^* \rangle = (x, \kappa_E^{-1} \varphi^* y^*)$$

Taking $y^* = \kappa_F y$ yields

$$(\varphi x, y) = (x, \kappa_E^{-1} \varphi^* \kappa_F y)$$

Therefore $\tilde{\varphi} = \kappa_E^{-1} \varphi^* \kappa_F$:

$$\begin{array}{ccc} E & \xleftarrow{\tilde{\varphi}} & F \\ \kappa_E \downarrow & & \downarrow \kappa_F \\ \overline{E} & \xleftarrow{\varphi^*} & \overline{F} \end{array}$$

This construction of the adjoint avoids using conjugations in E, F by using the conjugate spaces $\overline{E}, \overline{F}$ instead.

If $E = F$, $\Delta \neq 0$ is a determinant function in E , and Δ^* is the corresponding determinant function in $\overline{E}$ (11.6), then

$$\begin{aligned} \det \tilde{\varphi} \cdot \Delta(x_1, \dots, x_n) &= \Delta(\tilde{\varphi} x_1, \dots, \tilde{\varphi} x_n) \\ &= \Delta(\kappa_E^{-1} \varphi^* \kappa_F x_1, \dots, \kappa_E^{-1} \varphi^* \kappa_F x_n) \\ &= \overline{\Delta^*(\varphi^* \kappa_F x_1, \dots, \varphi^* \kappa_F x_n)} \\ &= \overline{\det \varphi^* \cdot \Delta^*(\kappa_F x_1, \dots, \kappa_F x_n)} \\ &= \overline{\det \varphi} \cdot \Delta(x_1, \dots, x_n) \end{aligned}$$

where $\kappa = \kappa_E$, so $\det \tilde{\varphi} = \overline{\det \varphi}$. This derivation of (11.21) similarly avoids using conjugations in E .

²²See problem 2(i) in § 2.

Remark. The mapping $\varphi \mapsto \tilde{\varphi}$ is conjugate-linear.

Remark. If $z \mapsto \bar{z}$ is a conjugation in E and Δ is a determinant function in E , then

$$\overline{\Delta}(z_1, \dots, z_n) = \overline{\Delta(\bar{z}_1, \dots, \bar{z}_n)}$$

is also a determinant function in E , called the *conjugate determinant function*. The proof of (11.21) in the book uses one of these.

Exercise (1). If E is a unitary space and $\varphi : E \rightarrow E$ is linear, then

$$\Phi(x, y) = (\varphi x, y)$$

is sesquilinear. Conversely, every sesquilinear function in E can be uniquely represented in this way. The adjoint $\tilde{\varphi}$ represents the adjoint $\tilde{\Phi}$.

Proof. The forward direction is trivial. If Φ is sesquilinear, then for any fixed vector x the function

$$y \mapsto \overline{\Phi(x, y)}$$

is linear, so by the Riesz theorem (11.5) there is a unique vector φx such that

$$\overline{\Phi(x, y)} = (y, \varphi x) = \overline{(\varphi x, y)}$$

and therefore

$$\Phi(x, y) = (\varphi x, y)$$

Clearly φ is linear and is uniquely determined by Φ . Also

$$(\tilde{\varphi}x, y) = \overline{(y, \tilde{\varphi}x)} = \overline{(\varphi y, x)} = \overline{\Phi(y, x)} = \tilde{\Phi}(x, y) \quad \square$$

Remark. This result shows that the linear transformation φ is naturally left-dual to the sesquilinear function Φ . Observe that Φ is Hermitian if and only if φ is Hermitian (self-adjoint) and Φ is skew-Hermitian if and only if φ is skew-Hermitian.

Chapter XII

§ 1

Remark. The pair $(\Gamma[t], t)$ is a universal (initial) object in the category of pointed algebras (A, a) , where A is an associative unital Γ -algebra with unit $e \in A$ and distinguished element $a \in A$. In this category, an arrow $\varphi : (A, a) \rightarrow (B, b)$ is a unital Γ -algebra homomorphism $\varphi : A \rightarrow B$ with $\varphi a = b$. This result follows from the existence and uniqueness of the evaluation homomorphism $\Gamma[t] \rightarrow A$.

Identifying the scalars $\alpha \in \Gamma$ with the elements $\alpha \cdot e \in A$, this result shows that $\Gamma[t]$ is universal as an algebra obtained by adjoining a single element to Γ , in the sense that every such algebra can be obtained from $\Gamma[t]$ in a unique way through evaluation.

Exercise (2). Define the linear mapping $\int : \Gamma[t] \rightarrow \Gamma[t]$ by

$$\int t^p = \frac{t^{p+1}}{p+1} \quad (p \geq 0)$$

Then $\int$ is homogeneous (of degree 1) with $d \circ \int = \iota$, and is unique with these properties.

Proof. If $\int$ satisfies the properties, then we must have

$$\int t^p = \frac{t^{p+1}}{p+1} + \alpha_p \quad (\alpha_p \in \Gamma)$$

since $d \circ \int = \iota$. But $\alpha_p = 0$ for all p since $\int$ is homogeneous. Finally, $\int$ is uniquely determined on $\Gamma[t]$ since $\int$ is linear and the t^p ($p \geq 0$) form a basis of $\Gamma[t]$. $\square$

Exercise (3). If $\int : \Gamma[t] \rightarrow \Gamma[t]$ is the integration operator, then $\int \circ d = \iota - \varrho$, where $\varrho : \Gamma[t] \rightarrow \Gamma$ is the scalar projection $\varrho f = f(0)$. It follows that

$$\int f g' = f g - \varrho(f g) - \int g f'$$

Proof. The first part is easily verified on the basis t^p ($p \geq 0$). The second part then follows since

$$f g - \varrho(f g) = \int (f g)' = \int (f' g + f g') = \int f' g + \int f g' \quad \square$$

§ 3

Remark. If $A = 0$, then the minimum polynomial of $a = 0$ is $\mu = 1$.

Remark. In the proof of Proposition I, to see why the elements $1, \bar{t}, \dots, \bar{t}^{r-1}$ form a basis of $\Gamma[t]/I_\mu$, first observe that every element of $\Gamma[t]/I_\mu$ is of the form $f(\bar{t})$ for some $f \in \Gamma[t]$. Write $f = g\mu + h$ where $h = 0$ or $\deg h < r$. Then

$$f(\bar{t}) = g(\bar{t})\mu(\bar{t}) + h(\bar{t}) = h(\bar{t})$$

since $\mu(\bar{t}) = 0$, and clearly $h(\bar{t})$ is in the span of $1, \bar{t}, \dots, \bar{t}^{r-1}$.

On the other hand, suppose

$$\alpha_0 + \alpha_1 \bar{t} + \dots + \alpha_{r-1} \bar{t}^{r-1} = 0$$

Define $f = \alpha_0 + \dots + \alpha_{r-1} t^{r-1}$. Then $f = 0$ or $\deg f < r$. But $f \in \ker \pi = I_\mu$, so $\mu \mid f$, which forces $f = 0$ and $\alpha_0 = \dots = \alpha_{r-1} = 0$. It follows that the elements $1, \bar{t}, \dots, \bar{t}^{r-1}$ are linearly independent.

Chapter XIII

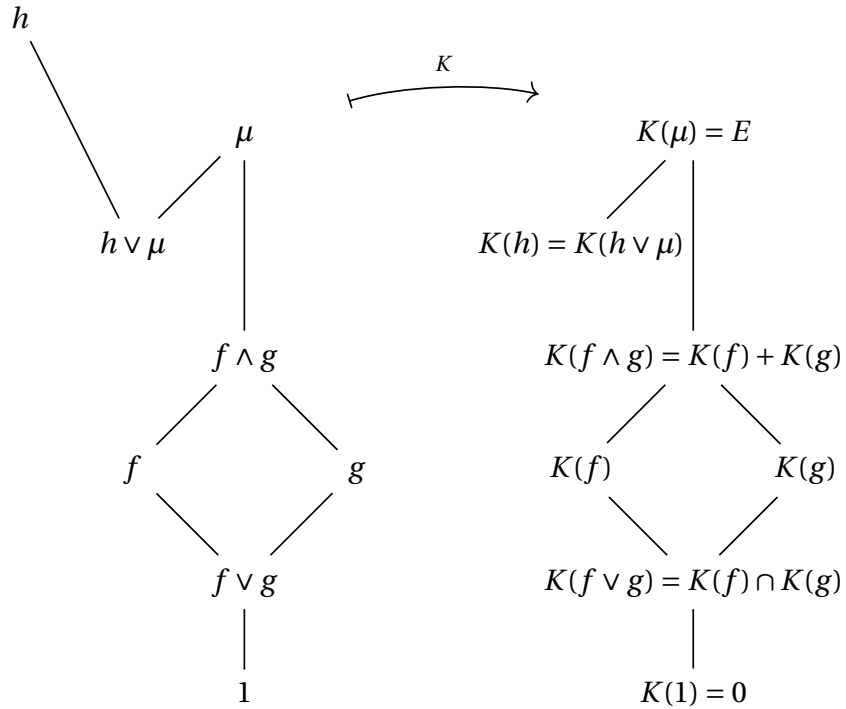
Remark. In parts of this chapter, it is implicitly assumed that the vector space E has dimension $n \geq 1$.

§ 1

In the following remarks, $f \in \Gamma[t]$.

Remark. Why is it natural to apply polynomials to a linear transformation? If we think of a linear transformation $\varphi : E \rightarrow E$ as being adjoined to the scalar transformations in $A(E; E)$, then we know that the structure of the resulting subalgebra is obtained from that of $\Gamma[t]$ in a unique way through evaluation $f \mapsto f(\varphi)$, by the universal property of $\Gamma[t]$.²³ This fact makes it natural to apply polynomials to φ in order to study the structure of φ .

Remark. The results of this section are summarized in the following diagram, which shows that K is a homomorphism from the lattice of polynomials in $\Gamma[t]$ under divisibility to the lattice of subspaces of E under inclusion:



²³See the remark in chapter XII, § 1 above.

Recall from chapter XII that $f \vee g$ denotes the greatest common divisor of f and g , while $f \wedge g$ denotes the least common multiple of f and g —notation inspired by the lattice of *ideals*.²⁴

Remark. If $\varphi : E \rightarrow E$ is linear and E_1 is stable under φ , then E_1 is also stable under $f(\varphi)$ and

$$f(\varphi)_{E_1} = f(\varphi_{E_1})$$

Dually,

$$\overline{f(\varphi)} = f(\overline{\varphi})$$

Proof. Stabilization of E_1 is preserved by addition, scalar multiplication, and composition, so is preserved by f . It follows that $f(\varphi)_{E_1}$ and $f(\varphi_{E_1})$ are both transformations of E_1 with the same action, so are equal. Finally,

$$\overline{f(\varphi)}(\overline{x}) = \overline{f(\varphi)(x)} = f(\overline{\varphi})(\overline{x}) \quad \square$$

Remark. It follows that if $K_1(f) = \ker f(\varphi_{E_1})$, then

$$K_1(f) = K(f) \cap E_1$$

Dually, if $\overline{K}(f) = \ker f(\overline{\varphi})$, then

$$\overline{K}(f) \supseteq \overline{K(f)}$$

The reverse inclusion does not hold in general. For a counterexample, consider $f = t$ with $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $\varphi(x, y) = (y, 0)$ and $E_1 = \mathbb{R} \times 0$.

Remark. If $\varphi_1 : E_1 \rightarrow E_1$ and $\varphi_2 : E_2 \rightarrow E_2$ are linear, then

$$f(\varphi_1 \oplus \varphi_2) = f(\varphi_1) \oplus f(\varphi_2)$$

Proof. Write $\varphi = \varphi_1 \oplus \varphi_2$, so $\varphi_{E_1} = \varphi_1$ and $\varphi_{E_2} = \varphi_2$. Then

$$f(\varphi) = f(\varphi)_{E_1} \oplus f(\varphi)_{E_2} = f(\varphi_1) \oplus f(\varphi_2) \quad \square$$

Remark. It follows that

$$r(f(\varphi)) = r(f(\varphi_1)) + r(f(\varphi_2))$$

where $r(\psi)$ denotes the rank of ψ .

²⁴See also problem 11.

Remark. If E, E^* are dual spaces and $\varphi : E \rightarrow E$ and $\varphi^* : E^* \leftarrow E^*$ are dual linear transformations, then

$$f(\varphi)^* = f(\varphi^*)$$

Proof. By the results of subsection 2.25. □

Remark. It follows that $f(\varphi) = 0$ if and only if $f(\varphi^*) = 0$, so $\mu_{\varphi^*} = \mu_{\varphi}$.

Remark. In particular, if E is an inner product space or a unitary space, then

$$\widetilde{f(\varphi)} = \overline{f(\tilde{\varphi})}$$

where $\overline{f}$ is obtained from f by conjugating the coefficients.²⁵

Proof. Let $\overline{E}$ be the conjugate space and $\kappa : E \rightarrow \overline{E}$ the conjugate-linear identity map. Then

$$\widetilde{f(\varphi)} = \kappa^{-1} f(\varphi)^* \kappa = \kappa^{-1} f(\varphi^*) \kappa = \overline{f}(\kappa^{-1} \varphi^* \kappa) = \overline{f}(\tilde{\varphi}) \quad \square$$

Remark. It follows that $f(\varphi) = 0$ if and only if $\overline{f}(\tilde{\varphi}) = 0$, so $\mu_{\tilde{\varphi}} = \overline{\mu_{\varphi}}$.

Remark. If $\varphi : E \rightarrow E$ and $\psi : F \rightarrow F$ are linear with

$$\psi = \alpha \circ \varphi \circ \alpha^{-1}$$

where $\alpha : E \cong F$, then

$$f(\psi) = \alpha \circ f(\varphi) \circ \alpha^{-1}$$

Proof. Conjugation by α is an algebra isomorphism $A(E; E) \cong A(F; F)$. □

Remark. It follows that $f(\psi) = 0$ if and only if $f(\varphi) = 0$, so $\mu_{\psi} = \mu_{\varphi}$.

Remark. In particular if $E = E_1 \oplus E_2$ and $\varphi : E \rightarrow E$ is given by $\varphi = \varphi_1 \oplus \varphi_2$ where $\varphi_1 : E_1 \rightarrow E_1$ and $\varphi_2 : E_2 \rightarrow E_2$, then $\mu_{\varphi_1} = \mu_{\overline{\varphi}}$ where $\overline{\varphi} : E/E_2 \rightarrow E/E_2$ is the induced transformation. Dually, $\mu_{\varphi_2} = \mu_{\overline{\varphi}}$ where $\overline{\varphi} : E/E_1 \rightarrow E/E_1$ is induced.²⁶

Remark. If $\varphi : E \rightarrow E$ is linear, then relative to a fixed basis of E

$$M(f(\varphi)) = f(M(\varphi))$$

Proof. M is an algebra isomorphism. □

²⁵In the case of an inner product space, $\overline{f} = f$.

²⁶See the conjugacy diagrams in chapter II, § 4 above.

Remark. Let $\varphi : E \rightarrow E$ be linear. If $\mu = \mu_\varphi = fg$ and

$$E = K(f) \oplus E_1 \quad (1)$$

where E_1 is stable under φ , then f and g are relatively prime and $E_1 = K(g)$.

Proof. We have

$$g(\varphi)E_1 \subseteq K(f) \cap E_1 = 0$$

so

$$E_1 \subseteq K(g) \quad (2)$$

It follows from (1) and (2) that $E = K(f \wedge g)$, so $\mu \mid f \wedge g$. This implies $fg = f \wedge g$ and therefore f and g are relatively prime. Now

$$E = K(f) \oplus K(g) \quad (3)$$

and it follows from (1), (2), and (3) that $E_1 = K(g)$. $\square$

Remark. This result is a converse of Corollary I to Proposition II. The proof is essentially the same as those of Proposition I and Theorem V in § 6.

Exercise (2). Consider the linear transformations $\varphi, \psi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$\varphi(x, y) = (y, 0) \quad \text{and} \quad \psi(x, y) = (0, y)$$

Then $\varphi\psi = \varphi$ while $\psi\varphi = 0$, so

$$\mu_{\varphi\psi} = t^2 \neq t = \mu_{\psi\varphi}$$

Exercise (3). If $\varphi_1 : E_1 \rightarrow E_1$ and $\varphi_2 : E_2 \rightarrow E_2$ are linear, then

$$\mu_{\varphi_1 \oplus \varphi_2} = \mu_{\varphi_1} \wedge \mu_{\varphi_2}$$

Proof. Write $\varphi = \varphi_1 \oplus \varphi_2$. By a remark above, $f(\varphi) = f(\varphi_1) \oplus f(\varphi_2)$, so $f(\varphi) = 0$ if and only if $f(\varphi_1) = 0$ and $f(\varphi_2) = 0$. It follows that μ_φ is the least common multiple of μ_{φ_1} and μ_{φ_2} . $\square$

Exercise (5). If $\varphi : E \rightarrow E$ stabilizes E_1 , and $\varphi_1 : E_1 \rightarrow E_1$ and $\bar{\varphi} : E/E_1 \rightarrow E/E_1$ are the induced transformations, then

$$\bar{\mu} \wedge \mu_1 \mid \mu \mid \bar{\mu}\mu_1$$

where $\mu = \mu_\varphi$, $\mu_1 = \mu_{\varphi_1}$, and $\bar{\mu} = \mu_{\bar{\varphi}}$.

Proof. By a remark above, $\mu(\varphi_1) = 0$ and $\mu(\overline{\varphi}) = 0$, so $\mu_1 \mid \mu$ and $\overline{\mu} \mid \mu$. On the other hand,

$$\overline{\mu}\mu_1(\varphi)(x) = \mu_1(\varphi)(\overline{\mu}(\varphi)(x))$$

But $\overline{\mu}(\varphi)(x)$ is in E_1 since

$$\overline{\overline{\mu}(\varphi)(x)} = \overline{\overline{\mu}(\varphi)(\overline{x})} = \overline{\mu(\overline{\varphi})(\overline{x})} = 0$$

so

$$\mu_1(\varphi)(\overline{\mu}(\varphi)(x)) = \mu_1(\varphi_1)(\overline{\mu}(\varphi)(x)) = 0$$

It follows that $\overline{\mu}\mu_1(\varphi) = 0$, so $\mu \mid \overline{\mu}\mu_1$. □

Exercise (6). The minimum polynomial μ of $\varphi : E \rightarrow E$ can be constructed in the following way: select $x_1 \in E$ and let k_1 be least such that the vectors

$$x_1, \varphi x_1, \dots, \varphi^{k_1} x_1 \tag{1}$$

are linearly dependent, with

$$\lambda_0 x_1 + \lambda_1 \varphi x_1 + \dots + \lambda_{k_1} \varphi^{k_1} x_1 = 0$$

where the λ_i are not all zero, so in particular $\lambda_{k_1} \neq 0$. Define the polynomial

$$f_1 = \lambda_0 + \lambda_1 t + \dots + \lambda_{k_1} t^{k_1}$$

If the vectors (1) do not span E , select $x_2 \in E$ not in their span and construct the corresponding polynomial f_2 . Repeat this procedure until E is exhausted. Then μ is the (monic) least common multiple of the f_i .

Proof. Let E_i be the span of the vectors

$$x_i, \varphi x_i, \dots, \varphi^{k_i-1} x_i \tag{2}$$

By construction, (2) is a basis of E_i with $\varphi^{k_i} x_i \in E_i$, and we have the direct sum $E = \sum_i E_i$. The minimum polynomial of the restriction $\varphi_i : E_i \rightarrow E_i$ is just $\lambda_{k_i}^{-1} f_i$, so the result follows from problem 3. □

Remark. The subspaces E_i are cyclic, so this procedure just produces a cyclic decomposition of E , from which the minimum polynomial is easily computed.²⁷

²⁷See Proposition IV and its corollary in § 3.

Exercise (8). If E is an inner product space and $\varphi : E \rightarrow E$ is a rotation, then the coefficients α_i of the minimum polynomial μ of φ satisfy

$$\alpha_i = \varepsilon \alpha_{k-i} \quad k = \deg \mu$$

where $\varepsilon = \pm 1$.

Proof. Since $\tilde{\varphi} = \varphi^{-1}$, we have $\mu_{\varphi^{-1}} = \mu$ by a remark above, so

$$\alpha_0 + \alpha_1 \varphi^{-1} + \cdots + \alpha_k \varphi^{-k} = \mu(\varphi^{-1}) = 0 \quad (1)$$

Multiplying both sides of (1) by φ^k yields

$$\alpha_0 \varphi^k + \alpha_1 \varphi^{k-1} + \cdots + \alpha_k = \mathfrak{u}(\varphi) = 0 \quad (2)$$

where $\mathfrak{u} = \sum_i \alpha_{k-i} t^i$. It follows from (2) that $\mu \mid \mathfrak{u}$, and in fact $\alpha_0 \mu = \mathfrak{u}$. Since $\alpha_0^2 = \alpha_k = 1$, the result holds with $\varepsilon = \alpha_0$. $\square$

Exercise (9). The minimum polynomial μ of a self-adjoint transformation of a unitary space has real coefficients.

Proof. By a remark above, $\bar{\mu} = \mu$. $\square$

Exercise (12). If $\varphi : E \rightarrow E$ is regular, then $\varphi^{-1} = f(\varphi)$ for some $f \in \Gamma[t]$.

Proof. Let μ be the minimum polynomial of φ and $\alpha = \mu(0)$. Since φ is regular, 0 is not an eigenvalue of φ , so $\alpha \neq 0$. Write

$$\alpha^{-1} \mu = 1 - t f \quad f \in \Gamma[t] \quad (1)$$

Evaluating both sides of (1) at φ yields $\varphi f(\varphi) = \iota$ and therefore $\varphi^{-1} = f(\varphi)$. $\square$

§ 2

Remark. If the minimum polynomial of $\varphi : E \rightarrow E$ has the prime decomposition

$$\mu = f_1^{k_1} \cdots f_r^{k_r}$$

and E has the generalized eigenspace decomposition

$$E = E_1 \oplus \cdots \oplus E_r$$

define

$$g_i = f_1^{k_1} \cdots \widehat{f_i^{k_i}} \cdots f_r^{k_r}$$

and

$$F_i = E_1 \oplus \cdots \oplus \widehat{E_i} \oplus \cdots \oplus E_r$$

Then $f_i^{k_i}(\varphi)$ kills E_i and induces an isomorphism on F_i , while $g_i(\varphi)$ kills F_i and induces an isomorphism on E_i .

More generally if $K \subseteq \{1, \dots, r\}$, define

$$f_K = \prod_{i \in K} f_i^{k_i}$$

and

$$E_K = \sum_{i \in K} E_i \quad E_{\widehat{K}} = \sum_{i \notin K} E_i$$

Then $f_K(\varphi)$ kills E_K and induces an isomorphism on $E_{\widehat{K}}$.

Remark. In Proposition I, if π_i and π_i^* are the projection operators associated with the generalized eigenspace decompositions in E and E^* , then

$$(\pi_i)^* = \pi_i^*$$

Proof. From (13.24), it follows that

$$\langle \pi_i^* x^*, x \rangle = \langle \pi_i^* x^*, \pi_i x \rangle = \langle x^*, \pi_i x \rangle \quad \square$$

Remark. If $g_i \in \Gamma[t]$ with $g_i(\varphi) = \pi_i$ (subsection 13.5), then

$$g_i(\varphi^*) = g_i(\varphi)^* = (\pi_i)^* = \pi_i^*$$

Alternatively, the fact that $g_i(\varphi^*) = \pi_i^*$ follows from the fact that g_i depends only on the common minimum polynomial of φ^* and φ , and the argument above can be run in reverse to obtain (13.24).²⁸

Exercise (1). The minimum polynomial μ of $\varphi : E \rightarrow E$ is completely reducible if and only if every nonzero subspace of E stable under φ contains an eigenvector of φ .

²⁸See problems 7 and 8.

Proof. If $F \neq 0$ is stable, then by (13.22)

$$F = \sum_i F \cap E_i$$

where the E_i are the generalized eigenspaces of φ . Fix i with $F \cap E_i \neq 0$. If μ is completely reducible, then $E_i = K((t - \lambda)^k)$ for some $\lambda \in \Gamma$ and $k \geq 1$. Choose $x \in F \cap E_i$ with $x \neq 0$ and let $1 \leq l \leq k$ be least with $(\varphi - \lambda t)^l x = 0$. Then it follows that $(\varphi - \lambda t)^{l-1} x \in F$ is an eigenvector.

Conversely, suppose f is monic irreducible with $\deg f \geq 1$ and $f \mid \mu$. Let $F = K(f)$. Then $F \neq 0$ is stable and so contains an eigenvector by assumption. If λ is the associated eigenvalue, then $(t - \lambda) \mid \mu_f = f$, so $f = t - \lambda$. It follows that μ is completely reducible. $\square$

Remark. This is true in particular if Γ is algebraically closed (for example $\Gamma = \mathbb{C}$).

Exercise (2). If the minimum polynomial μ of $\varphi : E \rightarrow E$ is completely reducible, then there is a basis of E with respect to which the matrix of φ is triangular:

$$\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ * & & \lambda_n \end{pmatrix}$$

Proof. By induction on $\dim E$.

Since $E \neq 0$ is stable, there is an eigenvector x_1 with $\varphi x_1 = \lambda_1 x_1$. Let $E_1 = \langle x_1 \rangle$ and consider $\bar{\varphi} : E/E_1 \rightarrow E/E_1$ with minimum polynomial $\bar{\mu}$. Since $\bar{\mu} \mid \mu$, $\bar{\mu}$ is completely reducible, and by induction there is a basis $\bar{x}_2, \dots, \bar{x}_n$ of E/E_1 with $\bar{\varphi}(\bar{x}_k) \in \langle \bar{x}_2, \dots, \bar{x}_k \rangle$ for $2 \leq k \leq n$. It follows that $x_1, \dots, x_n$ is a basis of E with $\varphi x_k \in \langle x_1, \dots, x_k \rangle$ for $1 \leq k \leq n$. $\square$

Remark. The diagonal entries of this matrix are just the eigenvalues of φ .

Exercise (9). Let F be a subspace of E stable under $\varphi : E \rightarrow E$ and consider the induced mappings $\varphi_F : F \rightarrow F$ and $\bar{\varphi} : E/F \rightarrow E/F$. Let $E = \sum_i E_i$ be the generalized eigenspace decomposition of E under φ .

(a) The generalized eigenspace decomposition of F under φ_F is

$$F = \sum_i F \cap E_i$$

(b) The generalized eigenspace decomposition of E/F under $\bar{\varphi}$ is

$$E/F = \sum_i E_i/F$$

Moreover, $E_i/F \cong E_i/(F \cap E_i)$.

(c) If π_i , π_i^F , and $\bar{\pi}_i$ are the projection operators in E , F , and E/F associated with the decompositions, then the diagram

$$\begin{array}{ccccc} F & \xrightarrow{j} & E & \xrightarrow{\rho} & E/F \\ \pi_i^F \downarrow & & \downarrow \pi_i & & \downarrow \bar{\pi}_i \\ F & \xrightarrow{j} & E & \xrightarrow{\rho} & E/F \end{array}$$

commutes, where j and ρ are the canonical injection and projection. Moreover,

$$\pi_i^F = (\pi_i)_F \quad \text{and} \quad \bar{\pi}_i = \overline{\pi_i}$$

so π_i^F and $\bar{\pi}_i$ are unique making the diagram commute.

(d) If $g_i \in \Gamma[t]$ with $g_i(\varphi) = \pi_i$, then

$$g_i(\varphi_F) = \pi_i^F \quad \text{and} \quad g_i(\bar{\varphi}) = \bar{\pi}_i$$

Proof. For (a), we already know from (13.22) that

$$F = \sum_i F \cap E_i \tag{1}$$

Let

$$F = \sum_i F_i \tag{2}$$

be the generalized eigenspace decomposition of F under φ_F . Since $\mu_{\varphi_F} \mid \mu_\varphi$, we may assume that $F_i = K_F(f_i^{j_i})$ and $E_i = K(f_i^{k_i})$ for some f_i with $j_i \leq k_i$. Then

$$F_i = K_F(f_i^{j_i}) = F \cap K(f_i^{j_i}) \subseteq F \cap K(f_i^{k_i}) = F \cap E_i \tag{3}$$

It follows from (1), (2), and (3) that $F_i = F \cap E_i$, establishing (a).

For (b), it follows from (1) that

$$E/F = \sum_i E_i/F \quad (4)$$

is a direct sum decomposition of E/F , and it is immediate that E_i/F is stable under $\bar{\varphi}$ since E_i is stable under φ . Let

$$E/F = \sum_i (E/F)_i \quad (5)$$

be the generalized eigenspace decomposition of E/F under $\bar{\varphi}$. Since $\mu_{\bar{\varphi}} \mid \mu_{\varphi}$, we may assume that $(E/F)_i = \bar{K}(f_i^{m_i})$ with $m_i \leq k_i$. Now

$$E_i/F = K(f_i^{k_i})/F \subseteq \bar{K}(f_i^{k_i})$$

so if μ_i denotes the minimum polynomial of $\bar{\varphi}_{E_i/F}$, then $\mu_i \mid f_i^{k_i}$. On the other hand, $\mu_i \mid \mu_{\bar{\varphi}}$, so we must have $\mu_i \mid f_i^{m_i}$ and therefore

$$E_i/F \subseteq \bar{K}(f_i^{m_i}) = (E/F)_i \quad (6)$$

It now follows from (4), (5), and (6) that $(E/F)_i = E_i/F$. Finally, the restriction to E_i of the canonical projection induces the isomorphism $E_i/F_i \cong E_i/F$.

For (c), commutativity of the left square in the diagram follows from (a) and commutativity of the right square follows from (b). The rest of (c) follows from the diagram.

Finally, (d) follows immediately from (c). $\square$

Exercise (10). If the minimum polynomial μ of $\varphi : E \rightarrow E$ is completely reducible, then $\mu \mid \chi$ where χ is the characteristic polynomial of φ .

Proof. First consider the case $\mu = (t - \lambda)^k$. By problem 2, there is a basis of E with respect to which the matrix of φ has the form

$$\begin{pmatrix} \lambda & & 0 \\ & \ddots & \\ * & & \lambda \end{pmatrix}$$

It follows immediately that $(\varphi - \lambda I)^n = 0$ where $n = \dim E$, so $\mu \mid (t - \lambda)^n$ and therefore $k \leq n$. Moreover, $\chi = (-1)^n(t - \lambda)^n$, so $\mu \mid \chi$.

In the general case, take the prime decomposition

$$\mu = (t - \lambda_1)^{k_1} \cdots (t - \lambda_r)^{k_r}$$

and let $E = \sum_i E_i$ be the corresponding generalized eigenspace decomposition. Let $\varphi_i : E_i \rightarrow E_i$ be the restriction with minimum polynomial $\mu_i = (t - \lambda_i)^{k_i}$ and characteristic polynomial χ_i . Then $\mu_i \mid \chi_i$ by the previous case, so

$$\mu = \mu_1 \cdots \mu_r \mid \chi_1 \cdots \chi_r = \chi \quad \square$$

Remark. This is just a special case of the Cayley-Hamilton theorem.²⁹

§ 3

Remark. In Proposition I, the vector a is intuitively “hard to kill”, and therefore generates a cyclic subspace that is as large as possible.³⁰

Remark. A vector space E is cyclic with generator $a \in E$ under $\varphi : E \rightarrow E$ if and only if E is the smallest subspace of E containing a and stable under φ .

Remark. To prove Proposition III, just recall from the proof of Proposition II that σ_a induces an isomorphism $\overline{\sigma}_a : \Gamma[t]/I_\mu \cong E$ defined by $\overline{\sigma}_a(f) = \sigma_a(f)$. Since $1, \bar{t}, \dots, \bar{t}^{m-1}$ is a basis of $\Gamma[t]/I_\mu$ and $\sigma_a(t^i) = \varphi^i(a)$, it follows that

$$a, \varphi(a), \dots, \varphi^{m-1}(a)$$

is a basis of E .

Remark. In the proof of the corollary to Proposition IV, to see that $\mu(\varphi) = 0$, first observe that

$$\mu(\varphi)(a_0) = \varphi^n(a_0) - \sum_{v=0}^{n-1} \alpha_v \varphi^v(a_0) = \varphi(a_{n-1}) - \sum_{v=0}^{n-1} \alpha_v a_v = 0$$

Now if $\mu(\varphi)(a_v) = 0$ for some $0 \leq v \leq n-2$, then

$$\mu(\varphi)(a_{v+1}) = \mu(\varphi)(\varphi(a_v)) = \varphi(\mu(\varphi)(a_v)) = \varphi(0) = 0$$

By induction, $\mu(\varphi)(a_v) = 0$ for all $0 \leq v \leq n-1$, so $\mu(\varphi) = 0$.

²⁹See Theorem II in § 5.

³⁰See Proposition V.

Remark. If E and E^* are dual spaces and $\varphi : E \rightarrow E$ and $\varphi^* : E^* \leftarrow E^*$ are dual transformations, then E is cyclic with respect to φ if and only if E^* is cyclic with respect to φ^* .

Proof. By Theorem I, if E is cyclic then

$$\dim E^* = \dim E = \deg \mu_\varphi = \deg \mu_{\varphi^*}$$

so E^* is cyclic. The converse follows by symmetry. $\square$

§ 4

Remark. If $E = \sum_i E_i$ is the generalized eigenspace decomposition of E and $E_i = \sum_j E_{ij}$ is a cyclic decomposition of E_i , then $E = \sum_{i,j} E_{ij}$ is an irreducible decomposition of E . Conversely, an irreducible decomposition of E yields an irreducible (and hence cyclic) decomposition of each E_i .³¹

Remark. In subsection 13.14, observe that

$$\begin{aligned} \sum_{v=0}^{p-1} \alpha_v f^{i-1}(\varphi) \varphi^v e + f(\varphi)^{i-1} \varphi^p e &= f(\varphi)^{i-1} \sum_{v=0}^p \alpha_v \varphi^v e \\ &= f(\varphi)^{i-1} f(\varphi) e \\ &= f(\varphi)^i e \end{aligned}$$

Remark. In subsection 13.17, observe from the formula for $r(\psi^j)$ that we can first determine N_k from $r(\psi^{k-1})$, then determine N_{k-1} from $r(\psi^{k-2})$ and N_k , then determine successively $N_{k-2}, \dots, N_1$.

Remark. Proposition II shows that two linear transformations are conjugate if and only if they have the same Jordan canonical matrix. The matrix is therefore a particularly simple representative for the conjugacy class.

The proof of case II can be simplified by noting that $\dim E_i = \dim F_i$ follows from the fact that $r(f_i(\varphi_i)^j) = r(f_i(\psi_i)^j)$ for all $j \geq 1$.

Remark. In the proof of Corollary II to Proposition II, it follows from the results of subsection 13.20 that the common characteristic polynomial of the induced restrictions of φ and ψ to E_i and F_i is just $f_i^{m_i}$.³²

³¹ See the remarks at the end of subsection 13.17.

³² See the remarks in § 5 below.

Exercise (5). If $\varphi : E \rightarrow E$ stabilizes E_1 , and $\varphi_1 : E_1 \rightarrow E_1$ and $\bar{\varphi} : E/E_1 \rightarrow E/E_1$ are the induced transformations, then E is cyclic if and only if (i) E_1 is cyclic, (ii) E/E_1 is cyclic, and (iii) $\mu = \mu_1 \bar{\mu}$, where $\mu = \mu_\varphi$, $\mu_1 = \mu_{\varphi_1}$, and $\bar{\mu} = \mu_{\bar{\varphi}}$.

Proof. If E is cyclic, then $\sigma_a : \Gamma[t] \rightarrow E$ is surjective for some $a \in E$. It follows that $\pi\sigma_a : \Gamma[t] \rightarrow E/E_1$ is also surjective, where $\pi : E \rightarrow E/E_1$ is the canonical projection. But

$$(\pi\sigma_a)(f) = \pi(f(\varphi)(a)) = f(\bar{\varphi})(\bar{a}) = \sigma_{\bar{a}}(f)$$

so $\sigma_{\bar{a}} = \pi\sigma_a$ is surjective, and therefore E/E_1 is cyclic. Now

$$\deg \mu_1 \leq \dim E_1 = \dim E - \dim E/E_1 = \deg \mu - \deg \bar{\mu} \leq \deg \mu_1$$

since $\mu \mid \mu_1 \bar{\mu}$. It follows that $\dim E_1 = \deg \mu_1$ so E_1 is cyclic, and $\deg \mu = \deg \mu_1 \bar{\mu}$ so $\mu = \mu_1 \bar{\mu}$.

Conversely, if E_1 and E/E_1 are cyclic and $\mu = \mu_1 \bar{\mu}$, then

$$\dim E = \dim E_1 + \dim E/E_1 = \deg \mu_1 + \deg \bar{\mu} = \deg \mu_1 \bar{\mu} = \deg \mu$$

so E is cyclic. □

Remark. If $\varphi = \varphi_1 \oplus \varphi_2$ with $\mu_2 = \mu_{\varphi_2}$, then $\bar{\mu} = \mu_2$, so if E is cyclic it follows that $\mu = \mu_1 \mu_2$. Since also $\mu = \mu_1 \wedge \mu_2$, this implies that μ_1 and μ_2 are relatively prime.

Exercise (6). Let $E = F_1 \oplus \cdots \oplus F_s$ and $\varphi = \varphi_1 \oplus \cdots \oplus \varphi_s$ where $\varphi_j : F_j \rightarrow F_j$ with $\mu = \mu_\varphi$ and $\mu_j = \mu_{\varphi_j}$.

- (a) E is cyclic if and only if each F_j is cyclic and the μ_j are relatively prime.
- (b) If E is cyclic, then each F_j is a sum of generalized eigenspaces for φ .
- (c) If E is cyclic and

$$a = a_1 + \cdots + a_s \quad a_j \in F_j$$

is any vector in E , then a generates E if and only if each a_j generates F_j .

Proof. For (a), if E is cyclic then by problem 5 each F_j is cyclic and (by induction on s)

$$\mu_1 \wedge \cdots \wedge \mu_s = \mu = \mu_1 \cdots \mu_s$$

so the μ_j are relatively prime. Conversely, if each F_j is cyclic and the μ_j are relatively prime, then

$$\begin{aligned}\dim E &= \dim F_1 + \cdots + \dim F_s \\ &= \deg \mu_1 + \cdots + \deg \mu_s \\ &= \deg \mu_1 \cdots \mu_s \\ &= \deg \mu_1 \wedge \cdots \wedge \mu_s \\ &= \deg \mu\end{aligned}$$

so E is cyclic.

For (b), since $\mu = \mu_1 \cdots \mu_s$ and the μ_j are relatively prime, the generalized eigenspaces of φ_j in F_j are generalized eigenspaces of φ in E . It follows that the projection operator associated with F_j is a polynomial $g_j(\varphi)$ in φ .

For (c), if a generates E and $x \in F_j$, then there is $f \in \Gamma[t]$ with

$$x = f(\varphi)a = f(\varphi_1)a_1 + \cdots + f(\varphi_s)a_s = f(\varphi_j)a_j$$

so a_j generates F_j . Conversely, if each a_j generates F_j and

$$x = x_1 + \cdots + x_s \quad x_j \in F_j$$

is a vector in E , there are $f_j \in \Gamma[t]$ with $x_j = f_j(\varphi_j)a_j$. Define $f = \sum f_j g_j$. Then

$$f(\varphi)a = \sum f_j(\varphi)g_j(\varphi)a = \sum f_j(\varphi)a_j = \sum x_j = x$$

so a generates E . □

Exercise (7). If $\varphi : E \rightarrow E$ stabilizes F , and $\varphi_F : F \rightarrow F$ and $\overline{\varphi} : E/F \rightarrow E/F$ are the induced transformations, then E is irreducible if and only if (i) E/F is irreducible, (ii) F is irreducible, and (iii) $\mu_{\varphi_F} = f^k$, $\mu_{\overline{\varphi}} = f^l$, and $\mu_{\varphi} = f^{k+l}$ where f is irreducible.

Proof. By Theorem I and problem 5. □

Exercise (8). Let E be irreducible with respect to φ , with $\mu_{\varphi} = f^k$ (f irreducible).

- (a) The k subspaces $K(f), \dots, K(f^k)$ are the only nontrivial stable subspaces of E .
- (b) $\text{Im } f(\varphi)^m = K(f^{k-m})$ for $0 \leq m \leq k$.

Proof. We may assume that f is monic with $\deg f = p$. For (a), we know that

$$0 = K(1) \subset K(f) \subset \cdots \subset K(f^k) = E$$

are stable subspaces of E . If $F \neq 0$ is a stable subspace of E , then $\mu_F = f^m$ for some $1 \leq m \leq k$ and hence $F \subseteq K(f^m)$. Since F and $K(f^m)$ are both cyclic,

$$\dim F = pm = \dim K(f^m)$$

so $F = K(f^m)$.

For (b), $\text{Im } f(\varphi)^m$ is stable and

$$\begin{aligned} \dim \text{Im } f(\varphi)^m &= \dim E - \dim \ker f(\varphi)^m \\ &= pk - pm \\ &= p(k - m) \\ &= \dim \ker f(\varphi)^{k-m} \end{aligned}$$

so we must have $\text{Im } f(\varphi)^m = K(f^{k-m})$ by (a). □

§ 5

Remark. The proof of Lemma I is essentially the same as the proof that the product of cyclic groups of orders m and n is cyclic if and only if m and n are relatively prime.

In the proof, if Y_i is the cyclic subspace generated by y_i , then $Y_i = Y \cap F_i$. The contradiction is that $Y = Y_1 \oplus Y_2$ is a cyclic subspace but the orders of y_1 and y_2 under $f_1(\varphi)$ are not relatively prime.³³

Remark. The minimum polynomial μ and the characteristic polynomial χ of a linear transformation $\varphi : E \rightarrow E$ have the same prime factors.

Proof. Let $E = \sum_i F_i$ be a cyclic decomposition of E and $\varphi_i : F_i \rightarrow F_i$ the induced restriction of φ with minimum polynomial μ_i and characteristic polynomial χ_i . Then

$$\mu = \bigwedge \mu_i$$

and

$$\chi = \prod \chi_i = \pm \prod \mu_i$$

by the cyclic case of the Cayley-Hamilton theorem (Lemma II). It follows that the prime factors of μ and χ are the prime factors of the μ_i .³⁴ □

³³See also problem 6 in § 4 above.

³⁴It also follows from this proof that $\mu \mid \chi$.

Remark. If $\varphi : E \rightarrow E$, let

$$\chi = f_1^{m_1} \cdots f_r^{m_r} \quad \text{and} \quad \mu = f_1^{k_1} \cdots f_r^{k_r} \quad (k_i \leq m_i)$$

be the prime decompositions of the characteristic and minimum polynomials, by the Cayley-Hamilton theorem. If

$$E = E_1 \oplus \cdots \oplus E_r$$

is the generalized eigenspace decomposition of E and $\varphi_i : E_i \rightarrow E_i$ the induced restriction of φ with minimum polynomial μ_i and characteristic polynomial χ_i , then

$$\chi_i = f_i^{m_i} \quad \text{and} \quad \mu_i = f_i^{k_i}$$

Proof. We already know that $\mu_i = f_i^{k_i}$, and it follows from the previous remark that f_i does not divide χ_j if $i \neq j$. On the other hand $\chi = \chi_1 \cdots \chi_r$, so we must have $\chi_i = f_i^{m_i}$. $\square$

Remark. This result shows that the algebraic multiplicity of an eigenvalue λ (the number of times it occurs as a root of the characteristic polynomial) is the dimension of the generalized eigenspace corresponding to λ , which is greater than or equal to the geometric multiplicity (the dimension of the eigenspace).

Remark. In subsection 13.20, we see that

$$\mu_\varphi(t) = \mu_{\varphi - \lambda I}(t - \lambda)$$

Substituting $t + \lambda$ for t , we obtain

$$\mu_{\varphi - \lambda I}(t) = \mu_\varphi(t + \lambda)$$

In particular, it follows that $\mu_\varphi(\lambda)$ is the constant term of $\mu_{\varphi - \lambda I}$, and $\mu_{\varphi - \lambda I}(-\lambda)$ is the constant term of μ_φ .

Remark. In subsections 13.21 and 13.22, we have

$$C^2(\varphi) = \Gamma(\varphi) \subseteq C(\varphi) = C(\Gamma(\varphi)) = C^3(\varphi)$$

Exercise (1). If $\varphi : E \rightarrow E$, let $\mu = f_1^{k_1} \cdots f_r^{k_r}$ be the prime decomposition of the minimum polynomial with $p_i = \deg f_i$ and $E = \sum_i E_i$ the generalized eigenspace

decomposition. Denote by N_{ij} the number of irreducible subspaces of E_i of dimension $p_i j$ for $1 \leq j \leq k_i$ and set

$$l_i = \sum_{j=1}^{k_i} j N_{ij}$$

Then the characteristic polynomial of φ is $\chi = f_1^{l_1} \cdots f_r^{l_r}$.

Proof. By a remark above, we know that the characteristic polynomial of the induced restriction $\varphi_i : E_i \rightarrow E_i$ is $\chi_i = f_i^{m_i}$ where

$$p_i m_i = \deg \chi_i = \dim E_i = \sum_{j=1}^{k_i} p_i j N_{ij} = p_i l_i$$

so $m_i = l_i$ and

$$\chi = \chi_1 \cdots \chi_r = f_1^{l_1} \cdots f_r^{l_r} \quad \square$$

Exercise (2). E is cyclic with respect to φ if and only if $\chi_\varphi = \pm \mu_\varphi$.

Proof. The forward direction follows from Lemma II, and the reverse from

$$\dim E = \deg \chi_\varphi = \deg \mu_\varphi \quad \square$$

Exercise (3). If $\varphi : E \rightarrow E$, and $E = \sum_j F_j$ is a decomposition of E as a direct sum of stable subspaces F_j , then each F_j is a sum of generalized eigenspaces for φ if and only if each F_j is stable under $C(\varphi)$.

Proof. If each F_j is a sum of generalized eigenspaces, then by Theorem I the projection operators ρ_j are in $\Gamma(\varphi)$. If $\psi \in C(\varphi)$, then ψ commutes with ρ_j , so

$$\psi F_j = \psi \rho_j F_j = \rho_j \psi F_j \subseteq F_j$$

Conversely, if each F_j is stable under $C(\varphi)$ and $\psi \in C(\varphi)$, then for $x = \sum_j x_j$ with $x_j \in F_j$, we have

$$\rho_j \psi x = \psi x_j = \psi \rho_j x$$

so ρ_j commutes with ψ . Since ψ was arbitrary, $\rho_j \in C^2(\varphi)$. But $C^2(\varphi) = \Gamma(\varphi)$ by Theorem III, so $\rho_j \in \Gamma(\varphi)$ and each F_j is a sum of generalized eigenspaces by Theorem I. $\square$

Exercise (4). Let $\varphi : E \rightarrow E$ and let P be the set of projection operators in $C(\varphi)$.

(a) $P = \{0, \iota\}$ if and only if E is irreducible.

(b) $P \subseteq C^2(\varphi)$ if and only if E is cyclic.

Proof. By Proposition I, P consists of the projection operators for stable direct sum decompositions of E . For (a), the condition is therefore equivalent to the existence of only trivial stable decompositions. For (b), since $C^2(\varphi) = \Gamma(\varphi)$ by Theorem III, the condition is equivalent to the existence of only generalized eigenspace decompositions by Theorem I, which is equivalent to E being cyclic by problem 6 in § 4 above. $\square$

Exercise (5). Let $\varphi : E \rightarrow E$.

(a) $C^3(\varphi) = C(\varphi)$

(b) $C^2(\varphi) = C(\varphi)$ if and only if E is cyclic.

Proof. By a remark above and problem 4. $\square$

§ 6

Remark. If $\varphi : E \rightarrow E$ is nilpotent of degree k and E is cyclic with respect to φ , then a generates E if and only if k is least with $\varphi^k a = 0$, which is true if and only if $\varphi^{k-1} a \neq 0$.

Proof. If a generates E , then the vectors

$$a, \varphi a, \dots, \varphi^{k-1} a \quad (1)$$

form a basis of E and are therefore nonzero, while $\varphi^k a = 0$. Conversely, if k is least with $\varphi^k a = 0$ and the vectors (1) are linearly dependent, there are scalars $\alpha_0, \dots, \alpha_{k-1}$ not all zero with

$$\alpha_0 a + \alpha_1 \varphi a + \dots + \alpha_{k-1} \varphi^{k-1} a = 0 \quad (2)$$

Let j be least with $\alpha_j \neq 0$. Applying φ^{k-j-1} to (2) yields

$$\alpha_j \varphi^{k-1} a = 0$$

which is a contradiction. $\square$

Remark. Semisimplicity of self-adjoint, skew, and isometric transformations in real and complex inner product spaces ultimately follows from the fact that if a transformation stabilizes a subspace, then the dual transformation stabilizes the orthogonal complement of the subspace.

Remark. In the proof of Theorem I, Proposition I is used as a lemma. On the other hand, Proposition I follows as a corollary of Theorem I by the remarks at the end of subsection 13.23.

Remark. In the proof of Theorem IV, in the reduction of the infinite case to the finite case, let S be a finite set of generators for A . Since S is finite, there is a finite set $T \subseteq \{\varphi_\alpha\}$ such that S is contained in the subalgebra $A' \subseteq A$ generated by T . By the finite case, T is semisimple, so A' is semisimple. It follows from Theorem V that S consists of semisimple transformations. By the finite case again, S is semisimple, so A is semisimple, so $\{\varphi_\alpha\}$ is semisimple.

Exercise (1). If $\varphi : E \rightarrow E$ is nilpotent and N_λ is the number of subspaces of dimension λ in an irreducible decomposition of E , then

$$\dim \ker \varphi = \sum_{\lambda} N_{\lambda}$$

Proof. From subsection 13.17,

$$r(\varphi) = \sum_{\lambda} (\lambda - 1) N_{\lambda} = \dim E - \sum_{\lambda} N_{\lambda}$$

so

$$\sum_{\lambda} N_{\lambda} = \dim E - \dim \operatorname{Im} \varphi = \dim \ker \varphi \quad \square$$

Exercise (3). Let $\varphi : E \rightarrow E$ and $\varphi^* : E^* \leftarrow E^*$ be dual. If φ is nilpotent of degree k and E is cyclic with respect to φ , then φ^* is nilpotent of degree k and E^* is cyclic with respect to φ^* . If a generates E , then a^* generates E^* if and only if

$$\langle a^*, \varphi^{k-1} a \rangle \neq 0$$

Proof. We know $\mu_{\varphi^*} = \mu_{\varphi} = t^k$, so φ^* is nilpotent of degree k , and

$$\dim E^* = \dim E = k$$

so E^* is cyclic with respect to φ^* .³⁵

³⁵See the remark in § 4 above.

By a remark above, a^* generates E^* if and only if $(\varphi^*)^{k-1}a^* \neq 0$. Since

$$\langle (\varphi^*)^{k-1}a^*, \varphi^j a \rangle = \langle a^*, \varphi^{k+j-1}a \rangle = 0 \quad (1)$$

for all $j \geq 1$ and the $(\varphi^j a)_{j \geq 0}$ form a basis of E , it follows that $(\varphi^*)^{k-1}a^* = 0$ if and only if (1) holds for $j = 0$. $\square$

Exercise (4). A linear transformation is diagonalizable if and only if it is semi-simple and its minimum polynomial is completely reducible, or equivalently if its minimum polynomial is the product of distinct monic linear factors.

Proof. A linear transformation is diagonalizable if and only if its generalized eigenspace decomposition is an eigenspace decomposition, which is true if and only if the minimum polynomial has the indicated form. The rest follows from Theorem I. $\square$

Exercise (5). If $\varphi, \psi : E \rightarrow E$ are commuting diagonalizable transformations, they are simultaneously diagonalizable (that is, there exists a basis of E which diagonalizes both φ and ψ).

Proof. Let

$$\sum_i E_i = E = \sum_j F_j$$

be the generalized eigenspace decompositions for φ and ψ respectively, which are eigenspace decompositions by diagonalizability of φ and ψ . It follows from commutativity of φ and ψ and the results of subsection 13.21 that there is an induced decomposition

$$E = \sum_{i,j} E_i \cap F_j$$

A basis of E obtained by joining bases of the subspaces $E_i \cap F_j$ diagonalizes both φ and ψ . $\square$

Remark. It follows from this problem and the previous problem that if φ and ψ are commuting semisimple transformations over the field of complex numbers (which is algebraically closed), then $\varphi + \psi$ and $\varphi\psi$ are semisimple.

Exercise (6). Let E be a complex vector space and $\varphi : E \rightarrow E$. Let $E = \sum_i E_i$ be the generalized eigenspace decomposition of E and $\pi_i : E \rightarrow E$ the corresponding projection operators. Assume that the minimum polynomial of the induced transformation $\varphi_i : E_i \rightarrow E_i$ is $\mu_i = (t - \lambda_i)^{k_i}$. Then the semisimple part of φ is

$$\varphi_S = \sum_i \lambda_i \pi_i$$

Proof. Let $\iota_i : E_i \rightarrow E_i$ be the identity map and write

$$\varphi_i = (\varphi_i - \lambda_i \iota_i) + \lambda_i \iota_i$$

Note that $\varphi_i - \lambda_i \iota_i$ is nilpotent, $\lambda_i \iota_i$ is semisimple, and the two transformations commute. Since

$$\varphi = \sum_i \varphi_i = \sum_i (\varphi_i - \lambda_i \iota_i) + \sum_i \lambda_i \iota_i$$

and direct sums preserve nilpotency, semisimplicity, and commutativity, the result now follows from the uniqueness of the Jordan-Chevalley decomposition (Theorem II). $\square$

Exercise (7). Let E be an n -dimensional complex vector space and $\varphi : E \rightarrow E$ have eigenvalues $\lambda_1, \dots, \lambda_n$ (not necessarily distinct). If $f \in \Gamma[t]$, then $f(\varphi)$ has the eigenvalues $f(\lambda_1), \dots, f(\lambda_n)$.

Proof. Fix a basis of E with

$$M(\varphi) = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ * & & \lambda_n \end{pmatrix}$$

By direct computation,

$$M(f(\varphi)) = f(M(\varphi)) = \begin{pmatrix} f(\lambda_1) & & 0 \\ & \ddots & \\ * & & f(\lambda_n) \end{pmatrix}$$

from which the result follows. $\square$

Exercise (8). Consider the transformations $\varphi, \psi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$\varphi(x, y) = (y, 0) \quad \text{and} \quad \psi(x, y) = (y, x)$$

The only 1-dimensional subspace stable under φ is the line $y = 0$, while the only 1-dimensional subspaces stable under ψ are the lines $y = \pm x$, so the set $\{\varphi, \psi\}$ is semisimple but φ is not. Note that φ and ψ do not commute since, for example

$$\varphi\psi(1, 0) = (1, 0) \neq (0, 0) = \psi\varphi(1, 0)$$

Remark. Conversely, let φ be reflection in the line $y = 0$ along the line $x = 0$, and let ψ be reflection in the line $y = 0$ along the line $y = x$. Then φ and ψ are semisimple but the set $\{\varphi, \psi\}$ is not.

§ 7

Remark. In Theorems I and II, the definition of *homothetic* must be loosened slightly to allow for possibly negative scalars.

In the proof of Theorem I, to see that τ is a *proper* rotation, let $E = \sum_i F_i$ be an irreducible decomposition. The minimum polynomial μ_i of the restriction $\varphi_i : F_i \rightarrow F_i$ must be $\mu_i = t^2 + \alpha t + \beta$, so the Jordan canonical matrix of φ_i is just

$$\begin{pmatrix} 0 & 1 \\ -\beta & -\alpha \end{pmatrix}$$

and therefore $\det \varphi_i = \beta > 0$. It follows that $\det \varphi > 0$ and $\det \tau > 0$.

Exercise (2). If E is an inner product space, then $\varphi : E \rightarrow E$ is normal if and only if $\tilde{\varphi} = f(\varphi)$ for some $f \in \mathbb{R}[t]$.

Proof. If $\tilde{\varphi} = f(\varphi)$, then $\tilde{\varphi}\varphi = \varphi\tilde{\varphi}$, so φ is normal. Conversely, if φ is normal let

$$E = E_1 \oplus \cdots \oplus E_r \tag{1}$$

be the orthogonal generalized eigenspace decomposition of E and $\varphi_i : E_i \rightarrow E_i$ the induced homothetic restriction (Theorem I). Write $\varphi_i = \lambda_i \tau_i$ where $\lambda_i \in \mathbb{R}$ and τ_i is a rotation. We know $\tau_i^{-1} = f_i(\tau_i)$ for some $f_i \in \mathbb{R}[t]$.³⁶ By subsection 8.5, we have the direct sum

$$\tilde{\varphi} = \sum \tilde{\varphi}_i = \sum \lambda_i \tilde{\tau}_i = \sum \lambda_i \tau_i^{-1} = \sum \lambda_i f_i(\tau_i) \tag{2}$$

Define $g_i \in \mathbb{R}[t]$ by

$$g_i(t) = \begin{cases} 0 & \text{if } \lambda_i = 0 \\ \lambda_i f_i(\lambda_i^{-1} t) & \text{if } \lambda_i \neq 0 \end{cases}$$

Let $\pi_i = h_i(\varphi)$ be the i -th projection operator for (1), where $h_i \in \mathbb{R}[t]$. It follows from (2) that

$$\tilde{\varphi} = \sum g_i(\varphi) h_i(\varphi)$$

Setting $f = \sum g_i h_i$, we have $\tilde{\varphi} = f(\varphi)$. □

Remark. In the case of a unitary space, φ and $\tilde{\varphi}$ are both diagonalizable by the spectral theorem³⁷ and hence simultaneously diagonalizable.³⁸ By Lagrange interpolation, there is a polynomial $f \in \mathbb{C}[t]$ with $f(M(\varphi)) = M(\tilde{\varphi})$, so $f(\varphi) = \tilde{\varphi}$.

³⁶See problem 12 in § 1 above.

³⁷See subsection 11.10.

³⁸See problem 5 in § 6 above.

Multilinear Algebra

Chapter 1

§ 1

Remark. If $\lambda^1 \lambda^4 - \lambda^2 \lambda^3 = 0$, we want $\xi^1, \xi^2, \eta^1, \eta^2$ with

$$\lambda^1 = \xi^1 \eta^1 \quad \lambda^2 = \xi^1 \eta^2 \quad \lambda^3 = \xi^2 \eta^1 \quad \lambda^4 = \xi^2 \eta^2$$

If $\lambda^1 = 0$, then $\lambda^2 \lambda^3 = \lambda^1 \lambda^4 = 0$, so $\lambda^2 = 0$ or $\lambda^3 = 0$.

- If $\lambda^2 = 0$, we take $\xi^1 = 0$, $\xi^2 = 1$, $\eta^1 = \lambda^3$, and $\eta^2 = \lambda^4$.
- If $\lambda^3 = 0$, we take $\xi^1 = \lambda^2$, $\xi^2 = \lambda^4$, $\eta^1 = 0$, and $\eta^2 = 1$.

If $\lambda^1 \neq 0$ and $\lambda^2 = 0$, then $\lambda^4 = 0$ and we take $\xi^1 = \lambda^1$, $\xi^2 = \lambda^3$, $\eta^1 = 1$, and $\eta^2 = 0$.

If $\lambda^1 \neq 0$ and $\lambda^2 \neq 0$, we take $\xi^1 = 1$, $\xi^2 = \lambda^3 / \lambda^1 = \lambda^4 / \lambda^2$, $\eta^1 = \lambda^1$, and $\eta^2 = \lambda^2$.

§ 2

Remark. In the construction of the first induced bilinear map $\tilde{\varphi}$, note that for each $y \in F$ the linear map $\varphi(-, y) : E \rightarrow G$ sends E_1 into G_1 and hence induces a linear map $\bar{\varphi}(-, y) : E/E_1 \rightarrow G/G_1$ by $\bar{\varphi}(\rho x, y) = \pi \varphi(x, y)$:

$$\begin{array}{ccc} E & \xrightarrow{\varphi(-, y)} & G \\ \rho \downarrow & & \downarrow \pi \\ E/E_1 & \xrightarrow{\bar{\varphi}(-, y)} & G/G_1 \end{array}$$

Since $\bar{\varphi}(-, y)$ depends linearly on y , $\bar{\varphi}$ is bilinear and we define $\tilde{\varphi} = \bar{\varphi}$.

In the construction of the second induced bilinear map $\tilde{\varphi}$, note that the linear map $y \mapsto \bar{\varphi}(-, y)$ kills F_1 , so it factors through σ :

$$\begin{array}{ccc} F & \xrightarrow{y \mapsto \bar{\varphi}(-, y)} & L(E/E_1; G/G_1) \\ \sigma \downarrow & \nearrow \sigma y \mapsto \bar{\varphi}(-, y) & \\ F/F_1 & & \end{array}$$

This allows us to define the bilinear map $\tilde{\varphi}(\rho x, \sigma y) = \overline{\varphi}(\rho x, y) = \pi\varphi(x, y)$.

§ 3

Remark. The claim in problem 5(b) is false because it implies, for example, that any two inner products in $\mathbb{R}^2$ agree on orthogonality, which is false. The claim holds if and only if ψ preserves linear relations satisfied by φ .

Exercise (7). If E, E^* are finite-dimensional dual spaces and $\Phi: E^* \times E \rightarrow \Gamma$ is a bilinear function such that

$$\Phi(\tau^{*-1}x^*, \tau x) = \Phi(x^*, x)$$

for every pair of dual automorphisms, then there is $\lambda \in \Gamma$ such that

$$\Phi(x^*, x) = \lambda \langle x^*, x \rangle$$

Proof. Let $\varphi: E \rightarrow E$ be the linear transformation with $\Phi(x^*, x) = \langle x^*, \varphi x \rangle$. If τ is an automorphism of E , then since $(\tau^{-1})^* = (\tau^*)^{-1}$,

$$\langle x^*, \tau^{-1}\varphi\tau(x) \rangle = \langle \tau^{*-1}x^*, \varphi\tau(x) \rangle = \Phi(\tau^{*-1}x^*, \tau x) = \Phi(x^*, x) = \langle x^*, \varphi x \rangle$$

It follows that $\tau^{-1}\varphi\tau = \varphi$. Since τ was arbitrary, $\varphi = \lambda \text{id}$ for some $\lambda \in \Gamma$, from which the result follows. $\square$

§ 4

Remark. The tensor product $E \otimes F$ is a universal (initial) object in the category of “vector spaces with bilinear maps of $E \times F$ into them”. In this category, the objects are bilinear maps $E \times F \rightarrow G$, and the arrows are linear maps $G \rightarrow H$ which respect the bilinear maps:

$$\begin{array}{ccc} E \times F & \xrightarrow{\quad} & H \\ \downarrow & \nearrow & \\ G & & \end{array}$$

Every object $E \times F \rightarrow G$ in this category can be obtained from the tensor product $\otimes: E \times F \rightarrow E \otimes F$ in a unique way. This is why $\otimes$ is said to satisfy the “universal property”. This is only possible because the elements of $E \otimes F$ satisfy only those relations required to make $E \otimes F$ into a vector space and to make $\otimes$ bilinear. By category theoretic abstract nonsense, $E \otimes F$ is unique up to isomorphism.

§ 5

Remark. Lemma 1.5.1 generalizes the consequence of linear independence to tensor products other than scalar multiplication.

We provide an alternative proof. For a linear function $g \in L(F)$, consider the bilinear map $E \times F \rightarrow E$ defined by $(x, y) \mapsto g(y)x$. By the universal property of the tensor product, there is a linear map $h : E \otimes F \rightarrow E$ with $h(x \otimes y) = g(y)x$. Now

$$0 = h\left(\sum a_i \otimes b_i\right) = \sum h(a_i \otimes b_i) = \sum g(b_i)a_i$$

By linear independence of the a_i , it follows that $g(b_i) = 0$ for all i . Since g was arbitrary, it follows that $b_i = 0$ for all i .

Note $h = \iota \otimes g : E \otimes F \rightarrow E \otimes \Gamma$ (see § 16).

Remark. Lemma 1.5.2 generalizes the existence and uniqueness of a representation relative to a basis.

Remark. If $z = \sum_{i=1}^n x_i \otimes y_i$ with $x_i \in E$ and $y_i \in F$, and $\{x_1, \dots, x_m\}$ is a maximal linearly independent subset of $\{x_1, \dots, x_n\}$ of size $m \leq n$, then

$$z = \sum_{i=1}^m x_i \otimes y'_i \quad \text{where} \quad y'_i - y_i \in \langle y_{m+1}, \dots, y_n \rangle$$

Proof. Write

$$x_j = \sum_{i=1}^m \lambda_{ji} x_i \quad j = m+1, \dots, n \quad (\lambda_{ji} \in \Gamma)$$

Then

$$\begin{aligned} z &= \sum_{i=1}^m x_i \otimes y_i + \sum_{j=m+1}^n \left(\sum_{i=1}^m \lambda_{ji} x_i \right) \otimes y_j \\ &= \sum_{i=1}^m x_i \otimes y_i + \sum_{i=1}^m x_i \otimes \left(\sum_{j=m+1}^n \lambda_{ji} y_j \right) \\ &= \sum_{i=1}^m x_i \otimes \left(y_i + \sum_{j=m+1}^n \lambda_{ji} y_j \right) \end{aligned}$$

Now take $y'_i = y_i + \sum_{j=m+1}^n \lambda_{ji} y_j$. □

Remark. Lemma 1.5.3 follows from this remark, which makes clear that a tensor representation of minimal length must consist of linearly independent vectors.

§ 8

Remark. In the proof of Proposition 1.8.1, note that if f is injective, it has a left inverse $g : G \rightarrow E \otimes F$ with $g \circ f = \iota$. If $\psi : E \times F \rightarrow K$ is bilinear, there is a linear map $h : E \otimes F \rightarrow K$ with

$$\psi = h \circ \otimes = h \circ \iota \circ \otimes = h \circ g \circ f \circ \otimes = h \circ g \circ \varphi$$

So ψ factors through φ . Since ψ was arbitrary, φ satisfies $\otimes_2$.

Exercise (3). If S and T are sets, then $C(S \times T) \cong C(S) \otimes C(T)$.

Proof. By the universal property of the free space, there is a unique linear map $\varphi : C(S \times T) \rightarrow C(S) \otimes C(T)$ with $\varphi(s, t) = s \otimes t$ for $s \in S$ and $t \in T$:

$$\begin{array}{ccc} S \times T & \xrightarrow{i \times i} & C(S) \times C(T) \\ \downarrow i & & \downarrow \otimes \\ C(S \times T) & \xrightarrow{\varphi} & C(S) \otimes C(T) \end{array}$$

Observe that φ is injective since

$$0 = \varphi\left(\sum_i \lambda_i (s_i, t_i)\right) = \sum_i \lambda_i s_i \otimes t_i$$

implies $\lambda_i = 0$ by linear independence of the distinct s_i in $C(S)$ and t_i in $C(T)$ (1.5.1). Also φ is surjective since

$$\left(\sum_i \lambda_i s_i\right) \otimes \left(\sum_j \mu_j t_j\right) = \sum_{i,j} \lambda_i \mu_j s_i \otimes t_j = \sum_{i,j} \lambda_i \mu_j \varphi(s_i, t_j)$$

and the elements on the left generate $C(S) \otimes C(T)$. □

Exercise (4). If $a \otimes b \neq 0$, then $a \otimes b = a' \otimes b'$ if and only if $a' = \lambda a$ and $b' = \lambda^{-1} b$ for some $\lambda \neq 0$.

Proof. For the forward direction, note all the vectors are nonzero by bilinearity of $\otimes$. Now

$$a \otimes b + a' \otimes (-b') = 0$$

so we must have $a' = \lambda a$ for some $\lambda \neq 0$ (1.5.1). It follows that

$$a \otimes (b - \lambda b') = 0$$

so $b - \lambda b' = 0$ and $b' = \lambda^{-1} b$.

The reverse direction follows from bilinearity of $\otimes$. □

§ 10

Remark. To obtain the tensor product of E/E_1 and F/F_1 , just take the tensor product of E and F and then kill off all the products by elements of E_1 and by elements of F_1 .

§ 11

Remark. The tensor product operation is bilinear on *spaces*:

$$\left(\bigoplus_{\alpha} E_{\alpha}\right) \otimes \left(\bigoplus_{\beta} F_{\beta}\right) = \bigoplus_{\alpha, \beta} E_{\alpha} \otimes F_{\beta}$$

In particular, $E \otimes 0 = 0 = 0 \otimes E$.

§ 12

Remark. In the proof of (1.2), the idea is that $E \cong \bigoplus_{\alpha} E_{\alpha}$ and $F \cong \bigoplus_{\beta} F_{\beta}$, so

$$E \otimes F \cong \left(\bigoplus_{\alpha} E_{\alpha}\right) \otimes \left(\bigoplus_{\beta} F_{\beta}\right) = \bigoplus_{\alpha, \beta} E_{\alpha} \otimes F_{\beta}$$

by the result for external direct sums in the previous subsection. Observe that $h = f \otimes g$ (see § 16).

§ 15

Remark. By (1.7), *the intersection of tensor products is the tensor product of the intersections*. Observe that (1.4) and (1.5) are special cases of (1.7).

In the proof of (1.4), $u_{\beta} = v_{\beta}$ follows immediately from Lemma 1.5.2. In the proof of (1.5), if $z \in (E_1 \otimes F) \cap (E \otimes F_1)$, then in particular $z = x + y$ with $x \in E_1 \otimes F_1$ and $y \in E_1 \otimes F'$. Now $y = z - x \in E \otimes F_1$, so

$$y \in (E \otimes F_1) \cap (E \otimes F') = E \otimes (F_1 \cap F') = E \otimes 0 = 0$$

by (1.4). Hence $y = 0$ and $z = x \in E_1 \otimes F_1$. Note that this argument makes (1.6) superfluous.

Remark. If $\sum_{i=1}^r x_i \otimes y_i = \sum_{j=1}^s x'_j \otimes y'_j$ and the x_i are linearly independent, then the y_i are in the span of the y'_j .

Proof. By induction on s . If the vectors in $\{x_i\} \cup \{x'_j\}$ are linearly independent, then $y_i = 0$ for all i (1.5.1), so the result holds trivially. Otherwise, since the x_i are linearly independent, we may assume (relabeling if necessary) that

$$x'_s = \sum_{i=1}^r \lambda_i x_i + \sum_{j=1}^{s-1} \mu_j x'_j \quad (\lambda_i, \mu_j \in \Gamma)$$

By bilinearity of the tensor product, it follows that

$$\sum_{i=1}^r x_i \otimes (y_i - \lambda_i y'_s) = \sum_{j=1}^{s-1} x'_j \otimes (y'_j + \mu_j y'_s)$$

By induction, the $y_i - \lambda_i y'_s$ are in the span of the $y'_j + \mu_j y'_s$, so the y_i are in the span of the y'_j . $\square$

Remark. If $\sum_{i=1}^r x_i \otimes y_i = \sum_{j=1}^s x'_j \otimes y'_j$ and the x_i and y_i are respectively linearly independent, then $r \leq s$.

Proof. By the previous remark and the elementary fact that the size of a linearly independent set is at most the size of a spanning set in a subspace. $\square$

Exercise (1). If $\sum_{i=1}^r x_i \otimes y_i = \sum_{j=1}^s x'_j \otimes y'_j$ and the x_i , y_i , x'_j , and y'_j are each respectively linearly independent, then $r = s$.

Proof. By the previous remark, $r \leq s$ and $s \leq r$. $\square$

Remark. It follows from this result and the proof of Lemma 1.5.3 that the tensor representations of minimal length are precisely the representations by linearly independent vectors. These representations are *not* unique, as already seen in problem 1.8.4.

Exercise (2). A bilinear mapping $\varphi : E \times F \rightarrow G$ satisfies $\otimes_2$ if and only if the vectors $\varphi(x_\alpha, y_\beta)$ are linearly independent whenever the vectors $x_\alpha \in E$ and $y_\beta \in F$ are linearly independent.

Proof. If $f : E \otimes F \rightarrow G$ is the induced linear map with $\varphi = f \circ \otimes$, then φ satisfies $\otimes_2$ if and only if f is injective (1.8.1)—that is, if and only if f preserves linear independence. But f preserves linear independence if and only if φ does, since $\otimes$ does (1.5.1). $\square$

Exercise (3). If $A \neq 0$ is a finite-dimensional algebra forming a tensor product under the algebra multiplication, then $\dim A = 1$.

Proof. By (1.3) $\dim A = (\dim A)^2$, and $\dim A \neq 0$, so $\dim A = 1$. $\square$

Exercise (5). If E, E^* and F, F^* are pairs of dual spaces of finite dimension and $\beta : E \times F \rightarrow B(E^*, F^*)$ is the bilinear map given by

$$\beta_{x,y}(x^*, y^*) = \langle x^*, x \rangle \langle y^*, y \rangle$$

then $(B(E^*, F^*), \beta)$ is the tensor product of E and F .

Proof. Let $x_1, \dots, x_n$ be a basis in E with dual basis $x^{*1}, \dots, x^{*n}$ in E^* and let $y_1, \dots, y_m$ be a basis in F with dual basis $y^{*1}, \dots, y^{*m}$ in F^* . Let $\varphi^{*kl} : E^* \times F^* \rightarrow \Gamma$ be the basis function in $B(E^*, F^*)$ defined by

$$\varphi^{*kl}(x^{*i}, y^{*j}) = \delta_i^k \delta_j^l$$

Then

$$\beta_{x_k, y_l}(x^{*i}, y^{*j}) = \langle x^{*i}, x_k \rangle \langle y^{*j}, y_l \rangle = \delta_i^k \delta_j^l = \varphi^{*kl}(x^{*i}, y^{*j})$$

so $\beta_{x_k, y_l} = \varphi^{*kl}$. It follows that $\text{Im } \beta = B(E^*, F^*)$, so β satisfies $\otimes_1$.

If $\varphi : E \times F \rightarrow G$ is bilinear, define $f : B(E^*, F^*) \rightarrow G$ by $f(\varphi^{*kl}) = \varphi(x_k, y_l)$. Then $\varphi(x_k, y_l) = f(\beta_{x_k, y_l})$, so $\varphi = f\beta$. It follows that β satisfies $\otimes_2$. $\square$

Remark. This result allows us to view an element $\sum x_i \otimes y_i$ of the tensor product as a bilinear function: given linear scalar substitutions $x_i \rightarrow \lambda_i$ and $y_i \rightarrow \mu_i$ as inputs, it produces the scalar $\sum \lambda_i \mu_i$ as output. This is analogous to viewing a vector as a linear function. Indeed, if we identify x with $\langle -, x \rangle$ and y with $\langle -, y \rangle$, then we can identify $x \otimes y$ with $\langle -, x \rangle \langle -, y \rangle$.

§ 16

Remark. The tensor product operation is a bifunctor in the category of vector spaces. It maps the objects E and F to the object $E \otimes F$ and the arrows $\varphi : E \rightarrow E'$ and $\psi : F \rightarrow F'$ to the arrow $\varphi \otimes \psi : E \otimes F \rightarrow E' \otimes F'$.

Remark. In the proof of Proposition 1.16.1, taking $x = a$ and y arbitrary in (1.9), it follows from a remark in § 5 above and Lemma 1.5.1 that the $\psi_i(y)$ are linearly dependent. But the linear dependence relation is independent of the choice of y , so the ψ_i are linearly dependent, contrary to the assumption. Here we are essentially “lifting” the remark from vectors to maps.

Remark. Corollary III to Proposition 1.16.1 is not established in subsection 1.27, where it is also assumed that E' and F' are finite-dimensional. However, it is true. In fact, for fixed bases in E, E', F, F' , the tensor products of the induced basis maps in $L(E; E')$ and $L(F; F')$ are the basis maps in $L(E \otimes F, E' \otimes F')$ induced by the tensor products of the basis vectors.³⁹

§ 17

Remark. In the example presented in this subsection, intuitively the left null space of a linear function in the image of β is “large” in the sense of having finite codimension in an infinite dimensional space, while this is not true for all linear functions in the codomain of β .⁴⁰

§ 19

Remark. The proof of (1.12) is captured in the following commutative diagram:

$$\begin{array}{ccc}
 E \otimes F & \xrightarrow{\varphi \otimes \psi} & E' \otimes F' \\
 \pi \searrow & & \nearrow \chi \\
 \pi_1 \otimes \pi_2 \downarrow & & \uparrow \\
 \overline{E} \otimes \overline{F} & \xrightarrow{g} & \overline{E} \otimes \overline{F} \\
 \overline{\varphi} \otimes \overline{\psi} \nearrow & & \searrow
 \end{array}$$

Remark. The “only if” part of the claim in problem 1(a) requires $E \neq 0$ and $F \neq 0$. For example, if $E \neq 0$ and $F = 0$ and $\varphi = 0$ and $\psi = 0$, then $E \otimes F = 0$ and $\varphi \otimes \psi = 0$ is injective despite the fact that φ is not.

Exercise (2). If $\varphi : E \rightarrow E$ and $\psi : F \rightarrow F$ are linear with $\dim E = n$ and $\dim F = m$, then

$$\text{tr}(\varphi \otimes \psi) = \text{tr} \varphi \cdot \text{tr} \psi$$

³⁹See problem 1.19.4.

⁴⁰See also the remark in chapter II, § 1 of [1] above.

and

$$\det(\varphi \otimes \psi) = (\det \varphi)^m \cdot (\det \psi)^n$$

Proof. Let $e_1, \dots, e_n$ be a basis of E and $f_1, \dots, f_m$ be a basis of F . Write

$$\begin{aligned}\varphi e_i &= \sum_{k=1}^n \alpha_i^k e_k \\ \psi f_j &= \sum_{l=1}^m \beta_j^l f_l\end{aligned}$$

Then $e_1 \otimes f_1, \dots, e_n \otimes f_m$ is a basis of $E \otimes F$ and

$$(\varphi \otimes \psi)(e_i \otimes f_j) = \varphi e_i \otimes \psi f_j = \sum_{k,l} \alpha_i^k \beta_j^l e_k \otimes f_l$$

It follows that

$$\operatorname{tr}(\varphi \otimes \psi) = \sum_{i,j} \alpha_i^i \beta_j^j = \left(\sum_i \alpha_i^i \right) \left(\sum_j \beta_j^j \right) = \operatorname{tr} \varphi \cdot \operatorname{tr} \psi$$

For the determinant, observe that

$$\varphi \otimes \psi = (\varphi \otimes \iota_F) \circ (\iota_E \otimes \psi)$$

so

$$\det(\varphi \otimes \psi) = \det(\varphi \otimes \iota_F) \cdot \det(\iota_E \otimes \psi)$$

But $M(\varphi \otimes \iota_F) = (\alpha_i^k \delta_j^l)$ is block diagonal with m blocks each equal to $M(\varphi)$, so $\det(\varphi \otimes \iota_F) = (\det \varphi)^m$. Similarly $\det(\iota_E \otimes \psi) = (\det \psi)^n$. $\square$

Exercise (3). If $\alpha, \beta : E \rightarrow E$ are linear with $\dim E = n$, and $\Phi : L(E; E) \rightarrow L(E; E)$ is defined by

$$\Phi \sigma = \alpha \circ \sigma \circ \beta$$

then

$$\operatorname{tr} \Phi = \operatorname{tr} \alpha \cdot \operatorname{tr} \beta$$

and

$$\det \Phi = \det(\alpha \circ \beta)^n$$

Proof. Let $e_1, \dots, e_n$ be a basis of E and let $\psi_{ij} : E \rightarrow E$ be the induced basis transformation defined by $\psi_{ij} e_k = \delta_k^i e_j$. The isomorphism $\Psi : E \otimes E \rightarrow L(E; E)$ defined by $\Psi(e_i \otimes e_j) = \psi_{ij}$ induces an isomorphism

$$\widehat{\Psi} : L(E \otimes E; E \otimes E) \cong L(L(E; E); L(E; E))$$

by $\widehat{\Psi}(\sigma \otimes \tau) = \Psi \circ (\sigma \otimes \tau) \circ \Psi^{-1}$:

$$\begin{array}{ccc} E \otimes E & \xrightarrow{\Psi} & L(E; E) \\ \sigma \otimes \tau \downarrow & & \downarrow \widehat{\Psi}(\sigma \otimes \tau) \\ E \otimes E & \xrightarrow{\Psi} & L(E; E) \end{array}$$

It is easy to verify that

$$\widehat{\Psi}(\beta^* \otimes \alpha) = \Phi$$

where β^* is the transpose of β . Therefore by the previous exercise,

$$\text{tr } \Phi = \text{tr}(\beta^* \otimes \alpha) = \text{tr}(\beta^*) \cdot \text{tr } \alpha = \text{tr } \alpha \cdot \text{tr } \beta$$

and

$$\det \Phi = \det(\beta^* \otimes \alpha) = (\det \beta^*)^n (\det \alpha)^n = \det(\alpha \circ \beta)^n \quad \square$$

Remark. It is more natural to do this problem using the composition algebra (see § 26). Indeed, if $\alpha = a^* \otimes b$, $\beta = c^* \otimes d$, and $\sigma = x^* \otimes y$, then

$$\begin{aligned} \alpha \circ \sigma \circ \beta &= (a^* \otimes b) \circ (x^* \otimes y) \circ (c^* \otimes d) \\ &= (a^* \otimes b) \circ (\langle x^*, d \rangle (c^* \otimes y)) \\ &= \langle x^*, d \rangle \langle a^*, y \rangle (c^* \otimes b) \\ &= (\langle x^*, d \rangle c^*) \otimes (\langle a^*, y \rangle b) \\ &= \beta^* x^* \otimes \alpha y \\ &= (\beta^* \otimes \alpha) \sigma \end{aligned}$$

so $\Phi = \beta^* \otimes \alpha$, the tensor product of the maps β^* and α , where $\beta^* = d \otimes c^*$.⁴¹ The general case follows by multilinearity. Alternately, we could make use of the isomorphism Ω from Proposition 1.29.1.

⁴¹See problem 1.30.1.

§ 20

Remark. A linear map is universal if and only if it is an isomorphism. In fact, it satisfies $\otimes_1$ if and only if it is surjective, and it satisfies $\otimes_2$ if and only if it is injective (has a left inverse). In this sense, a tensor product map is a multilinear analogue of an isomorphism.

Remark. To see that there is a unique isomorphism

$$f : (E_1 \otimes \cdots \otimes E_p) \otimes (E_{p+1} \otimes \cdots \otimes E_{p+q}) \rightarrow E_1 \otimes \cdots \otimes E_{p+q}$$

with

$$f((x_1 \otimes \cdots \otimes x_p) \otimes (x_{p+1} \otimes \cdots \otimes x_{p+q})) = x_1 \otimes \cdots \otimes x_{p+q} \quad (1)$$

observe that for each p -tuple $(x_1, \dots, x_p) \in E_1 \times \cdots \times E_p$, there is a q -linear map $\varphi_{x_1 \dots x_p} : E_{p+1} \times \cdots \times E_{p+q} \rightarrow E_1 \otimes \cdots \otimes E_{p+q}$ given by

$$\varphi_{x_1 \dots x_p}(x_{p+1}, \dots, x_{p+q}) = x_1 \otimes \cdots \otimes x_{p+q}$$

By the universal property of $E_{p+1} \otimes \cdots \otimes E_{p+q}$, it follows that there is a linear map $f_{x_1 \dots x_p} : E_{p+1} \otimes \cdots \otimes E_{p+q} \rightarrow E_1 \otimes \cdots \otimes E_{p+q}$ with

$$f_{x_1 \dots x_p}(x_{p+1} \otimes \cdots \otimes x_{p+q}) = x_1 \otimes \cdots \otimes x_{p+q}$$

Now the mapping $(x_1, \dots, x_p) \mapsto f_{x_1 \dots x_p}$ is p -linear, so by the universal property of $E_1 \otimes \cdots \otimes E_p$, there is a linear map

$$\widehat{f} : E_1 \otimes \cdots \otimes E_p \rightarrow L(E_{p+1} \otimes \cdots \otimes E_{p+q}; E_1 \otimes \cdots \otimes E_{p+q})$$

with $\widehat{f}(x_1 \otimes \cdots \otimes x_p) = f_{x_1 \dots x_p}$. Define

$$\varphi : (E_1 \otimes \cdots \otimes E_p) \times (E_{p+1} \otimes \cdots \otimes E_{p+q}) \rightarrow E_1 \otimes \cdots \otimes E_{p+q}$$

by $\varphi(x, y) = \widehat{f}(x)(y)$. Then φ is bilinear, so by the universal property of $(E_1 \otimes \cdots \otimes E_p) \otimes (E_{p+1} \otimes \cdots \otimes E_{p+q})$, there is a linear map f satisfying $f \circ \otimes = \varphi$, which implies (1).

By the universal property of $E_1 \otimes \cdots \otimes E_{p+q}$, the $(p+q)$ -linear map

$$\psi : E_1 \times \cdots \times E_{p+q} \rightarrow (E_1 \otimes \cdots \otimes E_p) \otimes (E_{p+1} \otimes \cdots \otimes E_{p+q})$$

defined by

$$\psi(x_1, \dots, x_{p+q}) = (x_1 \otimes \cdots \otimes x_p) \otimes (x_{p+1} \otimes \cdots \otimes x_{p+q})$$

induces a linear map

$$g : E_1 \otimes \cdots \otimes E_{p+q} \rightarrow (E_1 \otimes \cdots \otimes E_p) \otimes (E_{p+1} \otimes \cdots \otimes E_{p+q})$$

with

$$g(x_1 \otimes \cdots \otimes x_{p+q}) = (x_1 \otimes \cdots \otimes x_p) \otimes (x_{p+1} \otimes \cdots \otimes x_{p+q})$$

By universal properties again, $f \circ g = \iota$ and $g \circ f = \iota$ so f is an isomorphism, and f is uniquely determined by (1).

Remark. If the vectors a_v^i are linearly independent in E_i ($i = 1, \dots, p$), then their tensor products $a_{v_1}^1 \otimes \cdots \otimes a_{v_p}^p$ are linearly independent in $E_1 \otimes \cdots \otimes E_p$. Indeed, if $p = 2$, this follows from Lemma 1.5.1. If $p > 2$, we use the natural isomorphism

$$E_1 \otimes \cdots \otimes E_p \cong (E_1 \otimes \cdots \otimes E_{p-1}) \otimes E_p$$

If

$$\sum_{v_1, \dots, v_p} \lambda_{v_1 \dots v_p} a_{v_1}^1 \otimes \cdots \otimes a_{v_p}^p = 0$$

then

$$\sum_{v_1, \dots, v_p} (\lambda_{v_1 \dots v_p} a_{v_1}^1 \otimes \cdots \otimes a_{v_{p-1}}^{p-1}) \otimes a_{v_p}^p = 0$$

Since the $a_{v_p}^p$ are linearly independent in E_p , it follows from Lemma 1.5.1 that

$$\lambda_{v_1 \dots v_p} a_{v_1}^1 \otimes \cdots \otimes a_{v_{p-1}}^{p-1} = 0$$

for all $v_1, \dots, v_p$. But $a_{v_1}^1 \otimes \cdots \otimes a_{v_{p-1}}^{p-1} \neq 0$ since $a_v^i \neq 0$ for all i, v (see problem 1), so $\lambda_{v_1 \dots v_p} = 0$ for all $v_1, \dots, v_p$.

If the vectors a_v^i span E_i ($i = 1, \dots, p$), then clearly their tensor products span $E_1 \otimes \cdots \otimes E_p$. It follows that if the vectors a_v^i form a basis of E_i ($i = 1, \dots, p$), then their tensor products form a basis of $E_1 \otimes \cdots \otimes E_p$.

Remark. To see that for $\varphi_i : E_i \rightarrow F_i$ the map $(\varphi_1, \dots, \varphi_p) \mapsto \varphi_1 \otimes \cdots \otimes \varphi_p$ induces an injection

$$L(E_1; F_1) \otimes \cdots \otimes L(E_p; F_p) \rightarrow L(E_1 \otimes \cdots \otimes E_p; F_1 \otimes \cdots \otimes F_p)$$

proceed by induction on $p \geq 2$. For $p = 2$, this is just Proposition 1.16.1. For $p > 2$, if we make the appropriate identifications we have

$$\begin{aligned} L(E_1; F_1) \otimes \cdots \otimes L(E_p; F_p) &= (L(E_1; F_1) \otimes \cdots \otimes L(E_{p-1}; F_{p-1})) \otimes L(E_p; F_p) \\ &\subseteq L(E_1 \otimes \cdots \otimes E_{p-1}; F_1 \otimes \cdots \otimes F_{p-1}) \otimes L(E_p; F_p) \\ &\subseteq L((E_1 \otimes \cdots \otimes E_{p-1}) \otimes E_p; (F_1 \otimes \cdots \otimes F_{p-1}) \otimes F_p) \\ &= L(E_1 \otimes \cdots \otimes E_p; F_1 \otimes \cdots \otimes F_p) \end{aligned}$$

Exercise (1). In $E_1 \otimes \cdots \otimes E_p$:

- (a) $x_1 \otimes \cdots \otimes x_p = 0$ if and only if at least one $x_i = 0$.
- (b) If $x_1 \otimes \cdots \otimes x_p \neq 0$, then

$$x_1 \otimes \cdots \otimes x_p = y_1 \otimes \cdots \otimes y_p$$

if and only if $y_i = \lambda_i x_i$ with $\lambda_1 \cdots \lambda_p = 1$.

Proof. The reverse directions follow from multilinearity of the tensor product. The forward directions follow by induction on p using the natural isomorphism

$$E_1 \otimes \cdots \otimes E_p \cong (E_1 \otimes \cdots \otimes E_{p-1}) \otimes E_p$$

- (a) If $p = 2$, the result follows from Lemma 1.5.1. If $p > 2$, then $(x_1 \otimes \cdots \otimes x_{p-1}) \otimes x_p = 0$, so by Lemma 1.5.1 either $x_1 \otimes \cdots \otimes x_{p-1} = 0$ and $x_i = 0$ for some $i = 1, \dots, p-1$ by induction, or else $x_p = 0$.
- (b) If $p = 2$, the result follows from problem 1.8.4. If $p > 2$, then

$$(x_1 \otimes \cdots \otimes x_{p-1}) \otimes x_p = (y_1 \otimes \cdots \otimes y_{p-1}) \otimes y_p$$

so by the same problem,

$$y_1 \otimes \cdots \otimes y_{p-1} = \lambda_p^{-1} x_1 \otimes \cdots \otimes x_{p-1} \quad \text{and} \quad y_p = \lambda_p x_p$$

for some $\lambda_p \neq 0$. By induction, we may write $y_1 = \mu_1 \lambda_p^{-1} x_1$ and $y_i = \mu_i x_i$ for $i = 2, \dots, p-1$ with $\mu_1 \cdots \mu_{p-1} = 1$. Setting $\lambda_1 = \mu_1 \lambda_p^{-1}$ and $\lambda_i = \mu_i$ for $i = 2, \dots, p-1$, it follows that $y_i = \lambda_i x_i$ for $i = 1, \dots, p$ and $\lambda_1 \cdots \lambda_p = 1$. $\square$

§ 21

Remark. For $p, q \geq 1$, let

$$\varphi_i : \prod_{j=1}^p E_j^i \rightarrow E_{p+1}^i \quad (i = 1, \dots, q)$$

be a family of q p -linear maps. Then there is a unique p -linear map

$$\varphi = \varphi_1 \otimes \cdots \otimes \varphi_q : \prod_{j=1}^p \left(\bigotimes_{i=1}^q E_j^i \right) \rightarrow \bigotimes_{i=1}^q E_{p+1}^i$$

with

$$\varphi(x_1^1 \otimes \cdots \otimes x_1^q, \dots, x_p^1 \otimes \cdots \otimes x_p^q) = \varphi_1(x_1^1, \dots, x_p^1) \otimes \cdots \otimes \varphi_q(x_1^q, \dots, x_p^q)$$

Moreover,

$$\ker_j \varphi = \sum_{i=1}^q E_j^1 \otimes \cdots \otimes \ker_j \varphi_i \otimes \cdots \otimes E_j^q \quad (j = 1, \dots, p)$$

In particular, if each φ_i is nondegenerate, then φ is nondegenerate.

§ 22

Remark. If Φ is bilinear in $E \times E'$ and Ψ is bilinear in $F \times F'$, then nondegeneracy of $\Phi \otimes \Psi$ does not imply nondegeneracy of Φ and Ψ unless all the spaces are nonzero, contrary to what the book says. For example, if $E \neq 0$ and $E' = 0$, $F = F' = 0$, and $\Phi = 0$ and $\Psi = 0$, then Φ is degenerate, but $E \otimes F = 0$ and $E' \otimes F' = 0$, so $\Phi \otimes \Psi = 0$ is nondegenerate.⁴²

§ 26

Remark. Let $S : (E^* \otimes E) \otimes (E^* \otimes E) \rightarrow (E^* \otimes E^*) \otimes (E \otimes E)$ be the linear isomorphism defined by

$$(x^* \otimes x) \otimes (y^* \otimes y) \mapsto (x^* \otimes y^*) \otimes (y \otimes x)$$

Let $f : (E^* \otimes E^*) \otimes (E \otimes E) \rightarrow \Gamma \otimes (E^* \otimes E)$ be the linear map corresponding to the tensor product of the bilinear maps $\langle -, - \rangle : E^* \times E \rightarrow \Gamma$ and $\otimes : E^* \times E \rightarrow E^* \otimes E$. Let $g : \Gamma \otimes (E^* \otimes E) \rightarrow E^* \otimes E$ be the linear map defined by $g(\lambda \otimes z) = \lambda z$. Then the bilinear map $\circ$ corresponding to gfS satisfies

$$(x^* \otimes x) \circ (y^* \otimes y) = \langle x^*, y \rangle (y^* \otimes x)$$

Remark. If $\dim E < \infty$, one way to discover the composition algebra is (see § 28)

$$L(E; E) \cong L(E \otimes \Gamma; \Gamma \otimes E) \cong L(E; \Gamma) \otimes L(\Gamma; E) \cong L(E) \otimes E$$

Remark. If $\dim E < \infty$, the fact that

$$\langle T(x^* \otimes x), T(y^* \otimes y) \rangle = \langle x^* \otimes x, y^* \otimes y \rangle$$

follows from (1.29) in § 28. This fact implies that T is an isometry, so $T^* = T^{-1}$.

⁴²This is essentially the same error as in problem 1.19.1(a) above.

Exercise (1). For the bilinear map

$$\gamma : L(E, E'; E'') \times L(F, F'; F'') \rightarrow L(E \otimes F, E' \otimes F'; E'' \otimes F'')$$

with

$$\gamma(\varphi, \psi) : (x \otimes y, x' \otimes y') \mapsto \varphi(x, x') \otimes \psi(y, y')$$

the pair $(\text{Im } \gamma, \gamma)$ is the tensor product of $L(E, E'; E'')$ and $L(F, F'; F'')$.

Proof. By transfer of Corollary II of Proposition 1.16.1, using the isomorphism between linear and bilinear maps induced by the tensor product.

In detail, consider the following commutative diagram:

$$\begin{array}{ccc}
 L(E \otimes E'; E'') \otimes L(F \otimes F'; F'') & \xrightarrow{f} & L((E \otimes E') \otimes (F \otimes F'); E'' \otimes F'') \\
 \uparrow \cong & & \downarrow \cong \\
 & & L((E \otimes F) \otimes (E' \otimes F'); E'' \otimes F'') \\
 & & \downarrow \cong \\
 L(E, E'; E'') \otimes L(F, F'; F'') & \xrightarrow{g} & L(E \otimes F, E' \otimes F'; E'' \otimes F'') \\
 \uparrow \otimes & \nearrow \gamma & \\
 L(E, E'; E'') \times L(F, F'; F'') & &
 \end{array}$$

Note g is injective since f is injective (1.16.1), so γ satisfies $\otimes_2$ (1.8.1). Since γ also satisfies $\otimes_1$, it follows that γ is the tensor product. $\square$

Exercise (2). If E, E^* and F, F^* are pairs of dual spaces with $E_1 \subseteq E$ and $F_1 \subseteq F$ subspaces, then a scalar product is induced between

$$(E^* \otimes F^*) / (E_1^\perp \otimes F^* + E^* \otimes F_1^\perp) \quad \text{and} \quad E_1 \otimes F_1$$

by the scalar product between $E^* \otimes F^*$ and $E \otimes F$. In particular,

$$(E_1 \otimes F_1)^\perp = E_1^\perp \otimes F^* + E^* \otimes F_1^\perp$$

Proof. A scalar product is induced between $E^*/E_1^\perp$ and E_1 by $\langle \overline{x^*}, x \rangle = \langle x^*, x \rangle$, and between $F^*/F_1^\perp$ and F_1 by $\langle \overline{y^*}, y \rangle = \langle y^*, y \rangle$. Therefore a scalar product is induced between $(E^*/E_1^\perp) \otimes (F^*/F_1^\perp)$ and $E_1 \otimes F_1$ by

$$\langle \overline{x^*} \otimes \overline{y^*}, x \otimes y \rangle = \langle x^*, x \rangle \langle y^*, y \rangle = \langle x^* \otimes y^*, x \otimes y \rangle$$

where the scalar product on the right is between $E^* \otimes F^*$ and $E \otimes F$. However,

$$(E^* \otimes F^*) / (E_1^\perp \otimes F^* + E^* \otimes F_1^\perp) \cong (E^* / E_1^\perp) \otimes (F^* / F_1^\perp)$$

where $\overline{x^* \otimes y^*} \mapsto \overline{x^*} \otimes \overline{y^*}$, so there is a scalar product

$$\langle \overline{x^* \otimes y^*}, x \otimes y \rangle = \langle x^* \otimes y^*, x \otimes y \rangle$$

as required. □

Exercise (3). If E, E^* are dual spaces and $\varphi : E \rightarrow E$ and $\varphi^* : E^* \leftarrow E^*$ are dual transformations, then $\varphi \otimes \varphi^*$ is self-dual.

Proof. Recall that $E \otimes E^*$ is self-dual under the scalar product

$$\langle x \otimes x^*, y \otimes y^* \rangle = \langle x^*, y \rangle \langle y^*, x \rangle$$

Now

$$\begin{aligned} \langle (\varphi \otimes \varphi^*)(x \otimes x^*), y \otimes y^* \rangle &= \langle \varphi x \otimes \varphi^* x^*, y \otimes y^* \rangle \\ &= \langle \varphi^* x^*, y \rangle \langle y^*, \varphi x \rangle \\ &= \langle x^*, \varphi y \rangle \langle \varphi^* y^*, x \rangle \\ &= \langle x \otimes x^*, \varphi y \otimes \varphi^* y^* \rangle \\ &= \langle x \otimes x^*, (\varphi \otimes \varphi^*)(y \otimes y^*) \rangle \end{aligned}$$

so $(\varphi \otimes \varphi^*)^* = \varphi \otimes \varphi^*$. □

§ 28

In this subsection, all vector spaces are finite-dimensional.

Remark. If $\varphi : E \rightarrow F$ is linear, then $T^{-1}(\varphi)$ is decomposable if and only if φ has rank at most 1.

Proof. If $T^{-1}(\varphi) = a^* \otimes b$, then $\text{Im } \varphi = \langle b \rangle$. Conversely if $\text{Im } \varphi = \langle b \rangle$, there is a linear function $f \in L(E)$ with $\varphi(x) = f(x)b$ for all $x \in E$. If $a^* \in E^*$ corresponds to f , then $T^{-1}(\varphi) = a^* \otimes b$. $\square$

Remark. If $\varphi : E \rightarrow F$ is linear and

$$T^{-1}(\varphi) = \sum_{i=1}^r a^{*i} \otimes b_i \quad (a^{*i} \in E^*, b_i \in F)$$

where the a^{*i} and the b_i are linearly independent, then φ has rank r . Thus the rank of φ is equal to the tensor rank of $T^{-1}(\varphi)$.

Proof. It is sufficient to prove that $\text{Im } \varphi = \langle b_i \rangle$. Clearly $\text{Im } \varphi \subseteq \langle b_i \rangle$. Conversely if $y = \sum_{i=1}^r \beta_i b_i$ and $a_j \in E$ with $\langle a^{*i}, a_j \rangle = \delta_{ij}$ for all $1 \leq i, j \leq r$, then for the vector $x = \sum_{j=1}^r \beta_j a_j$ we have $\varphi(x) = y$. $\square$

Remark. Fix dual bases e_i, e^{*i} and f_j, f^{*j} of E, E^* and F, F^* . If $z \in E^* \otimes F$ and $\varphi = T(z)$, write

$$z = \sum_{\mu, \nu} (z)_{\nu}^{\mu} e^{*\nu} \otimes f_{\mu}$$

Then

$$\varphi(e_{\lambda}) = \sum_{\mu, \nu} (z)_{\nu}^{\mu} \langle e^{*\nu}, e_{\lambda} \rangle f_{\mu} = \sum_{\mu} (z)_{\lambda}^{\mu} f_{\mu}$$

so

$$(z)_{\lambda}^{\kappa} = \langle f^{*\kappa}, \varphi(e_{\lambda}) \rangle$$

This shows that the components of the tensor z are precisely the matrix entries of the map φ , with respect to the chosen bases.

Remark. By (1.30), if we view $a^* \otimes a$ as a linear transformation, then its trace is just $\langle a^*, a \rangle$. This extends by linearity to a natural (coordinate-free) definition of the trace which yields a slick proof that $\text{tr}(\varphi \otimes \psi) = \text{tr } \varphi \cdot \text{tr } \psi$. Indeed, if $\varphi = a^* \otimes a$ and $\psi = b^* \otimes b$, then we can write $\varphi \otimes \psi = (a^* \otimes b^*) \otimes (a \otimes b)$ since

$$\begin{aligned} (\varphi \otimes \psi)(x \otimes y) &= \varphi x \otimes \psi y \\ &= (\langle a^*, x \rangle a) \otimes (\langle b^*, y \rangle b) \\ &= \langle a^*, x \rangle \langle b^*, y \rangle (a \otimes b) \\ &= \langle a^* \otimes b^*, x \otimes y \rangle (a \otimes b) \end{aligned}$$

Therefore

$$\text{tr}(\varphi \otimes \psi) = \langle a^* \otimes b^*, a \otimes b \rangle = \langle a^*, a \rangle \langle b^*, b \rangle = \text{tr } \varphi \cdot \text{tr } \psi$$

The general case follows by multilinearity.

Remark. Let E be an inner product space⁴³ and let $\sigma : E \rightarrow E^*$ be the Riesz isomorphism. Define the isomorphism $Q : E^* \otimes E \rightarrow E^* \otimes E$ by

$$Q(a^* \otimes a) = \sigma a \otimes \sigma^{-1} a^*$$

If $\varphi : E \rightarrow E$ is linear, then $\tilde{\varphi} = TQT^{-1}\varphi$:

$$\begin{array}{ccc} E^* \otimes E & \xrightarrow[\cong]{T} & L(E; E) \\ Q \downarrow & & \downarrow \widetilde{(-)} \\ E^* \otimes E & \xrightarrow[\cong]{T} & L(E; E) \end{array}$$

Proof. If $T^{-1}(\varphi) = a^* \otimes a$, then

$$(\varphi x, y) = \langle a^*, x \rangle (a, y) = (\sigma^{-1} a^*, x) \langle \sigma a, y \rangle = (x, TQT^{-1}(\varphi)y)$$

The general case follows by multilinearity. □

This shows that Q computes the adjoint tensor, so can be used to study things like symmetry and skew-symmetry.⁴⁴

§ 29

In this subsection, all vector spaces are finite-dimensional.

Remark. In the proof of Proposition 1.29.1, note that if $\alpha = a^* \otimes b$ and $\beta = c^* \otimes d$, then

$$F(\alpha \otimes \beta)(x^* \otimes y) = \langle a^* \otimes b, x^* \otimes y \rangle (c^* \otimes d) = \langle x^*, b \rangle \langle a^*, y \rangle (c^* \otimes d)$$

Comparing with the calculation after problem 1.19.3 above, we immediately see that Q , which swaps b and d , satisfies

$$F(Q(\alpha \otimes \beta)) = \beta^* \otimes \alpha = \Omega(\alpha \otimes \beta)$$

where $\beta^* \otimes \alpha$ is the tensor product of the maps β^* and α . These results show that we can view $L(A; A)$ as a tensor product in multiple ways.

⁴³See subsection 1.25.

⁴⁴See subsection 8.3.

Remark. Let E, E^* be dual spaces. The algebras $A = A(E; E)$ and $A^* = A(E^*; E^*)$ are dual under the scalar product

$$\langle \psi, \varphi \rangle = \text{tr}(\psi^* \circ \varphi) \quad (\psi \in A^*, \varphi \in A)$$

The isomorphism $T : A^* \otimes A \rightarrow L(A; A)$ satisfies

$$T(\alpha^* \otimes \beta)\varphi = \langle \alpha^*, \varphi \rangle \beta = \text{tr}(\alpha \circ \varphi) \beta$$

Let $S : A^* \otimes A \rightarrow A^* \otimes A$ be the isomorphism

$$(b \otimes a^*) \otimes (c^* \otimes d) \mapsto (b \otimes c^*) \otimes (a^* \otimes d)$$

Then

$$TS(\alpha^* \otimes \beta) = \alpha^* \otimes \beta$$

where the right-hand side is the tensor product of the maps α^* and β . Since S is not the identity map, we must be careful when using the notation $\alpha^* \otimes \beta$ to denote an element of $L(A; A)$.

§ 30

In this subsection, all vector spaces are finite-dimensional.

Exercise (1). Let E, E^* and F, F^* be pairs of dual spaces. If $a^* \in E^*$ and $b \in F$, then

$$(a^* \otimes b)^* = b \otimes a^*$$

Proof. Recall $a^* \otimes b : E \rightarrow F$ is defined by $x \mapsto \langle a^*, x \rangle b$ and $b \otimes a^* : F^* \rightarrow E^*$ is defined by $y^* \mapsto \langle y^*, b \rangle a^*$. For $y^* \in F^*$ and $x \in E$,

$$\langle (b \otimes a^*)y^*, x \rangle = \langle y^*, b \rangle \langle a^*, x \rangle = \langle y^*, (a^* \otimes b)x \rangle \quad \square$$

Exercise (2). If $E, F \neq 0$ are Euclidean spaces with $\varphi : E \rightarrow E$ and $\psi : F \rightarrow F$, then $\varphi \otimes \psi : E \otimes F \rightarrow E \otimes F$ is a rotation if and only if $\varphi = \lambda \tau_E$ and $\psi = \lambda^{-1} \tau_F$ where τ_E and τ_F are rotations and $\lambda \neq 0$.

Proof. If $\varphi = \lambda \tau_E$ and $\psi = \lambda^{-1} \tau_F$, then $\varphi \otimes \psi = \tau_E \otimes \tau_F$, and

$$\begin{aligned} \widetilde{\tau_E \otimes \tau_F} \circ (\tau_E \otimes \tau_F) &= (\widetilde{\tau_E} \otimes \widetilde{\tau_F}) \circ (\tau_E \otimes \tau_F) \\ &= (\widetilde{\tau_E} \circ \tau_E) \otimes (\widetilde{\tau_F} \circ \tau_F) \\ &= \iota_E \otimes \iota_F \\ &= \iota_{E \otimes F} \end{aligned}$$

so $\widetilde{\varphi \otimes \psi} = (\varphi \otimes \psi)^{-1}$ and $\varphi \otimes \psi$ is a rotation.

Conversely if $\varphi \otimes \psi$ is a rotation, then

$$\widetilde{\varphi} \otimes \widetilde{\psi} = \widetilde{\varphi \otimes \psi} = (\varphi \otimes \psi)^{-1} = \varphi^{-1} \otimes \psi^{-1}$$

It follows that $\widetilde{\varphi} = \mu\varphi^{-1}$ and $\widetilde{\psi} = \mu^{-1}\psi^{-1}$ for some $\mu \neq 0$.⁴⁵ We may assume $\mu > 0$ (otherwise consider $-\varphi$ and $-\psi$). Set $\lambda = \sqrt{\mu} > 0$ and define $\tau_E = \lambda^{-1}\varphi$ and $\tau_F = \lambda\psi$. Then $\widetilde{\tau_E} \circ \tau_E = \iota_E$, so $\widetilde{\tau_E} = \tau_E^{-1}$ and τ_E is a rotation, and similarly for τ_F . $\square$

Remark. The proofs of the other parts of this problem in the book are similar.

Exercise (3). Let E be a real vector space with $\dim E = n$ and $\varphi, \psi : E \rightarrow E$ be regular transformations.

- (a) If n is even, then $\varphi \otimes \psi$ preserves orientation.
- (b) If n is odd, then $\varphi \otimes \psi$ preserves orientation if and only if φ and ψ either both preserve orientation or both reverse orientation.

Proof. This follows from $\det(\varphi \otimes \psi) = (\det \varphi \cdot \det \psi)^n$.⁴⁶ $\square$

⁴⁵See problem 1.8.4.

⁴⁶See problem 1.19.2.

Chapter 2

Remark. In the sections of this chapter, the tensor product operation is seen to be a bifunctor in various subcategories of the category of vector spaces.

§ 2

Remark. The multiplication in the canonical tensor product $A \otimes B$ of algebras A and B is just the tensor product of the multiplications in A and B , as defined in subsection 1.21.

Remark. Let A and B be nonzero associative unital algebras. There are injective unital algebra homomorphisms

$$A \xrightarrow{i_1} A \otimes B \xleftarrow{i_2} B$$

given by $i_1(a) = a \otimes 1$ and $i_2(b) = 1 \otimes b$, whose images commute:

$$i_1(a)i_2(b) = (a \otimes 1)(1 \otimes b) = a \otimes b = (1 \otimes b)(a \otimes 1) = i_2(b)i_1(a)$$

Moreover, given any unital algebra homomorphisms $f : A \rightarrow C$ and $g : B \rightarrow C$ whose images commute in an associative unital algebra C , there is a unique unital algebra homomorphism $h : A \otimes B \rightarrow C$ with $h \circ i_1 = f$ and $h \circ i_2 = g$:

$$\begin{array}{ccccc} A & \xrightarrow{i_1} & A \otimes B & \xleftarrow{i_2} & B \\ & \searrow f & \downarrow h & \swarrow g & \\ & & C & & \end{array}$$

Indeed, if such an h exists it must satisfy

$$h(a \otimes b) = h((a \otimes 1)(1 \otimes b)) = h(a \otimes 1)h(1 \otimes b) = f(a)g(b)$$

and this uniquely determines it by linearity. On the other hand, the mapping $(a, b) \mapsto f(a)g(b)$ is bilinear, so induces a linear map h with $h(a \otimes b) = f(a)g(b)$. This map is a homomorphism since it is linear and

$$\begin{aligned} h((a_1 \otimes b_1)(a_2 \otimes b_2)) &= h(a_1 a_2 \otimes b_1 b_2) \\ &= f(a_1 a_2)g(b_1 b_2) \\ &= f(a_1)f(a_2)g(b_1)g(b_2) \\ &= f(a_1)g(b_1)f(a_2)g(b_2) \\ &= h(a_1 \otimes b_1)h(a_2 \otimes b_2) \end{aligned}$$

Moreover,

$$h(a \otimes 1) = f(a)g(1) = f(a) \quad \text{and} \quad h(1 \otimes b) = f(1)g(b) = g(b)$$

since f and g are unital, so the diagram above commutes and h is also unital.

This argument shows that $A \otimes B$ is a universal (initial) object in the category of “associative unital algebras with unital algebra homomorphisms of A and B into them whose images commute”. By category theoretic abstract nonsense, $A \otimes B$ is therefore unique up to isomorphism. If A and B are also *commutative*, then $A \otimes B$ is just the coproduct of A and B in the subcategory of algebras that are also commutative.

§ 3

Remark. If $\varphi_1 : A_1 \rightarrow B_1$ and $\varphi_2 : A_2 \rightarrow B_2$ are algebra homomorphisms and $\varphi = \varphi_1 \otimes \varphi_2$ is the tensor product of the underlying *linear maps*, then

$$\begin{aligned} \varphi((x_1 \otimes x_2)(y_1 \otimes y_2)) &= \varphi(x_1 y_1 \otimes x_2 y_2) \\ &= \varphi_1(x_1 y_1) \otimes \varphi_2(x_2 y_2) \\ &= (\varphi_1 x_1 \varphi_1 y_1) \otimes (\varphi_2 x_2 \varphi_2 y_2) \\ &= (\varphi_1 x_1 \otimes \varphi_2 x_2)(\varphi_1 y_1 \otimes \varphi_2 y_2) \\ &= \varphi(x_1 \otimes x_2) \varphi(y_1 \otimes y_2) \end{aligned}$$

Since φ is linear, it follows that φ is an algebra homomorphism. This proof avoids the use of structure maps.

§ 5

Remark. The canonical tensor product $A \otimes B$ of G -graded algebras $A = \sum_{\alpha \in G} A_\alpha$ and $B = \sum_{\beta \in G} B_\beta$ is a *graded algebra* under the simple gradation in (2.15). In fact, if $x_i \in A_{\alpha_i}$ and $y_i \in B_{\beta_i}$ for $i = 1, 2$, then

$$(x_1 \otimes y_1)(x_2 \otimes y_2) = x_1 x_2 \otimes y_1 y_2$$

Since A and B are graded algebras, $x_1 x_2 \in A_{\alpha_1 + \alpha_2}$ and $y_1 y_2 \in B_{\beta_1 + \beta_2}$, so by (2.15) $x_1 x_2 \otimes y_1 y_2$ is homogeneous with

$$\deg(x_1 x_2 \otimes y_1 y_2) = \alpha_1 + \alpha_2 + \beta_1 + \beta_2 = \deg(x_1 \otimes y_1) + \deg(x_2 \otimes y_2)$$

The result now follows from bilinearity of the algebra multiplication.

Remark. If x and y are distinct indeterminates, then

$$\Gamma[x] \otimes \Gamma[y] \cong \Gamma[x, y]$$

is an isomorphism of graded algebras.

§ 8

Remark. Let $A = \sum_p A_p$ and $B = \sum_q B_q$ be skew graded algebras, so $A \hat{\otimes} B$ is also a skew graded algebra. Define the skew flip operator $\hat{f} : A \hat{\otimes} B \rightarrow B \hat{\otimes} A$ by

$$\hat{f}(a \otimes b) = (-1)^{pq} b \otimes a \quad (a \in A_p, b \in B_q)$$

Then $\hat{f}$ is an algebra isomorphism. Indeed, it is a linear isomorphism and if $a_1 \in A_p$, $a_2 \in A_q$, $b_1 \in B_r$, and $b_2 \in B_s$, then

$$\begin{aligned} \hat{f}[(a_1 \otimes b_1)(a_2 \otimes b_2)] &= (-1)^{qr} \hat{f}(a_1 a_2 \otimes b_1 b_2) \\ &= (-1)^{qr+(p+q)(r+s)} b_1 b_2 \otimes a_1 a_2 \\ &= (-1)^{pr+qs} (b_1 \otimes a_1)(b_2 \otimes a_2) \\ &= \hat{f}(a_1 \otimes b_1) \hat{f}(a_2 \otimes b_2) \end{aligned}$$

The result now follows by multilinearity.

Note the regular flip operator $a \otimes b \mapsto b \otimes a$ is *not* an algebra isomorphism from $A \hat{\otimes} B$ to $B \hat{\otimes} A$. This shows that $\hat{f}$ is the appropriate flip operator to use when working with skew products of skew graded algebras.⁴⁷

⁴⁷See for example subsection 6.18.

Chapter 3

In this chapter, we see three ways of defining a tensor:

- As an object in a tensor algebra characterized by a universal property.
- As a multilinear function.
- As a multidimensional array of scalars which transforms according to a certain rule under change of basis.

The first way is the most elegant but perhaps the hardest to learn; the second, easier to get started with but messier in the long run; the third, quick and dirty.

§ 2

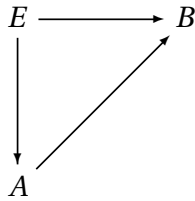
Remark. In this section, the underlying vector space of the tensor algebra $\otimes E$ is constructed bottom up as a direct sum

$$\otimes E = \sum_{p=0}^{\infty} \otimes^p E$$

from tensorial powers $\otimes^p E$, each of which is constructed as quotient space. The multiplication operation is then defined in $\otimes E$ to make it an algebra.

§ 3

Remark. The tensor algebra $\otimes E$ is a universal (initial) object in the category of “associative unital algebras with linear maps of E into them”. In this category, the objects are linear maps $E \rightarrow A$, for associative unital algebras A , and the arrows are unital algebra homomorphisms $A \rightarrow B$ which preserve the units and respect the linear maps from E :



Every object in this category can be obtained from the tensor algebra $\otimes E$ in a unique way. This is why $\otimes E$ is said to satisfy the “universal property”. This is only possible because the elements of $\otimes E$ satisfy only those properties that are required to make $\otimes E$ into an associative unital algebra containing E . By category theoretic abstract nonsense, $\otimes E$ is unique up to isomorphism (§ 4).

§ 5

Remark. We have a functor from the category of vector spaces into the category of associative unital algebras, which sends vector spaces E and F to the tensor algebras $\otimes E$ and $\otimes F$, and which sends a linear map $\varphi : E \rightarrow F$ to the unital algebra homomorphism $\varphi_{\otimes} : \otimes E \rightarrow \otimes F$.

Remark. For a linear map $\varphi : E \rightarrow F$, the following diagram commutes:

$$\begin{array}{ccc} E & \xrightarrow{\varphi} & F \\ i \downarrow & & \downarrow j \\ \otimes E & \xrightarrow{\varphi_{\otimes}} & \otimes F \end{array}$$

This shows that the canonical injection $i : E \rightarrow \otimes E$ is a natural transformation from the identity functor to the tensor algebra functor (in the category of vector spaces).

Remark. By a remark from § 21 of chapter 1 above,

$$\ker \varphi_{\otimes} = \sum_{p=0}^{\infty} \sum_{i=1}^p E \otimes \cdots \otimes \ker \varphi \otimes \cdots \otimes E$$

where $\ker \varphi$ is in position i of p on the right.

Remark. For vector spaces E and F , the canonical injections

$$E \xrightarrow{i_1} E \oplus F \xleftarrow{i_2} F$$

induce the homomorphisms

$$\otimes E \xrightarrow{(i_1)_{\otimes}} \otimes(E \oplus F) \xleftarrow{(i_2)_{\otimes}} \otimes F$$

which in turn induce a linear map $f : (\otimes E) \otimes (\otimes F) \rightarrow \otimes(E \oplus F)$ homogeneous of degree zero with

$$f(u \otimes v) = (i_1)_{\otimes}(u) \otimes (i_2)_{\otimes}(v) \quad (u \in \otimes E, v \in \otimes F)$$

However f is *not* an algebra homomorphism if $E \neq 0$ and $F \neq 0$. Indeed, if $x \in E$ with $x \neq 0$ and $y \in F$ with $y \neq 0$, then x and y are linearly independent

in $E \oplus F$, so $x \otimes y$ and $y \otimes x$ are linearly independent in $\otimes(E \oplus F)$. If f were a homomorphism, we would have

$$y \otimes x = f(1 \otimes y) \otimes f(x \otimes 1) = f((1 \otimes y)(x \otimes 1)) = f(x \otimes y) = x \otimes y$$

which is a contradiction.

In the other direction, the sum of the injections

$$E \longrightarrow \otimes E \longrightarrow (\otimes E) \otimes (\otimes F)$$

given by $x \mapsto x \otimes 1$ and

$$F \longrightarrow \otimes F \longrightarrow (\otimes E) \otimes (\otimes F)$$

given by $y \mapsto 1 \otimes y$ induces a homomorphism $h: \otimes(E \oplus F) \rightarrow (\otimes E) \otimes (\otimes F)$. If

$$u = x_1 \otimes \cdots \otimes x_p \in \otimes^p E \quad \text{and} \quad v = y_1 \otimes \cdots \otimes y_q \in \otimes^q F$$

then

$$\begin{aligned} (h \circ f)(u \otimes v) &= h(x_1 \otimes \cdots \otimes x_p \otimes y_1 \otimes \cdots \otimes y_q) \\ &= h(x_1) \cdots h(x_p) h(y_1) \cdots h(y_q) \\ &= (x_1 \otimes 1) \cdots (x_p \otimes 1) (1 \otimes y_1) \cdots (1 \otimes y_q) \\ &= (x_1 \otimes \cdots \otimes x_p) \otimes (y_1 \otimes \cdots \otimes y_q) \\ &= u \otimes v \end{aligned}$$

By multilinearity, it follows that $h \circ f = \iota$, so f is injective and h is surjective. However for $x \in E$ and $y \in F$,

$$h(y \otimes x) = (1 \otimes y)(x \otimes 1) = x \otimes y = (x \otimes 1)(1 \otimes y) = h(x \otimes y)$$

so h is not injective and f is not surjective if $E \neq 0$ and $F \neq 0$.

Intuitively it makes sense that $(\otimes E) \otimes (\otimes F)$ can be viewed as a subspace of $\otimes(E \oplus F)$ since elements in the former can be represented in the latter, but not a subalgebra since factors from E commute with factors from F in products in the former but not in the latter.

Similar remarks apply to $(\otimes E) \hat{\otimes} (\otimes F)$ and $\otimes(E \oplus F)$.

§ 6

Remark. For a linear map $\varphi : E \rightarrow E$, the following diagram commutes:

$$\begin{array}{ccc} E & \xrightarrow{\varphi} & E \\ i \downarrow & & \downarrow i \\ \otimes E & \xrightarrow{\theta_{\otimes}(\varphi)} & \otimes E \end{array}$$

Remark. There is a connection between the derivation and the trace of a linear transformation. If E is an n -dimensional vector space and $\varphi : E \rightarrow E$ is linear, let $\Delta : E^n \rightarrow \Gamma$ be a nonzero determinant function in E . Recall

$$\mathrm{tr} \varphi \cdot \Delta(x_1, \dots, x_n) = \sum_{i=1}^n \Delta(x_1, \dots, \varphi x_i, \dots, x_n)$$

By the universal property of the tensor product $\otimes^n E$, there is an induced linear function $\Delta_{\otimes} : \otimes^n E \rightarrow \Gamma$ with $\Delta_{\otimes}(x_1 \otimes \cdots \otimes x_n) = \Delta(x_1, \dots, x_n)$. Now

$$\begin{aligned} \mathrm{tr} \varphi \cdot \Delta_{\otimes}(x_1 \otimes \cdots \otimes x_n) &= \sum_{i=1}^n \Delta_{\otimes}(x_1 \otimes \cdots \otimes \varphi x_i \otimes \cdots \otimes x_n) \\ &= \Delta_{\otimes}\left(\sum_{i=1}^n x_1 \otimes \cdots \otimes \varphi x_i \otimes \cdots \otimes x_n\right) \\ &= \Delta_{\otimes}(\theta_{\otimes}(\varphi)(x_1 \otimes \cdots \otimes x_n)) \end{aligned}$$

It follows that

$$\Delta_{\otimes} \circ (\mathrm{tr} \varphi \cdot \iota) = \mathrm{tr} \varphi \cdot \Delta_{\otimes} = \Delta_{\otimes} \circ \theta_{\otimes}(\varphi)$$

Since $\theta_{\otimes}$ is linear by (3.7), it follows that the trace is also linear. If $\psi : E \rightarrow E$ is another linear transformation, then it follows from (3.8) that

$$\mathrm{tr}(\varphi\psi) - \mathrm{tr}(\psi\varphi) = \mathrm{tr}(\varphi\psi - \psi\varphi) = \mathrm{tr} \varphi \cdot \mathrm{tr} \psi - \mathrm{tr} \psi \cdot \mathrm{tr} \varphi = 0$$

so $\mathrm{tr}(\varphi\psi) = \mathrm{tr}(\psi\varphi)$.⁴⁸

⁴⁸This is what problem 3.7.3 was supposed to be.

§ 7

Exercise (1). If $u_1 = a_1 \otimes b_1 \neq 0$ and $u_2 = a_2 \otimes b_2$ are decomposable tensors, then $u_1 + u_2$ is decomposable if and only if $a_2 = \lambda a_1$ or $b_2 = \mu b_1$ for some $\lambda, \mu \in \Gamma$.

Proof. By the remark after problem 1.15.1 above. $\square$

§ 11

Remark. Define

$$M_q^p(E^*, E) = \underbrace{E^* \times \cdots \times E^*}_p \times \underbrace{E \times \cdots \times E}_q$$

Consider the $(p+q)$ -linear map $\otimes_q^p : M_q^p(E^*, E) \rightarrow \otimes_q^p(E^*, E)$ defined by

$$\otimes_q^p(x_1^*, \dots, x_p^*; x_1, \dots, x_q) = (x_1^* \otimes \cdots \otimes x_p^*) \otimes (x_1 \otimes \cdots \otimes x_q)$$

It is easy to verify that $\otimes_q^p$ satisfies the following universal property:

If $\varphi : M_q^p(E^*, E) \rightarrow F$ is any $(p+q)$ -linear map, then there is a unique linear map $f : \otimes_q^p(E^*, E) \rightarrow F$ with $f \circ \otimes_q^p = \varphi$:

$$\begin{array}{ccc} M_q^p(E^*, E) & \xrightarrow{\varphi} & F \\ \otimes_q^p \downarrow & \nearrow f & \\ \otimes_q^p(E^*, E) & & \end{array}$$

By category theoretic abstract nonsense, this property characterizes $\otimes_q^p(E^*, E)$ uniquely up to isomorphism.

Remark. In the finite-dimensional case, the following are familiar types of mixed tensors in E^*, E :

Bidegree	Type
(0,0)	Scalar
(0,1)	Vector
(1,0)	Linear function
(1,1)	Linear transformation

§ 12

Remark. The mixed tensor algebra $\otimes(E^*, E)$ can be constructed in two ways:

- Top down, as $\otimes(E^*, E) = (\otimes E^*) \otimes (\otimes E)$ — as in this section.
- Bottom up, as

$$\otimes(E^*, E) = \sum_{p,q} \otimes_q^p(E^*, E)$$

An element of this algebra is a finite sum of decomposable homogeneous mixed tensors. As an example of multiplication, if $x^* \otimes x$ and $y^* \otimes y$ are (1,1)-tensors, then

$$(x^* \otimes x)(y^* \otimes y) = x^* \otimes y^* \otimes x \otimes y$$

is a (2,2)-tensor.

Remark. By a remark in § 2 of chapter 2 above, $\otimes(E^*, E)$ satisfies a universal property as an associative unital algebra and is unique up to isomorphism.

Remark. By a remark in § 5 of chapter 2 above, $\otimes(E^*, E)$ is a *graded algebra* under the simple gradation in (2.15).

Remark. By (1.25), $\otimes(E^*, E)$ is dual to itself and to $\otimes(E, E^*)$.

§ 14

Remark. Let $\alpha^\otimes$ denote the restriction of $\alpha^\otimes$ to $\otimes^p E^*$ and $\alpha_\otimes$ the restriction of $\alpha_\otimes$ to $\otimes^q E$. Then $\alpha^\otimes$ and $\alpha_\otimes$ are automorphisms, and

$$T_\alpha = (\alpha^\otimes)^{-1} \otimes \alpha_\otimes$$

is an automorphism of $\otimes_q^p(E^*, E)$ which extends to $(\alpha^\otimes)^{-1} \otimes \alpha_\otimes$ on $\otimes(E^*, E)$.

Remark. The mapping $T : GL(E) \rightarrow GL(\otimes_q^p(E^*, E))$ given by $\alpha \mapsto T_\alpha$ is a group homomorphism. The map T_α is tensorial if α is in the center of $GL(E)$.

Remark. We have $T_\alpha^* = T_{\alpha^{-1}}$. Indeed, since $(\alpha^\otimes)^{-1} = (\alpha^{-1})^\otimes$,

$$\begin{aligned} \langle y^* \otimes y, T_\alpha(x^* \otimes x) \rangle &= \langle y^* \otimes y, (\alpha^\otimes)^{-1} x^* \otimes \alpha_\otimes x \rangle \\ &= \langle y^*, \alpha_\otimes x \rangle \langle (\alpha^\otimes)^{-1} x^*, y \rangle \\ &= \langle \alpha^\otimes y^*, x \rangle \langle x^*, (\alpha^{-1})_\otimes y \rangle \\ &= \langle \alpha^\otimes y^* \otimes (\alpha^{-1})_\otimes y, x^* \otimes x \rangle \\ &= \langle T_{\alpha^{-1}}(y^* \otimes y), x^* \otimes x \rangle \end{aligned}$$

Remark. Writing $\otimes_q^p(E^*, E) \otimes \otimes_q^p(E^*, E) = \otimes_q^p(E^* \otimes E^*, E \otimes E)$ (see problem 10), we have $T_\alpha \otimes T_\beta = T_{\alpha \otimes \beta}$. Writing $\otimes_q^p(E^*, E) \otimes \otimes_s^r(E^*, E) = \otimes_{q+s}^{p+r}(E^*, E)$, we have $T_\alpha \otimes T_\alpha = T_\alpha$.

Remark. If e_i, e^{*i} and $\bar{e}_i, \bar{e}^{*i}$ are pairs of dual bases in E, E^* and α is the change of basis transformation $e_i \mapsto \bar{e}_i$ in E , then α^{*-1} is the corresponding change of basis transformation $e^{*i} \mapsto \bar{e}^{*i}$ in E^* .⁴⁹ Therefore in $\otimes_q^p(E^*, E)$,

$$T_\alpha(e_{\mu_1 \dots \mu_q}^{v_1 \dots v_p}) = \bar{e}_{\mu_1 \dots \mu_q}^{v_1 \dots v_p}$$

In other words, T_α is the induced change of basis transformation in $\otimes_q^p(E^*, E)$. It follows that a mapping is tensorial if and only if its matrix is invariant under this type of change of basis.

Remark. For $\alpha = \lambda \iota$ with $\lambda \neq 0$, $\alpha^{*-1} = \lambda^{-1} \iota$, so $T_\alpha = \lambda^{q-p} \iota$ on $\otimes_q^p(E^*, E)$.

Remark. If E is finite-dimensional and we identify $\otimes_1^1(E^*, E)$ with $L(E; E)$ in the natural way, then for any automorphism α of E ,

$$T_\alpha(a^* \otimes b) = \alpha \circ (a^* \otimes b) \circ \alpha^{-1}$$

Exercise (1). If e_v, e^{*v} and $\bar{e}_v, \bar{e}^{*v}$ are pairs of dual bases in E, E^* related by

$$\bar{e}_v = \sum_\lambda \alpha_v^\lambda e_\lambda \quad \text{and} \quad \bar{e}^{*v} = \sum_\lambda \beta_\lambda^v e^{*\lambda}$$

then the components of a mixed tensor $w \in \otimes_q^p(E^*, E)$ are related by

$$\bar{\zeta}_{v_1 \dots v_p}^{\mu_1 \dots \mu_q} = \sum_{(\kappa), (\lambda)} \alpha_{v_1}^{\lambda_1} \dots \alpha_{v_p}^{\lambda_p} \beta_{\kappa_1}^{\mu_1} \dots \beta_{\kappa_q}^{\mu_q} \zeta_{\lambda_1 \dots \lambda_p}^{\kappa_1 \dots \kappa_q}$$

Proof. In this proof, we use the Einstein summation convention to simplify the notation. Since $e_\kappa = \beta_\kappa^\mu \bar{e}_\mu$ and $e^{*\lambda} = \alpha_\nu^\lambda \bar{e}^{*\nu}$, we have

$$\begin{aligned} w &= \zeta_{\lambda_1 \dots \lambda_p}^{\kappa_1 \dots \kappa_q} e_{\kappa_1 \dots \kappa_q}^{\lambda_1 \dots \lambda_p} \\ &= \zeta_{\lambda_1 \dots \lambda_p}^{\kappa_1 \dots \kappa_q} e^{*\lambda_1} \otimes \dots \otimes e^{*\lambda_p} \otimes e_{\kappa_1} \otimes \dots \otimes e_{\kappa_q} \\ &= \zeta_{\lambda_1 \dots \lambda_p}^{\kappa_1 \dots \kappa_q} (\alpha_{v_1}^{\lambda_1} \bar{e}^{*v_1}) \otimes \dots \otimes (\alpha_{v_p}^{\lambda_p} \bar{e}^{*v_p}) \otimes (\beta_{\kappa_1}^{\mu_1} \bar{e}_{\mu_1}) \otimes \dots \otimes (\beta_{\kappa_q}^{\mu_q} \bar{e}_{\mu_q}) \\ &= \zeta_{\lambda_1 \dots \lambda_p}^{\kappa_1 \dots \kappa_q} \alpha_{v_1}^{\lambda_1} \dots \alpha_{v_p}^{\lambda_p} \beta_{\kappa_1}^{\mu_1} \dots \beta_{\kappa_q}^{\mu_q} \bar{e}_{\mu_1 \dots \mu_q}^{v_1 \dots v_p} \end{aligned}$$

Therefore

$$\bar{\zeta}_{v_1 \dots v_p}^{\mu_1 \dots \mu_q} = \alpha_{v_1}^{\lambda_1} \dots \alpha_{v_p}^{\lambda_p} \beta_{\kappa_1}^{\mu_1} \dots \beta_{\kappa_q}^{\mu_q} \zeta_{\lambda_1 \dots \lambda_p}^{\kappa_1 \dots \kappa_q}$$

□

⁴⁹See the remark in chapter III, § 3 of [1] above.

Remark. This relationship serves as the basis of an alternative definition of a *tensor* as a multidimensional array of scalars which transforms according to this rule under change of basis.

Exercise (3). For $u^* \in \otimes^p E^*$ and $u \in \otimes^p E$,

$$\langle u^*, u \rangle = (C_1^1)^p(u^* \otimes u)$$

Proof. By induction on p . For $p = 0$ the result is trivial, and for $p = 1$ it follows from the definition of C_1^1 . For $p > 1$, write $u^* = u^{*1} \otimes \tilde{u}^*$ and $u = u_1 \otimes \tilde{u}$. Then

$$\begin{aligned} (C_1^1)^p(u^* \otimes u) &= (C_1^1)^{p-1}(C_1^1(u^* \otimes u)) \\ &= \langle u^{*1}, u_1 \rangle (C_1^1)^{p-1}(\tilde{u}^* \otimes \tilde{u}) \\ &= \langle u^{*1}, u_1 \rangle \langle \tilde{u}^*, \tilde{u} \rangle \\ &= \langle u^*, u \rangle \end{aligned}$$

□

Exercise (4). If the mappings

$$\Phi : \otimes_q^p(E^*, E) \rightarrow \otimes_s^r(E^*, E) \quad \text{and} \quad \Phi^* : \otimes_p^q(E^*, E) \leftarrow \otimes_r^s(E^*, E)$$

are dual and Φ is tensorial, then Φ^* is tensorial.

Proof. If α is an automorphism of E , then by a remark above

$$\Phi^* \circ T_\alpha = \Phi^* \circ T_{\alpha^{-1}}^* = (T_{\alpha^{-1}} \circ \Phi)^* = (\Phi \circ T_{\alpha^{-1}})^* = T_{\alpha^{-1}}^* \circ \Phi^* = T_\alpha \circ \Phi^* \quad \square$$

Exercise (5). The sum, composite, and tensor product of tensorial mappings are tensorial.

Proof. If Φ and Ψ are tensorial, then it is obvious that $\Phi + \Psi$ and $\Phi \circ \Psi$ are also tensorial. By a remark above,

$$\begin{aligned} (\Phi \otimes \Psi) \circ T_\alpha &= (\Phi \otimes \Psi) \circ (T_\alpha \otimes T_\alpha) \\ &= (\Phi \circ T_\alpha) \otimes (\Psi \circ T_\alpha) \\ &= (T_\alpha \circ \Phi) \otimes (T_\alpha \circ \Psi) \\ &= (T_\alpha \otimes T_\alpha) \circ (\Phi \otimes \Psi) \\ &= T_\alpha \circ (\Phi \otimes \Psi) \end{aligned}$$

so $\Phi \otimes \Psi$ is tensorial. □

Exercise (6). Let Γ have characteristic zero. If $\Phi : \otimes_q^p(E^*, E) \rightarrow \otimes_s^r(E^*, E)$ is a nonzero tensorial mapping, then $r - p = s - q$.

Proof. By a remark above with $\lambda = 2$, $2^{q-p}\Phi = 2^{s-r}\Phi$ since Φ is tensorial. It follows that $2^{q-p+r-s} = 1$ since $\Phi \neq 0$, so $q - p + r - s = 0$. $\square$

Exercise (7). If E is finite-dimensional and for $a \in E^* \otimes E$ the linear map $\mu(a)$ is defined by $z \mapsto a \otimes z$ for $z \in E$, then $\mu(a)$ is tensorial if and only if $a = \lambda t$ where t is the unit tensor.

Proof. By direct computation, it is easily verified that $\mu(a)$ is tensorial if and only if $T_\alpha(a) = a$ for all automorphisms α of E . By a remark above, the latter is true if and only if a , when viewed as a transformation of E , is preserved under conjugation, which is true if and only if it is a scalar transformation. Since the unit tensor corresponds to the identity transformation, the result follows. $\square$

Exercise (8). If E is finite-dimensional, then every tensorial map $\Phi : E^* \otimes E \rightarrow \Gamma$ is of the form $\Phi = \lambda \cdot C$ where C is the contraction operator.

Proof. This follows from problem 1.3.7 applied to $\Phi \circ \otimes$. $\square$

Exercise (11). Let Γ have characteristic zero.

- (a) If $u \in \otimes_q^p(E^*, E)$ is a nonzero invariant tensor, then $p = q$.
- (b) If E has finite dimension, then $u \in E^* \otimes E$ is invariant if and only if $u = \lambda t$ where t is the unit tensor.
- (c) If E has infinite dimension and $u \in E^* \otimes E$ is invariant, then $u = 0$.
- (d) If E has finite dimension, then u is invariant if and only if the components of u are the same with respect to every pair of dual bases.

Proof. For (a), taking $\alpha = 2\iota$ we have $T_\alpha = 2^{q-p}\iota$ by a remark above, so $2^{q-p}u = u$ since u is invariant, which implies $2^{q-p} = 1$ since $u \neq 0$, which implies $q - p = 0$.

For (b) and (c), if we view u as a linear transformation of E , then u is invariant if and only if it is preserved under conjugation, which is true if and only if it is a scalar transformation. Now (b) follows since the unit tensor corresponds to the identity transformation when $\dim E < \infty$, and (c) follows since there is no unit tensor when $\dim E = \infty$.

For (d), since T_α is a change of basis transformation by a remark above. $\square$

Remark. Intuitively, an invariant tensor represents a geometric quantity whose description does not depend upon the choice of coordinate system.

§ 15

Remark. The induced inner products in $\otimes^p E$ and $\otimes E$ are just the induced scalar products from § 8 with $E^* = E$ under the inner product.

§ 17

Remark. The metric tensors encode all of the information required to compute inner products, and hence metric properties (length, angle, etc.), in E and E^* .

Remark. Since E and the orthonormal basis e_v are self-dual under the inner product, the metric tensor $g = \sum_v e_v \otimes e_v$ is just the unit tensor in $E \otimes E$. It follows that $g \otimes g$ is the unit tensor in $(E \otimes E) \otimes (E \otimes E)$, and hence is the metric tensor in $E \otimes E$, and more generally

$$\underbrace{g \otimes \cdots \otimes g}_p$$

is the metric tensor in $\otimes^p E$.⁵⁰

Remark. We have

$$\tau_{\otimes}(g) = \sum_v \tau e_v \otimes \tau e_v = \sum_v e^{*v} \otimes e^{*v} = g^*$$

so $\tau_{\otimes}^{-1}(g^*) = g$. This yields

$$\begin{aligned} (x^*, y^*) &= (\tau^{-1} x^*, \tau^{-1} y^*) \\ &= \langle g^*, \tau^{-1} x^* \otimes \tau^{-1} y^* \rangle \\ &= \langle g^*, \tau_{\otimes}^{-1}(x^* \otimes y^*) \rangle \\ &= \langle x^* \otimes y^*, \tau_{\otimes}^{-1} g^* \rangle \\ &= \langle x^* \otimes y^*, g \rangle \end{aligned}$$

Exercise (2). Let E, E^* be a pair of dual finite-dimensional Euclidean spaces with metric tensors $g \in E \otimes E$ and $g^* \in E^* \otimes E^*$.

(a) $C_2^1(g^* \otimes g)$ is the unit tensor in $E^* \otimes E$.

(b) $\underbrace{g \otimes \cdots \otimes g}_p$ is the metric tensor of $\otimes^p E$.

⁵⁰See problem 2(b) for another proof.

Proof. Let $e_\nu, e^{*\nu}$ be a pair of dual orthonormal bases in E, E^* . We know that $g = \sum_\nu e_\nu \otimes e_\nu$ and $g^* = \sum_\mu e^{*\mu} \otimes e^{*\mu}$, so

$$\begin{aligned} C_2^1(g^* \otimes g) &= \sum_{\mu, \nu} C_2^1(e^{*\mu} \otimes e^{*\mu} \otimes e_\nu \otimes e_\nu) \\ &= \sum_{\mu, \nu} \langle e^{*\mu}, e_\nu \rangle e^{*\mu} \otimes e_\nu \\ &= \sum_\nu e^{*\nu} \otimes e_\nu \end{aligned}$$

which is the unit tensor in $E^* \otimes E$.

Writing $\bigotimes^p(E \otimes E) = (\bigotimes^p E) \otimes (\bigotimes^p E)$, we have

$$\begin{aligned} \underbrace{g \otimes \cdots \otimes g}_p &= \left(\sum_{\nu_1} e_{\nu_1} \otimes e_{\nu_1} \right) \otimes \cdots \otimes \left(\sum_{\nu_p} e_{\nu_p} \otimes e_{\nu_p} \right) \\ &= \sum_{(\nu)} (e_{\nu_1} \otimes \cdots \otimes e_{\nu_p}) \otimes (e_{\nu_1} \otimes \cdots \otimes e_{\nu_p}) \end{aligned}$$

which is the metric tensor of $\bigotimes^p E$ since $e_{\nu_1} \otimes \cdots \otimes e_{\nu_p}$ is an orthonormal basis in $\bigotimes^p E$. $\square$

Exercise (4). Let E, E^* be a pair of dual finite-dimensional Euclidean spaces.

- (a) If $\varphi = a^* \otimes b$ is a linear transformation of E , then $\tilde{\varphi} = \tau b \otimes \tau^{-1} a^*$.
- (b) The inner product induced in $L(E; E)$ is given by

$$(\varphi, \psi) = \text{tr}(\varphi \circ \tilde{\psi})$$

Proof. For (a), if $\varphi = a^* \otimes b$ in $E^* \otimes E$ then $\varphi = \tau^{-1} a^* \otimes b$ in $E \otimes E$, so $\tilde{\varphi} = b \otimes \tau^{-1} a^*$ in $E \otimes E$ by problem 1.30.1 and $\tilde{\varphi} = \tau b \otimes \tau^{-1} a^*$ in $E^* \otimes E$.

For (b), if $\varphi = a^* \otimes b$ and $\psi = c^* \otimes d$, then

$$\begin{aligned} \varphi \circ \tilde{\psi} &= (a^* \otimes b) \circ (\tau d \otimes \tau^{-1} c^*) \\ &= \langle a^*, \tau^{-1} c^* \rangle (\tau d \otimes b) \\ &= (\tau^{-1} a^*, \tau^{-1} c^*) (\tau d \otimes b) \\ &= (a^*, c^*) (\tau d \otimes b) \end{aligned}$$

so

$$\begin{aligned}
\text{tr}(\varphi \circ \tilde{\psi}) &= (a^*, c^*) \langle \tau d, b \rangle \\
&= (a^*, c^*)(b, d) \\
&= (a^* \otimes b, c^* \otimes d) \\
&= (\varphi, \psi)
\end{aligned}$$

The general case follows by multilinearity. $\square$

Exercise (7). If E, E^* is a pair of dual finite-dimensional Euclidean spaces and $x_\nu, x^{*\nu}$ a pair of dual bases, then

$$g^* = \sum_{\nu, \mu} (x_\nu, x_\mu) x^{*\nu} \otimes x^{*\mu} \quad \text{and} \quad g = \sum_{\nu, \mu} (x^{*\nu}, x^{*\mu}) x_\nu \otimes x_\mu$$

Proof. Since $x_\nu \otimes x_\mu$ and $x^{*\nu} \otimes x^{*\mu}$ are dual bases of $E \otimes E$ and $E^* \otimes E^*$, we have

$$g^* = \sum_{\nu, \mu} g_{\nu\mu} x^{*\nu} \otimes x^{*\mu} \quad \text{and} \quad g = \sum_{\nu, \mu} g^{\nu\mu} x_\nu \otimes x_\mu$$

where

$$g_{\nu\mu} = \langle g^*, x_\nu \otimes x_\mu \rangle = (x_\nu, x_\mu)$$

and

$$g^{\nu\mu} = \langle x^{*\nu} \otimes x^{*\mu}, g \rangle = (x^{*\nu}, x^{*\mu}) \quad \square$$

Remark. It follows that the components of g^* vary in the same way as the basis vectors in E under a change of basis, while the components of g vary inversely. This is why g^* is called “covariant” and g is called “contravariant”.

§ 18

Remark. The multiplication in the algebra $T^*(E)$ is defined by

$$\Phi \cdot \Psi = \sum_{p, q} \Phi_p \cdot \Psi_q$$

where $\Phi = \sum_p \Phi_p$ with $\Phi_p \in T^p(E)$, $\Psi = \sum_q \Psi_q$ with $\Psi_q \in T^q(E)$, and $\Phi_p \cdot \Psi_q$ is given by (3.17).⁵¹ This definition makes $T^*(E)$ into a *graded algebra* where the elements of $T^p(E)$ are homogeneous of degree p .

⁵¹ Compare with the definition in subsection 3.2.

Remark. The function $f_1 \cdots f_p$ ($f_v \in T^1(E)$) is *decomposable* in $T^p(E)$.

Remark. For a given linear map $\varphi : E \rightarrow F$, to see that $\varphi^* : T^*(E) \leftarrow T^*(F)$ is a homomorphism homogeneous of degree zero, first note that $\varphi^* \Psi \in T^p(E)$ for $\Psi \in T^p(F)$ by the definition since φ is linear and Ψ is p -linear. Moreover,

$$\varphi^*(\lambda\Phi + \mu\Psi) = \lambda\varphi^*\Phi + \mu\varphi^*\Psi \quad (\Phi, \Psi \in T^p(F))$$

so $\varphi^* : T^p(E) \leftarrow T^p(F)$ is linear. This map extends via direct sum over all p to the linear map $\varphi^* : T^*(E) \leftarrow T^*(F)$ which is homogeneous of degree zero. For $\Phi \in T^p(F)$ and $\Psi \in T^q(F)$,

$$\begin{aligned} (\varphi^*(\Phi \cdot \Psi))(x_1, \dots, x_{p+q}) &= (\Phi \cdot \Psi)(\varphi x_1, \dots, \varphi x_{p+q}) \\ &= \Phi(\varphi x_1, \dots, \varphi x_p) \cdot \Psi(\varphi x_{p+1}, \dots, \varphi x_{p+q}) \\ &= (\varphi^*\Phi)(x_1, \dots, x_p) \cdot (\varphi^*\Psi)(x_{p+1}, \dots, x_{p+q}) \\ &= ((\varphi^*\Phi) \cdot (\varphi^*\Psi))(x_1, \dots, x_{p+q}) \end{aligned}$$

so

$$\varphi^*(\Phi \cdot \Psi) = (\varphi^*\Phi) \cdot (\varphi^*\Psi)$$

This equation extends to all $\Phi, \Psi \in T^*(F)$ by bilinearity of the multiplication in $T^*(E)$ and $T^*(F)$ and linearity of φ^* , so φ^* is a homomorphism. Under the natural identifications $E^* = T^1(E)$ and $F^* = T^1(F)$, φ^* is an extension of the dual map $\varphi^* : E^* \leftarrow F^*$ of φ and

$$\varphi^*(y_1^* \cdots y_p^*) = \varphi^* y_1^* \cdots \varphi^* y_p^* \quad (y_v^* \in F^*)$$

Remark. For a linear map $\varphi : E \rightarrow E$, the map $\theta^T(\varphi) : T^*(E) \leftarrow T^*(E)$ is seen to be a derivation homogeneous of degree zero by an argument similar to that in the previous remark. In this case for $\Phi \in T^p(E)$ and $\Psi \in T^q(E)$ with $p, q \geq 1$,

$$\begin{aligned} (\theta^T(\varphi)(\Phi \cdot \Psi))(x_1, \dots, x_{p+q}) &= \sum_{i=1}^{p+q} (\Phi \cdot \Psi)(x_1, \dots, \varphi x_i, \dots, x_{p+q}) \\ &= \sum_{i=1}^p \Phi(x_1, \dots, \varphi x_i, \dots, x_p) \cdot \Psi(x_{p+1}, \dots, x_{p+q}) \\ &\quad + \sum_{i=p+1}^{p+q} \Phi(x_1, \dots, x_p) \cdot \Psi(x_{p+1}, \dots, \varphi x_i, \dots, x_{p+q}) \\ &= (\theta^T(\varphi)(\Phi))(x_1, \dots, x_p) \cdot \Psi(x_{p+1}, \dots, x_{p+q}) \\ &\quad + \Phi(x_1, \dots, x_p) \cdot (\theta^T(\varphi)(\Psi))(x_{p+1}, \dots, x_{p+q}) \\ &= (\theta^T(\varphi)(\Phi) \cdot \Psi + \Phi \cdot \theta^T(\varphi)(\Psi))(x_1, \dots, x_{p+q}) \end{aligned}$$

so

$$\theta^T(\varphi)(\Phi \cdot \Psi) = \theta^T(\varphi)(\Phi) \cdot \Psi + \Phi \cdot \theta^T(\varphi)(\Psi)$$

Under the identification $E^* = T^1(E)$, $\theta^T(\varphi)$ extends the dual map $\varphi^* : E^* \leftarrow E^*$ and

$$\theta^T(\varphi)(x_1^* \cdots x_p^*) = \sum_{i=1}^p x_1^* \cdots \varphi^* x_i^* \cdots x_p^* \quad (x_v^* \in E^*)$$

§ 19

Remark. The definitions of the substitution operators $i_v(h)$, $i_A(h)$, and $i_S(h)$ in this subsection are stated on $T^p(E)$ and understood to extend via direct sum over all p to $T^*(E)$. The operator $i_v(h)$ acts as the identity on $T^p(E)$ for $p < v$. The definitions of $i_A(h)$ and $i_S(h)$ are inappropriate for use with the algebras of skew-symmetric and symmetric multilinear functions, respectively. See below.

Remark. The operator $i_A(h)$ is seen to be an antiderivation with respect to the canonical involution in $T^*(E)$, while $i_S(h)$ is seen to be a derivation in $T^*(E)$.

§ 21

Remark. By definition,

$$T_*(E) = \sum_p T_p(E) = \sum_p T^p(E^*) = T^*(E^*)$$

The multiplication makes $T_*(E)$ into a *graded algebra* where the elements of $T_p(E)$ are homogeneous of degree p .

Remark. For a linear map $\varphi : E \rightarrow F$, the dual map $\varphi^* : E^* \leftarrow F^*$ induces the homomorphism $\varphi_* = (\varphi^*)^* : T^*(E^*) \rightarrow T^*(F^*)$, homogeneous of degree zero. Under the natural identifications $E = T_1(E)$ and $F = T_1(F)$, φ_* extends φ and

$$\varphi_*(x_1 \cdots x_p) = \varphi x_1 \cdots \varphi x_p \quad (x_v \in E)$$

Remark. For a linear map $\varphi : E \rightarrow E$, the dual map $\varphi^* : E^* \leftarrow E^*$ induces the derivation $\theta_T(\varphi) = \theta^T(\varphi^*) : T^*(E^*) \rightarrow T^*(E^*)$ which is homogeneous of degree zero. Under the identification $E = T_1(E)$, $\theta_T(\varphi)$ extends φ and

$$\theta_T(\varphi)(x_1 \cdots x_p) = \sum_{i=1}^p x_1 \cdots \varphi x_i \cdots x_p \quad (x_v \in E)$$

§ 22

Remark. The scalar product between $T^p(E)$ and $T_p(E)$ is given by

$$\langle f_1 \cdots f_p, x_1 \cdots x_p \rangle = \langle f_1, x_1 \rangle \cdots \langle f_p, x_p \rangle \quad (f_v \in E^*, x_v \in E)$$

It follows that the isomorphisms $\otimes^p E^* \cong T^p(E)$ and $\otimes^p E \cong T_p(E)$ preserve scalar products.

Remark. A scalar product between $T^*(E)$ and $T_*(E)$ is induced by

$$\langle \Phi, \Psi \rangle = \sum_p \langle \Phi_p, \Psi_p \rangle$$

where $\Phi = \sum_p \Phi_p$ with $\Phi_p \in T^p(E)$ and $\Psi = \sum_p \Psi_p$ with $\Psi_p \in T_p(E)$. It follows from the previous remark that the isomorphisms $\otimes E^* \cong T^*(E)$ and $\otimes E \cong T_*(E)$ preserve scalar products.

Remark. For a linear map $\varphi : E \rightarrow F$, the homomorphisms $\varphi_* : T_*(E) \rightarrow T_*(F)$ and $\varphi^* : T^*(F) \leftarrow T^*(E)$ are dual. This can be proved by direct computation or by appealing to commutativity of the following diagrams:

$$\begin{array}{ccc} \otimes E^* & \xleftarrow{\varphi^\otimes} & \otimes F^* \\ \cong \downarrow & & \downarrow \cong \\ T^*(E) & \xleftarrow{\varphi^*} & T^*(F) \end{array} \quad \begin{array}{ccc} \otimes E & \xrightarrow{\varphi_\otimes} & \otimes F \\ \cong \downarrow & & \downarrow \cong \\ T_*(E) & \xrightarrow{\varphi_*} & T_*(F) \end{array}$$

Remark. For a linear map $\varphi : E \rightarrow E$, the derivations $\theta_T(\varphi) : T_*(E) \rightarrow T_*(E)$ and $\theta^T(\varphi) : T^*(E) \leftarrow T^*(E)$ are dual, from the following diagrams:

$$\begin{array}{ccc} \otimes E^* & \xleftarrow{\theta^\otimes(\varphi)} & \otimes E^* \\ \cong \downarrow & & \downarrow \cong \\ T^*(E) & \xleftarrow{\theta^T(\varphi)} & T^*(E) \end{array} \quad \begin{array}{ccc} \otimes E & \xrightarrow{\theta_\otimes(\varphi)} & \otimes E \\ \cong \downarrow & & \downarrow \cong \\ T_*(E) & \xrightarrow{\theta_T(\varphi)} & T_*(E) \end{array}$$

§ 23

Remark. We have

$$T_q^p(E) \cong (\otimes^p E^*) \otimes (\otimes^q E) = \otimes_q^p(E^*, E)$$

and

$$T(E) \cong (\otimes E^*) \otimes (\otimes E) = \otimes(E^*, E)$$

It follows from a remark in § 12 above that $T(E)$ is a *graded algebra* under the simple gradation in (2.15). Also $T_q^p(E)$ is dual to $T_p^q(E)$, and $T(E)$ is dual to itself and to $T(E^*)$.

Remark. The results in this subsection show that it is possible to develop tensor algebra using multilinear function spaces instead of tensor product spaces (at least in the finite-dimensional case). While it is easier to get started this way, it is messier in the long run.

Chapter 4

§ 2

Remark. The factor $1/p!$ in the definition of the alternator π_A on $\otimes^p E$ ensures that π_A is a projection operator ($\pi_A^2 = \pi_A$), which is necessary to establish the “associativity” property (4.11). This factor also creeps into the scalar product between dual spaces of skew-symmetric tensors in (4.8).

Remark. By (4.4) and (4.5), we have the following commutative diagram:

$$\begin{array}{ccc} \otimes^p E & \xrightarrow{\pi_A} & X^p(E) \\ \pi \downarrow & \nearrow \cong & \\ \otimes^p E / N^p(E) & & \end{array}$$

Therefore $\otimes^p E / N^p(E)$ can be viewed as a top down construction of the space of skew-symmetric tensors of degree p , where we start with tensors of degree p and kill off tensors with equal factors. This is the tensor equivalent of the result that a p -linear function is skew-symmetric if and only if it is alternating. In fact, the proof of (4.4) (including the proof of lemma (4.1)) is just like the proof of the latter result.⁵²

§ 3

Remark. The duality of $\pi_A : \otimes^p E \rightarrow \otimes^p E$ and $\pi^A : \otimes^p E^* \leftarrow \otimes^p E^*$ induces a duality between the restriction $\pi_A : \otimes^p E \rightarrow X^p(E)$ and the quotient

$$\overline{\pi^A} : \otimes^p E^* \leftarrow \otimes^p E^* / N^p(E^*)$$

since

$$N^p(E^*) = \ker \pi^A = (\operatorname{Im} \pi_A)^\perp = X^p(E)^\perp$$

On the other hand, $\otimes^p E^* / N^p(E^*) \cong X^p(E^*)$, so $\pi_A : \otimes^p E \rightarrow X^p(E)$ is dual to the restriction $\pi^A : \otimes^p(E^*) \leftarrow X^p(E^*)$.

⁵²See Proposition I in chapter IV, § 1 of [1] and chapter 5, § 1 of [2].

§ 4

Remark. We see from (4.10) that the skew-symmetric part of a product depends only on the skew-symmetric parts of the factors.

Exercise (1). The map $\sigma : \otimes^p E \rightarrow \otimes^p E$ is tensorial. If E is finite-dimensional, then σ is generated by $\mu(t)$ and C_j^i , where t is the unit tensor.

Proof. If α is an automorphism of E and $u = x_1 \otimes \cdots \otimes x_p \in \otimes^p E$, then

$$\sigma T_\alpha u = \alpha x_{\sigma^{-1}(1)} \otimes \cdots \otimes \alpha x_{\sigma^{-1}(p)} = T_\alpha \sigma u$$

so $\sigma T_\alpha = T_\alpha \sigma$ and hence σ is tensorial.

For the second claim, we may assume that $\sigma = (12 \cdots j)$ where $1 < j$. Write $t = \sum_v e^{*v} \otimes e_v$ where e_v, e^{*v} is any pair of dual bases in E, E^* . Then

$$\begin{aligned} C_{j+1}^1(t \otimes u) &= \sum_v C_{j+1}^1(e^{*v} \otimes e_v \otimes x_1 \otimes \cdots \otimes x_p) \\ &= \sum_v \langle e^{*v}, x_j \rangle e_v \otimes x_1 \otimes \cdots \otimes x_{j-1} \otimes x_{j+1} \otimes \cdots \otimes x_p \\ &= x_j \otimes x_1 \otimes \cdots \otimes x_{j-1} \otimes x_{j+1} \otimes \cdots \otimes x_p \\ &= \sigma u \end{aligned}$$

so $C_{j+1}^1 \circ \mu(t) = \sigma$. □

§ 5

Remark. In the following subsections, an algebra of skew-symmetric tensors is constructed in two ways:

- Top down, starting with the tensor algebra $\otimes E$ and killing off tensors in $N(E)$ to form $\otimes E / N(E)$. The multiplication in $\otimes E / N(E)$ is induced from that in $\otimes E$.
- Bottom up, by collecting skew-symmetric tensors in $\otimes E$ to form $X(E)$. The multiplication in $X(E)$ is defined using the alternator as in problem 1 of § 8 below.

The two constructions are essentially equivalent by (4.17), although the scalar product between dual algebras is defined slightly differently in § 8—specifically, the product for the top down construction is induced by that for the bottom up construction, but the factor $1/p!$ in (4.8) is killed off in (4.18) to obtain (4.20).

§ 6

Remark. If we make the appropriate identifications, we have

$$\pi(\otimes^p E) = \otimes^p E / (N(E) \cap \otimes^p E) = \otimes^p E / N^p(E) = X^p(E)$$

so we already know

$$\otimes E / N(E) = \sum_p \pi(\otimes^p E) = \sum_p X^p(E) = X(E)$$

Remark. To see how (4.16) follows from (4.13), write $u = \pi(a)$ and $v = \pi(b)$ with $a \in \otimes^p E$ and $b \in \otimes^q E$. By (4.15), (4.17), and (4.13),

$$uv = \pi a \cdot \pi b = \pi(a \otimes b) = \rho \pi_A(a \otimes b) = (-1)^{pq} \rho \pi_A(b \otimes a) = (-1)^{pq} vu$$

§ 7

Remark. $X(E)$ is not a subalgebra of $\otimes E$, but can be made into an algebra as in problem 1 of § 8 below.

Remark. The inverse of ρ is the induced isomorphism $\overline{\pi_X} : \otimes E / N(E) \cong X(E)$:

$$\begin{array}{ccc} \otimes E & \xrightarrow{\pi_X} & X(E) \\ \pi \downarrow & \nearrow \rho^{-1} = \overline{\pi_X} & \\ \otimes E / N(E) & & \end{array}$$

This diagram generalizes the diagram in § 2 above.

§ 8

Exercise (1). If we define a product in $X(E)$ by

$$u \cdot v = \pi_A(u \otimes v) \quad (u, v \in X(E))$$

then $X(E)$ becomes a graded algebra and ρ an algebra isomorphism, where the elements of $X^p(E)$ are homogeneous of degree p . Moreover,

$$\pi_A u \cdot \pi_A v = \pi_A(u \otimes v) \quad (u, v \in \otimes E)$$

Proof. Clearly the product makes $X(E)$ into a graded algebra. By (4.17) and (4.15),

$$\rho(u \cdot v) = \rho \pi_A(u \otimes v) = \pi(u \otimes v) = \pi u \cdot \pi v = \rho u \cdot \rho v$$

for all $u, v \in X(E)$, so ρ is an algebra isomorphism. By (4.10),

$$\pi_A u \cdot \pi_A v = \pi_A(\pi_A u \otimes \pi_A v) = \pi_A(u \otimes v)$$

for all $u, v \in \otimes E$. □

Remark. The proof also shows that the product is uniquely determined by the requirement that ρ be an algebra isomorphism.

§ 9

Remark. Remarks analogous to those made above for skew-symmetry can be made for symmetry.

Chapter 5

§ 1

Remark. For $\sigma \in S_p$,

$$\sigma(f_1 \cdots f_p) = f_{\sigma^{-1}(1)} \cdots f_{\sigma^{-1}(p)} \quad (f_v \in T^1(E))$$

since

$$\begin{aligned} (\sigma(f_1 \cdots f_p))(x_1, \dots, x_p) &= (f_1 \cdots f_p)(x_{\sigma(1)}, \dots, x_{\sigma(p)}) \\ &= f_1(x_{\sigma(1)}) \cdots f_p(x_{\sigma(p)}) \\ &= f_{\sigma^{-1}(1)}(x_1) \cdots f_{\sigma^{-1}(p)}(x_p) \\ &= (f_{\sigma^{-1}(1)} \cdots f_{\sigma^{-1}(p)})(x_1, \dots, x_p) \end{aligned}$$

If E is finite-dimensional, it follows that these diagrams commute:

$$\begin{array}{ccc} \otimes^p E^* & \xrightarrow{\sigma} & \otimes^p E^* \\ \cong \downarrow & & \downarrow \cong \\ T^p(E) & \xrightarrow{\sigma} & T^p(E) \end{array} \quad \begin{array}{ccc} \otimes^p E^* & \xrightarrow{\pi^A} & \otimes^p E^* \\ \cong \downarrow & & \downarrow \cong \\ T^p(E) & \xrightarrow{A} & T^p(E) \end{array}$$

As with π_A , the factor $1/p!$ in the definition of A on $T^p(E)$ ensures that A is a projection operator ($A^2 = A$), which is necessary to establish the “associativity” property (5.81).

Exercise (1). Let $L_p(E; F)$ denote the space of p -linear mappings $E^p \rightarrow F$, and $A_p(E; F)$ the subspace of skew-symmetric mappings. If $T : L_p(E; F) \rightarrow L_p(E; F)$ is linear such that $T\varphi = \varphi$ for all $\varphi \in A_p(E; F)$, and $T(\sigma\varphi) = \varepsilon_\sigma T\varphi$ for all $\sigma \in S_p$ and $\varphi \in L_p(E; F)$, then $T = A$ (the alternator).

Proof. By assumption, $TA = A$ and

$$(TA)\varphi = T(A\varphi) = \frac{1}{p!} \sum_{\sigma} \varepsilon_\sigma T(\sigma\varphi) = \frac{1}{p!} \sum_{\sigma} T\varphi = T\varphi$$

so $TA = T$. □

Remark. This result shows that the alternator is the unique “skew-symmetric” transformation of p -linear mappings which fixes skew-symmetric mappings.

§ 3

Remark. The p -th exterior power $\wedge^p E$ of E is a universal (initial) object in the category of “vector spaces with skew-symmetric p -linear maps of E into them”. In this category, the objects are skew-symmetric p -linear maps $E^p \rightarrow F$, and the arrows are linear maps $F \rightarrow G$ which respect the p -linear maps:

$$\begin{array}{ccc} E^p & \xrightarrow{\quad} & G \\ \downarrow & \nearrow & \\ F & & \end{array}$$

Every object $E^p \rightarrow F$ in this category can be obtained from the exterior product $\wedge^p : E^p \rightarrow \wedge^p E$ in a unique way. This is why $\wedge^p$ is said to satisfy the “universal property”. This is only possible because the elements of $\wedge^p E$ satisfy only those relations required to make $\wedge^p$ skew-symmetric and p -linear. By category theoretic abstract nonsense, $\wedge^p E$ is unique up to isomorphism.

Remark. The p -th exterior power $\wedge^p E$ can be constructed in two ways:

- Top down, as $\wedge^p E = \otimes^p E / N^p(E)$.
- Bottom up, as $\wedge^p E = X^p(E)$.

Remark. Geometrically, the p -vector

$$\wedge^p(x_1, \dots, x_p) = x_1 \wedge \cdots \wedge x_p$$

represents the oriented (signed) volume of the p -dimensional parallelepiped determined by $x_1, \dots, x_p$. It can also be thought of as a family of p -dimensional parallelepipeds with the same orientation and volume.

Remark. If E is n -dimensional and $\Delta \neq 0$ is a determinant function in E , then since $\wedge^n$ is skew-symmetric and n -linear in E ,

$$x_1 \wedge \cdots \wedge x_n = \Delta(x_1, \dots, x_n) \cdot e_1 \wedge \cdots \wedge e_n$$

where $\Delta(e_1, \dots, e_n) = 1$. Here $e_1 \wedge \cdots \wedge e_n$ represents the unit volume under Δ . On the other hand, since Δ is skew-symmetric and n -linear, there is a linear function $\Delta_\wedge : \wedge^n E \rightarrow \Gamma$ with

$$\Delta_\wedge(x_1 \wedge \cdots \wedge x_n) = \Delta(x_1, \dots, x_n)$$

Note $\Delta_\wedge$ is inverse to $e_1 \wedge \cdots \wedge e_n \cdot \iota$, so it is an isomorphism. Also

$$\Delta_\wedge(x_1 \wedge \cdots \wedge x_n) = \Delta_\otimes(x_1 \otimes \cdots \otimes x_n)$$

These results show that the universality of determinant functions is a special case of the universality of the exterior product.

§ 4

Remark. The projection $\pi : \otimes E \rightarrow \wedge E$ constructed in this subsection and the canonical projection $\rho : \otimes E \rightarrow \otimes E / N(E)$ are connected by this commutative diagram:

$$\begin{array}{ccc} \otimes E & \xrightarrow{\pi} & \wedge E \\ \rho \downarrow & \nearrow f & \\ \otimes E / N(E) & & \end{array}$$

For $u, v \in \wedge E$, if we write $f^{-1}u = \rho \tilde{u}$ and $f^{-1}v = \rho \tilde{v}$ for $\tilde{u}, \tilde{v} \in \otimes E$, then $u = \pi \tilde{u}$ and $v = \pi \tilde{v}$ and

$$u \wedge v = f(\rho \tilde{u} \cdot \rho \tilde{v}) = f(\rho(\tilde{u} \otimes \tilde{v})) = \pi(\tilde{u} \otimes \tilde{v})$$

This multiplication makes $\wedge E$ into a *graded algebra* where the elements of $\wedge^p E$ are homogeneous of degree p . It also makes π into an algebra homomorphism.

Remark. The factor $1/k!$ in the definition of u^k kills off coefficients arising in

$$\underbrace{u \wedge \cdots \wedge u}_k$$

For example if $u = e_1 \wedge e_2 + e_3 \wedge e_4$ where $e_i \in E$, then

$$u \wedge u = 2e_1 \wedge e_2 \wedge e_3 \wedge e_4$$

so

$$u^2 = \frac{1}{2} u \wedge u = e_1 \wedge e_2 \wedge e_3 \wedge e_4$$

Remark. If $u \in \wedge E$, then

$$u^{p+q} = \frac{1}{(p+q)!} \underbrace{u \wedge \cdots \wedge u}_p \wedge \underbrace{u \wedge \cdots \wedge u}_q = \frac{p!q!}{(p+q)!} u^p \wedge u^q \quad (p, q \geq 0)$$

In particular,

$$u \wedge u^k = u^k \wedge u = (k+1)u^{k+1} \quad (k \geq 0)$$

Remark. If $u \in \wedge^p E$ and $v \in \wedge^q E$ with pq even, then

$$\begin{aligned} (u+v)^k &= \frac{1}{k!} \underbrace{(u+v) \wedge \cdots \wedge (u+v)}_k \\ &= \frac{1}{k!} \sum_{i+j=k} \frac{k!}{i!j!} \underbrace{u \wedge \cdots \wedge u}_i \wedge \underbrace{v \wedge \cdots \wedge v}_j \\ &= \sum_{i+j=k} \frac{1}{i!j!} (i!u^i) \wedge (j!v^j) \\ &= \sum_{i+j=k} u^i \wedge v^j \end{aligned}$$

Remark. If $u, v \in \wedge E$ and at least one of u, v is homogeneous, then $u \wedge v = 0$ if and only if $v \wedge u = 0$, but this is not true in general. For example if e_1, e_2, e_3, e_4 are linearly independent vectors in E , let

$$u = e_1 + e_3 \wedge e_4 \quad \text{and} \quad v = e_2 \wedge e_3 \wedge e_4 + e_1 \wedge e_2$$

Then it is easy to verify that $u \wedge v \neq 0$ while $v \wedge u = 0$.

§ 5

Remark. The exterior algebra $\wedge E$ is a universal (initial) object in the category of “associative unital algebras with linear maps of E into them whose squares are zero”. In this category, the objects are linear maps $\varphi : E \rightarrow A$ for associative unital algebras A with $\varphi^2 = 0$, and the arrows are unital algebra homomorphisms $A \rightarrow B$ which preserve the units and respect the linear maps:

$$\begin{array}{ccc} E & \xrightarrow{\quad} & B \\ \downarrow & \nearrow & \\ A & & \end{array}$$

Every object in this category can be obtained from the exterior algebra $\wedge E$ in a unique way. This is why $\wedge E$ is said to satisfy the “universal property”. This is only possible because the elements of $\wedge E$ satisfy only those properties that are required to make $\wedge E$ into an associative unital algebra containing E with skew-symmetric multiplication in E . By category theoretic abstract nonsense, $\wedge E$ is unique up to isomorphism.

Remark. The exterior algebra $\wedge E$ can be constructed in two ways:

- Top down, as $\wedge E = \otimes E / N(E)$.
- Bottom up, as $\wedge E = \sum_p \wedge^p E$, where $\wedge^p E$ is constructed either top down or bottom up as remarked above.

In § 4, both methods are used—in fact, the underlying vector space of $\wedge E$ is initially constructed bottom up while the multiplication and scalar product are induced from the top down.

Remark. It is not necessary to establish (5.9), since the skew-symmetry of α follows from the result in subsection 5.1.

§ 7

Remark. If $v_1 < \dots < v_p$ and $\mu_1 < \dots < \mu_p$, then

$$\det(\delta_{\mu_j}^{v_i}) = \begin{cases} 1 & \text{if } (v_1, \dots, v_p) = (\mu_1, \dots, \mu_p) \\ 0 & \text{otherwise} \end{cases}$$

Proof. If $(v_1, \dots, v_p) = (\mu_1, \dots, \mu_p)$, then $\delta_{\mu_i}^{v_i} = 1$ for all i and $\delta_{\mu_j}^{v_i} = 0$ for all $i \neq j$ since $v_i \neq v_j = \mu_j$, so $\det(\delta_{\mu_j}^{v_i}) = 1$. Conversely if k is least with $v_k \neq \mu_k$, say $v_k < \mu_k$, then $\delta_{\mu_j}^{v_k} = 0$ for all j , so $\det(\delta_{\mu_j}^{v_i}) = 0$. $\square$

Remark. If E is n -dimensional, then $\wedge^p E = 0$ for $p > n$. Geometrically, this just reflects the fact that you can't fit a p -dimensional parallelepiped into the n -dimensional space.

Remark. Let E, E^* be a pair of dual n -dimensional vector spaces with dual bases $e_1, \dots, e_n$ and $e^{*1}, \dots, e^{*n}$. For $p \leq n$, write

$$x_i = \sum_{v=1}^n \xi_i^v e_v \quad (i = 1, \dots, p)$$

and

$$x_1 \wedge \cdots \wedge x_p = \sum_{<} \xi^{v_1 \cdots v_p} e_{v_1} \wedge \cdots \wedge e_{v_p}$$

By (5.13),

$$\xi^{v_1 \cdots v_p} = \langle e^{*v_1} \wedge \cdots \wedge e^{*v_p}, x_1 \wedge \cdots \wedge x_p \rangle = \det \langle e^{*v_j}, x_i \rangle = \det(\xi_i^{v_j})$$

This shows that the components of a decomposable p -vector are just the $p \times p$ minor determinants of the matrix of components of the p factors.

Remark. Let E, E^* be a pair of dual n -dimensional vector spaces and e^* a basis vector of $\wedge^n E^*$. Let Δ be the determinant function in E corresponding to e^* ,⁵³ so that

$$\Delta(x_1, \dots, x_n) = \langle e^*, x_1 \wedge \cdots \wedge x_n \rangle$$

By a Laplace expansion (5.11), for any $x^* \in E^*$ and $x \in E$ we have

$$\begin{aligned} 0 &= \langle x^* \wedge e^*, x \wedge x_1 \wedge \cdots \wedge x_n \rangle \\ &= \langle x^*, x \rangle \langle e^*, x_1 \wedge \cdots \wedge x_n \rangle - \sum_{i=1}^n (-1)^{i-1} \langle x^*, x_i \rangle \langle e^*, x \wedge x_1 \wedge \cdots \wedge \widehat{x}_i \wedge \cdots \wedge x_n \rangle \end{aligned}$$

so

$$\langle x^*, \Delta(x_1, \dots, x_n) x \rangle = \langle x^*, \sum_{i=1}^n (-1)^{i-1} \Delta(x, x_1, \dots, \widehat{x}_i, \dots, x_n) x_i \rangle$$

By nondegeneracy of the scalar product, it follows that⁵⁴

$$\Delta(x_1, \dots, x_n) x = \sum_{i=1}^n (-1)^{i-1} \Delta(x, x_1, \dots, \widehat{x}_i, \dots, x_n) x_i$$

Remark. If E is a Euclidean space and $x_1, \dots, x_p \in E$, then

$$|x_1 \wedge \cdots \wedge x_p| = \sqrt{\det(x_i, x_j)}$$

is the volume of the p -dimensional parallelepiped determined by $x_1, \dots, x_p$.

Remark (Cartan). Let E be a vector space with basis $e_1, \dots, e_n$. If $x_1, \dots, x_p \in E$ with $p \leq n$ and

$$\sum_{i=1}^p x_i \wedge e_i = 0$$

then we can write $x_i = \sum_{j=1}^p \xi_i^j e_j$ where $\xi_i^j = \xi_j^i$ for all $i, j = 1, \dots, p$.

⁵³See subsection 5.33.

⁵⁴This is (4.6) in [1]. See also the remark in chapter IV, § 1 of [1] above.

Proof. Write $x_i = \sum_{j=1}^n \xi_i^j e_j$ for $i = 1, \dots, p$. Then by hypothesis

$$0 = \sum_{i=1}^p x_i \wedge e_i = \sum_{1 \leq i < j \leq p} (\xi_i^j - \xi_j^i) e_i \wedge e_j + \sum_{1 \leq i \leq p < j \leq n} \xi_i^j e_i \wedge e_j$$

The result follows since the $e_i \wedge e_j$ are basis vectors of $\wedge^2 E$. $\square$

Exercise (1). Let E, E^* be a pair of finite-dimensional dual vector spaces and $A \subseteq L(E^*; E)$ be the subspace of linear maps $\varphi : E^* \rightarrow E$ for which $\varphi^* = -\varphi$. If $\varphi : E \times E \rightarrow A$ is the bilinear map defined by

$$\varphi_{a,b}(x^*) = \langle x^*, a \rangle b - \langle x^*, b \rangle a$$

then (A, φ) is a second exterior power of E .

Proof. Identifying $E \otimes E$ with $L(E^*; E)$ under the natural isomorphism

$$a \otimes b \mapsto (x^* \mapsto \langle x^*, a \rangle b)$$

A is just the subspace of skew-symmetric tensors in $E \otimes E$, and

$$\varphi_{a,b} = a \otimes b - b \otimes a$$

is (up to a factor of 2) just the skew-symmetric part of $a \otimes b$. $\square$

Exercise (4 - Lagrange).

$$\begin{vmatrix} \sum_{v=1}^n \xi_1^v \eta_v^1 & \cdots & \sum_{v=1}^n \xi_1^v \eta_v^p \\ \vdots & \ddots & \vdots \\ \sum_{v=1}^n \xi_p^v \eta_v^1 & \cdots & \sum_{v=1}^n \xi_p^v \eta_v^p \end{vmatrix} = \sum_{<} \begin{vmatrix} \xi_1^{v_1} & \cdots & \xi_1^{v_p} \\ \vdots & \ddots & \vdots \\ \xi_p^{v_1} & \cdots & \xi_p^{v_p} \end{vmatrix} \begin{vmatrix} \eta_{v_1}^1 & \cdots & \eta_{v_p}^1 \\ \vdots & \ddots & \vdots \\ \eta_{v_1}^p & \cdots & \eta_{v_p}^p \end{vmatrix} \quad (1)$$

Proof. Let L denote the left-hand side of (1). Let E, E^* be a pair of dual vector spaces with dual bases $e_1, \dots, e_n$ and $e^{*1}, \dots, e^{*n}$. Define

$$x_i = \sum_{v=1}^n \xi_i^v e_v \quad \text{and} \quad y^{*j} = \sum_{\mu=1}^n \eta_{\mu}^j e^{*\mu} \quad (i, j = 1, \dots, p)$$

Since

$$\langle y^{*j}, x_i \rangle = \sum_{v=1}^n \xi_i^v \eta_v^j$$

it follows from (5.10) and (5.13) that

$$\begin{aligned}
L &= \langle y^{*1} \wedge \cdots \wedge y^{*p}, x_1 \wedge \cdots \wedge x_p \rangle \\
&= \sum_{(v), (\mu)} \xi_1^{v_1} \cdots \xi_p^{v_p} \eta_{\mu_1}^1 \cdots \eta_{\mu_p}^p \langle e^{*\mu_1} \wedge \cdots \wedge e^{*\mu_p}, e_{v_1} \wedge \cdots \wedge e_{v_p} \rangle \\
&= \sum_{(v), (\mu)} \xi_1^{v_1} \cdots \xi_p^{v_p} \eta_{\mu_1}^1 \cdots \eta_{\mu_p}^p \det(\delta_{v_i}^{\mu_j})
\end{aligned}$$

Now

$$\det(\delta_{v_i}^{\mu_j}) = \sum_{\sigma} \varepsilon_{\sigma} \delta_{v_{\sigma(1)}}^{\mu_1} \cdots \delta_{v_{\sigma(p)}}^{\mu_p}$$

so $\det(\delta_{v_i}^{\mu_j}) = \varepsilon_{\tau}$ if the v_i are distinct and $\mu_i = v_{\tau(i)}$ for some permutation τ , otherwise $\det(\delta_{v_i}^{\mu_j}) = 0$. The v_i are distinct if and only if $v_{\sigma(1)} < \cdots < v_{\sigma(p)}$ for some permutation σ , so we have

$$\begin{aligned}
L &= \sum_{v_1 < \cdots < v_p} \sum_{\sigma} \xi_1^{v_{\sigma(1)}} \cdots \xi_p^{v_{\sigma(p)}} \sum_{\tau} \varepsilon_{\tau} \eta_{v_{\tau(1)}}^1 \cdots \eta_{v_{\tau(p)}}^p \\
&= \sum_{v_1 < \cdots < v_p} \sum_{\sigma} \varepsilon_{\sigma} \xi_1^{v_{\sigma(1)}} \cdots \xi_p^{v_{\sigma(p)}} \sum_{\tau} \varepsilon_{\tau\sigma} \eta_{v_{\tau\sigma(1)}}^1 \cdots \eta_{v_{\tau\sigma(p)}}^p
\end{aligned}$$

Since $\tau\sigma$ runs over all permutations as τ does, we have

$$L = \sum_{<} \left(\sum_{\sigma} \varepsilon_{\sigma} \xi_1^{v_{\sigma(1)}} \cdots \xi_p^{v_{\sigma(p)}} \right) \left(\sum_{\varrho} \varepsilon_{\varrho} \eta_{v_{\varrho(1)}}^1 \cdots \eta_{v_{\varrho(p)}}^p \right)$$

which is just (1). □

Exercise (6). Let $(\wedge E, \wedge)$ be an exterior algebra over E . If we define

$$u \tilde{\wedge} v = \frac{(p+q)!}{p!q!} u \wedge v \quad (u \in \wedge^p E, v \in \wedge^q E)$$

and extend the definition to $\wedge E$ through bilinearity, then the resulting algebra $(\tilde{\wedge} E, \tilde{\wedge})$ is also an exterior algebra over E .

Proof. Define $f: \wedge E \rightarrow \tilde{\wedge} E$ by $f(u) = p!u$ for $u \in \wedge^p E$. Then clearly f is a linear isomorphism homogeneous of degree zero, and

$$f(u \wedge v) = (p+q)! u \wedge v = \frac{(p+q)!}{p!q!} (p!u) \wedge (q!v) = f(u) \tilde{\wedge} f(v)$$

for $u \in \wedge^p E$ and $v \in \wedge^q E$, so f is a graded algebra isomorphism. It follows that $(\tilde{\wedge} E, \tilde{\wedge})$ satisfies the universal property of an exterior algebra over E . □

Remark. If E, E^* are dual spaces, then $\tilde{\wedge} E, \tilde{\wedge} E^*$ are dual graded algebras under the scalar product induced by f , which is just the scalar product in (5.10) with $\wedge$ replaced by $\tilde{\wedge}$.

§ 8

Remark. We have a functor from the category of vector spaces into the category of associative unital algebras, which sends vector spaces E and F to the exterior algebras $\wedge E$ and $\wedge F$, and which sends a linear map $\varphi : E \rightarrow F$ to the unital algebra homomorphism $\varphi_\wedge : \wedge E \rightarrow \wedge F$.

Remark. For a linear map $\varphi : E \rightarrow F$, the following diagram commutes:

$$\begin{array}{ccc} \otimes E & \xrightarrow{\varphi_\otimes} & \otimes F \\ \pi_E \downarrow & & \downarrow \pi_F \\ \wedge E & \xrightarrow{\varphi_\wedge} & \wedge F \end{array}$$

This shows that the projection $\pi_E : \otimes E \rightarrow \wedge E$ is a natural transformation from the tensor algebra functor to the exterior algebra functor.

Remark. For a linear map $\varphi : E \rightarrow F$, the following diagram commutes:

$$\begin{array}{ccc} \otimes E & \xrightarrow{\varphi_\otimes} & \otimes F \\ \pi_A \downarrow & & \downarrow \pi_A \\ \otimes E & \xrightarrow{\varphi_\otimes} & \otimes F \end{array}$$

This shows that $\varphi_\otimes$ restricts to $X(E), X(F)$ and in fact this diagram commutes:⁵⁵

$$\begin{array}{ccc} \wedge E & \xrightarrow{\varphi_\wedge} & \wedge F \\ \cong \downarrow & & \downarrow \cong \\ X(E) & \xrightarrow{\varphi_\otimes} & X(F) \end{array}$$

Remark. Let E be an n -dimensional space and Δ a determinant function in E .

⁵⁵See subsection 5.4 for the isomorphism.

If $\varphi : E \rightarrow E$ is linear, then

$$\begin{aligned}
\Delta_{\wedge} \circ \varphi_{\wedge}(x_1 \wedge \cdots \wedge x_n) &= \Delta_{\wedge}(\varphi x_1 \wedge \cdots \wedge \varphi x_n) \\
&= \Delta(\varphi x_1, \dots, \varphi x_n) \\
&= \det \varphi \cdot \Delta(x_1, \dots, x_n) \\
&= \det \varphi \cdot \Delta_{\wedge}(x_1 \wedge \cdots \wedge x_n) \\
&= \Delta_{\wedge} \circ (\det \varphi \cdot \iota)(x_1 \wedge \cdots \wedge x_n)
\end{aligned}$$

for all $x_1, \dots, x_n \in E$. Since $\Delta_{\wedge}$ is injective, it follows that

$$\wedge^n \varphi = \det \varphi \cdot \iota$$

where $\wedge^n \varphi$ is the restriction of $\varphi_{\wedge}$ to $\wedge^n E$. Equivalently

$$\text{tr}(\wedge^n \varphi) = \det \varphi$$

This provides another natural (coordinate-free) definition of the determinant of a linear map.

Remark. Let E, E^* be a pair of dual n -dimensional spaces with dual bases e_i, e^{*i} and $\varphi : E \rightarrow E$ linear. For $0 \leq p \leq n$, if the induced basis of $\wedge^p E$ is ordered lexically, then for

$$1 \leq \nu_1 < \cdots < \nu_p \leq n \quad \text{and} \quad 1 \leq \mu_1 < \cdots < \mu_p \leq n$$

the (ν, μ) entry of the $\binom{n}{p} \times \binom{n}{p}$ matrix of $\wedge^p \varphi$ with respect to the basis is

$$\langle e^{*\mu_1} \wedge \cdots \wedge e^{*\mu_p}, \varphi e_{\nu_1} \wedge \cdots \wedge \varphi e_{\nu_p} \rangle = \det \langle e^{*\mu_j}, \varphi e_{\nu_i} \rangle$$

Since $\langle e^{*\mu_j}, \varphi e_{\nu_i} \rangle$ is the (ν_i, μ_j) entry of the matrix of φ with respect to the e_i , this shows that the matrix of $\wedge^p \varphi$ consists of the $p \times p$ minor determinants of the matrix of φ . In particular the 1×1 matrix of $\wedge^n \varphi$ consists of the single element $\det \varphi$, consistent with the previous remark.

It follows from the determinantal characterization of the rank of a linear map⁵⁶ that the rank of φ is the largest r for which $\wedge^r \varphi \neq 0$. Intuitively, the rank is the highest-dimensional volume that φ “preserves” in the sense of mapping to another volume of the same dimension and not lower dimension.

The functoriality condition (5.20) is just Cauchy-Binet.

⁵⁶See subsection 4.16 of [1].

§ 10

Remark. For a linear map $\varphi : E \rightarrow E$, the following diagram commutes:

$$\begin{array}{ccc} \otimes E & \xrightarrow{\theta_{\otimes}(\varphi)} & \otimes E \\ \pi_E \downarrow & & \downarrow \pi_E \\ \wedge E & \xrightarrow{\theta_{\wedge}(\varphi)} & \wedge E \end{array}$$

Remark. Let E be an n -dimensional space and Δ a determinant function in E . If $\varphi : E \rightarrow E$ is linear, then

$$\begin{aligned} \Delta_{\wedge} \circ \theta_{\wedge}(\varphi)(x_1 \wedge \cdots \wedge x_n) &= \sum_{i=1}^n \Delta_{\wedge}(x_1 \wedge \cdots \wedge \varphi x_i \wedge \cdots \wedge x_n) \\ &= \sum_{i=1}^n \Delta(x_1, \dots, \varphi x_i, \dots, x_n) \\ &= \text{tr } \varphi \cdot \Delta(x_1, \dots, x_n) \\ &= \text{tr } \varphi \cdot \Delta_{\wedge}(x_1 \wedge \cdots \wedge x_n) \\ &= \Delta_{\wedge} \circ (\text{tr } \varphi \cdot \iota)(x_1 \wedge \cdots \wedge x_n) \end{aligned}$$

for all $x_1, \dots, x_n \in E$. Since $\Delta_{\wedge}$ is injective, it follows that

$$\theta_{\wedge}^{(n)}(\varphi) = \text{tr } \varphi \cdot \iota$$

where $\theta_{\wedge}^{(n)}(\varphi)$ is the restriction of $\theta_{\wedge}(\varphi)$ to $\wedge^n E$. Equivalently

$$\text{tr } \theta_{\wedge}^{(n)}(\varphi) = \text{tr } \varphi$$

This is also equivalent to the result in subsection 3.6 above, by the following commutative diagram:

$$\begin{array}{ccccc} & & E^n & & \\ & & \downarrow \otimes^n & \searrow \Delta & \\ \otimes^n E & \xrightarrow{\theta_{\otimes}(\varphi)} & \otimes^n E & \xrightarrow{\Delta_{\otimes}} & \Gamma \\ \pi \downarrow & & \downarrow \pi & \nearrow \Delta_{\wedge} & \\ \wedge^n E & \xrightarrow{\theta_{\wedge}^{(n)}(\varphi)} & \wedge^n E & & \end{array}$$

This provides another natural (coordinate-free) definition of the trace.

§ 11

Remark. If $\omega = \iota_\wedge$ and $\varphi : E \rightarrow E$ is linear, then $\Omega(\varphi) = \theta_\wedge(\varphi)$. This shows that the construction in § 10 is just a special case of the construction in this subsection.

§ 12

Remark. From (5.36), we see that the following diagram commutes:

$$\begin{array}{ccc} \wedge E & \xrightarrow{\Omega_E(\varphi)} & \wedge E \\ \alpha_\wedge \downarrow & & \downarrow \alpha_\wedge \\ \wedge F & \xrightarrow{\Omega_F(\psi)} & \wedge F \end{array}$$

Exercise (2). Let $a \in \wedge^k E$ where k is odd. Define $\theta : \wedge E \rightarrow \wedge E$ by

$$\theta u = \begin{cases} a \wedge u & \text{if } u \in \wedge^p E, p \text{ odd} \\ 0 & \text{if } u \in \wedge^p E, p \text{ even} \end{cases}$$

Then θ is a derivation homogeneous of degree k in $\wedge E$.

Proof. It is clear that θ is linear and homogeneous of degree k . We must prove

$$\theta(u \wedge v) = \theta u \wedge v + u \wedge \theta v \quad (u, v \in \wedge E) \quad (1)$$

By multilinearity, we may assume $u \in \wedge^p E$ and $v \in \wedge^q E$, so $u \wedge v \in \wedge^{p+q} E$. We proceed by cases on the parities of p and q .

If p and q are even, then $p + q$ is even and both sides of (1) are zero.

If p is even and q is odd, then $p + q$ is odd and

$$\begin{aligned} \theta u \wedge v + u \wedge \theta v &= 0 \wedge v + u \wedge a \wedge v \\ &= a \wedge u \wedge v \\ &= \theta(u \wedge v) \end{aligned}$$

If p is odd and q is even, then the proof is similar to the previous case.

If p and q are odd, then $p + q$ is even and pk is odd, so

$$\begin{aligned}
 \theta u \wedge v + u \wedge \theta v &= a \wedge u \wedge v + u \wedge a \wedge v \\
 &= a \wedge u \wedge v - a \wedge u \wedge v \\
 &= 0 \\
 &= \theta(u \wedge v)
 \end{aligned}
 \quad \square$$

Remark. It follows from Proposition 5.27.2 that every derivation of odd degree in $\wedge E$ has this form.

Exercise (6). Let Ω be an antiderivation of degree -1 in $\wedge E$ (with respect to the canonical involution). If $\tilde{\Omega}$ is defined by $\tilde{\Omega}u = p^{-1} \cdot \Omega u$ for $u \in \wedge^p E$, where $p^{-1} = 0$ for $p = 0$, then $\tilde{\Omega}$ is an antiderivation in $\tilde{\wedge} E$.⁵⁷

Proof. Clearly $\tilde{\Omega}$ is linear. We must prove

$$\tilde{\Omega}(u \tilde{\wedge} v) = \tilde{\Omega}u \tilde{\wedge} v + (-1)^p u \tilde{\wedge} \tilde{\Omega}v \quad (u \in \wedge^p E, v \in \wedge^q E)$$

We have

$$\begin{aligned}
 \tilde{\Omega}(u \tilde{\wedge} v) &= \frac{1}{p+q} \Omega(u \tilde{\wedge} v) \\
 &= \frac{(p+q-1)!}{p!q!} \Omega(u \wedge v) \\
 &= \frac{(p+q-1)!}{p!q!} (\Omega u \wedge v + (-1)^p u \wedge \Omega v)
 \end{aligned}$$

On the other hand

$$\tilde{\Omega}u \tilde{\wedge} v = \frac{1}{p} \Omega u \tilde{\wedge} v = \frac{(p+q-1)!}{p!q!} \Omega u \wedge v$$

and

$$u \tilde{\wedge} \tilde{\Omega}v = \frac{1}{q} u \tilde{\wedge} \Omega v = \frac{(p+q-1)!}{p!q!} u \wedge \Omega v$$

so the result follows. $\square$

Remark. As an example, if E, E^* are dual spaces, then the insertion operator $i(h)$ on $\wedge E^*$ with $h \in E$ must be redefined as in this problem to be an antiderivation on $\tilde{\wedge} E^*$.⁵⁸

⁵⁷See problem 5.7.6 for the definition of $\tilde{\wedge} E$.

⁵⁸See subsection 5.14.

Exercise (8). If $\psi : E \rightarrow E$ is a linear transformation with $\dim E = n$, then there exist linear transformations $\psi_i^{(n)} : \wedge^n E \rightarrow \wedge^n E$ unique such that

$$(\psi - \lambda \iota)_\wedge^{(n)} = \sum_{i=0}^n \psi_i^{(n)} \lambda^{n-i} \quad \lambda \in \Gamma$$

Moreover, $\text{tr} \psi_i^{(n)} = \alpha_i$ where α_i is the i -th characteristic coefficient of ψ .

Proof. By an argument similar to that given in subsection 4.19 of [1],

$$(\psi - \lambda \iota)_\wedge^{(n)} = \sum_{i=0}^n \tilde{S}_i$$

where

$$\tilde{S}_i(x_1 \wedge \cdots \wedge x_n) = \frac{(-\lambda)^{n-i}}{i!(n-i)!} \sum_{\sigma} \varepsilon_{\sigma} \varphi x_{\sigma(1)} \wedge \cdots \wedge \varphi x_{\sigma(i)} \wedge x_{\sigma(i+1)} \wedge \cdots \wedge x_{\sigma(n)}$$

This uniquely determines $\psi_i^{(n)}$ with $\tilde{S}_i = \psi_i^{(n)} \lambda^{n-i}$. By (4.47) and (4.48) of [1],

$$\Delta_{\wedge} \circ \tilde{S}_i = \alpha_i \lambda^{n-i} \Delta_{\wedge} = \Delta_{\wedge} \circ (\alpha_i \lambda^{n-i} \iota)$$

where $\Delta \neq 0$ is a determinant function in E . Since $\Delta_{\wedge}$ is injective, this implies

$$\psi_i^{(n)} \lambda^{n-i} = \tilde{S}_i = \alpha_i \lambda^{n-i} \iota$$

so $\psi_i^{(n)} = \alpha_i \iota$ and $\text{tr} \psi_i^{(n)} = \alpha_i$. □

Remark. We have $\psi_0^{(n)} = (-1)^n \iota$, $\psi_1^{(n)} = (-1)^{n-1} \theta_{\wedge}^{(n)}(\psi)$, and $\psi_n^{(n)} = \psi_{\wedge}^{(n)}$.

§ 13

Remark. The multiplication map $\mu : \wedge E \rightarrow L(\wedge E; \wedge E)$ is linear, and is an algebra homomorphism by (5.37). Dually, the insertion map $i : \wedge E \rightarrow L(\wedge E^*; \wedge E^*)$ is an algebra antihomomorphism by (5.38).

Remark. The insertion operator $i(a) : \wedge E^* \leftarrow \wedge E^*$ generalizes the substitution operator on skew-symmetric functions if E is finite-dimensional. In particular

if $h \in E$, then the following diagram commutes:⁵⁹

$$\begin{array}{ccc} \wedge E^* & \xleftarrow{i(h)} & \wedge E^* \\ \cong \downarrow & & \downarrow \cong \\ A^*(E) & \xleftarrow{i_A(h)} & A^*(E) \end{array}$$

The results of this subsection show that insertion is dual to multiplication.

§ 14

Remark. The fact that $i(h)^2 = 0$ is equivalent to the fact that skew-symmetric functions are alternating, when E is finite-dimensional.

Remark. The operator $i(h)$ preserves decomposability.

Proof. Let

$$u^* = x_1^* \wedge \cdots \wedge x_p^* \quad (x_i^* \in E^*)$$

If $u^* = 0$ or $x_1^*, \dots, x_p^* \in \langle h \rangle^\perp$, then $i(h)u^* = 0$ is decomposable. Otherwise, there exist $y_1^*, \dots, y_{p-1}^* \in \langle h \rangle^\perp$ and $y_p^* \in E^*$ with⁶⁰

$$u^* = (y_1^* \wedge \cdots \wedge y_{p-1}^*) \wedge y_p^*$$

By Corollary III to Proposition 5.14.1 with $F = \langle h \rangle$ and $a = h$,

$$i(h)u^* = (-1)^{p-1} y_1^* \wedge \cdots \wedge y_{p-1}^* \wedge i(h)y_p^* = \pm \langle y_p^*, h \rangle y_1^* \wedge \cdots \wedge y_{p-1}^* \quad \square$$

Remark. Let E, E^* be a pair of dual n -dimensional vector spaces. If $v^* \in \wedge^n E^*$ and $v^* \neq 0$, then for any $u^* \in \wedge^{n-1} E^*$ there is a unique vector $h \in E$ with

$$i(h)v^* = u^*$$

Proof. If $u^* = 0$, we take $h = 0$. If $u^* \neq 0$, then since u^* has degree $n-1$ it follows from Poincaré duality⁶¹ that u^* is decomposable and we may write

$$u^* = x_1^* \wedge \cdots \wedge x_{n-1}^* \quad (x_i^* \in E^*)$$

⁵⁹See subsection 5.33.

⁶⁰See the remarks in § 26 below.

⁶¹See subsection 6.13.

Fix $h^* \in E^*$ with $v^* = h^* \wedge u^*$. Let F^* be the span of $x_1^*, \dots, x_{n-1}^*$, and let h be the basis vector of $(F^*)^\perp$ in E with

$$i(h)h^* = \langle h^*, h \rangle = 1$$

Since $i(h)$ is an antiderivation with respect to the canonical involution in $\wedge E^*$, it follows that

$$i(h)v^* = i(h)(h^* \wedge u^*) = i(h)h^* \wedge u^* - h^* \wedge i(h)u^* = u^*$$

The uniqueness of h is clear. □

Remark. This result is equivalent to the result in problem 5.1.2(b), which is also covered in the remarks on chapter IV, § 1 of [1] above, by the isomorphism $\wedge E^* \cong A^\bullet(E)$ of § 33 below.

Exercise (1). If $u^* \in \wedge^p E^*$ and $i(a)u^* = 0$ for all $a \in \wedge^k E$ for some $k \leq p$, then $u^* = 0$.

Proof. We claim $\langle u^*, v \rangle = 0$ for all $v \in \wedge^p E$, from which it follows that $u^* = 0$. Since $k \leq p$, we may write

$$v = \sum_{\nu} a_{\nu} \wedge v_{\nu} \quad a_{\nu} \in \wedge^k E, v_{\nu} \in \wedge^{p-k} E$$

Then

$$\langle u^*, v \rangle = \sum_{\nu} \langle u^*, a_{\nu} \wedge v_{\nu} \rangle = \sum_{\nu} \langle i(a_{\nu})u^*, v_{\nu} \rangle = 0 \quad \square$$

Remark. This result generalizes Proposition 5.14.2, which in turn generalizes the familiar fact that if a vector is zero under all linear functions, then the vector must be zero.

Exercise (2). Let E, E^* be a pair of dual n -dimensional vector spaces and $e_{\nu}, e^{*\nu}$ a pair of dual bases. If $\varphi : E \rightarrow E$ is linear, then

$$\theta_{\wedge}(\varphi) = \sum_{\nu} \mu(\varphi e_{\nu}) i(e^{*\nu}) \quad (1)$$

$$\theta^{\wedge}(\varphi) = \sum_{\nu} \mu(e^{*\nu}) i(\varphi e_{\nu}) \quad (2)$$

Proof. By Corollary II to Proposition 5.14.1,

$$i(e^{*v})(x_1 \wedge \cdots \wedge x_p) = \sum_{\mu=1}^p (-1)^{\mu-1} \langle e^{*v}, x_\mu \rangle x_1 \wedge \cdots \wedge \widehat{x_\mu} \wedge \cdots \wedge x_p$$

It follows that

$$\mu(\varphi e_v) i(e^{*v})(x_1 \wedge \cdots \wedge x_p) = \sum_{\mu=1}^p x_1 \wedge \cdots \wedge \varphi(\langle e^{*v}, x_\mu \rangle e_v) \wedge \cdots \wedge x_p$$

and so

$$\begin{aligned} \sum_{v=1}^n \mu(\varphi e_v) i(e^{*v})(x_1 \wedge \cdots \wedge x_p) &= \sum_{\mu=1}^p x_1 \wedge \cdots \wedge \varphi\left(\sum_{v=1}^n \langle e^{*v}, x_\mu \rangle e_v\right) \wedge \cdots \wedge x_p \\ &= \sum_{\mu=1}^p x_1 \wedge \cdots \wedge \varphi x_\mu \wedge \cdots \wedge x_p \\ &= \theta_\wedge(\varphi)(x_1 \wedge \cdots \wedge x_p) \end{aligned}$$

This establishes (1). Now (2) follows from (1) by duality since

$$\theta^\wedge(\varphi) = \theta_\wedge(\varphi)^* = \sum_{v=1}^n i(e^{*v})^* \mu(\varphi e_v)^* = \sum_{v=1}^n \mu(e^{*v}) i(\varphi e_v) \quad \square$$

Exercise (3). Let E, E^* be a pair of dual n -dimensional vector spaces and e_v, e^{*v} a pair of dual bases.

(a) For $x \in E$ and $x^* \in E^*$,

$$i(x^*)\mu(x) + \mu(x)i(x^*) = \langle x^*, x \rangle \iota$$

(b) For $u \in \wedge^p E$,

$$\sum_v \mu(e_v) i(e^{*v}) u = pu$$

(c) For $u \in \wedge^p E$,

$$\sum_v i(e^{*v}) \mu(e_v) u = (n-p)u$$

Proof. For (a), dualize Corollary I to Proposition 5.14.1; for (b), apply problem 2 with $\varphi = \iota$; for (c), apply (a) and (b) to obtain

$$\sum_{v=1}^n i(e^{*v}) \mu(e_v) u = \sum_{v=1}^n \langle e^{*v}, e_v \rangle u - \sum_{v=1}^n \mu(e_v) i(e^{*v}) u = nu - pu \quad \square$$

Exercise (6). If $a \in \wedge^p E$ and $p \leq q$, then

$$i(a)(x_1^* \wedge \cdots \wedge x_q^*) = \sum_{v_1 < \cdots < v_p} (-1)^{\sum_{i=1}^p (v_i - i)} \langle x_{v_1}^* \wedge \cdots \wedge x_{v_p}^*, a \rangle x_{v_{p+1}}^* \wedge \cdots \wedge x_{v_q}^*$$

where $(v_{p+1}, \dots, v_q)$ is the ordered tuple complementary to $(v_1, \dots, v_p)$.

Proof. By induction on p . For $p = 0$ the result is trivial, and for $p = 1$ it is just Corollary II to Proposition 5.14.1. We illustrate the induction step for $p = 2$, where we may assume that $a = a_1 \wedge a_2$ and write $u^* = x_1^* \wedge \cdots \wedge x_q^*$:

$$\begin{aligned} i(a)u^* &= i(a_2)i(a_1)(x_1^* \wedge \cdots \wedge x_q^*) \\ &= i(a_2) \sum_{v_1} (-1)^{v_1-1} \langle x_{v_1}^*, a_1 \rangle x_1^* \wedge \cdots \wedge \widehat{x_{v_1}^*} \wedge \cdots \wedge x_q^* \\ &= \sum_{v_1} (-1)^{v_1-1} \langle x_{v_1}^*, a_1 \rangle i(a_2)(x_1^* \wedge \cdots \wedge \widehat{x_{v_1}^*} \wedge \cdots \wedge x_q^*) \\ &= \sum_{v_1 < v_2} (-1)^{(v_1-1)+(v_2-2)} C_{v_1, v_2} x_1^* \wedge \cdots \wedge \widehat{x_{v_1}^*} \wedge \cdots \wedge \widehat{x_{v_2}^*} \wedge \cdots \wedge x_q^* \end{aligned}$$

where

$$C_{v_1, v_2} = \sum_{\sigma} \varepsilon_{\sigma} \langle x_{v_{\sigma(1)}}^*, a_1 \rangle \langle x_{v_{\sigma(2)}}^*, a_2 \rangle = \det(\langle x_{v_i}^*, a_j \rangle) = \langle x_{v_1}^* \wedge x_{v_2}^*, a_1 \wedge a_2 \rangle \quad \square$$

§ 15

Remark. The elements of $\wedge(E \oplus F)$ are sums of wedge products with factors from $E \oplus F$, which by bilinearity of the wedge product expand to sums of wedge products with factors from E and F . By anticommutativity of the wedge product, the factors from E in any product can be moved to the left of the factors from F with at most a change in sign, resulting in a wedge product with a factor from $\wedge E$ and a factor from $\wedge F$. This leads to the natural isomorphism

$$\wedge(E \oplus F) \cong \wedge E \widehat{\otimes} \wedge F$$

The anticommutative tensor product ensures it is an algebra isomorphism, in fact a *graded algebra* isomorphism. It follows that

$$\wedge^k(E \oplus F) \cong \sum_{p+q=k} \wedge^p E \otimes \wedge^q F$$

In particular if $\dim E = n$ and $\dim F = m$, then

$$\wedge^{n+m}(E \oplus F) \cong \wedge^n E \otimes \wedge^m F$$

since $\wedge^p E = 0$ for $p > n$ and $\wedge^q F = 0$ for $q > m$.

More generally⁶²

$$\wedge(E_1 \oplus \cdots \oplus E_r) \cong \wedge E_1 \widehat{\otimes} \cdots \widehat{\otimes} \wedge E_r$$

and if $d_i = \dim E_i < \infty$ for $1 \leq i \leq r$ and $d = d_1 + \cdots + d_r$, then

$$\wedge^d(E_1 \oplus \cdots \oplus E_r) \cong \wedge^{d_1} E_1 \otimes \cdots \otimes \wedge^{d_r} E_r$$

Remark. In the proof of Theorem 5.15.1, it follows from (5.42) that f is homogeneous of degree zero. It follows from (5.51) that f preserves scalar products.

§ 16

Remark. The formula (5.45) expresses commutativity of the following diagram:

$$\begin{array}{ccc} \wedge(E \oplus F) & \xrightarrow{(\varphi \oplus \psi)_\wedge} & \wedge(E' \oplus F') \\ \downarrow h \cong & & \cong \downarrow h' \\ \wedge E \widehat{\otimes} \wedge F & \xrightarrow{\varphi_\wedge \otimes \psi_\wedge} & \wedge E' \widehat{\otimes} \wedge F' \end{array}$$

For $x \in E$ and $y \in F$,

$$\begin{aligned} (h' \circ (\varphi \oplus \psi)_\wedge)(x + y) &= h'(\varphi x + \psi y) \\ &= \varphi x \otimes 1 + 1 \otimes \psi y \\ &= (\varphi_\wedge \otimes \psi_\wedge)(x \otimes 1 + 1 \otimes y) \\ &= ((\varphi_\wedge \otimes \psi_\wedge) \circ h)(x + y) \end{aligned}$$

It follows that the diagram commutes since $\wedge(E \oplus F)$ is generated as an algebra by the elements of $E \oplus F$ and the scalar 1.

⁶²See subsection 5.20.

Remark. If $\varphi : E \rightarrow E$ and $\psi : F \rightarrow F$ are linear transformations with $\dim E = n$ and $\dim F = m$, then by a remark above

$$\wedge^{n+m}(\varphi \oplus \psi) = \wedge^n \varphi \otimes \wedge^m \psi$$

It follows that

$$\begin{aligned} \det(\varphi \oplus \psi) &= \text{tr}(\wedge^{n+m}(\varphi \oplus \psi)) \\ &= \text{tr}(\wedge^n \varphi \otimes \wedge^m \psi) \\ &= \text{tr}(\wedge^n \varphi) \text{tr}(\wedge^m \psi) \\ &= \det(\varphi) \det(\psi) \end{aligned}$$

§ 18

Remark. To prove (5.48), for $u \in \wedge^p E$ and $v \in \wedge F$ observe that

$$\begin{aligned} \mu(a \otimes b)(u \otimes v) &= (a \otimes b) \wedge (u \otimes v) \\ &= (-1)^{pq} (a \wedge u) \otimes (b \wedge v) \\ &= (a \wedge (-1)^{pq} u) \otimes (b \wedge v) \\ &= ((\mu(a) \circ \omega^q) u) \otimes (\mu(b) v) \\ &= ((\mu(a) \circ \omega^q) \otimes \mu(b))(u \otimes v) \end{aligned}$$

The result now follows by multilinearity.

Remark. We provide an alternative proof of (5.51).⁶³ For $u = x_1 \wedge \cdots \wedge x_p \in \wedge^p E$, $v = y_1 \wedge \cdots \wedge y_r \in \wedge^r F$, $u^* = x_1^* \wedge \cdots \wedge x_p^* \in \wedge^p E^*$, and $v^* = y_1^* \wedge \cdots \wedge y_r^* \in \wedge^r F^*$, let

$$z_i = \begin{cases} x_i & \text{for } 1 \leq i \leq p \\ y_{i-p} & \text{for } p < i \leq p+r \end{cases}$$

and dually for z_j^* . Then

$$\langle\langle u^* \otimes v^*, u \otimes v \rangle\rangle = \langle z_1^* \wedge \cdots \wedge z_{q+s}^*, z_1 \wedge \cdots \wedge z_{p+r} \rangle$$

If $p+r \neq q+s$, then the right-hand side is zero and (5.51) holds, otherwise

$$\langle z_1^* \wedge \cdots \wedge z_{q+s}^*, z_1 \wedge \cdots \wedge z_{p+r} \rangle = \det \langle z_j^*, z_i \rangle$$

⁶³This proof is similar to that of Proposition 6.18.1.

Now $\langle z_j^*, z_i \rangle = 0$ if $1 \leq i \leq p$ and $q < j \leq q + s$, and also if $p < i \leq p + r$ and $1 \leq j \leq q$, so it follows that $\det \langle z_j^*, z_i \rangle = 0$ unless $p = q$ and $r = s$, in which case the determinant is block diagonal with

$$\det \langle z_j^*, z_i \rangle = \det \langle x_j^*, x_i \rangle \cdot \det \langle y_j^*, y_i \rangle$$

Since $\det \langle x_j^*, x_i \rangle = \langle u^*, u \rangle$ and $\det \langle y_j^*, y_i \rangle = \langle v^*, v \rangle$, (5.51) holds. The general case follows by multilinearity.

§ 21

Exercise (1). In the exterior algebra $\wedge E$,

$$u \wedge v = \pi_\wedge(u \otimes v) \quad (u, v \in \wedge E)$$

where π_1 and π_2 are the canonical projections of $E \oplus E$ onto E and $\pi = \pi_1 + \pi_2$.

Proof. By (5.52), interchanging E and a dual space E^* and noting that

$$\Delta^\wedge = (\Delta^*)_\wedge = (i_1^* + i_2^*)_\wedge = (\pi_1 + \pi_2)_\wedge = \pi_\wedge \quad \square$$

Remark. This shows that $\pi_\wedge$ is the structure map for $\wedge E$. If μ is the structure map for $\otimes E$, then the following diagram commutes:

$$\begin{array}{ccc} \otimes E \hat{\otimes} \otimes E & \xrightarrow{\mu} & \otimes E \\ \pi \otimes \pi \downarrow & & \downarrow \pi \\ \wedge E \hat{\otimes} \wedge E & \xrightarrow{\pi_\wedge} & \wedge E \end{array}$$

Also $\pi_\wedge$ is isomorphic to π_A on $X(E)$.⁶⁴

Exercise (2). Let $E = E_1 + E_2$ and $E_{12} = E_1 \cap E_2$. For $\overline{E} = E/E_{12}$, $\overline{E}_1 = E_1/E_{12}$, and $\overline{E}_2 = E_2/E_{12}$, the following diagram commutes:

$$\begin{array}{ccccc} \wedge E_1 \hat{\otimes} \wedge E_2 & \xrightarrow[\cong]{f} & \wedge(E_1 \oplus E_2) & \xrightarrow{\varphi_\wedge} & \wedge E \\ (\rho_1)_\wedge \otimes (\rho_2)_\wedge \downarrow & & & & \downarrow \rho_\wedge \\ \wedge \overline{E}_1 \hat{\otimes} \wedge \overline{E}_2 & \xrightarrow[\cong]{g} & \wedge(\overline{E}_1 \oplus \overline{E}_2) & \xrightarrow[\cong]{\psi_\wedge} & \wedge \overline{E} \end{array}$$

⁶⁴See problem 1 in chapter 4, § 8 above.

Here f and g are defined as in (5.42),

$$\varphi = [i_1, i_2] : E_1 \oplus E_2 \rightarrow E$$

where $i_1 : E_1 \rightarrow E$ and $i_2 : E_2 \rightarrow E$ are the inclusions,

$$\psi = [\bar{i}_1, \bar{i}_2] : \bar{E}_1 \oplus \bar{E}_2 \rightarrow \bar{E}$$

is the induced isomorphism described in chapter II, § 4 of [1] above, and the ρ 's are canonical projections.

Proof. By definition of ψ , this diagram of canonical injections commutes:

$$\begin{array}{ccccc} \bar{E}_1 & \xrightarrow{j_1} & \bar{E}_1 \oplus \bar{E}_2 & \xleftarrow{j_2} & \bar{E}_2 \\ & \searrow \bar{i}_1 & \downarrow \psi & \swarrow \bar{i}_2 & \\ & & \bar{E} & & \end{array}$$

Hence for $u_1 \in \wedge E_1$ and $u_2 \in \wedge E_2$,

$$\begin{aligned} (\rho_{\wedge} \circ \varphi_{\wedge} \circ f)(u_1 \otimes u_2) &= \rho_{\wedge}((i_1)_{\wedge}(u_1) \wedge (i_2)_{\wedge}(u_2)) \\ &= (\rho \circ i_1)_{\wedge}(u_1) \wedge (\rho \circ i_2)_{\wedge}(u_2) \\ &= (\bar{i}_1 \circ \rho_1)_{\wedge}(u_1) \wedge (\bar{i}_2 \circ \rho_2)_{\wedge}(u_2) \\ &= (\psi \circ j_1 \circ \rho_1)_{\wedge}(u_1) \wedge (\psi \circ j_2 \circ \rho_2)_{\wedge}(u_2) \\ &= \psi_{\wedge}[(j_1)_{\wedge}((\rho_1)_{\wedge}(u_1)) \wedge (j_2)_{\wedge}((\rho_2)_{\wedge}(u_2))] \\ &= \psi_{\wedge}[g((\rho_1)_{\wedge}(u_1) \otimes (\rho_2)_{\wedge}(u_2))] \\ &= [\psi_{\wedge} \circ g \circ ((\rho_1)_{\wedge} \otimes (\rho_2)_{\wedge})](u_1 \otimes u_2) \end{aligned}$$

The result now follows by linearity. $\square$

§ 23

Remark. By (5.54) and a remark in § 15 above, the elements of I_{E_1} are sums of wedge products with *at least one factor from* E_1 followed by zero or more factors from E_2 , so (5.55) follows. Since any element of $\wedge E$ is equal to such an element plus a sum of wedge products with factors from E_2 (zero factors from E_1), (5.56) follows.

Remark. By (5.63), we have

$$I_{\wedge^p E_1} = I_{\wedge^{p+1} E_1} \oplus (\wedge^p E_1 \hat{\otimes} \wedge E_2)$$

Taking $p = 0$ in this equation yields (5.56).

§ 25

Remark. In the proof of Proposition 5.25.2, we see that if $u \in \wedge^r E$, then we may take $v \in \wedge^{n-r} E$.

Remark. In the proof of the second part of the corollary to Proposition 5.25.2, if E has infinite dimension, then $I^q \neq 0$ for each q (since for example there are q linearly independent vectors in E , so $\wedge^q E \neq 0$). If there were a minimal ideal I in $\wedge E$, then we would have $0 \neq I \subseteq I^q$ for all q , hence $0 \neq I \subseteq \bigcap_q I^q$. Since $\bigcap_q I^q = 0$ (which is equally true when E has finite dimension), it follows that there is no such minimal ideal I .

§ 26

Remark. We prove that

$$N(E) = \begin{cases} 0 & \text{if } \dim E = \infty \\ \wedge^n E & \text{if } \dim E = n \end{cases}$$

First if $u \in N(E)$, then $u = \sum_p u_p$ with $u_p \in N(E) \cap \wedge^p E$ since $N(E)$ is a graded ideal. We claim that $u_p = 0$ for all $p < \dim E$. We know that $u_p \in \wedge^p F \subseteq \wedge^p E$ for some m -dimensional subspace F of E with $p < m$. Let $e_1, \dots, e_m$ be a basis of F and write

$$u_p = \sum_{<} \lambda^{v_1, \dots, v_p} e_{v_1} \wedge \dots \wedge e_{v_p}$$

For any given $1 \leq \mu_1 < \dots < \mu_p \leq m$, fix $1 \leq \mu \leq m$ with $\mu \neq \mu_i$ for all $1 \leq i \leq p$. Since $u_p \in N(E)$,

$$0 = e_\mu \wedge u_p = \sum_{<} \lambda^{v_1, \dots, v_p} e_\mu \wedge e_{v_1} \wedge \dots \wedge e_{v_p}$$

Now the $e_\mu \wedge e_{v_1} \wedge \dots \wedge e_{v_p}$ that are nonzero are linearly independent in $\wedge^{p+1} F$. These include $e_\mu \wedge e_{\mu_1} \wedge \dots \wedge e_{\mu_p}$, so we must have $\lambda^{\mu_1, \dots, \mu_p} = 0$. Since the μ_i were arbitrary, it follows that $u_p = 0$. Now if $\dim E = \infty$ this implies $u = 0$, and if $\dim E = n$ this implies $u = u_n \in \wedge^n E$. Conversely in the latter case, if we take $F = E$ and $x = \sum_i \lambda^i e_i$ arbitrary, then

$$x \wedge e_1 \wedge \dots \wedge e_n = \sum_i \lambda^i e_i \wedge e_1 \wedge \dots \wedge e_n = 0$$

so $\wedge^n E \subseteq N(E)$, completing the proof.

The fact that $N(E) = \wedge^n E$ if $\dim E = n$ also follows from Proposition 5.26.1 by taking $F = E$ and $p = n$.

Remark. If E is n -dimensional, then we have the following inclusions:

$$\begin{array}{ccccccc} N(\wedge^0 E) & \subseteq & N(\wedge^1 E) & \subseteq & \cdots & \subseteq & N(\wedge^n E) \subseteq N(\wedge^{n+1} E) \\ \parallel & & \parallel & & & & \parallel \\ I_{\wedge^{n+1} E} & \subseteq & I_{\wedge^n E} & \subseteq & \cdots & \subseteq & I_{\wedge^1 E} \subseteq I_{\wedge^0 E} \end{array}$$

Remark. In the corollary to Proposition 5.26.1, note that $N(y) = I_y$, so it follows that u is divisible by y if and only if $y \wedge u = 0$ for *any* $u \in \wedge E$.

Remark. If X, Y are finite-dimensional subspaces of E , then $N(X) \subseteq N(Y)$ if and only if $Y \subseteq X$.

Proof. If $Y \subseteq X$, then clearly $N(X) \subseteq N(Y)$. Conversely if $N(X) \subseteq N(Y)$, let

$$u = x_1 \wedge \cdots \wedge x_p \quad \text{and} \quad v = y_1 \wedge \cdots \wedge y_q$$

where $x_1, \dots, x_p$ is a basis of X and $y_1, \dots, y_q$ is a basis of Y . By the corollary to Proposition 5.26.1,

$$I_u = N(X) \subseteq N(Y) = I_v$$

so v divides u , which by the same corollary implies

$$I_Y = N(v) \subseteq N(u) = I_X$$

so

$$Y = I_Y \cap E \subseteq I_X \cap E = X \quad \square$$

Remark. For vectors $x_1, \dots, x_p$ and $y_1, \dots, y_p$ in E ,

$$x_1 \wedge \cdots \wedge x_p = y_1 \wedge \cdots \wedge y_p \neq 0 \quad (1)$$

if and only if $x_1, \dots, x_p$ are linearly independent and

$$y_j = \sum_{i=1}^p \alpha_j^i x_i \quad (1 \leq j \leq p) \quad (2)$$

where $\det(\alpha_j^i) = 1$.

Proof. If $x_1, \dots, x_p$ are linearly independent and (2) holds, then

$$y_1 \wedge \cdots \wedge y_p = \det(\alpha_j^i) x_1 \wedge \cdots \wedge x_p = x_1 \wedge \cdots \wedge x_p \neq 0 \quad (3)$$

so (1) holds. Conversely if (1) holds, let X be the span of $x_1, \dots, x_p$ and Y be the span of $y_1, \dots, y_p$. By the corollary to Proposition 5.26.1,

$$N(X) = I_{x_1 \wedge \cdots \wedge x_p} = I_{y_1 \wedge \cdots \wedge y_p} = N(Y)$$

so $X = Y$ by the previous remark. It follows that (2) holds for some scalars α_j^i , and $\det(\alpha_j^i) = 1$ by (3). $\square$

Geometrically, this result shows that (1) holds if and only if the p -dimensional parallelepipeds determined by $x_1, \dots, x_p$ and $y_1, \dots, y_p$ have the same oriented (signed) volume.

Remark. For a homogeneous element $u \in \wedge E$, define the *annihilated subspace*

$$E_u = E \cap N(u) = \{x \in E \mid u \wedge x = 0\}$$

Then $E_0 = E$, and if $u = x_1 \wedge \cdots \wedge x_p \neq 0$, then $E_u = \langle x_1, \dots, x_p \rangle$. In fact, the mapping $u \mapsto E_u$ is an isomorphism from the lattice of nonzero decomposable elements of $\wedge E$ under divisibility to the lattice of finite-dimensional subspaces of E under inclusion. This means $E_u \subseteq E_v$ if and only if u divides v and

$$E_{(u,v)} = E_u \cap E_v \quad \text{and} \quad E_{[u,v]} = E_u + E_v$$

where (u, v) denotes the greatest common divisor of u and v , and $[u, v]$ denotes the least common multiple of u and v .⁶⁵ This isomorphism establishes a bridge between algebraic facts in $\wedge E$ and geometric facts in E .

Remark. If $u, v \neq 0$ are decomposable elements of $\wedge E$, then $u \wedge v \neq 0$ if and only if $(u, v) = 1$ (that is, u and v are relatively prime), which is true if and only if $[u, v] = u \wedge v$.

Proof. By the previous remark. $\square$

⁶⁵See subsections 6.27 and 6.28 of [3].

§ 27

Remark. If $\varphi : E \rightarrow \wedge E$ is linear and F is a finite-dimensional subspace of E such that

$$y \wedge \varphi y = 0 \quad (y \in F)$$

then there is $v \in \wedge E$ with

$$\varphi y = y \wedge v \quad (y \in F)$$

If φ is homogeneous of degree k ($\text{Im } \varphi \subseteq \wedge^{k+1} E$), then we may take $v \in \wedge^k E$.

Proof. By induction on $m = \dim F$. If $m = 0$, then we can take $v = 0$, say. If $m \geq 1$ and the result holds for subspaces of dimension $m-1$, fix $a \in F$ with $a \neq 0$. Since $a \wedge \varphi a = 0$, it follows from the corollary to Proposition 5.26.1 that there is $c \in \wedge E$ with $\varphi a = a \wedge c$.

Define $\sigma : E \rightarrow \wedge E$ by $\sigma x = \varphi x - x \wedge c$. Then $\sigma a = 0$, $y \wedge \sigma y = 0$ for all $y \in F$, and $a \wedge \sigma y = 0$ for all $y \in F$, as in (5.70).

Write $E = \langle a \rangle \oplus E'$ and $F = \langle a \rangle \oplus F'$. Since $\wedge E = \wedge \langle a \rangle \hat{\otimes} \wedge E'$, there are linear maps $\sigma_i : E \rightarrow \wedge E'$ ($i = 0, 1$) with

$$\sigma x = 1 \otimes \sigma_0 x + a \otimes \sigma_1 x \quad (x \in E)$$

Taking $x = y \in F$ and multiplying both sides by a , it follows that $\sigma_0 y = 0$ for all $y \in F$, so $\sigma y = a \otimes \sigma_1 y$ for all $y \in F$. A simple computation then shows that $y \wedge \sigma_1 y = 0$ for all $y \in F$, and hence in particular for all $y \in F'$.

By induction, there is $v' \in \wedge E$ with $\sigma_1 y' = y' \wedge v'$ for all $y' \in F'$. For $y = \lambda a + y'$ with $\lambda \in \Gamma$ and $y' \in F'$, it follows that $\sigma y = -y \wedge (a \wedge v')$, as in (5.73). Therefore

$$\varphi y = y \wedge c + \sigma y = y \wedge (c - a \wedge v') \quad (y \in F)$$

The result now follows by taking $v = c - a \wedge v'$.

If $\text{Im } \varphi \subseteq \wedge^{k+1} E$, the refinement follows by projecting onto $\wedge^{k+1} E$ as in the proof of the corollary to Proposition 5.27.1. $\square$

Exercise (4). For a subspace F of E , there is a natural isomorphism of graded algebras $\wedge(E/F) \cong \wedge E / I_F$.

Proof. If $\pi : E \rightarrow E/F$ denotes the canonical projection, then $\pi_\wedge : \wedge E \rightarrow \wedge(E/F)$ is a surjective homomorphism homogeneous of degree zero with

$$\ker \pi_\wedge = I_{\ker \pi} = I_F$$

by (5.57). It follows that there is an induced isomorphism $\overline{\pi_\wedge} : \wedge E / I_F \rightarrow \wedge(E/F)$ which is also homogeneous of degree zero. $\square$

§ 28

Exercise (2). Let E, E^* be a pair of dual finite-dimensional vector spaces. If F is a subspace of E , then

$$\wedge F = \bigcap_{u^* \in I_{F^\perp}} \ker i(u^*)$$

Proof. The right-hand side is a nonzero graded subalgebra of $\wedge E$ which has grade 1 subspace F and is closed under $i(h^*)$ for all $h^* \in E^*$ by (5.38), so the result follows from Proposition 5.28.2. $\square$

§ 29

Remark. The formula (5.81) follows from (4.11) and a diagram in § 1 above.

§ 30

Remark. Why is the factor $\frac{(p+q)!}{p!q!}$ in the definition of the Grassmann product? From problem 5.7.6, we know it is not needed algebraically, although it may be more natural geometrically in certain applications. However, the factorial in the definition of the alternator *is* needed for associativity of the Grassmann product.⁶⁶

The presence of the factorials has the following consequences:

- The substitution operator $i_A(h)$ must be rescaled in § 32 (see below).
- The isomorphism $\otimes E^* \rightarrow T^*(E)$ must be rescaled when it is restricted to obtain the isomorphism $X(E^*) \rightarrow A^*(E)$ in § 33.
- The scalar product between $T^p(E)$ and $T_p(E)$ must be rescaled to obtain the scalar product between $A^p(E)$ and $A_p(E)$ in § 34.
- The definitions and proofs are uglier, and do not generalize beyond fields of characteristic zero.

⁶⁶See (5.81) and the proof of Proposition 5.30.1(B).

§ 32

Remark. In this subsection, the substitution operator $i_A(h)$ should be redefined on $T^p(E)$ as

$$i_A(h) = \frac{1}{p} \sum_{v=1}^p (-1)^{v-1} i_v(h)$$

Then for $\Phi \in A^p(E)$,

$$\begin{aligned} p \cdot (i_A(h)\Phi)(x_1, \dots, x_{p-1}) &= \sum_{v=1}^p (-1)^{v-1} (i_v(h)\Phi)(x_1, \dots, x_{p-1}) \\ &= \sum_{v=1}^p (-1)^{v-1} \Phi(x_1, \dots, x_{v-1}, h, x_{v+1}, \dots, x_{p-1}) \\ &= \sum_{v=1}^p \Phi(h, x_1, \dots, x_{p-1}) \\ &= p \cdot \Phi(h, x_1, \dots, x_{p-1}) \\ &= p \cdot (i_1(h)\Phi)(x_1, \dots, x_{p-1}) \end{aligned}$$

so

$$i_A(h)\Phi = i_1(h)\Phi$$

as desired. This redefinition is required to make $i_A(h)$ an antiderivation in the algebra $A^*(E)$ of skew-symmetric multilinear functions (under the definition of Grassmann product in subsection 5.30), although under this redefinition $i_A(h)$ is not an antiderivation in the algebra $T^*(E)$ of multilinear functions.⁶⁷

§ 33

Remark. The algebra isomorphism $\beta \circ \eta : \wedge E^* \rightarrow A^*(E)$ satisfies

$$f_1^* \wedge \cdots \wedge f_p^* \mapsto f_1^* \wedge \cdots \wedge f_p^* = p! A(f_1^* \cdots f_p^*) \quad f_i^* \in E^*$$

The product on the left is the exterior product and the product on the right is the Grassmann product, where f_i^* has been identified with $\langle f_i^*, - \rangle$.

§ 34

Remark. We see that the algebra isomorphisms $\wedge E^* \cong A^*(E)$ and $\wedge E \cong A_*(E)$ preserve scalar products.

⁶⁷See also the corrected version of problem 5.12.6 above.

Chapter 6

In this chapter, E and E^* are dual vector spaces over Γ . In §6–16, they are n -dimensional.

Products

This table summarizes the way in which products are transformed by maps for a *finite-dimensional* vector space E :

Map	Source product	Target product
T_E	[Skew] mixed exterior [Skew] mixed intersection Composition Inner	[Skew] box [Skew] intersection Composition Inner
D_E	[Skew] mixed exterior [Skew] mixed intersection Composition Inner	[Skew] mixed intersection [Skew] mixed exterior Composition (opposite) Inner

§ 1

Remark. Recall that

$$M_q^p(E^*, E) = \underbrace{E^* \times \cdots \times E^*}_p \times \underbrace{E \times \cdots \times E}_q$$

Let $\wedge_q^p(E^*, E) = \wedge^p E^* \otimes \wedge^q E$. The map $\varphi: M_q^p(E^*, E) \rightarrow \wedge_q^p(E^*, E)$ defined by

$$\varphi(x_1^*, \dots, x_p^*; x_1, \dots, x_q) = (x_1^* \wedge \cdots \wedge x_p^*) \otimes (x_1 \wedge \cdots \wedge x_q)$$

is skew-symmetric of type (p, q) , and Proposition 6.1.1 shows that φ is *universal* with this property in that every skew-symmetric map of type (p, q) factors in a unique way through φ : if $\psi: M_q^p(E^*, E) \rightarrow H$ is any such map, then there is a unique linear map $f: \wedge_q^p(E^*, E) \rightarrow H$ with $f \circ \varphi = \psi$:

$$\begin{array}{ccc} M_q^p(E^*, E) & \xrightarrow{\psi} & H \\ \varphi \downarrow & \nearrow f & \\ \wedge_q^p(E^*, E) & & \end{array}$$

In the proof in the book, observe that $N^p(E^*) \subseteq N_1(\beta)$ and $N^q(E) \subseteq N_2(\beta)$, so β induces γ as in problem 1.3.5(a).

As an alternative proof, for fixed covectors $x_1^*, \dots, x_p^* \in E^*$ observe that the map defined by

$$(x_1, \dots, x_q) \mapsto \psi(x_1^*, \dots, x_p^*; x_1, \dots, x_q)$$

is skew-symmetric, so there is a linear map $g_{x_1^* \dots x_p^*} : \wedge^q E \rightarrow H$ with

$$g_{x_1^* \dots x_p^*}(x_1 \wedge \dots \wedge x_q) = \psi(x_1^*, \dots, x_p^*; x_1, \dots, x_q)$$

The map $(x_1^*, \dots, x_p^*) \mapsto g_{x_1^* \dots x_p^*}$ is also skew-symmetric, so there is a linear map $\wedge^p E^* \rightarrow L(\wedge^q E, H)$ which corresponds to the bilinear map $\gamma : \wedge^p E^* \times \wedge^q E \rightarrow H$ with

$$\begin{aligned} \gamma(x_1^* \wedge \dots \wedge x_p^*, x_1 \wedge \dots \wedge x_q) &= g_{x_1^* \dots x_p^*}(x_1 \wedge \dots \wedge x_q) \\ &= \psi(x_1^*, \dots, x_p^*; x_1, \dots, x_q) \end{aligned}$$

The linear map $f : \wedge_q^p(E^*, E) \rightarrow H$ induced by γ satisfies $f \circ \varphi = \psi$ and is unique with this property.

Remark. Let $\pi_1 : \otimes^p E^* \rightarrow \wedge^p E^*$ and $\pi_2 : \otimes^q E \rightarrow \wedge^q E$ denote the canonical projections. The map

$$\pi_1 \otimes \pi_2 : \otimes_q^p(E^*, E) \rightarrow \wedge_q^p(E^*, E)$$

is surjective with

$$\begin{aligned} \ker(\pi_1 \otimes \pi_2) &= \ker \pi_1 \otimes \otimes^q E + \otimes^p E^* \otimes \ker \pi_2 \\ &= N^p(E^*) \otimes \otimes^q E + \otimes^p E^* \otimes N^q(E) \\ &= T(N^p(E^*), N^q(E)) \end{aligned}$$

so it induces the canonical isomorphism

$$\frac{\otimes_q^p(E^*, E)}{T(N^p(E^*), N^q(E))} \cong \wedge_q^p(E^*, E)$$

Remark. If E, E^* are finite-dimensional, then the space $\wedge_q^p(E^*, E)$ is isomorphic to the space of skew-symmetric functions of type (p, q) on E, E^* .

§ 2

Remark. There is a canonical isomorphism

$$\frac{\otimes(E^*, E)}{T(N(E^*), N(E))} \cong \wedge(E^*, E)$$

Remark. If $w \in \wedge(E^*, E)$, then

$$w^{p+q} = \frac{1}{(p+q)!} \underbrace{w \cdots w}_p \cdot \underbrace{w \cdots w}_q = \frac{p!q!}{(p+q)!} w^p \cdot w^q \quad (p, q \geq 0)$$

In particular,

$$w \cdot w^k = w^k \cdot w = (k+1)w^{k+1} \quad (k \geq 0)$$

Remark. The inner product in (6.2), which is just a special case of (1.25), is compatible with the bigradation in $\wedge(E^*, E)$ and therefore restricts to a scalar product between $\wedge_q^p(E^*, E)$ and $\wedge_p^q(E^*, E)$. In particular it restricts to an inner product in $\wedge_p^p(E^*, E)$.

The flip isomorphism used in the definition of (1.25) induces flips in many results in the mixed exterior algebra—for example, in various results related to the bigradation, and results like

$$i(u^* \otimes u) = i(u) \otimes i(u^*) \quad (u^* \in \wedge E^*, u \in \wedge E)$$

and (6.4).

Remark. If $z \in \wedge_q^p(E^*, E)$ and $w \in \wedge_s^r(E^*, E)$, then $\mu(z)w \in \wedge_{q+s}^{p+r}(E^*, E)$, so $\mu(z)$ is homogeneous of bidegree (p, q) . It follows from this that $i(z)$ is homogeneous of bidegree $(-q, -p)$ (note the flip!).

Remark. If $u^* \in \wedge E^*$, $u \in \wedge E$, $x^* \in E^*$, and $x \in E$, then

$$\langle x^* \otimes u, u^* \otimes x \rangle = \langle \mu(x^*)u^*, \mu(x)u \rangle + \langle i(x)u^*, i(x^*)u \rangle$$

Proof. By Corollary I to Proposition 5.14.1,

$$\begin{aligned} \langle x^* \otimes u, u^* \otimes x \rangle &= \langle x^*, x \rangle \langle u^*, u \rangle \\ &= \langle \langle x^*, x \rangle u^*, u \rangle \\ &= \langle i(x)\mu(x^*)u^* + \mu(x^*)i(x)u^*, u \rangle \\ &= \langle i(x)\mu(x^*)u^*, u \rangle + \langle \mu(x^*)i(x)u^*, u \rangle \\ &= \langle \mu(x^*)u^*, \mu(x)u \rangle + \langle i(x)u^*, i(x^*)u \rangle \end{aligned}$$

□

Remark. The flip operator $Q_E : \wedge(E^*, E) \rightarrow \wedge(E, E^*)$ restricts to operators

$$Q_E^{p,q} : \wedge_q^p(E^*, E) \rightarrow \wedge_p^q(E, E^*)$$

In particular $Q_E^{p,p} : \Delta_p E \rightarrow \Delta_p E^*$. It follows that $Q_{E^*}^{q,p} \circ Q_E^{p,q} = \iota$, so the operators are linear isomorphisms. They also preserve scalar products. Finally, $Q_E^* = Q_{E^*}$ and $(Q_E^{p,q})^* = Q_{E^*}^{p,q}$.

Remark. Interchanging the roles of E and E^* , we obtain the map

$$T_{E^*} : \wedge(E, E^*) \rightarrow L(\wedge E^*; \wedge E^*)$$

given by

$$T_{E^*}(b \otimes a^*)v^* = \langle v^*, b \rangle a^* \quad (v^* \in \wedge E^*)$$

Note $T_{E^*}(b \otimes a^*)$ is dual to $T_E(a^* \otimes b)$.

Remark. If Φ is a skew-symmetric function of type (p, q) on E, E^* and Ψ is a skew-symmetric function of type (r, s) on E, E^* , then the *mixed Grassmann product* of Φ and Ψ is the skew-symmetric function of type $(p+r, q+s)$ on E, E^* defined by

$$\begin{aligned} (\Phi \cdot \Psi)(x_1, \dots, x_{p+r}; x_1^*, \dots, x_{q+s}^*) \\ = \varepsilon_{pr} \varepsilon_{qs} \sum_{\sigma, \tau} \varepsilon_{\sigma} \varepsilon_{\tau} \Phi(x_{\sigma(1)}, \dots, x_{\sigma(p)}; x_{\tau(1)}^*, \dots, x_{\tau(q)}^*) \\ \Psi(x_{\sigma(p+1)}, \dots, x_{\sigma(p+r)}; x_{\tau(q+1)}^*, \dots, x_{\tau(q+s)}^*) \end{aligned}$$

where

$$\varepsilon_{pr} = \frac{(p+r)!}{p! r!} \quad \text{and} \quad \varepsilon_{qs} = \frac{(q+s)!}{q! s!}$$

If E, E^* are finite-dimensional, then the mixed exterior algebra $\wedge(E^*, E)$ is isomorphic to the algebra of skew-symmetric functions on E, E^* under the mixed Grassmann product, which is called the *mixed Grassmann algebra* on E, E^* .⁶⁸ It is therefore possible to develop mixed exterior algebra using function spaces (at least in the finite-dimensional case), although it is messier.

⁶⁸See subsection 8.14 of [3].

§ 3

Remark. The p -ary box product, which generalizes the p -th exterior power to more than one linear transformation, is p -linear and symmetric, so factors through the p -th symmetric power of $L(E; E)$:

$$\begin{array}{ccc} L(E; E)^p & \xrightarrow{\square^p} & L(\wedge^p E; \wedge^p E) \\ \downarrow \vee^p & \nearrow & \\ \vee^p L(E; E) & & \end{array}$$

If E is finite-dimensional, then the induced linear map is *surjective*. In fact, since $\vee^p L(E; E)$ is spanned by elements of the form $\varphi \vee \cdots \vee \varphi$,⁶⁹ it follows that $L(\wedge^p E; \wedge^p E)$ is spanned by transformations of the form

$$\underbrace{\varphi \square \cdots \square \varphi}_p = p! \wedge^p \varphi$$

In this sense, the p -th exterior power is fully general, and the p -ary box product is no more general.

Remark. If E is finite-dimensional and $\varphi_1, \dots, \varphi_p$ are linear transformations of E , then

$$(\varphi_1 \square \cdots \square \varphi_p)^* = \varphi_1^* \square \cdots \square \varphi_p^*$$

Proof. By multilinearity, we may assume without loss of generality that

$$\varphi_i = T_E(a_i^* \otimes a_i) \quad (a_i^* \in E^*, a_i \in E)$$

Then

$$\varphi_i^* = T_{E^*}(a_i \otimes a_i^*)$$

and

$$\begin{aligned} \varphi_1^* \square \cdots \square \varphi_p^* &= T_{E^*}[(a_1 \otimes a_1^*) \cdots (a_p \otimes a_p^*)] \\ &= T_{E^*}[(a_1 \wedge \cdots \wedge a_p) \otimes (a_1^* \wedge \cdots \wedge a_p^*)] \\ &= T_E[(a_1^* \wedge \cdots \wedge a_p^*) \otimes (a_1 \wedge \cdots \wedge a_p)]^* \\ &= T_E[(a_1^* \otimes a_1) \cdots (a_p^* \otimes a_p)]^* \\ &= (\varphi_1 \square \cdots \square \varphi_p)^* \end{aligned}$$

□

⁶⁹See the remark in chapter 9, § 10 below.

Remark. Let E, E^* be a pair of dual n -dimensional spaces with dual bases e_i, e^{*i} and let $\varphi_1, \dots, \varphi_p$ be linear transformations of E with $1 \leq p \leq n$. If the induced basis of $\wedge^p E$ is ordered lexically, then for

$$1 \leq \nu_1 < \dots < \nu_p \leq n \quad \text{and} \quad 1 \leq \mu_1 < \dots < \mu_p \leq n$$

the (ν, μ) entry of the $\binom{n}{p} \times \binom{n}{p}$ matrix of $\varphi_1 \square \dots \square \varphi_p$ with respect to the basis is

$$\sum_{\sigma, \tau \in S_p} \varepsilon_\sigma \varepsilon_\tau \langle e^{*\mu_{\tau(1)}}, \varphi_1 e_{\nu_{\sigma(1)}} \rangle \dots \langle e^{*\mu_{\tau(p)}}, \varphi_p e_{\nu_{\sigma(p)}} \rangle$$

For $n \times n$ matrices $A_1, \dots, A_p$, we define the $\binom{n}{p} \times \binom{n}{p}$ box product matrix by

$$(A_1 \square \dots \square A_p)_\nu^\mu = \sum_{\sigma, \tau \in S_p} \varepsilon_\sigma \varepsilon_\tau (A_1)_{\nu_{\sigma(1)}}^{\mu_{\tau(1)}} \dots (A_p)_{\nu_{\sigma(p)}}^{\mu_{\tau(p)}}$$

It then follows that

$$M(\varphi_1 \square \dots \square \varphi_p) = M(\varphi_1) \square \dots \square M(\varphi_p)$$

with respect to the given bases.

Remark. If E is finite-dimensional, we can generalize the definition of the box product. For linear transformations φ and ψ of $\wedge E$, we define

$$\varphi \square \psi = T_E(T_E^{-1} \varphi \cdot T_E^{-1} \psi)$$

and similarly for more than two transformations. Then T_E becomes an algebra isomorphism from the mixed exterior algebra $\wedge(E^*, E)$ to a box product algebra in $L(\wedge E; \wedge E)$. The box product is associative and unital, and is *commutative* on the diagonal subalgebra in $L(\wedge E; \wedge E)$. We also define

$$\wedge^p \varphi = \frac{1}{p!} \underbrace{\varphi \square \dots \square \varphi}_p$$

when it is understood to be a linear transformation of $\wedge E$. By Proposition 6.3.1, these definitions agree with the ones previously given for transformations of E when $L(\wedge^p E; \wedge^p E)$ is viewed as a subspace of $L(\wedge E; \wedge E)$.⁷⁰

⁷⁰See the remark in chapter II, § 4 of [1] above.

For fixed φ , there is a left multiplication operator $\mu_{\square}(\varphi)$ given by

$$\mu_{\square}(\varphi)\psi = \varphi \square \psi$$

and a dual insertion operator $i_{\square}(\varphi)$ satisfying

$$\langle i_{\square}(\varphi)\chi, \psi \rangle = \langle \chi, \varphi \square \psi \rangle = \langle \chi, \mu_{\square}(\varphi)\psi \rangle$$

For $z \in \wedge(E^*, E)$, the following diagrams commute:

$$\begin{array}{ccc} \wedge(E^*, E) & \xrightarrow[\cong]{T_E} & L(\wedge E; \wedge E) \\ \mu(z) \downarrow & & \downarrow \mu_{\square}(T_E z) \\ \wedge(E^*, E) & \xrightarrow[\cong]{T_E} & L(\wedge E; \wedge E) \end{array} \quad \begin{array}{ccc} \wedge(E^*, E) & \xrightarrow[\cong]{T_E} & L(\wedge E; \wedge E) \\ i(z) \downarrow & & \downarrow i_{\square}(T_E z) \\ \wedge(E^*, E) & \xrightarrow[\cong]{T_E} & L(\wedge E; \wedge E) \end{array}$$

An argument similar to that given in the previous remark shows that

$$(\varphi \square \psi)^* = \varphi^* \square \psi^*$$

§ 4

Remark. The diagonal subspace $\Delta E = \sum_p \Delta_p E$ is closed under the composition product. In fact, if $w = \sum_p w_p$ with $w_p \in \Delta_p E$ and $z = \sum_q z_q$ with $z_q \in \Delta_q E$, then

$$w \circ z = \sum_{p,q} w_p \circ z_q$$

But $w_p \circ z_q = 0$ if $p \neq q$ by (6.10), and $w_p \circ z_p \in \Delta_p E$ by (6.11), so

$$w \circ z = \sum_p w_p \circ z_p \in \Delta E$$

For this reason ΔE is a subalgebra under the composition product, although it is not commutative like it is under the mixed exterior product if $\dim E > 1$.

Remark. If $u, v, w \in \wedge(E^*, E)$, then

$$\langle u \circ v, w \rangle = \langle u, v \circ w \rangle$$

In other words, left composition is dual to right composition.⁷¹

⁷¹This is equation (7') in [4].

Proof. By multilinearity, we may assume without loss of generality that

$$u = a^* \otimes a \quad v = b^* \otimes b \quad w = c^* \otimes c$$

Then

$$u \circ v = \langle a^*, b \rangle b^* \otimes a$$

and

$$v \circ w = \langle b^*, c \rangle c^* \otimes b$$

so

$$\langle u \circ v, w \rangle = \langle a^*, b \rangle \langle b^*, c \rangle \langle c^*, a \rangle = \langle u, v \circ w \rangle$$

□

§ 5

Remark (Greub-Vanstone⁷²). If $h_1, \dots, h_p, k_1, \dots, k_p \in E^* \otimes E$, then

$$\begin{aligned} (h_1 \cdots h_p) \circ (k_1 \cdots k_p) &= \sum_{\sigma \in S_p} (h_{\sigma(1)} \circ k_1) \cdots (h_{\sigma(p)} \circ k_p) \\ &= \sum_{\tau \in S_p} (h_1 \circ k_{\tau(1)}) \cdots (h_p \circ k_{\tau(p)}) \\ &= \frac{1}{p!} \sum_{\sigma, \tau \in S_p} (h_{\sigma(1)} \circ k_{\tau(1)}) \cdots (h_{\sigma(p)} \circ k_{\tau(p)}) \end{aligned}$$

Proof. It is sufficient to prove the first equality, since the others then follow by commutativity of the mixed exterior product in the diagonal subalgebra. By multilinearity, we may assume without loss of generality that

$$h_i = x_i^* \otimes x_i \quad \text{and} \quad k_i = y_i^* \otimes y_i \quad (x_i^*, y_i^* \in E^*, \quad x_i, y_i \in E)$$

Then

$$\begin{aligned} (h_1 \cdots h_p) \circ (k_1 \cdots k_p) &= \det \langle x_i^*, y_j \rangle y_1^* \wedge \cdots \wedge y_p^* \otimes x_1 \wedge \cdots \wedge x_p \\ &= \sum_{\sigma \in S_p} \langle x_{\sigma(1)}^*, y_1 \rangle \cdots \langle x_{\sigma(p)}^*, y_p \rangle \\ &\quad y_1^* \wedge \cdots \wedge y_p^* \otimes x_{\sigma(1)} \wedge \cdots \wedge x_{\sigma(p)} \\ &= \sum_{\sigma \in S_p} (h_{\sigma(1)} \circ k_1) \cdots (h_{\sigma(p)} \circ k_p) \end{aligned}$$

□

⁷²This is the theorem in [4].

This fundamental identity helps us convert problems in the *non-commutative* composition algebra in ΔE into problems in the *commutative* mixed exterior algebra in ΔE . Because many of the results of classical linear algebra have to do with the composition product, this is an important and powerful technique. Proposition 6.5.1 follows from this identity.⁷³

Remark. In the Corollary to Proposition 6.5.1,

$$\begin{aligned}
p! i(z)(w^p) &= i(z)(\underbrace{w \cdots w}_p) \\
&= \sum_v \langle z, w \rangle \underbrace{w \cdots w}_{p-1} - \sum_{v < \mu} 2(w \circ z \circ w) \underbrace{w \cdots w}_{p-2} \\
&= p \cdot (p-1)! \langle z, w \rangle w^{p-1} - \frac{p(p-1)}{2} \cdot 2 \cdot (p-2)! (w \circ z \circ w) w^{p-2} \\
&= p! [\langle z, w \rangle w^{p-1} - (w \circ z \circ w) w^{p-2}]
\end{aligned}$$

Dividing both sides by $p!$ yields the desired result.

§ 6

Remark. It follows from (6.12) that T_E is an isometry, so $T_E^* = T_E^{-1}$.

§ 7

Remark. It follows from (6.14) and a remark in § 2 above that

$$\mathfrak{t} \cdot \mathfrak{t}_p = \mathfrak{t}_p \cdot \mathfrak{t} = (p+1) \mathfrak{t}_{p+1} \quad (0 \leq p \leq n-1)$$

Remark. In (6.17), we see that the unit tensor $\mathfrak{t}_p$ in $\bigwedge_p^p(E^*, E)$ encodes all of the information about the scalar product between $\bigwedge^p E^*$ and $\bigwedge^p E$, just like a metric tensor does for an inner product.⁷⁴

Remark. In the proof of Proposition 6.7.1, if (6.19) holds for some $q \geq 1$, then

⁷³See equation (23) in [4].

⁷⁴See the remarks in chapter 3, § 17 above.

for any $p \geq q + 1$ we have

$$\begin{aligned}
(q+1)i(\mathfrak{t}_{q+1})\mathfrak{t}_p &= i(\mathfrak{t}_q \cdot \mathfrak{t})\mathfrak{t}_p \\
&= i(\mathfrak{t})(i(\mathfrak{t}_q)\mathfrak{t}_p) \\
&= \binom{n-p+q}{q} i(\mathfrak{t})\mathfrak{t}_{p-q} \\
&= \binom{n-p+q}{q} (n-p+q+1)\mathfrak{t}_{p-(q+1)}
\end{aligned}$$

Since

$$\binom{n-p+q}{q} \frac{n-p}{q+1} = \binom{n-p+q}{q+1}$$

and

$$\binom{n-p+q}{q+1} + \binom{n-p+q}{q} = \binom{n-p+q+1}{q+1}$$

it follows that

$$i(\mathfrak{t}_{q+1})\mathfrak{t}_p = \binom{n-p+q+1}{q+1} \mathfrak{t}_{p-(q+1)}$$

so (6.19) holds for $q + 1$.

§ 8

Remark. It follows from Theorem 6.8.1 parts (2) and (3) that

$$D_p^{-1} = (D^p)^* \quad \text{and} \quad (D^p)^{-1} = D_p^*$$

Remark. Since $D_e = \sum_p D_p$, it follows from Theorem 6.8.1(2) that

$$D_e^* = \sum_p D_p^* = \sum_p (-1)^{p(n-p)} D_{n-p} = \sum_p (-1)^{p(n-p)} D_p = \omega_{E^*} \circ D_e$$

where $\omega_{E^*}(u^*) = (-1)^{p(n-p)} u^*$ for $u^* \in \wedge^p E^*$. Similarly $(D^e)^* = \omega_E \circ D^e$.

It follows from Theorem 6.8.1(3) that

$$D^e \circ D_e = \sum_p (D^{n-p} \circ D_p) = \sum_p (-1)^{p(n-p)} \iota = \omega_E$$

Since $\omega_E^2 = \iota$, this implies

$$D_e^{-1} = \omega_E \circ D^e = (D^e)^*$$

Similarly

$$(D^e)^{-1} = \omega_{E^*} \circ D_e = D_e^*$$

Remark. Let $\Delta, \Delta^* \neq 0$ be determinant functions in E, E^* and let e^*, e be the corresponding (not necessarily dual) vectors in $\wedge^n E^*, \wedge^n E$ with $\alpha = \langle e^*, e \rangle \neq 0$. Then

$$\Delta^*(x_1^*, \dots, x_n^*) = \langle x_1^* \wedge \dots \wedge x_n^*, e \rangle = i(x_1^* \wedge \dots \wedge x_n^*)e = D^e(x_1^* \wedge \dots \wedge x_n^*)$$

and

$$\Delta(x_1, \dots, x_n) = \langle e^*, x_1 \wedge \dots \wedge x_n \rangle = i(x_1 \wedge \dots \wedge x_n)e^* = \alpha D_e(x_1 \wedge \dots \wedge x_n)$$

By Theorem 6.8.1(1),

$$\begin{aligned} \Delta^*(x_1^*, \dots, x_n^*) \Delta(x_1, \dots, x_n) &= \alpha \langle D_e(x_1 \wedge \dots \wedge x_n), D^e(x_1^* \wedge \dots \wedge x_n^*) \rangle \\ &= \alpha \langle x_1^* \wedge \dots \wedge x_n^*, x_1 \wedge \dots \wedge x_n \rangle \\ &= \alpha \det \langle x_j^*, x_i \rangle \end{aligned}$$

This is (4.21) in [1], which was proved there using the equivalent of the universal property of $\wedge_n^n(E^*, E)$. This shows that Δ, Δ^* are dual if and only if e^*, e are dual.

§ 9

Remark. Let $\mathbf{V}$ denote the category of vector spaces and linear maps over Γ . Let $\mathbf{C}$ denote the category of objects (E, e) where E is an n -dimensional vector space over Γ and e is a basis vector of $\wedge^n E$, and arrows $\varphi : (E, e) \rightarrow (F, f)$ where $\varphi : E \rightarrow F$ is a linear isomorphism with $\varphi_\wedge e = f$.

Define $F : \mathbf{C} \rightarrow \mathbf{V}$ by

$$F(E, e) = \wedge E \quad \text{and} \quad F(\varphi) = \varphi_\wedge$$

and define $G : \mathbf{C} \rightarrow \mathbf{V}$ by

$$G(E, e) = \wedge L(E) \quad \text{and} \quad G(\varphi) = (\varphi^\wedge)^{-1}$$

Then F and G are functors, and the diagrams in this section show that the maps $D_{(-)} : F \rightarrow G$ and $D^{(-)} : G \rightarrow F$ are natural transformations.

§ 10

Remark. By (6.20),

$$D_E(u^* \otimes u) = D_e u \otimes D^e u^* = Q_{E^*}(D^e u^* \otimes D_e u) = [Q_{E^*} \circ (D^e \otimes D_e)](u^* \otimes u)$$

so it follows that

$$D_E = Q_{E^*} \circ (D^e \otimes D_e)$$

Remark. For the proof of Theorem 6.10.1(2), we have by (6.21) and results above

$$\begin{aligned} D_E^* &= Q_E^* \circ (D_e \otimes D^e)^* \\ &= Q_E^* \circ [(\omega_E \circ D^e) \otimes (\omega_{E^*} \circ D_e)] \\ &= Q_E^* \circ (\omega_E \otimes \omega_{E^*}) \circ (D^e \otimes D_e) \\ &= (\omega_{E^*} \otimes \omega_E) \circ Q_E^* \circ (D^e \otimes D_e) \\ &= \Omega_E \circ D_E \end{aligned}$$

Remark. It follows from Theorem 6.10.1 that $D_E^* = D_E^{-1}$. We have

$$D_E^{-1} = [(D^e)^{-1} \otimes (D_e)^{-1}] \circ Q_E = Q_{E^*} \circ [(D_e)^{-1} \otimes (D^e)^{-1}]$$

Remark. The operator D_E restricts to operators

$$D_E^{p,q} : \Lambda_q^p(E^*, E) \rightarrow \Lambda_{n-p}^{n-q}(E^*, E)$$

In particular $D_E^{p,p} : \Delta_p E \rightarrow \Delta_{n-p} E$ and $D_\Delta = \sum_p D_E^{p,p}$. The operators are linear isomorphisms and the scalar product, dual, and composition properties follow from Theorem 6.10.1.

Remark. For $w, z \in \Lambda(E^*, E)$,

$$D_E(w \circ z) = D_E z \circ D_E w$$

Proof. We may assume that $w = u^* \otimes u$ and $z = v^* \otimes v$, so by (6.7)

$$w \circ z = \langle u^*, v \rangle v^* \otimes u$$

and therefore by (6.20) and Theorem 6.8.1(1),

$$\begin{aligned} D_E(w \circ z) &= \langle u^*, v \rangle D_E(v^* \otimes u) \\ &= \langle D_e v, D^e u^* \rangle D_e u \otimes D^e v^* \\ &= (D_e v \otimes D^e v^*) \circ (D_e u \otimes D^e u^*) \\ &= D_E(v^* \otimes v) \circ D_E(u^* \otimes u) \\ &= D_E z \circ D_E w \end{aligned}$$

□

Remark. Interchanging the roles of E and E^* , we obtain the map

$$D_{E^*} : \wedge(E, E^*) \rightarrow \wedge(E, E^*)$$

satisfying

$$D_{E^*} = (D^e \otimes D_e) \circ Q_{E^*} = Q_E \circ (D_e \otimes D^e)$$

§ 11

Remark. Let $\mathbf{C}$ denote the category of finite-dimensional vector spaces over Γ and the linear isomorphisms between them. For an object $E \in \mathbf{C}$, define

$$\alpha(E) = \wedge(L(E), E)$$

and for an arrow $\varphi : E \rightarrow F$ in $\mathbf{C}$, define

$$\alpha(\varphi) = \alpha_\varphi = (\varphi^\wedge)^{-1} \otimes \varphi_\wedge : \alpha(E) \rightarrow \alpha(F)$$

Then $\alpha : \mathbf{C} \rightarrow \mathbf{C}$ is a functor. Indeed, α maps objects to objects and arrows to arrows, respecting domains and codomains and identity arrows. If $\psi : F \rightarrow G$ is another arrow in $\mathbf{C}$, then

$$\begin{aligned} \alpha(\psi \circ \varphi) &= ((\psi \circ \varphi)^\wedge)^{-1} \otimes (\psi \circ \varphi)_\wedge \\ &= ((\psi^\wedge)^{-1} \circ (\varphi^\wedge)^{-1}) \otimes (\psi_\wedge \circ \varphi_\wedge) \\ &= ((\psi^\wedge)^{-1} \otimes \psi_\wedge) \circ ((\varphi^\wedge)^{-1} \otimes \varphi_\wedge) \\ &= \alpha(\psi) \circ \alpha(\varphi) \end{aligned}$$

so α also respects composites.

The diagram in this section shows that D_E is a natural isomorphism of the functor α :

$$\begin{array}{ccc} \alpha(E) & \xrightarrow[\cong]{\alpha_\varphi} & \alpha(F) \\ D_E \downarrow \cong & & \cong \downarrow D_F \\ \alpha(E) & \xrightarrow[\alpha_\varphi]{\cong} & \alpha(F) \end{array}$$

To see that the diagram actually commutes, let e and e^* be dual basis vectors of $\wedge^n E$ and $\wedge^n E^*$ and let $f = \varphi_\wedge e$ and $f^* = (\varphi^\wedge)^{-1} e^*$ be the corresponding dual basis vectors of $\wedge^n F$ and $\wedge^n F^*$. Then

$$\begin{aligned}
D_F \circ \alpha_\varphi &= (D_f \otimes D^f) \circ Q_F \circ ((\varphi^\wedge)^{-1} \otimes \varphi_\wedge) \\
&= (D_f \otimes D^f) \circ (\varphi_\wedge \otimes (\varphi^\wedge)^{-1}) \circ Q_E \\
&= [(D_f \circ \varphi_\wedge) \otimes (D^f \circ (\varphi^\wedge)^{-1})] \circ Q_E \\
&= [((\varphi^\wedge)^{-1} \circ D_e) \otimes (\varphi_\wedge \circ D^e)] \circ Q_E \\
&= ((\varphi^\wedge)^{-1} \otimes \varphi_\wedge) \circ (D_e \otimes D^e) \circ Q_E \\
&= \alpha_\varphi \circ D_E
\end{aligned}$$

§ 12

Remark. If $u, v \neq 0$ are decomposable elements of $\wedge E$, then the intersection product $u \cap v$ is also decomposable and $u \cap v \neq 0$ if and only if the annihilated subspaces E_u and E_v are in general position ($E = E_u + E_v$), in which case

$$E_{u \cap v} = E_u \cap E_v$$

This is why $u \cap v$ is called the “intersection product”.⁷⁵

Remark. The intersection product is associative since the wedge product is: if $u, v, w \in \wedge E$, then writing $D = D^e$ we have

$$\begin{aligned}
(u \cap v) \cap w &= D[D^{-1}(u \cap v) \wedge D^{-1}w] \\
&= D[(D^{-1}u \wedge D^{-1}v) \wedge D^{-1}w] \\
&= D[D^{-1}u \wedge (D^{-1}v \wedge D^{-1}w)] \\
&= D[D^{-1}u \wedge D^{-1}(v \cap w)] \\
&= u \cap (v \cap w)
\end{aligned}$$

Therefore we may simply write $u \cap v \cap w$.

Remark. We have

$$u \cap v = i[(D^e)^{-1}v]u = i(D_e^*v)u$$

⁷⁵See the remarks in chapter 5, § 26 above and § 13 below, as well as subsection 6.30 of [3].

In particular if $u \in \wedge^p E$ and $v \in \wedge^{n-p} E$, then $u \cap v$ is a scalar and

$$\begin{aligned} u \cap v &= \langle u \cap v, 1 \rangle \\ &= \langle i(D_e^* v)u, 1 \rangle \\ &= \langle u, D_e^* v \rangle \\ &= \langle D_e u, v \rangle \\ &= \langle i(u)e^*, v \rangle \\ &= \langle e^*, u \wedge v \rangle \end{aligned}$$

Remark. Let $u^*, v^* \in \wedge E^*$. Taking $u = D^e u^*$ and $v = D^e v^*$ in the definition of the intersection product for $\wedge E$, we obtain

$$D^e(u^* \wedge v^*) = D^e u^* \cap D^e v^*$$

so D^e is an algebra isomorphism from the exterior algebra to the intersection algebra. Applying $D_e^* = (D^e)^{-1}$ to both sides of the definition, we obtain

$$D_e^*(u \cap v) = D_e^* u \wedge D_e^* v$$

so D_e^* is an algebra isomorphism from the intersection algebra to the exterior algebra. This all strongly suggests that D_e should be an algebra isomorphism, which a computation involving sign factors verifies.

Remark. The intersection product in $\wedge E^*$ is given by

$$u^* \cap v^* = D_e[(D_e)^{-1} u^* \wedge (D_e)^{-1} v^*] \quad (u^*, v^* \in \wedge E^*)$$

Remark. For $w, z \in \wedge(E^*, E)$, we define the *mixed intersection product* by

$$w \cap z = D_E(D_E^{-1} w \cdot D_E^{-1} z)$$

If $w \in \wedge_q^p(E^*, E)$ and $z \in \wedge_s^r(E^*, E)$, then $w \cap z \in \wedge_{q+s-n}^{p+r-n}(E^*, E)$.

This definition makes $\wedge(E^*, E)$ into an algebra called the *mixed intersection algebra* which is associative, unital (with unit ϵ_n), and anticommutative. It also makes ΔE a *commutative* subalgebra. We have

$$w \cap z = i(D_E^{-1} z)w$$

If $w, z \in \Delta E$, then

$$w \cap z = z \cap w = i(D_\Delta^{-1} w)z = i(D_\Delta w)z$$

It is immediate from the definition that

$$D_E(w \cdot z) = D_E w \cap D_E z$$

so D_E is an *algebra* isomorphism from the mixed exterior algebra to the mixed intersection algebra.

For $u, v \in \wedge E$ and $u^*, v^* \in \wedge E^*$, we have

$$\begin{aligned} (u^* \otimes u) \cap (v^* \otimes v) &= D_E[(D^e)^{-1}u \otimes (D_e)^{-1}u^*] \cdot [(D^e)^{-1}v \otimes (D_e)^{-1}v^*] \\ &= D_E[(D^e)^{-1}u \wedge (D^e)^{-1}v] \otimes [(D_e)^{-1}u^* \wedge (D_e)^{-1}v^*] \\ &= D_e[(D_e)^{-1}u^* \wedge (D_e)^{-1}v^*] \otimes D^e[(D^e)^{-1}u \wedge (D^e)^{-1}v] \\ &= (u^* \cap v^*) \otimes (u \cap v) \end{aligned}$$

It follows that

$$\begin{aligned} D_E[(u^* \otimes u) \cap (v^* \otimes v)] &= D_e(u \cap v) \otimes D^e(u^* \cap v^*) \\ &= (D_e u \wedge D_e v) \otimes (D^e u^* \wedge D^e v^*) \\ &= (D_e u \otimes D^e u^*) \cdot (D_e v \otimes D^e v^*) \\ &= D_E(u^* \otimes u) \cdot D_E(v^* \otimes v) \end{aligned}$$

By multilinearity, it follows that

$$D_E(w \cap z) = D_E w \cdot D_E z$$

for all $w, z \in \wedge(E^*, E)$, so D_E is also an algebra isomorphism from the mixed intersection algebra to the mixed exterior algebra.

§ 13

Remark. Let e_i, e^{*i} be a pair of dual bases in E, E^* with $e_1 \wedge \cdots \wedge e_n = e$. Write

$$e_{\mu_1 \cdots \mu_q} = e_{\mu_1} \wedge \cdots \wedge e_{\mu_q} \quad \text{and} \quad e^{v_1 \cdots v_p} = e^{*v_1} \wedge \cdots \wedge e^{*v_p}$$

for ordered tuples $1 \leq \mu_1 < \cdots < \mu_q \leq n$ and $1 \leq v_1 < \cdots < v_p \leq n$. Then

$$e_{\mu_1 \cdots \mu_q}^{v_1 \cdots v_p} = e^{v_1 \cdots v_p} \otimes e_{\mu_1 \cdots \mu_q}$$

form a basis of $\wedge(E^*, E)$ and for $z \in \wedge(E^*, E)$ we write

$$z = \sum_{<} (z)_{v_1 \cdots v_p}^{\mu_1 \cdots \mu_q} e_{\mu_1 \cdots \mu_q}^{v_1 \cdots v_p}$$

The components $(z)_{v_1 \dots v_p}^{\mu_1 \dots \mu_q}$ are also the matrix entries of $T_E z$ with respect to the lexically ordered basis $e_{v_1 \dots v_p}$ of $\wedge E$.⁷⁶ In particular if $z \in E^* \otimes E$, then $(z^p)_{v_1 \dots v_p}^{\mu_1 \dots \mu_p}$ is the (v, μ) entry of the matrix of $\wedge^p T_E z$, which is the $p \times p$ minor determinant of the matrix of $T_E z$ with rows $v_1, \dots, v_p$ and columns $\mu_1, \dots, \mu_p$.⁷⁷

By (6.20) and (6.23),

$$\begin{aligned} D_E(e_{\mu_1 \dots \mu_q}^{v_1 \dots v_p}) &= D_e(e_{\mu_1 \dots \mu_q}) \otimes D^e(e^{v_1 \dots v_p}) \\ &= [(-1)^{\sum_{j=1}^q (\mu_j - j)} e^{\mu_{q+1} \dots \mu_n}] \otimes [(-1)^{\sum_{i=1}^p (v_i - i)} e_{v_{p+1} \dots v_n}] \\ &= (-1)^{\sum_{i=1}^p (v_i - i) + \sum_{j=1}^q (\mu_j - j)} e_{v_{p+1} \dots v_n}^{\mu_{q+1} \dots \mu_n} \end{aligned}$$

For $z \in \wedge(E^*, E)$, it follows that

$$\begin{aligned} D_E z &= \sum_{<} (z)_{v_1 \dots v_p}^{\mu_1 \dots \mu_q} D_E(e_{\mu_1 \dots \mu_q}^{v_1 \dots v_p}) \\ &= \sum_{<} (-1)^{\sum_{i=1}^p (v_i - i) + \sum_{j=1}^q (\mu_j - j)} (z)_{v_1 \dots v_p}^{\mu_1 \dots \mu_q} e_{v_{p+1} \dots v_n}^{\mu_{q+1} \dots \mu_n} \end{aligned}$$

so⁷⁸

$$(D_E z)_{\mu_{q+1} \dots \mu_n}^{v_{p+1} \dots v_n} = (-1)^{\sum_{i=1}^p (v_i - i) + \sum_{j=1}^q (\mu_j - j)} (z)_{v_1 \dots v_p}^{\mu_1 \dots \mu_q}$$

In particular if $z \in E^* \otimes E$, then $(D_E z^{n-p})_{\mu_1 \dots \mu_p}^{v_1 \dots v_p}$ is the signed $(n-p) \times (n-p)$ minor determinant of the matrix of $T_E z$ with rows complementary to $v_1, \dots, v_p$ and columns complementary to $\mu_1, \dots, \mu_p$.

Remark. Every $u \in \wedge^{n-1} E$ is decomposable.

Proof. If $u = 0$, the result is trivial, otherwise $v_1^* = (D^e)^{-1} u$ can be extended to a basis $v_1^*, \dots, v_n^*$ in E^* with dual basis $v_1, \dots, v_n$ in E . Let $v = v_1 \wedge \dots \wedge v_n$. Then $v = \lambda e$ for some $\lambda \neq 0$ and $D^v = \lambda D^e$. Now

$$u = D^e (D^e)^{-1} u = \lambda^{-1} D^v v_1^* = \pm \lambda^{-1} v_2 \wedge \dots \wedge v_n \quad \square$$

In fact the Poincaré isomorphisms D_e and D^e are anti-isomorphisms between the lattices of nonzero decomposable elements of $\wedge E$ and $\wedge E^*$ under divisibility,⁷⁹ so this result follows from surjectivity of $D^1 : E^* \rightarrow \wedge^{n-1} E$.

⁷⁶See the remark in chapter 1, § 28 above.

⁷⁷See the remark in chapter 5, § 8 above.

⁷⁸This is problem 6.15.10.

⁷⁹See subsection 6.30 of [3].

If $u \in \wedge E$ and $u^* \in \wedge E^*$ are nonzero decomposable elements, then we have the following relations between annihilated subspaces of E and E^* :⁸⁰

$$E_{D_e u}^* = E_u^\perp \quad \text{and} \quad E_{D^e u^*}^\perp = E_{u^*}^*$$

§ 14

Remark. The external product on E makes the following diagram commute:

$$\begin{array}{ccc} E^{n-1} & \xrightarrow{[-]} & E^* \\ \wedge^{n-1} \downarrow & \nearrow D_{n-1} & \\ \wedge^{n-1} E & & \end{array}$$

Remark. We define the *mixed external product* $M_{n-1}^{n-1}(E^*, E) \rightarrow E^* \otimes E$ by

$$[x_1^*, \dots, x_{n-1}^*, x_1, \dots, x_{n-1}] = D_E[(x_1^* \otimes x_1) \cdots (x_{n-1}^* \otimes x_{n-1})]$$

This product is skew-symmetric of type $(n-1, n-1)$. We have

$$\begin{aligned} [x_1^*, \dots, x_{n-1}^*, x_1, \dots, x_{n-1}] &= D_E[(x_1^* \wedge \cdots \wedge x_{n-1}^*) \otimes (x_1 \wedge \cdots \wedge x_{n-1})] \\ &= D_e(x_1 \wedge \cdots \wedge x_{n-1}) \otimes D^e(x_1^* \wedge \cdots \wedge x_{n-1}^*) \\ &= [x_1, \dots, x_{n-1}] \otimes [x_1^*, \dots, x_{n-1}^*] \end{aligned}$$

It follows that

$$\langle [x_1^*, \dots, x_{n-1}^*, x_1, \dots, x_{n-1}], [x_1^*, \dots, x_{n-1}^*, x_1, \dots, x_{n-1}] \rangle = (\det \langle x_j^*, x_i \rangle)^2$$

§ 15

Remark. In the case of a finite-dimensional oriented Euclidean space, Poincaré duality reduces to Hodge duality and the Poincaré map is just the Hodge star.

Exercise (1). For an isomorphism $\varphi : E \rightarrow E$,

$$\alpha_\varphi t^p = t^p \quad (0 \leq p \leq n)$$

where $\alpha_\varphi = (\varphi^\wedge)^{-1} \otimes \varphi_\wedge$.

⁸⁰See the remark in chapter 5, § 26 above and subsection 6.30 of [3].

Proof. Let e_i, e^{*i} be a pair of dual bases in E, E^* . Then

$$t^p = \sum_{\prec} e_{v_1 \dots v_p}^{v_1 \dots v_p}$$

If $\bar{e}_i = \varphi e_i$ and $\bar{e}^{*i} = (\varphi^*)^{-1} e^{*i}$, then $\bar{e}_i, \bar{e}^{*i}$ is another pair of dual bases in E, E^* and

$$\alpha_\varphi t^p = \sum_{\prec} \alpha_\varphi (e_{v_1 \dots v_p}^{v_1 \dots v_p}) = \sum_{\prec} \bar{e}_{v_1 \dots v_p}^{v_1 \dots v_p} = t^p \quad \square$$

Remark. This result shows that unit tensors in the mixed exterior algebra are *invariant* in that their descriptions do not depend upon the choice of basis.

Exercise (3).

$$\begin{aligned} D_e i(v^*) u &= (-1)^{q(p-q)} v^* \wedge D_e u & u \in \wedge^p E, v^* \in \wedge^q E^* \\ D^e i(v) u^* &= (-1)^{p(q-p)} v \wedge D^e u^* & v \in \wedge^p E, u^* \in \wedge^q E^* \\ D_E i(z) w &= (-1)^{p(s-p)+q(r-q)} z \cdot D_E w & z \in \wedge_q^p(E^*, E), w \in \wedge_s^r(E^*, E) \end{aligned}$$

Proof. For any $w^* \in \wedge^{p-q} E^*$,

$$\begin{aligned} \langle D_e i(v^*) u, D^e w^* \rangle &= \langle w^*, i(v^*) u \rangle \\ &= \langle v^* \wedge w^*, u \rangle \\ &= \langle D_e u, D^e(v^* \wedge w^*) \rangle \\ &= (-1)^{q(p-q)} \langle D_e u, D^e(w^* \wedge v^*) \rangle \\ &= (-1)^{q(p-q)} \langle D_e u, i(v^*) D^e w^* \rangle \\ &= \langle (-1)^{q(p-q)} v^* \wedge D_e u, D^e w^* \rangle \end{aligned}$$

The first formula follows since D^e is an isomorphism, and the second formula follows from the first by symmetry of duality. Finally if $z = v^* \otimes v$ and $w = u^* \otimes u$, then

$$\begin{aligned} D_E i(z) w &= D_E [i(v) u^* \otimes i(v^*) u] \\ &= D_e i(v^*) u \otimes D^e i(v) u^* \\ &= (-1)^{p(s-p)+q(r-q)} (v^* \wedge D_e u) \otimes (v \wedge D^e u^*) \\ &= (-1)^{p(s-p)+q(r-q)} (v^* \otimes v) \cdot (D_e u \otimes D^e u^*) \\ &= (-1)^{p(s-p)+q(r-q)} z \cdot D_E w \end{aligned}$$

The third formula follows by multilinearity. $\square$

Remark. This result shows that multiplication is the dual of insertion under Poincaré duality.

Exercise (4).

$$D_\Delta \mu(z) = i(z) D_\Delta \quad (z \in \Delta E)$$

Proof. For $z = v^* \otimes v$ and $w = u^* \otimes u$ in ΔE ,

$$\begin{aligned} D_\Delta \mu(z) w &= D_\Delta(z \cdot w) \\ &= D_\Delta(w \cdot z) \\ &= D_\Delta[(u^* \wedge v^*) \otimes (u \wedge v)] \\ &= D_e(u \wedge v) \otimes D^e(u^* \wedge v^*) \\ &= [i(v) D_e u] \otimes [i(v^*) D^e u^*] \\ &= i(v^* \otimes v)(D_e u \otimes D^e u^*) \\ &= i(z) D_\Delta w \end{aligned}$$

The formula follows by multilinearity. $\square$

Remark. This result shows that insertion is the dual of multiplication under Poincaré duality in the diagonal subalgebra.

Exercise (7). Let E be an oriented 3-dimensional Euclidean space. Then

$$x \times y = D_e(x \wedge y) = [x, y] \quad (x, y \in E)$$

It follows that

$$(x_1 \times y_1, x_2 \times y_2) = (x_1, x_2)(y_1, y_2) - (x_1, y_2)(x_2, y_1)$$

and

$$(x \times y) \times z = (x, z)y - (y, z)x$$

Proof. Let Δ be the normed determinant function in E which represents the orientation. Then for any $z \in E$,

$$(D_e(x \wedge y), z) = ([x, y], z) = (e, x \wedge y \wedge z) = \Delta(x, y, z) = (x \times y, z)$$

so the first equation follows by definiteness of the inner product. Now by the Lagrange identity,

$$\begin{aligned} (x_1 \times y_1, x_2 \times y_2) &= ([x_1, y_1], [x_2, y_2]) \\ &= \begin{vmatrix} (x_1, x_2) & (x_1, y_2) \\ (y_1, x_2) & (y_1, y_2) \end{vmatrix} \\ &= (x_1, x_2)(y_1, y_2) - (x_1, y_2)(x_2, y_1) \end{aligned}$$

Finally,

$$\begin{aligned}
(x \times y) \times z &= D_e[D_e(x \wedge y) \wedge z] \\
&= i(z)D_e[D_e(x \wedge y)] \\
&= i(z)(x \wedge y) \\
&= (x, z)y - (y, z)x
\end{aligned}
\quad \square$$

Remark. This result shows that the cross product is dual to the wedge product under Poincaré duality, and is just a special case of the external product.

Exercise (8). Let e_i, e^{*i} be a pair of dual bases in E, E^* with $e_1 \dots e_n = e$. Then

$$D_e(x_1 \wedge \dots \wedge x_p) = \sum_{v_1 < \dots < v_p} (-1)^{\sum_{i=1}^p (v_i - i)} \det \langle e^{*v_j}, x_i \rangle e^{v_{p+1} \dots v_n}$$

where $(v_1, \dots, v_p)$ and $(v_{p+1}, \dots, v_n)$ are complementary ordered tuples.

Proof. We know

$$x_1 \wedge \dots \wedge x_p = \sum_{<} \langle e^{v_1 \dots v_p}, x_1 \wedge \dots \wedge x_p \rangle e_{v_1 \dots v_p} = \sum_{<} \det \langle e^{*v_j}, x_i \rangle e_{v_1 \dots v_p}$$

so

$$\begin{aligned}
D_e(x_1 \wedge \dots \wedge x_p) &= \sum_{<} \det \langle e^{*v_j}, x_i \rangle D_e(e_{v_1 \dots v_p}) \\
&= \sum_{<} (-1)^{\sum_{i=1}^p (v_i - i)} \det \langle e^{*v_j}, x_i \rangle e^{v_{p+1} \dots v_n}
\end{aligned}
\quad \square$$

Remark. This result generalizes (6.23).

Exercise (11). Let E, E^* be a pair of dual 3-dimensional vector spaces with $a \in E$ and $b^* \in E^*$. If $a \neq 0$, then there is $x \in E$ with $[a, x] = b^*$ if and only if $\langle b^*, a \rangle = 0$. If $x_0 \in E$ is a particular solution, then the general solution is given by $x = x_0 + \lambda a$ for $\lambda \in \Gamma$.

Proof. If $[a, x] = b^*$, then

$$\langle b^*, a \rangle = \langle [a, x], a \rangle = \langle e^*, a \wedge x \wedge a \rangle = 0$$

Conversely if $\langle b^*, a \rangle = 0$ and $a^* \in E^*$ is chosen with $\langle a^*, a \rangle = -1$, then the vector $x = [a^*, b^*]$ satisfies

$$\begin{aligned} [a, x] &= D_e(a \wedge D^e(a^* \wedge b^*)) \\ &= -i(a)(D_e(D^e(a^* \wedge b^*))) \\ &= -i(a)(a^* \wedge b^*) \\ &= \langle b^*, a \rangle a^* - \langle a^*, a \rangle b^* \\ &= b^* \end{aligned}$$

If $[a, x_0] = b^*$, then

$$[a, x_0 + \lambda a] = [a, x_0] + \lambda[a, a] = b^*$$

If also $[a, x] = b^*$, then

$$[a, x - x_0] = [a, x] - [a, x_0] = b^* - b^* = 0$$

so $a \wedge (x - x_0) = 0$, which implies $x - x_0 = \lambda a$ for some $\lambda \in \Gamma$.⁸¹ □

Remark. If E is an oriented 3-dimensional Euclidean space, this result shows that for $a \neq 0$ there is x with $a \times x = b$ if and only if $(a, b) = 0$.

Exercise (12). If $\varphi : E^{n-1} \rightarrow F$ is an $(n-1)$ -linear skew-symmetric map, then there is a unique linear map $\chi : E^* \rightarrow F$ with

$$\varphi(x_1, \dots, x_{n-1}) = \chi[x_1, \dots, x_{n-1}] \quad (x_i \in E)$$

Proof. If there is such a χ , then by definition of the external product

$$\chi D_{n-1}(x_1 \wedge \dots \wedge x_{n-1}) = \varphi(x_1, \dots, x_{n-1}) \quad (x_i \in E)$$

so $f = \chi \circ D_{n-1} : \wedge^{n-1} E \rightarrow F$ is the linear map induced by φ on $\wedge^{n-1} E$:

$$\begin{array}{ccc} E^{n-1} & \xrightarrow{\varphi} & F \\ \wedge^{n-1} \downarrow & \nearrow f & \\ \wedge^{n-1} E & & \end{array}$$

It follows that $\chi = f \circ D_{n-1}^{-1}$, so χ is uniquely determined. On the other hand, this map satisfies the property. □

⁸¹See the corollary to Proposition 5.26.1.

Remark. This is the universal property of the external product, which is dual to the universal property of the wedge product under Poincaré duality.

Exercise (13). If $\varphi : E \rightarrow E$ is an isomorphism, then

$$[\varphi x_1, \dots, \varphi x_{n-1}] = \det \varphi \cdot (\varphi^{-1})^* [x_1, \dots, x_{n-1}] \quad (x_i \in E)$$

Proof. For any $x_n \in E$,

$$\begin{aligned} \langle [\varphi x_1, \dots, \varphi x_{n-1}], \varphi x_n \rangle &= \langle e^*, \varphi x_1 \wedge \dots \wedge \varphi x_n \rangle \\ &= \det \varphi \cdot \langle e^*, x_1 \wedge \dots \wedge x_n \rangle \\ &= \det \varphi \cdot \langle [x_1, \dots, x_{n-1}], \varphi^{-1} \varphi x_n \rangle \\ &= \langle \det \varphi \cdot (\varphi^{-1})^* [x_1, \dots, x_{n-1}], \varphi x_n \rangle \end{aligned}$$

The result follows by nondegeneracy of the scalar product. □

Remark. Substituting $(\varphi^{-1})^*$ for φ , we obtain

$$\varphi[x_1^*, \dots, x_{n-1}^*] = \det \varphi \cdot [(\varphi^{-1})^* x_1^*, \dots, (\varphi^{-1})^* x_{n-1}^*] \quad (x_i^* \in E^*)$$

In particular if E is an oriented 3-dimensional Euclidean space, we have⁸²

$$\varphi x \times \varphi y = \det \varphi \cdot \tilde{\varphi}^{-1}(x \times y) \quad \text{and} \quad \varphi(x \times y) = \det \varphi \cdot (\tilde{\varphi}^{-1} x \times \tilde{\varphi}^{-1} y)$$

Exercise (14). Let e_i, e^{*i} be a pair of dual bases in E, E^* with $e_1 \dots e_n = e$ and

$$x_i = \sum_{v=1}^n \xi_i^v e_v \quad (1 \leq i \leq n-1, \xi_i^v \in \Gamma)$$

Then

$$[x_1, \dots, x_{n-1}] = \sum_{v=1}^n \eta_v e^{*v}$$

where

$$\eta_v = (-1)^{n-v} \begin{vmatrix} \xi_1^1 & \dots & \widehat{\xi_1^v} & \dots & \xi_1^n \\ \vdots & & & & \vdots \\ \xi_{n-1}^1 & \dots & \widehat{\xi_{n-1}^v} & \dots & \xi_{n-1}^n \end{vmatrix}$$

Proof. By the corrected version of problem 8 above. □

⁸²See the remark in chapter VIII, § 4 of [1] above.

Exercise (15 - Lagrange).

$$\left| \begin{array}{ccc} \sum_{v=1}^n \xi_1^v \eta_v^1 & \cdots & \sum_{v=1}^n \xi_1^v \eta_v^{n-1} \\ \vdots & \ddots & \vdots \\ \sum_{v=1}^n \xi_{n-1}^v \eta_v^1 & \cdots & \sum_{v=1}^n \xi_{n-1}^v \eta_v^{n-1} \end{array} \right| = \sum_{v=1}^n \left| \begin{array}{ccc} \xi_1^1 & \cdots & \widehat{\xi}_1^v & \cdots & \xi_1^n \\ \vdots & & \vdots & & \vdots \\ \xi_{n-1}^1 & \cdots & \widehat{\xi}_{n-1}^v & \cdots & \xi_{n-1}^n \end{array} \right| \left| \begin{array}{ccc} \eta_1^1 & \cdots & \widehat{\eta}_1^v & \cdots & \eta_1^n \\ \vdots & & \vdots & & \vdots \\ \eta_{n-1}^1 & \cdots & \widehat{\eta}_{n-1}^v & \cdots & \eta_{n-1}^n \end{array} \right|$$

Proof. Let e_i, e^{*i} be a pair of dual bases in E, E^* with $e_1 \dots e_n = e$ and

$$x_i = \sum_{v=1}^n \xi_i^v e_v \quad x^{*j} = \sum_{\mu=1}^n \eta_{\mu}^j e^{*\mu} \quad (1 \leq i, j \leq n-1)$$

By the Lagrange identity, the left-hand side of the equation is

$$\det \langle x^{*j}, x_i \rangle = \langle [x_1, \dots, x_{n-1}], [x^{*1}, \dots, x^{*n-1}] \rangle$$

which equals the right-hand side of the equation by problem 14. $\square$

§ 16

Remark. The material in this subsection really belongs in chapter 1, as none of it is specific to the mixed exterior algebra. Some of it is also redundant, as noted below.

Remark. Let $T_1 : E^* \otimes E \rightarrow L(E; E)$ denote the isomorphism from subsection 1.26 (using the notation from subsection 6.4) given by

$$T_1(b^* \otimes a) = \langle b^*, - \rangle a \quad (a \in E, b^* \in E^*)$$

Let e_i, e^{*i} be a pair of dual bases in E, E^* . Then

$$\begin{aligned} T(T_1(b^* \otimes a)) &= (\iota \otimes T_1(b^* \otimes a)) \left(\sum_v e^{*v} \otimes e_v \right) \\ &= \sum_v e^{*v} \otimes \langle b^*, e_v \rangle a \\ &= \left(\sum_v \langle b^*, e_v \rangle e^{*v} \right) \otimes a \\ &= b^* \otimes a \end{aligned}$$

It follows that $T = T_1^{-1} = T_1^*$.

Remark. (6.28) follows from (1.30), or from (6.16) with $p = 1$, since $T = T_1^{-1}$.

Remark. T is an *algebra* isomorphism since T_1 is, so

$$T(\psi\varphi) = T(\psi)T(\varphi) = (\iota \otimes \psi)T(\varphi)$$

This is the correct version of (6.30).

Remark. We know that T and $\tilde{T}$ are isometries since T_1 is. It is thus unnecessary to prove that they are linear isomorphisms with the aid of (6.28) and (6.30), and also circular given how nondegeneracy of the trace form is established in subsection 1.28!

§ 17

Remark. (6.31) follows from the fact that the skew tensor product of two skew graded algebras is skew.⁸³ In detail, if

$$w_1 = u^* \otimes u \in \bigwedge_q^p(E^*, E) \quad \text{and} \quad w_2 = v^* \otimes v \in \bigwedge_s^r(E^*, E)$$

then

$$\begin{aligned} w_1 \wedge w_2 &= (u^* \otimes u) \wedge (v^* \otimes v) \\ &= (-1)^{qr} (u^* \wedge v^*) \otimes (u \wedge v) \\ &= (-1)^{qr+pr+qs} (v^* \wedge u^*) \otimes (v \wedge u) \\ &= (-1)^{qr+pr+qs+ps} (v^* \otimes v) \wedge (u^* \otimes u) \\ &= (-1)^{(p+q)(r+s)} w_2 \wedge w_1 \end{aligned}$$

Remark. It follows from (6.33) and Proposition 6.3.1 that

$$T_E(z_1 \wedge \cdots \wedge z_k) = (-1)^{k(k-1)/2} T_E z_1 \square \cdots \square T_E z_k \quad (z_i \in E^* \otimes E)$$

In particular

$$T_E(z^{\widehat{k}}) = (-1)^{k(k-1)/2} \bigwedge^k T_E z \quad (z \in E^* \otimes E)$$

Remark. If E is *finite-dimensional*, we can define the *skew box product* of linear transformations φ and ψ of $\bigwedge E$ by

$$\varphi \widehat{\square} \psi = T_E(T_E^{-1} \varphi \wedge T_E^{-1} \psi)$$

so T_E becomes an *algebra* isomorphism from the skew mixed exterior algebra to a skew box product algebra.

We can also define the *skew mixed intersection product* by

$$w \widehat{\cap} z = D_E(D_E^{-1} w \wedge D_E^{-1} z)$$

⁸³See subsection 2.8.

§ 18

Remark. The inner product $\langle\langle -, - \rangle\rangle$ in $\wedge E^* \hat{\otimes} \wedge E$ is obtained from the scalar product between $\wedge E^* \hat{\otimes} \wedge E$ and $\wedge E \hat{\otimes} \wedge E^*$ using the *skew* flip operator⁸⁴

$$\hat{f}: \wedge E^* \hat{\otimes} \wedge E \rightarrow \wedge E \hat{\otimes} \wedge E^*$$

defined by

$$\hat{f}(u^* \otimes u) = (-1)^{pq} u \otimes u^* \quad (u^* \in \wedge^p E^*, u \in \wedge^q E)$$

This introduces the sign factors into the inner product, which are preserved by the canonical isomorphism

$$f: \wedge E^* \hat{\otimes} \wedge E \rightarrow \wedge(E^* \oplus E)$$

as seen in Proposition 6.18.1.

Remark. The proof of Proposition 6.18.1 is similar to the alternative proof of (5.51) we provide in chapter 5, § 18 above.

⁸⁴See the remark in chapter 2, § 8 above.

Chapter 7

In this chapter, E and E^* are dual n -dimensional vector spaces over Γ .

§ 1

Remark. The map $D_L = T_E \circ D_E \circ T_E^{-1}$ is an isometry since it is a composite of isometries. We have

$$D_L^* = (T_E^{-1})^* \circ D_E^* \circ T_E^* = T_E \circ \Omega_E \circ D_E \circ T_E^{-1} = \Omega_L \circ D_L$$

and

$$D_L^2 = T_E \circ D_E^2 \circ T_E^{-1} = T_E \circ \Omega_E \circ T_E^{-1} = \Omega_L$$

Remark. In Proposition 7.1.1, (1) is essentially the correlate of (6.20) for D_L , and (2) also follows from the corresponding result about D_E .⁸⁵ It follows from (1) that

$$\det(D_L \alpha) = \det \alpha$$

We also have

$$D_L^{-1} \alpha = D_e^{-1} \circ \alpha^* \circ D_e$$

Remark. When $L(\wedge^p E; \wedge^q E)$ is viewed as a subspace of $L(\wedge E; \wedge E)$,⁸⁶ then the diagonal subalgebra of $L(\wedge E; \wedge E)$ is

$$H(\wedge E; \wedge E) = \sum_{p=0}^n L(\wedge^p E; \wedge^p E)$$

where $H(\wedge E; \wedge E)$ consists of the transformations homogeneous of degree zero.⁸⁷

Remark. For linear transformations α and β of $\wedge E$, we define the *intersection product* in terms of the generalized box product⁸⁸ by

$$\alpha \cap \beta = D_L(D_L^{-1} \alpha \square D_L^{-1} \beta)$$

This definition makes $L(\wedge E; \wedge E)$ into an algebra called the *intersection algebra*, and makes T_E an *algebra* isomorphism from the mixed intersection algebra⁸⁹

⁸⁵See the remark in chapter 6, § 10 above.

⁸⁶See the remark in chapter II, § 4 of [1] above.

⁸⁷See problem 7.9.13.

⁸⁸See the remark in chapter 6, § 3 above.

⁸⁹See the remark in chapter 6, § 12 above.

since for $w, z \in \wedge(E^*, E)$,

$$\begin{aligned} T_E(w \cap z) &= T_E D_E(D_E^{-1} w \cdot D_E^{-1} z) \\ &= D_L(T_E D_E^{-1} w \square T_E D_E^{-1} z) \\ &= D_L(D_L^{-1} T_E w \square D_L^{-1} T_E z) \\ &= T_E w \cap T_E z \end{aligned}$$

It follows that the intersection algebra is associative, unital (with unit $\iota_{\wedge^n E}$), and anticommutative. It also follows that D_L is an *algebra* isomorphism from the box product algebra to the intersection algebra *and vice versa* since

$$\begin{aligned} D_L(\alpha \cap \beta) &= D_L T_E(T_E^{-1} \alpha \cap T_E^{-1} \beta) \\ &= T_E(D_E T_E^{-1} \alpha \cdot D_E T_E^{-1} \beta) \\ &= D_L \alpha \square D_L \beta \end{aligned}$$

The diagonal subspace $H(\wedge E; \wedge E)$ forms a *commutative* subalgebra, and for α, β in this subalgebra we have

$$\alpha \cap \beta = i_{\square}(D_L^{\Delta} \alpha) \beta$$

For a linear transformation φ of E , viewed as a linear transformation of $\wedge E$, we define

$$\cap^p \varphi = \frac{1}{p!} \underbrace{\varphi \cap \cdots \cap \varphi}_p$$

so

$$D_L(\wedge^p \varphi) = \cap^p D_L \varphi \quad \text{and} \quad D_L(\cap^p \varphi) = \wedge^p D_L \varphi$$

If ψ is another linear transformation of E , then

$$\wedge^p \varphi \cap \wedge^q \psi = i_{\square}(D_L(\wedge^p \varphi))(\wedge^q \psi)$$

Remark. Interchanging the roles of E and E^* , we obtain the map

$$D_{L^*} : L(\wedge E^*; \wedge E^*) \rightarrow L(\wedge E^*; \wedge E^*)$$

given by⁹⁰

$$D_{L^*} = T_{E^*} \circ D_{E^*} \circ T_{E^*}^{-1}$$

⁹⁰See the remarks in chapter 6, § 2,10 above.

Remark. If α is a linear transformation of $\wedge E$, then by Proposition 7.1.1(1),

$$(D_L \alpha)^* = (D^e)^{-1*} \circ \alpha \circ (D^e)^* = D_e \circ \alpha \circ D_e^{-1} = D_{L^*} \alpha^*$$

and by a remark above

$$(D_L^{-1} \alpha)^* = D_{L^*}^{-1} \alpha^*$$

In particular if $\varphi_1, \dots, \varphi_p, \varphi$ are linear transformations of E , then

$$(D_L(\varphi_1 \square \dots \square \varphi_p))^* = D_{L^*}(\varphi_1^* \square \dots \square \varphi_p^*)$$

and

$$(D_L(\wedge^p \varphi))^* = D_{L^*}(\wedge^p \varphi^*)$$

If β is also a linear transformation of $\wedge E$, then

$$(\alpha \cap \beta)^* = \alpha^* \cap \beta^*$$

Remark. The map

$$\tilde{D}_L : L(\wedge E; \wedge E) \rightarrow L(\wedge E^*; \wedge E^*)$$

defined by

$$\tilde{D}_L \alpha = D_{L^*} \alpha^*$$

is an algebra isomorphism making the following diagram commute:⁹¹

$$\begin{array}{ccc} L(\wedge E; \wedge E) & \xrightarrow{T} & \wedge(E^*, E) \\ \tilde{D}_L \downarrow & & \downarrow D_E \\ L(\wedge E^*; \wedge E^*) & \xrightarrow{\tilde{T}} & \wedge(E^*, E) \end{array}$$

Interchanging the roles of E and E^* , we obtain the map

$$\tilde{D}_{L^*} : L(\wedge E^*; \wedge E^*) \rightarrow L(\wedge E; \wedge E)$$

defined by

$$\tilde{D}_{L^*} \alpha^* = D_L \alpha$$

It follows that

$$\tilde{D}_L^{-1} = \tilde{D}_L^* = \Omega_L \circ \tilde{D}_{L^*} \quad \text{and} \quad \tilde{D}_{L^*}^{-1} = \tilde{D}_{L^*}^* = \Omega_{L^*} \circ \tilde{D}_L$$

In particular $\tilde{D}_L$ and $\tilde{D}_{L^*}$ are isometries. Finally, $\tilde{D}_L$ and $\tilde{D}_{L^*}$ interchange box and intersection products.

⁹¹See subsection 6.16.

§ 2

Remark (Laplace). In the corollary to Proposition 7.2.1, summing over p on both sides of (1) yields

$$\wedge \varphi \circ D_L(\wedge \varphi) = \det \varphi \cdot \iota_{\wedge E} \quad (1')$$

Conversely, restricting to $\wedge^p E$ in (1') yields (1), so this is an equivalent form of the Laplace formula (1).

Taking duals on both sides of (1') yields

$$D_{L^*}(\wedge \varphi^*) \circ \wedge \varphi^* = \det \varphi^* \cdot \iota_{\wedge E^*}$$

Substituting E for E^* and φ for φ^* yields

$$D_L(\wedge \varphi) \circ \wedge \varphi = \det \varphi \cdot \iota_{\wedge E} \quad (2')$$

which is equivalent to (2). Conversely, taking duals and making substitutions in (2') yields (1').

Remark (Laplace). Let $z \in E^* \otimes E$, $\varphi = T_E z$, and $M = M(\varphi)$ the matrix of φ with respect to some basis of E . The matrix $M(\wedge^p \varphi)$ with respect to the induced basis of $\wedge^p E$ consists of the $p \times p$ minors of M .⁹² On the other hand, since

$$D_L(\wedge^{n-p} \varphi) = D_L T_E z^{n-p} = T_E D_E z^{n-p}$$

the matrix $M(D_L(\wedge^{n-p} \varphi))$ consists of the components of $D_E z^{n-p}$, so is the transpose of the matrix of signed $(n-p) \times (n-p)$ minors of M complementary to the minors in $M(\wedge^p \varphi)$.⁹³ By part (1) of the corollary to Proposition 7.2.1,

$$M(\wedge^p \varphi) M(D_L(\wedge^{n-p} \varphi)) = \det M \cdot I$$

which, by the above results, expresses the classical Laplace expansion of $\det M$ by minors from any p rows of M and their corresponding cofactors.⁹⁴ Part (2) of the corollary expresses the Laplace expansion from columns of M .

Remark. It follows from Proposition 7.2.2 that⁹⁵

$$i_{\square}(D_L(\wedge^p \varphi))(\wedge^q \varphi) = \binom{2n-p-q}{n-p} \det \varphi \cdot \wedge^{p+q-n} \varphi \quad (1)$$

⁹²See the remark in chapter 5, § 8 above.

⁹³See the remark in chapter 6, § 13 above.

⁹⁴See also problem 7.9.9.

⁹⁵See the remark in chapter 6, § 3 above.

and

$$D_L(\wedge^p \varphi) \square D_L(\wedge^q \varphi) = \binom{2n-p-q}{n-p} \det \varphi \cdot D_L(\wedge^{p+q-n} \varphi) \quad (2)$$

Therefore

$$\wedge^p \varphi \cap \wedge^q \varphi = \binom{2n-p-q}{n-p} \det \varphi \cdot \wedge^{p+q-n} \varphi$$

and dually

$$\cap^p \varphi \square \cap^q \varphi = \binom{2n-p-q}{n-p} \det \varphi \cdot \cap^{p+q-n} \varphi$$

In particular

$$\varphi \cap \wedge^{n-1} \varphi = \wedge^{n-1} \varphi \cap \varphi = n \cdot \det \varphi \cdot \iota_{\wedge^0 E}$$

§ 3

Remark. If $z_1, \dots, z_n \in E^* \otimes E$, then⁹⁶

$$\begin{aligned} D_E(z_1 \cdots z_n) \mathfrak{t} &= \sum_{i=1}^n z_i \circ D_E(z_1 \cdots \widehat{z}_i \cdots z_n) \\ &= \sum_{i=1}^n D_E(z_1 \cdots \widehat{z}_i \cdots z_n) \circ z_i \end{aligned}$$

In other words

$$\begin{aligned} D_E(z_1 \cdots z_n) \mathfrak{t} &= \sum_{i=1}^n z_i \circ \text{Ad}(z_1, \dots, \widehat{z}_i, \dots, z_n) \\ &= \sum_{i=1}^n \text{Ad}(z_1, \dots, \widehat{z}_i, \dots, z_n) \circ z_i \end{aligned}$$

Taking $z_1 = \cdots = z_n = z$ yields

$$\det T_E(z) \mathfrak{t} = z \circ \text{ad}(z) = \text{ad}(z) \circ z$$

Remark. We have

$$\text{Ad}(x_1^* \otimes x_1, \dots, x_{n-1}^* \otimes x_{n-1}) = [x_1^*, \dots, x_{n-1}^*, x_1, \dots, x_{n-1}]$$

where the product on the right is the mixed external product.⁹⁷

⁹⁶See equation (21) in [4].

⁹⁷See the remark in chapter 6, § 14 above.

Remark. For linear transformations $\varphi_1, \dots, \varphi_{n-1}$ of E , it is possible to *define*

$$\text{Ad}(\varphi_1, \dots, \varphi_{n-1}) = T_E \text{Ad}(T_E^{-1} \varphi_1, \dots, T_E^{-1} \varphi_{n-1})$$

Then

$$\begin{aligned} \text{Ad}(\varphi_1, \dots, \varphi_{n-1}) &= T_E D_E(T_E^{-1} \varphi_1 \cdots T_E^{-1} \varphi_{n-1}) \\ &= D_L(\varphi_1 \square \cdots \square \varphi_{n-1}) \\ &= D_L \varphi_1 \cap \cdots \cap D_L \varphi_{n-1} \end{aligned}$$

For a linear transformation φ of E , it is also possible to *define*

$$\text{ad } \varphi = \frac{1}{(n-1)!} \text{Ad}(\varphi, \dots, \varphi)$$

Then

$$\text{ad } \varphi = D_L(\wedge^{n-1} \varphi) = \cap^{n-1} D_L \varphi$$

It follows that

$$\varphi \circ \text{ad}(\varphi) = \text{ad}(\varphi) \circ \varphi = \det \varphi \cdot \iota$$

This provides an alternative approach to the classical adjoint.⁹⁸

Remark. It follows from Proposition 7.2.2(1) that for $z \in E^* \otimes E$,

$$i(\text{ad } z) z^{n-p} = (p+1) \det(T_E z) z^{n-(p+1)} \quad (0 \leq p \leq n-1)$$

Remark. We provide an alternative inductive proof of the Jacobi identity (7.6). The identity is certainly true for $p = 1$, and if it is true for p then by the previous remark

$$\begin{aligned} (p+1) D_E[(\text{ad } z)^{p+1}] &= D_E[(\text{ad } z)^p \cdot \text{ad } z] \\ &= i(\text{ad } z) D_E[(\text{ad } z)^p] \\ &= (\det T_E z)^{p-1} i(\text{ad } z) z^{n-p} \\ &= (p+1) (\det T_E z)^p z^{n-(p+1)} \end{aligned}$$

Dividing both sides by $p+1$ shows that the identity holds for $p+1$.

⁹⁸Compare with subsection 7.4.

Remark. Applying T_E to both sides of the Jacobi identity (7.6) with $\varphi = T_E z$ yields

$$D_L(\wedge^p \text{ad } \varphi) = (\det \varphi)^{p-1} \wedge^{n-p} \varphi$$

or equivalently, by applying D_L to both sides of this,

$$\wedge^p \text{ad } \varphi = (\det \varphi)^{p-1} D_L(\wedge^{n-p} \varphi)$$

In terms of matrices, this identity relates the minors of a matrix to the minors of its adjoint matrix.⁹⁹ An alternative proof of this identity uses the intersection product and a remark in § 2 above to show that

$$D_L(\wedge^p \text{ad } \varphi) = \cap^p D_L(\text{ad } \varphi) = \cap^p \wedge^{n-1} \varphi = (\det \varphi)^{p-1} \wedge^{n-p} \varphi$$

§ 4

Remark. To see that Ω_Δ is skew-symmetric, observe that it is multilinear and for $\tau \in S_n$

$$\begin{aligned} (\tau \Omega_\Delta)(x_1, \dots, x_n)h &= \Omega_\Delta(x_{\tau(1)}, \dots, x_{\tau(n)})h \\ &= \sum_{\sigma} \varepsilon_\sigma \Delta(\varphi_1 x_{\tau\sigma(1)}, \dots, \varphi_{n-1} x_{\tau\sigma(n-1)}, h) x_{\tau\sigma(n)} \\ &= \varepsilon_\tau \sum_{\sigma} \varepsilon_{\tau\sigma} \Delta(\varphi_1 x_{\tau\sigma(1)}, \dots, \varphi_{n-1} x_{\tau\sigma(n-1)}, h) x_{\tau\sigma(n)} \\ &= \varepsilon_\tau \Omega_\Delta(x_1, \dots, x_n)h \end{aligned}$$

It follows that $\tau \Omega_\Delta = \varepsilon_\tau \Omega_\Delta$.

Remark. $\text{Ad}(\varphi_1, \dots, \varphi_{n-1})$ does not depend upon the choice of Δ , and therefore only depends upon $\varphi_1, \dots, \varphi_{n-1}$. Since Ad is $(n-1)$ -linear and symmetric, it factors through the $(n-1)$ -th symmetric power of $L(E; E)$.

Remark. We have¹⁰⁰

$$\Omega_\Delta(x_1, \dots, x_n)h = \sum_{\sigma} \Delta(\varphi_{\sigma^{-1}(1)} x_1, \dots, h, \dots, \varphi_{\sigma^{-1}(n)} x_n) x_{\sigma(n)}$$

⁹⁹See problem 7.9.11.

¹⁰⁰Compare with subsection 6.3.

where h is in the $\sigma(n)$ -th position of Δ . For $\varphi_1 = \dots = \varphi_{n-1} = \varphi$ this simplifies to

$$\begin{aligned}\Omega_\Delta(x_1, \dots, x_n)h &= \sum_{\sigma} \Delta(\varphi x_1, \dots, h, \dots, \varphi x_n) x_{\sigma(n)} \\ &= (n-1)! \sum_{i=1}^n \Delta(\varphi x_1, \dots, h, \dots, \varphi x_n) x_i \\ &= (n-1)! \sum_{i=1}^n (-1)^{i-1} \Delta(h, \varphi x_1, \dots, \widehat{\varphi x_i}, \dots, \varphi x_n) x_i\end{aligned}$$

Therefore

$$\Delta(x_1, \dots, x_n) \operatorname{ad}(\varphi)h = \sum_{i=1}^n (-1)^{i-1} \Delta(h, \varphi x_1, \dots, \widehat{\varphi x_i}, \dots, \varphi x_n) x_i$$

which agrees with (4.12) in [1].

Remark. By Proposition 7.4.1,

$$T_E \operatorname{Ad}(z_1, \dots, z_{n-1}) = \operatorname{Ad}(T_E(z_1), \dots, T_E(z_{n-1}))$$

In particular

$$T_E \operatorname{ad}(z) = \operatorname{ad}(T_E z)$$

Remark. By Proposition 7.4.1 and remarks above,

$$\begin{aligned}\operatorname{Ad}(\varphi_1, \dots, \varphi_{n-1})^* &= D_L(\varphi_1 \square \dots \square \varphi_{n-1})^* \\ &= D_{L^*}(\varphi_1^* \square \dots \square \varphi_{n-1}^*) \\ &= \operatorname{Ad}(\varphi_1^*, \dots, \varphi_{n-1}^*)\end{aligned}$$

In particular

$$\operatorname{ad}(\varphi)^* = \operatorname{ad}(\varphi^*)$$

Remark. Let e, e^* be dual basis vectors in $\wedge^n E, \wedge^n E^*$. By Proposition 7.4.1(2),

$$\wedge^{n-1} \varphi^* D_e h = D_e [D_e^{-1} \wedge^{n-1} \varphi^* D_e h] = D_e [\operatorname{ad}(\varphi)h]$$

It follows that

$$\Delta(h, \varphi x_1, \dots, \varphi x_{n-1}) = \Delta(\operatorname{ad}(\varphi)h, x_1, \dots, x_{n-1})$$

for all $h, x_i \in E$. This identity characterizes $\operatorname{ad}(\varphi)$.¹⁰¹ More generally,

$$\sum_{\tau \in S_{n-1}} \Delta(h, \varphi_{\tau(1)} x_1, \dots, \varphi_{\tau(n-1)} x_{n-1}) = \Delta(\operatorname{Ad}(\varphi_1, \dots, \varphi_{n-1})h, x_1, \dots, x_{n-1})$$

for all $h, x_i \in E$, and this identity characterizes $\operatorname{Ad}(\varphi_1, \dots, \varphi_{n-1})$.

¹⁰¹See also the remark in chapter IV, § 2 of [1] above.

§ 5

Remark. Since the function C_p is p -linear and symmetric, it factors through the p -th symmetric power of $L(E; E)$.

Remark. If φ is a linear transformation of E , then

$$C_p(\varphi) = \frac{1}{p!(n-p)!} C_n(\underbrace{\varphi, \dots, \varphi}_p, \underbrace{t, \dots, t}_{n-p})$$

Proof. By (7.9) and (7.10), for $z = T_E^{-1}\varphi$ we have

$$\begin{aligned} C_p(\varphi) &= \langle z^p, t^p \rangle \\ &= \langle z^p, i(t^{n-p}) t^p \rangle \\ &= \langle z^p \cdot t^{n-p}, t^p \rangle \\ &= \frac{1}{p!(n-p)!} \langle \underbrace{z \cdots z}_p \cdot \underbrace{t \cdots t}_{n-p}, t^p \rangle \\ &= \frac{1}{p!(n-p)!} C_n(\underbrace{\varphi, \dots, \varphi}_p, \underbrace{t, \dots, t}_{n-p}) \end{aligned} \quad \square$$

Remark. If φ is a linear transformation of E , then

$$C_p(\varphi) x_1 \wedge \cdots \wedge x_n = \sum_{v_1 < \cdots < v_p} x_1 \wedge \cdots \wedge \varphi x_{v_1} \wedge \cdots \wedge \varphi x_{v_p} \wedge \cdots \wedge x_n$$

where φx_{v_i} is in the i -th position on the right for all $1 \leq i \leq p$.

Proof. By the previous remark and the action of the transformation

$$\underbrace{\varphi \square \cdots \square \varphi}_p \underbrace{t \square \cdots \square t}_{n-p} \quad \square$$

Remark. If φ and ψ are linear transformations of E , then we have the binomial formula

$$\wedge^n(\varphi + \psi) = \sum_{p+q=n} \wedge^p \varphi \square \wedge^q \psi \quad (1)$$

Taking the trace of both sides yields

$$\det(\varphi + \psi) = \sum_{p+q=n} \text{tr}(\wedge^p \varphi \square \wedge^q \psi) \quad (2)$$

or equivalently

$$\det(\varphi + \psi) = \sum_{p+q=n} \frac{1}{p!q!} \text{Det}(\underbrace{\varphi, \dots, \varphi}_p, \underbrace{\psi, \dots, \psi}_q)$$

which is the correct version of the formula in the book.

Taking $\psi = \lambda \iota$ in (2) yields Proposition 7.5.1.

Remark. Let Δ be a determinant function in E . By definition of the box product,

$$\text{Det}(\varphi_1, \dots, \varphi_n) \cdot \Delta(x_1, \dots, x_n) = \sum_{\sigma \in S_n} \Delta(\varphi_{\sigma(1)} x_1, \dots, \varphi_{\sigma(n)} x_n)$$

for all $x_i \in E$. This identity characterizes $\text{Det}(\varphi_1, \dots, \varphi_n)$.

Remark. By a remark above,

$$\begin{aligned} \text{Det}(\varphi_1, \dots, \varphi_n) \iota &= \sum_{i=1}^n \varphi_i \circ \text{Ad}(\varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n) \\ &= \sum_{i=1}^n \text{Ad}(\varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n) \circ \varphi_i \end{aligned}$$

This generalizes Corollary I to Proposition 7.4.1.

Remark (Sylvester-Weinstein-Aronszajn). If $\varphi : E \rightarrow F$ and $\psi : F \rightarrow E$ are linear, then

$$\det(\varphi\psi + \iota) = \det(\psi\varphi + \iota)$$

Proof. Let $n = \dim E$ and $m = \dim F$, and assume $m \geq n$. By remarks above,

$$\begin{aligned} \det(\varphi\psi + \iota) &= \sum_{p=0}^m \text{tr}(\wedge^p \varphi\psi \square \wedge^{m-p} \iota) \\ &= \sum_{p=0}^m \text{tr}(\wedge^p \varphi\psi) \\ &= \sum_{p=0}^n \text{tr}(\wedge^p \psi\varphi) \\ &= \sum_{p=0}^n \text{tr}(\wedge^p \psi\varphi \square \wedge^{n-p} \iota) \\ &= \det(\psi\varphi + \iota) \end{aligned}$$

□

§ 6

Remark. In the proof of Proposition 7.6.1, (7.11) is obtained by induction on p . For $p = 0$ the result is certainly true, and if it is true for $p - 1 < n$, then it is also true for p since

$$\begin{aligned} A_p(\varphi) &= C_p(\varphi) \cdot \iota - \varphi \circ A_{p-1}(\varphi) \\ &= C_p(\varphi) \varphi^0 - \sum_{v=0}^{p-1} (-1)^{v+1} C_{p-(v+1)}(\varphi) \varphi^{v+1} \\ &= \sum_{v=0}^p (-1)^v C_{p-v}(\varphi) \varphi^v \end{aligned}$$

Remark. It follows from the binomial formula that

$$\wedge^{n-1}(\varphi + \lambda \iota) = \sum_{p=0}^{n-1} \wedge^p \varphi \square \wedge^{n-p-1} \iota \cdot \lambda^{n-p-1}$$

so

$$\begin{aligned} \text{ad}(\varphi + \lambda \iota) &= D_L(\wedge^{n-1}(\varphi + \lambda \iota)) \\ &= \sum_{p=0}^{n-1} D_L(\wedge^p \varphi) \cap \wedge^{p+1} \iota \cdot \lambda^{n-p-1} \end{aligned}$$

and therefore

$$A_p(\varphi) = D_L(\wedge^p \varphi) \cap \wedge^{p+1} \iota = i_{\square}(\wedge^p \varphi) \iota_{\wedge^{p+1} E} \quad (0 \leq p \leq n-1)$$

This is equivalent to Proposition 7.3.1.

Remark. If φ is invertible, then

$$\varphi^{-1} = (\det \varphi)^{-1} \text{ad} \varphi$$

by Corollary I to Proposition 7.4.1, so it follows from (7.14) that

$$C_p(\varphi^{-1}) = (\det \varphi)^{-p} C_p(\text{ad} \varphi) = (\det \varphi)^{-1} C_{n-p}(\varphi) \quad (0 \leq p \leq n)$$

or equivalently

$$C_p(\varphi) = \det \varphi \cdot C_{n-p}(\varphi^{-1}) \quad (0 \leq p \leq n)$$

In particular if $n = 2m$, then¹⁰²

$$C_m(\varphi) = \det \varphi \cdot C_m(\varphi^{-1})$$

¹⁰²See also problem 7.9.8(c).

Remark. Let $\varphi_1, \dots, \varphi_n$ be *commuting* linear transformations of E . Define

$$\chi_{\varphi_1 \dots \varphi_n}^+(\lambda_1, \dots, \lambda_n) = \text{Det}(\varphi_1 + \lambda_1 \iota, \dots, \varphi_n + \lambda_n \iota) \quad (\lambda_i \in \Gamma)$$

Then

$$\chi_{\varphi_1 \dots \varphi_n}^+(\lambda_1, \dots, \lambda_n) = \sum_{p \in 2^n} C_{p_1 \dots p_n}(\varphi_1, \dots, \varphi_n) \lambda_1^{p_1} \dots \lambda_n^{p_n} \quad (1)$$

where

$$C_{p_1 \dots p_n}(\varphi_1, \dots, \varphi_n) = \text{Det}(\varphi_1^{1-p_1}, \dots, \varphi_n^{1-p_n})$$

Similarly

$$\begin{aligned} \text{Ad}(\varphi_1 + \lambda_1 \iota, \dots, \widehat{\varphi_i + \lambda_i \iota}, \dots, \varphi_n + \lambda_n \iota) = \\ \sum_{\widehat{p} \in 2^{n-1}} A_{p_1 \dots \widehat{p_i} \dots p_n}(\varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n) \lambda_1^{p_1} \dots \widehat{\lambda_i^{p_i}} \dots \lambda_n^{p_n} \end{aligned} \quad (2)$$

where $\widehat{p} = (p_1, \dots, \widehat{p_i}, \dots, p_n)$ and

$$A_{p_1 \dots \widehat{p_i} \dots p_n}(\varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n) = \text{Ad}(\varphi_1^{1-p_1}, \dots, \widehat{\varphi_i^{1-p_i}}, \dots, \varphi_n^{1-p_n})$$

By a remark above,

$$\begin{aligned} C_{p_1 \dots p_n}(\varphi_1, \dots, \varphi_n) \iota &= \sum_{i=1}^n \varphi_i^{1-p_i} \circ A_{p_1 \dots \widehat{p_i} \dots p_n}(\varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n) \\ &= \sum_{i=1}^n A_{p_1 \dots \widehat{p_i} \dots p_n}(\varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n) \circ \varphi_i^{1-p_i} \end{aligned} \quad (3)$$

The *mixed characteristic polynomial* of $\varphi_1, \dots, \varphi_n$ is defined by

$$\chi_{\varphi_1 \dots \varphi_n}(\lambda_1, \dots, \lambda_n) = \chi_{\varphi_1 \dots \varphi_n}^+(-\lambda_1, \dots, -\lambda_n)$$

It follows from (1) and (3) that

$$\begin{aligned} \chi_{\varphi_1 \dots \varphi_n}(\lambda_1, \dots, \lambda_n) \iota &= \sum_{p \in 2^n} \sum_{i=1}^n (-1)^{\sum_{j=1}^n p_j} \varphi_i^{1-p_i} \circ A_{p_1 \dots \widehat{p_i} \dots p_n}(\varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n) \lambda_1^{p_1} \dots \lambda_n^{p_n} \\ &= \sum_{p \in 2^n} \sum_{i=1}^n (-1)^{\sum_{j=1}^n p_j} A_{p_1 \dots \widehat{p_i} \dots p_n}(\varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n) \circ \varphi_i^{1-p_i} \lambda_1^{p_1} \dots \lambda_n^{p_n} \end{aligned}$$

Substituting $\varphi_1, \dots, \varphi_n$ for $\lambda_1, \dots, \lambda_n$ we obtain

$$\begin{aligned}
\chi_{\varphi_1 \cdots \varphi_n}(\varphi_1, \dots, \varphi_n) &= \sum_{i=1}^n \sum_{p \in 2^n} (-1)^{\sum_{j=1}^n p_j} A_{p_1 \cdots \widehat{p_i} \cdots p_n}(\varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n) \varphi_1^{p_1} \cdots \varphi_i \cdots \varphi_n^{p_n} \\
&= \sum_{i=1}^n \sum_{\widehat{p} \in 2^{n-1}} (-1)^{\sum_{j \neq i} p_j} \left(A_{p_1 \cdots \widehat{p_i} \cdots p_n}(\varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n) \varphi_1^{p_1} \cdots \varphi_i \cdots \varphi_n^{p_n} \right. \\
&\quad \left. - A_{p_1 \cdots \widehat{p_i} \cdots p_n}(\varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n) \varphi_1^{p_1} \cdots \varphi_i \cdots \varphi_n^{p_n} \right) \\
&= 0
\end{aligned}$$

This is the mixed Cayley-Hamilton theorem for commuting transformations, which reduces to the classical Cayley-Hamilton theorem when $\varphi_1 = \cdots = \varphi_n$.

§ 7

Remark. Since the function Tr_p is p -linear and symmetric, it factors through the p -th symmetric power of $L(E; E)$.

Remark. It would be more consistent with other definitions in this chapter to define

$$\text{Tr}_p(\varphi_1, \dots, \varphi_p) = \sum_{\sigma} \text{tr}(\varphi_{\sigma(1)} \circ \cdots \circ \varphi_{\sigma(p)})$$

and

$$\text{Tr}_p(\varphi) = \frac{1}{p!} \text{Tr}_p(\varphi, \dots, \varphi)$$

Remark. Since

$$C_p(\varphi) = \text{tr}(\wedge^p \varphi) \quad \text{and} \quad \text{Tr}_p(\varphi) = \text{tr}(\varphi^p)$$

Proposition 7.7.1 establishes a relationship between the exterior powers and the composition powers of a linear transformation. This is a relationship between the box product and the composition product in the diagonal subspace of $L(\wedge E; \wedge E)$, and equivalently between the mixed exterior product and the composition product in ΔE .¹⁰³

Question. What is the relationship between $C_p(\varphi_1, \dots, \varphi_p)$ and $\text{Tr}_p(\varphi_1, \dots, \varphi_p)$?

¹⁰³See the remark in chapter 6, § 5 above.

§ 9

Remark. Throughout this subsection, the definition of Δ should be

$$\Delta = \frac{i^n}{2^n} \Delta_E \wedge \overline{\Delta}_E$$

where the product on the right-hand side is a generalization of the Grassmann product¹⁰⁴ for $\mathbb{C}$ -valued skew-symmetric $\mathbb{R}$ -multilinear functions. This gives

$$\Delta(x_1, \dots, x_{n+n}) = \frac{i^n}{2^n (n!)^2} \sum_{\rho \in S_{n+n}} \varepsilon_\rho \Delta_E(x_{\rho(1)}, \dots, x_{\rho(n)}) \overline{\Delta}_E(x_{\rho(n+1)}, \dots, x_{\rho(n+n)}) \quad (1)$$

Taking $x_k = a_k$ and $x_{n+k} = i a_k$ for $1 \leq k \leq n$, what are the nonzero terms in the sum on the right-hand side of (1)? In such a term the subscripts associated with the a 's and $i a$'s (not the x 's) occurring within Δ_E must be distinct, lest there be a linear dependence relation over $\mathbb{C}$ which causes Δ_E , and hence the term, to be zero. Similarly for $\overline{\Delta}_E$.

If the subscripts within each of Δ_E and $\overline{\Delta}_E$ are strictly increasing (we call this an *increasing term*), then the term is nonzero and for each $1 \leq k \leq n$ we must have either

$$\rho(k) = k \quad \text{and} \quad \rho(n+k) = n+k \quad (2)$$

or

$$\rho(k) = n+k \quad \text{and} \quad \rho(n+k) = k \quad (3)$$

Note there are 2^n such permutations ρ , and $\varepsilon_\rho = (-1)^p$ where p is the number of k 's for which (3) holds, which is just the number of i 's occurring within Δ_E . These i 's can be moved over to $\overline{\Delta}_E$ by multiplying by $(-1)^p = \varepsilon_\rho$, since Δ_E is $\mathbb{C}$ -multilinear and $\overline{\Delta}_E$ is conjugate $\mathbb{C}$ -multilinear.

Now every nonzero term can be obtained from a unique increasing term by applying unique signed permutations $\sigma, \tau \in S_n$ to Δ_E and $\overline{\Delta}_E$, respectively. So

¹⁰⁴See subsection 5.30.

we obtain

$$\begin{aligned}
\Delta(a_1, \dots, a_n, i a_1, \dots, i a_n) &= \frac{i^n}{2^n (n!)^2} \cdot 2^n \cdot \sum_{\sigma, \tau \in S_n} \varepsilon_\sigma \varepsilon_\tau \Delta_E(a_{\sigma(1)}, \dots, a_{\sigma(n)}) \bar{\Delta}_E(i a_{\tau(1)}, \dots, i a_{\tau(n)}) \\
&= \frac{i^n}{(n!)^2} \cdot (n!)^2 \cdot \Delta_E(a_1, \dots, a_n) \bar{\Delta}_E(i a_1, \dots, i a_n) \\
&= \Delta_E(a_1, \dots, a_n) \bar{\Delta}_E(a_1, \dots, a_n) \\
&= |\Delta_E(a_1, \dots, a_n)|^2 > 0
\end{aligned}$$

If Δ_E is replaced by $\lambda \Delta_E$, Δ still becomes $|\lambda|^2 \Delta$, so $E_{\mathbb{R}}$ has a natural orientation. It is possible to give a more abstract proof of this result which avoids the tedious combinatorics.¹⁰⁵

Exercise (4). If $z \in E^* \otimes E$ and $\varphi = T_E z$, then

$$\langle z^p, (\text{ad } z)^p \rangle = \binom{n}{p} \det(\varphi)^p \quad (1 \leq p \leq n)$$

Proof. By the Jacobi identity (7.6) and Proposition 7.2.2,

$$\begin{aligned}
\langle z^p, (\text{ad } z)^p \rangle &= \langle D_E z^p, D_E (\text{ad } z)^p \rangle \\
&= \det(\varphi)^{p-1} \langle D_E z^p, z^{n-p} \rangle \\
&= \det(\varphi)^{p-1} i (D_E z^p) z^{n-p} \\
&= \binom{n}{p} \det(\varphi)^p \quad \square
\end{aligned}$$

Exercise (5 - Cauchy). If $\varphi : E \rightarrow E$ is linear, then

$$\det(\wedge^p \varphi) \cdot \det(\wedge^{n-p} \varphi) = \det(\varphi)^{\binom{n}{p}} \quad (0 \leq p \leq n)$$

Proof. By taking the determinant on both sides of the Laplace formula, using the fact that $\det D_L(\wedge^{n-p} \varphi) = \det(\wedge^{n-p} \varphi)$. $\square$

Remark. In particular if $n = 2m$, then taking $p = m$ we obtain

$$\det(\wedge^m \varphi)^2 = \det(\varphi)^{\binom{2m}{m}}$$

¹⁰⁵See <https://math.stackexchange.com/q/4565568>.

This also follows from the Sylvester-Franke theorem

$$\det(\wedge^p \varphi) = \det(\varphi)^{\binom{n-1}{p-1}}$$

Exercise (6). Let A be an $n \times n$ matrix. The coefficient of λ^{n-p} in $\det(A - \lambda I)$ is $(-1)^{n-p}$ times the sum of the principal $p \times p$ minors of A .

Proof. Write $A = M(\varphi)$, so $\det(A - \lambda I) = \det(\varphi - \lambda \iota)$. By Proposition 7.5.1, the coefficient of λ^{n-p} in this determinant is

$$(-1)^{n-p} C_p(\varphi) = (-1)^{n-p} \operatorname{tr}(\wedge^p \varphi)$$

which is just the sum of the principal $p \times p$ minors of A by a remark in chapter 5, § 8 above. $\square$

Exercise (9, 12 - Laplace). Let A be an $n \times n$ matrix. For any $1 \leq \lambda_1 < \dots < \lambda_p \leq n$,

$$\det A = \sum_{\nu_1 < \dots < \nu_p} (-1)^{\sum_{i=1}^p \lambda_i + \nu_i} \det A_{\lambda_1 \dots \lambda_p}^{\nu_1 \dots \nu_p} \cdot \det A_{\lambda_{p+1} \dots \lambda_n}^{\nu_{p+1} \dots \nu_n}$$

where $(\lambda_{p+1}, \dots, \lambda_n)$ and $(\nu_{p+1}, \dots, \nu_n)$ are the ordered $(n-p)$ -tuples complementary to $(\lambda_1, \dots, \lambda_p)$ and $(\nu_1, \dots, \nu_p)$, respectively.

Proof. By a remark in § 2 above.

Alternatively, write $A = (\alpha_i^j)$. Assume without loss of generality that $\lambda_i = i$ for all $1 \leq i \leq p$. Let e_i, e^{*i} be a pair of dual bases in E, E^* with $e = e_1 \dots e_n$ and $e^* = e^{1 \dots n}$ and let Δ be the determinant function in E with $\Delta(e_1, \dots, e_n) = 1$. Set

$$x_i = \sum_{j=1}^n \alpha_i^j e_j \quad (1 \leq i \leq n)$$

Then

$$\begin{aligned} \det A &= \Delta(x_1, \dots, x_n) \\ &= \langle e^*, x_1 \wedge \dots \wedge x_n \rangle \\ &= \langle e^*, (x_1 \wedge \dots \wedge x_p) \wedge (x_{p+1} \wedge \dots \wedge x_n) \rangle \\ &= \langle D_e(x_1 \wedge \dots \wedge x_p), x_{p+1} \wedge \dots \wedge x_n \rangle \end{aligned} \quad (1)$$

Now we know from the corrected version of problem 8 in chapter 6, § 15 above that

$$D_e(x_1 \wedge \dots \wedge x_p) = \sum_{\nu_1 < \dots < \nu_p} (-1)^{\sum_{i=1}^p i + \nu_i} \det A_{1 \dots p}^{\nu_1 \dots \nu_p} \cdot e^{\nu_{p+1} \dots \nu_n} \quad (2)$$

and we also know from a remark in chapter 5, § 7 above that

$$\langle e^{v_{p+1} \cdots v_n}, x_{p+1} \wedge \cdots \wedge x_n \rangle = \det A_{p+1 \cdots n}^{v_{p+1} \cdots v_n} \quad (3)$$

The result follows from (1), (2), and (3). $\square$

Exercise (10). If $n \geq 2$, the rank of the adjoint transformation is given by

$$r(\text{ad } \varphi) = \begin{cases} n & \text{if } r(\varphi) = n \\ 1 & \text{if } r(\varphi) = n - 1 \\ 0 & \text{if } r(\varphi) \leq n - 2 \end{cases}$$

Proof. By the Jacobi identity (7.6),

$$\wedge^p(\text{ad } \varphi) = (\det \varphi)^{p-1} D_L(\wedge^{n-p} \varphi)$$

The result now follows from the fact that the rank of a transformation ψ is the largest r for which $\wedge^r \psi \neq 0$. $\square$

Chapter 8

In this chapter, E is an n -dimensional vector space over Γ .

§ 3

Remark. Recall that a skew transformation $\varphi : E \rightarrow E$ naturally corresponds to a skew bilinear form $\Phi : E \times E \rightarrow \Gamma$ given by

$$\Phi(x, y) = (x, \varphi y) \quad (1)$$

If we “curry” Φ , we obtain the corresponding linear map $\hat{\Phi} : E \rightarrow E^*$ given by

$$\langle \hat{\Phi}x, y \rangle = \Phi(x, y) \quad (2)$$

The graph of φ , when φ is viewed as a form in this way, is naturally represented as a subspace of $E^* \oplus E$:

$$\text{Graph}(\varphi) = \{ \hat{\Phi}x \oplus x \mid x \in E \} \quad (3)$$

Now by (1) and (2), skew-symmetry of φ , and the definition σ ,

$$\langle \hat{\Phi}x, y \rangle = (x, \varphi y) = (-\varphi x, y) = \langle -\sigma\varphi x, y \rangle \quad (4)$$

By nondegeneracy of the scalar product, (4) implies

$$\hat{\Phi} = -\sigma\varphi \quad \text{and} \quad \varphi = -\sigma^{-1}\hat{\Phi} \quad (5)$$

By the definition of τ , and (5),

$$\begin{aligned} \tau(\hat{\Phi}x \oplus x) &= (\sigma x + \hat{\Phi}x) \oplus (x - \sigma^{-1}\hat{\Phi}x) \\ &= (\sigma x - \sigma\varphi x) \oplus (x + \varphi x) \\ &= \sigma(\iota - \varphi)x \oplus (\iota + \varphi)x \end{aligned} \quad (6)$$

Since φ is skew, $\iota + \varphi$ is invertible, and writing $v = (\iota - \varphi)x$ and $u = (\iota + \varphi)x$, we have

$$v = (\iota - \varphi)(\iota + \varphi)^{-1}u = \text{Cay}(\varphi)u \quad (7)$$

where $\text{Cay}(\varphi) : E \rightarrow E$ is the Cayley transform of φ . If we define

$$\text{Graph}(\text{Cay}(\varphi)) = \{ \sigma(\text{Cay}(\varphi)u) \oplus u \mid u \in E \} \quad (8)$$

then it follows from (3) and (6)–(8) that

$$\boxed{\tau(\text{Graph}(\varphi)) = \text{Graph}(\text{Cay}(\varphi))} \quad (9)$$

So τ is an intrinsic, coordinate-free, geometric version of the Cayley transform operating on graphs in the space $E^* \oplus E$.

Remark. In addition to the natural inner product in $E^* \oplus E$, there is another inner product which is induced by the inner product in E :¹⁰⁶

$$\langle\langle x^* \oplus x, y^* \oplus y \rangle\rangle = (x, y) - (x^*, y^*)$$

We have

$$\begin{aligned} \langle\langle \tau(x^* \oplus x), \tau(y^* \oplus y) \rangle\rangle &= \langle\langle \tau^2(x^* \oplus x), y^* \oplus y \rangle\rangle \\ &= 2\langle\langle \sigma x \oplus -\sigma^{-1}x^*, y^* \oplus y \rangle\rangle \\ &= 2[\langle \sigma x, y \rangle - \langle y^*, \sigma^{-1}x^* \rangle] \\ &= 2[(x, y) - (x^*, y^*)] \\ &= 2\langle\langle x^* \oplus x, y^* \oplus y \rangle\rangle \end{aligned}$$

Therefore τ is (up to the constant factor) an *isometry* between the natural and induced inner products in $E^* \oplus E$.

If $\varphi : E \rightarrow E$ is skew, then $\text{Graph}(\varphi)$ is totally isotropic under the natural inner product since

$$\langle\langle \hat{\Phi}x \oplus x, \hat{\Phi}x \oplus x \rangle\rangle = 2\langle \hat{\Phi}x, x \rangle = 2(\varphi x, x) = 0$$

Since $\text{Cay}(\varphi)$ is orthogonal, $\text{Graph}(\text{Cay}(\varphi))$ is also totally isotropic under the induced inner product since

$$\langle\langle \sigma(\text{Cay}(\varphi)u) \oplus u, \sigma(\text{Cay}(\varphi)u) \oplus u \rangle\rangle = (u, u) - (\text{Cay}(\varphi)u, \text{Cay}(\varphi)u) = 0$$

Remark. For a skew transformation $\varphi : E \rightarrow E$, the map $\tau_{\wedge}$ provides a “bridge” between the tensor representations of φ in $E^* \otimes E$ and $\wedge^2 E$.

In the proof of Theorem 8.3.1, this bridge is crossed—from (8.9) to (8.13)—to obtain more detailed information about the characteristic coefficients of a skew transformation.

¹⁰⁶See subsection 1.25 for the definition of *inner product*.

§ 4

Remark. In the proof of Proposition 8.4.1, we see that the following diagram commutes for an isometry τ :

$$\begin{array}{ccc} \text{Sk}(E) & \xrightarrow[\cong]{\Psi_E} & \wedge^2 E \\ \tau \circ (-) \circ \tau^{-1} \downarrow \cong & & \downarrow \cong \tau_\wedge \\ \text{Sk}(E) & \xrightarrow[\Psi_E]{\cong} & \wedge^2 E \end{array}$$

This shows that $\tau_\wedge$ performs conjugation by τ for tensor representations of skew transformations.

Question. For skew transformations, what is the relationship between mixed characteristic coefficients and mixed Pfaffians?

§ 5

Remark. We see that the following diagrams commute:

$$\begin{array}{ccc} \text{Sk}(E) & \xrightarrow[\cong]{\Psi_E} & \wedge^2 E \\ i_E \circ (-) \circ \pi_E \downarrow & & \downarrow (i_E)_\wedge \\ \text{Sk}(E \oplus F) & \xrightarrow[\Psi_{E \oplus F}]{\cong} & \wedge^2(E \oplus F) \end{array} \quad \begin{array}{ccc} \text{Sk}(F) & \xrightarrow[\cong]{\Psi_F} & \wedge^2 F \\ i_F \circ (-) \circ \pi_F \downarrow & & \downarrow (i_F)_\wedge \\ \text{Sk}(E \oplus F) & \xrightarrow[\Psi_{E \oplus F}]{\cong} & \wedge^2(E \oplus F) \end{array}$$

These show that $(i_E)_\wedge$ and $(i_F)_\wedge$ perform injection into direct sums for tensor representations of skew transformations.

In (8.14), we then see that *the tensor representation of a (direct) sum of skew transformations is the sum of the tensor representations of the transformations:*

$$\begin{array}{ccc} \text{Sk}(E) \times \text{Sk}(F) & \xrightarrow{\Psi_E \times \Psi_F} & \wedge^2 E \times \wedge^2 F \xrightarrow{(i_E)_\wedge \times (i_F)_\wedge} \wedge^2(E \oplus F) \times \wedge^2(E \oplus F) \\ \oplus \downarrow & & \downarrow + \\ \text{Sk}(E \oplus F) & \xrightarrow[\Psi_{E \oplus F}]{\cong} & \wedge^2(E \oplus F) \end{array}$$

§ 7

Remark. If E is complex and $\varphi : E_{\mathbb{R}} \rightarrow E_{\mathbb{R}}$ is $\mathbb{R}$ -linear, then the anticommutator $\{i_{\mathbb{R}}, \varphi\} : E \rightarrow E$ defined by

$$\{i_{\mathbb{R}}, \varphi\} = i_{\mathbb{R}} \circ \varphi + \varphi \circ i_{\mathbb{R}}$$

is $\mathbb{C}$ -linear since it is $\mathbb{R}$ -linear and

$$\{i_{\mathbb{R}}, \varphi\} \circ i_{\mathbb{R}} = i_{\mathbb{R}} \circ \{i_{\mathbb{R}}, \varphi\}$$

The commutator $[i_{\mathbb{R}}, \varphi] : E \rightarrow E$ defined by

$$[i_{\mathbb{R}}, \varphi] = i_{\mathbb{R}} \circ \varphi - \varphi \circ i_{\mathbb{R}}$$

is conjugate $\mathbb{C}$ -linear. Note

$$\varphi = \frac{1}{2i} (\{i_{\mathbb{R}}, \varphi\} + [i_{\mathbb{R}}, \varphi])$$

If $\varphi : E \rightarrow E$ is $\mathbb{C}$ -linear, then $\{i_{\mathbb{R}}, \varphi_{\mathbb{R}}\} = 2i\varphi$ and $[i_{\mathbb{R}}, \varphi_{\mathbb{R}}] = 0$.

Remark. If E, E^* are complex dual spaces with scalar product $\langle -, - \rangle : E^* \times E \rightarrow \mathbb{C}$, then $E_{\mathbb{R}}, (E^*)_{\mathbb{R}}$ are dual with scalar product $\langle -, - \rangle_{\mathbb{R}} : (E^*)_{\mathbb{R}} \times E_{\mathbb{R}} \rightarrow \mathbb{R}$ given by

$$\langle x^*, x \rangle_{\mathbb{R}} = \operatorname{Re} \langle x^*, x \rangle$$

Indeed, clearly $\langle -, - \rangle_{\mathbb{R}}$ is well-defined and $\mathbb{R}$ -bilinear. If $\langle x^*, - \rangle_{\mathbb{R}} = 0$, then for any $x \in E_{\mathbb{R}}$,

$$\operatorname{Re} \langle x^*, x \rangle = \langle x^*, x \rangle_{\mathbb{R}} = 0$$

and also

$$\operatorname{Im} \langle x^*, x \rangle = -\langle x^*, i_{\mathbb{R}} x \rangle_{\mathbb{R}} = 0$$

so $\langle x^*, x \rangle = 0$. Hence $\langle x^*, - \rangle = 0$ and $x^* = 0$. By symmetry, if $\langle -, x \rangle_{\mathbb{R}} = 0$ then $x = 0$. Therefore $\langle -, - \rangle_{\mathbb{R}}$ is nondegenerate. We write

$$E_{\mathbb{R}}^* = (E_{\mathbb{R}})^* = (E^*)_{\mathbb{R}}$$

Remark. If E, E^* and F, F^* are pairs of complex dual spaces and $\varphi : E \rightarrow F$ and $\varphi^* : E^* \leftarrow F^*$ are complex dual maps, then $\varphi_{\mathbb{R}} : E_{\mathbb{R}} \rightarrow F_{\mathbb{R}}$ and $(\varphi^*)_{\mathbb{R}} : E_{\mathbb{R}}^* \leftarrow F_{\mathbb{R}}^*$ are dual since

$$\langle (\varphi^*)_{\mathbb{R}} y^*, x \rangle_{\mathbb{R}} = \operatorname{Re} \langle \varphi^* y^*, x \rangle = \operatorname{Re} \langle y^*, \varphi x \rangle = \langle y^*, \varphi_{\mathbb{R}} x \rangle_{\mathbb{R}}$$

We write

$$\varphi_{\mathbb{R}}^* = (\varphi_{\mathbb{R}})^* = (\varphi^*)_{\mathbb{R}}$$

Remark. If E is complex with Hermitian inner product $(-, -)_H : E \times E \rightarrow \mathbb{C}$ (that is, a unitary space), and $E_{\mathbb{R}}$ has the inner product $(-, -) : E_{\mathbb{R}} \times E_{\mathbb{R}} \rightarrow \mathbb{R}$ given by

$$(x, y) = \operatorname{Re}(x, y)_H$$

then if $\varphi : E \rightarrow E$ is skew-Hermitian, also $\varphi_{\mathbb{R}} : E_{\mathbb{R}} \rightarrow E_{\mathbb{R}}$ is skew-symmetric since

$$\widetilde{\varphi_{\mathbb{R}}} = \widetilde{\varphi}_{\mathbb{R}} = (-\varphi)_{\mathbb{R}} = -(\varphi_{\mathbb{R}})$$

Therefore it makes sense to talk about $\Psi_{E_{\mathbb{R}}}(\varphi_{\mathbb{R}}) \in \wedge^2 E_{\mathbb{R}}$ —which in the book is confusingly denoted $\Psi_E(\varphi)$.

Remark. If E is again a unitary space, recall that there is a *conjugate-linear* Riesz isomorphism $\sigma_H : E \rightarrow E^*$ with

$$\langle \sigma_H x, y \rangle = \overline{(x, y)_H}$$

For $x, y \in E_{\mathbb{R}}$,

$$\overline{(x, y)_H} = \operatorname{Re}(x, y)_H - i \operatorname{Im}(x, y)_H = (x, y) - i \operatorname{Im}(x, y)_H$$

and

$$(x, i y)_H = -i(x, y)_H = \operatorname{Im}(x, y)_H - i \operatorname{Re}(x, y)_H$$

so that

$$(x, i_{\mathbb{R}} y) = \operatorname{Re}(x, i y)_H = \operatorname{Im}(x, y)_H$$

and hence

$$\langle \sigma_H x, y \rangle = (x, y) - i(x, i_{\mathbb{R}} y)$$

This shows that the map $\sigma : E_{\mathbb{R}} \rightarrow E^*$ in the book is given by $\sigma = \iota_{\mathbb{R}, \mathbb{C}} \circ (\sigma_H)_{\mathbb{R}}$, where $(\sigma_H)_{\mathbb{R}} : E_{\mathbb{R}} \rightarrow E_{\mathbb{R}}^*$ is the underlying real map of the Riesz isomorphism and $\iota_{\mathbb{R}, \mathbb{C}} : E_{\mathbb{R}}^* \rightarrow E^*$ is the identity map. Note

$$\langle (\sigma_H)_{\mathbb{R}} x, y \rangle_{\mathbb{R}} = (x, y)$$

so $(\sigma_H)_{\mathbb{R}}$ is also the Riesz isomorphism for the underlying real spaces. Note also that (8.16) follows from conjugate-linearity of σ_H .

The map σ is used in order to avoid the conjugate-linearity of σ_H , which does not play as well with the multilinearity of tensor and exterior algebra.

Remark. If E is again a unitary space and $\varphi : E \rightarrow E$ is skew-Hermitian, then for $z = \Psi_{E_{\mathbb{R}}}(\varphi_{\mathbb{R}})$ and $J = \Psi_{E_{\mathbb{R}}}(i_{\mathbb{R}})$ and $0 \leq r \leq n$,

$$\begin{aligned}
(z^r, z^r) &= C_{2r}(\varphi_{\mathbb{R}}) && \text{by Theorem 8.3.1} \\
&= \sum_{p+q=2r} C_p(\varphi) \overline{C_q(\varphi)} && \text{by Proposition 7.9.1} \\
&= \sum_{p+q=2r} i^p (z^p, J^p) \overline{i^q (z^q, J^q)} && \text{by Theorem 8.7.1} \\
&= (-1)^r \sum_{p+q=2r} (z^p, J^p) (z^q, (-J)^q)
\end{aligned}$$

In particular for $r = n$,

$$(z^n, z^n) = (z^n, J^n)^2$$

Since $\det(\varphi_{\mathbb{R}}) = (z^n, z^n)$, and $\text{Pf}(\varphi_{\mathbb{R}}) = (z^n, J^n)$ by the corollary to Theorem 8.7.1, this is consistent with

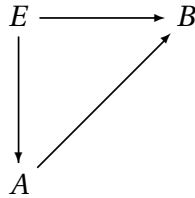
$$\det(\varphi_{\mathbb{R}}) = \text{Pf}(\varphi_{\mathbb{R}})^2 = (-1)^n \det(\varphi)^2$$

Chapter 9

Remarks analogous to many of those made above for exterior algebra can also be made for symmetric algebra. We highlight some differences.

§ 3

Remark. The symmetric algebra $\vee E$ is a universal (initial) object in the category of “associative unital algebras with linear maps of E into them whose image elements commute”. In this category, the objects are linear maps $\varphi : E \rightarrow A$ for associative unital algebras A with $\varphi x \cdot \varphi y = \varphi y \cdot \varphi x$, and the arrows are unital algebra homomorphisms $A \rightarrow B$ which preserve the units and respect the linear maps:



Every object in this category can be obtained from the symmetric algebra $\vee E$ in a unique way. This is why $\vee E$ is said to satisfy the “universal property”. This is only possible because the elements of $\vee E$ satisfy only those properties that are required to make $\vee E$ into an associative, unital, and *commutative* algebra containing E . By abstract nonsense, $\vee E$ is unique up to isomorphism.

§ 6

Remark. The multiplication map $\mu : \vee E \rightarrow L(\vee E; \vee E)$ is linear, and is an algebra homomorphism by (9.9). Dually, the insertion map $i : \vee E \rightarrow L(\vee E^*; \vee E^*)$ is also an algebra homomorphism by *commutativity*!

Remark. For $h, k \in E$, by the binomial formula for composition of commuting transformations,

$$i(h+k)^p = [i(h) + i(k)]^p = \sum_{r+s=p} \frac{p!}{r!s!} i(h)^r \circ i(k)^s$$

This can also be proved using the binomial formula (9.3) and the dual of (9.9).

§ 7

Remark. Informally, the lemma to Proposition 9.7.1 says that if an element in the symmetric algebra is always zero when “measured” by repeatedly inserting a single vector, then the element is zero. Since insertion of a single vector is a *derivation*, this is a powerful criterion.

In the proof by induction, given $p > 1$ and $u^* \in I_p$ satisfying $i(h)^p u^* = 0$ for all $h \in E$, it is natural to fix $k \in E$ and consider $i(k)u^* \in I_{p-1}$. In order to leverage the induction hypothesis, we need $i(h)^{p-1} i(k)u^* = 0$ for all $h \in E$. Since $i(h)$ and $i(k)$ commute, we know we will find terms of the form $i(h)^r i(k)^s u^*$ in the expansion of $i(h+k)^p u^*$.¹⁰⁷ To ensure that all of them are zero, we introduce a scalar parameter λ and consider the polynomial function

$$f(\lambda) = i(\lambda h + k)^p u^* = \sum_{r+s=p} \frac{p!}{r!s!} \lambda^r i(h)^r i(k)^s u^*$$

with coefficients in the symmetric algebra. By assumption $f(\lambda) = 0$ for all λ , so (since there are infinitely many λ) the coefficients of f must be zero, which implies $i(h)^r i(k)^s u^* = 0$ for all r, s with $r+s = p$, in particular $r = p-1$ and $s = 1$. By induction $i(k)u^* = 0$, and since k was arbitrary it follows that $u^* = 0$.

§ 10

Remark. It follows from the lemma to Proposition 9.7.1 that $\bigvee^p E$ has a basis of elements of the form h^p for $h \in E$. Indeed if this were not the case, then the orthogonal complement in $\bigvee^p E^*$ of the span of such elements would contain an element $u^* \neq 0$ such that

$$i(h)^p u^* = p! i(h^p) u^* = p! \langle u^*, h^p \rangle = 0$$

for all $h \in E$, contradicting the lemma.

§ 12

Remark. To see that $H(E) = \sum_p H^p(E)$ is a *direct* sum, suppose that $0 = \sum_{p=0}^r h_p$ for $h_p \in H^p(E)$. Substituting λx on both sides we obtain

$$0 = \sum_{p=0}^r \lambda^p h_p(x)$$

¹⁰⁷ See the remark in § 6 above.

Viewing this as a polynomial function in the scalar λ with coefficients $h_p(x)$, it follows (since there are infinitely many λ) that $h_p(x) = 0$ for all $0 \leq p \leq r$, and since x was arbitrary $h_p = 0$ for all $0 \leq p \leq r$. Therefore the zero function is uniquely represented in $H(E)$, and $H(E)$ is a direct sum.

Remark. $H(E)$ is a *graded algebra* where the elements of $H^p(E)$ are (in the sense of gradation) homogeneous of degree p , and $\tau : S^*(E) \rightarrow H(E)$ is homogeneous of degree zero.

Remark. The function h_Φ should be redefined on $S^p(E)$ as

$$h_\Phi(x) = \frac{1}{p!} \Phi(x, \dots, x)$$

§ 13

Remark. Γ_n is a *graded algebra* where the elements of Γ_n^p are (in the sense of gradation) homogeneous of degree p , and the (non-natural) isomorphisms to the symmetric algebra of an n -dimensional vector space are homogeneous of degree zero.

§ 15

Remark. $S^*(E)$ is a *graded algebra* where the elements of $S^p(E)$ are (in the sense of gradation) homogeneous of degree p , and dually for $S_*(E)$ and $S_p(E)$.

§ 16

Remark. In this subsection, the substitution operator $i_S(h)$ should be redefined on $T^p(E)$ as

$$i_S(h) = \frac{1}{p} \sum_{v=1}^p i_v(h)$$

Then for $\Phi \in S^p(E)$,

$$i_S(h)\Phi = i_1(h)\Phi$$

as desired. This redefinition is required to make $i_S(h)$ a derivation in $S^*(E)$ (for the definition of symmetric product in subsection 9.15), although under this redefinition $i_S(h)$ is not a derivation in $T^*(E)$.

If h_Φ is also redefined as in the remark in § 12 above, then

$$h_\Phi(x) = i_S(x)^p \Phi$$

as desired.

§ 18

Remark. The (non-natural) isomorphisms $\Gamma_n \cong S^*(E)$ where $n = \dim E$ do *not* show that all polynomial functions are symmetric! Under these isomorphisms, polynomial variables correspond to dual basis functions on E .

References

- [1] Greub, W. *Linear Algebra*, 4th ed. Springer, 1975.
- [2] Greub, W. *Multilinear Algebra*, 2nd ed. Springer, 1978.
- [3] Greub, W. *Multilinear Algebra*, 1st ed. Springer, 1967.
- [4] Greub, W. and Vanstone, J. "A Basic Identity in Mixed Exterior Algebra." *Linear and Multilinear Algebra*. Gordon and Breach, 1987.